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REVIEWER

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CELL NERVE MUSCLE PHYSIO

Cell Physiology:

- i. Cell:
 - a. Centriole – direct spindle fibers for mitosis, important for the reproduction of cell.
 - b. Plasma membrane, divides outer cell (ECF) and inner cell(ICF)
 - i. Semi-permeable
 - ii. Fluid, shift in position and shape, Not Solid
 - iii. Charge: -25mV
 - iv. Phospholipid bilayer
 - o Center fat soluble (hydrophobic)/ more permeable
 - CO₂, O₂, Alcohol
 - o Outer water soluble/ less permeable
 - v. Proteins
 - o Integral protein – serves as a water channel
 - o Peripheral protein
 - c. Endocytosis – bulk uptake of materials when it comes in contact with the cell membrane
 - vi. Pinocytosis
 - vii. Phagocytosis
 - viii. Receptor mediated – coated pits
 - o Uptakes Cholesterol and growth factor
 - d. Cytosol – clear fluid where organelles are suspended. Crystalloid and colloids
 - e. Endoplasmic reticulum – bridge, connects the nucleus to cytoplasm
 - 2 types:
 - i. Rough ER – Protein synthesis
 - ii. Smooth ER – Lipid synthesis “Smooth madulas, Lipids/ oils”
 - a. Other functions:
 - i. Enzymes from glycogen breakdown
 - ii. Enzymes for detoxification
 - f. Golgi Apparatus “Highway of the cell”
 - a. Produces lysosomes
 - b. Produces carbohydrates from SER
 - g. Lysosomes vs Peroxisomes [Digestive system of the cell]

Lysosomes	Peroxisomes
Produced by Golgi Apparatus	Self Replicating
Contains hydrolase	Contains Oxidase
Uptakes bacteria and damaged cell parts	Uptakes Alcohol
 - h. Mitochondria (Power house of the cell)
 - a. Responsible in producing ATP

- i. Nucleus (Control center)
 - a. Contains DNA or genes
 - b. *Chromatin* dark staining in nucleus that have not undergone duplication > *Chromosome* (undergoing duplication)
 - c. Nucleoli – contains the RNA (Uses mRNA to duplicate/ provide the “recipe” from nucleus to cytoplasm, given to tRNA to transcribe the recipe to rRNA to RER to create protein)
- ii. Cell Division
 - a. Mitosis
 - i. Interphase: Before Mitosis or meiosis
 - 1. Preparation for cell division
 - 2. Duplication of DNA
 - ii. Prophase – chromosomes pair, lose nuclear envelope
 - iii. Metaphase – Aligning in the middle (equatorial plate)
 - iv. Anaphase – Chromosomes are separated; moves towards opposite poles
 - v. Telophase – 2 nuclei, cells cleaved into 2

MITOSIS

P – Preparation of nucleus

M – Middle chromosome

A – Apart

T – Talo

- b. Meiosis
 - i. Interphase
 - ii. P1M1A1T1 -> P2M2A2T2

	Mitosis	Meiosis
Time of DNA replication	Interphase	Interphase
Product	Diploid	Haploid F:(23) 1 egg cell, 3 polar bodies M:(23) 4 sperm cells
Purpose	Growth and repair	Reproduction

- iii. Ions:

	ICF	ECF
Cations	Potassium	Sodium
Anions	Organic	Chlorine

- iv. Carbohydrates

- a. Function: provide cellular nutrition
- b. Glucose: readily available; *Simplest form*
- c. Glycogen: storage form of glucose in the cell

Monosaccharides (frugalglu)		Disaccharides (SLaM)
Fructose	+ Glucose	=Sucrose
Galactose	+ Glucose	=Lactose
Glucose	+ Glucose	=Maltose

- v. Transport mechanisms:
 - a. Passive transport/ Diffusion
 - i. Does not need energy
 - ii. *Higher concentration to lower concentration*
 - iii. O₂, CO₂, Alcohol
 - iv. Water (Least common pathway for ions)
 - v. 3 types:
 - 1. Passive/ Simple diffusion (Without carrier protein)
 - 2. Facilitated diffusion
 - a. With carrier protein, (+) binding site
 - b. E.g. Glucose and Amino Acids
 - 3. Osmosis – Solvent moves region of higher concentration to lower concentration
 - a. Maintains the volume of the cell to prevent shrinking or bursting
 - b. Active transport
 - i. Requires the use of energy (ATP)
 - ii. Uphill transport, Lower to higher concentration
 - iii. 2 types:
 - 1. Primary
 - a. Uses direct ATP
 - b. Na-K pump (Na: 3 K: 2)
 - c. Ca pump
 - 2. Secondary
 - a. Uses indirect ATP (Uses ionic concentration from primary transport)
 - i. Co transport: Glucose and amino acid and Cl
 - ii. Counter transport : H⁺ and Ca⁺

Symport - 2 substances go in the same direction. E.g. Na + glucose transport in small intestine

Antiport - 2 substances go to the opposite direction. E.g. Na + Ca in heart muscles

Uniport - only one substance is transport

Nerve Physiology:

- vi. Nerves:
 - a. Action potential
 - b. RMP of muscle: $-90mV$
 - c. RMP of nerve: $-70mV$
 - d. Nerve cells at rest are positively charged outside and negatively charged inside (RMP)
 - e. Period of Action potentation prior to application to stimulus (RMP)
 - f. Upon application of stimulus: Depolarization
 - g. Hyperpolarization: Less than the RMP

- vii. Propagation of action potential
 - a. Direction of propagation
 - b. Na influx: Depolarization
 - c. K efflux: Repolarization
 - d. All or nothing principle: Maximum intensity above the threshold or resting state
 - e. Absolute refractory period:
 - i. Cell is no longer stimulated
 - ii. Peak of action potential
 - f. Relative Refractory period
 - i. 2/3 of repolarization
 - ii. Supramaximal intensity to stimulate the cell membrane
- viii. Sensory receptors
 - a. According to source of stimulus
 - i. Exteroreceptor
 - ii. Proprioceptor
 - iii. Enteroceptors/ visceroreceptors
 - b. According to modality
 - i. Nociceptors – pain A delta & C fiber
 - ii. Thermoreceptor – Ruffini & Krause
 - iii. Mechanoreceptor – Merkel, Meissner, Pacinian

“Never mind the Merkel light. Discrimination / light touch
Meissner (2) 2 point(s)
Pacini (vibration) – vibration”

Muscle Physiology

- ix. Muscle:
 - a. Sarcomere
 - i. Functional unit of the MS/ basic contractile units
 - ii. Myofilaments > Myofibrils > Muscle fiber > Fasciculus > Fasciculi > Muscle belly
 - iii. Myofilament: Actin & Myosin
 - iv. Sarcomere song

Danadanana!

I band: Actin only

A band: Myosin and Actin!

H band M line: myosin only

Z disc Z disc attachment of actin

Nag contract nag contract nawala!

	Skeletal	Cardiac	Smooth
Location	Bones	Heart	Viscera
Striation	(+)	(+)	(-)
Nucleation	Multi	Mono	Mono
Intercalated disc	(-)	(+)	(+)
Control	Voluntary	Involuntary	Involuntary

- x. Events of muscle contraction
- Action potential in nerve
 - Release of calcium from synaptic vesicle
 - Release of Ach at motor end plate
 - Binding of Ach to ligand gated sodium channel
 - Na⁺ influx of the muscle cell
 - Action potential in muscle
 - Calcium is released from the sarcoplasmic reticulum
 - Calcium attaches to Troponin C, to unravel troponin-tropomyosin complex and reveals actin binding sites

Troponin I – Binding site for actin

Troponin C – Calcium attachment

Troponin P – Myosin

- Formation of crossbridge between actin & myosin sliding of thin on thick filaments, producing shortening (power stroke).
- Ca is pumped back into sarcoplasmic reticulum

Fast twitch	Slow twitch
Large diameter	Small diameter
Glycolytic	Oxidative
Low mitochondria	More mitochondria
More sarcoplasmic reticulum	Endurance and posture
100m Dash	

- xi. Other concepts:
- Summation (Addition of muscle twitches)
 - Increase the force generated in a contraction
 - Multiple fiber summation (Size principle: Small muscles before larger muscle groups)
 - Spatial: ↑ the # of motor units being stimulated
 - Temporal: ↑ Frequency of the stimulation
 - Tetanic: 100x/ sec; 4x Greater than individual muscle twitch (Continuous)
 - Incomplete Tetanic: 10-30x/sec, with rest period
 - Trepe/ Staircase contraction: ↑ Frequency until plateau is reached

ENDOCRINE PHYSIO

- i. Endocrine system
 - a. 2nd great controlling system in the body
 - b. *Ductless glands*, and produce hormones in the blood or lymph system.
 - c. Hormones: To arouse, mediator molecule that will activate the activity of the cell in the body.
- ii. Control of release
 - a. Hormonal stimulus: Endocrine glands are stimulated by another hormone
 - b. Humoral stimulus: Blood borne ions -> chemicals
 - c. Neural hormones: Nerve fibers are stimulated by hormones (E.g. Epinephrine and Norepinephrine)
- iii. Hypothalamus – Major part of limbic system
 - a. Responsible for neuronal circuitry for emotional and motivational drive
 - b. Function:
 - i. Temperature regulation
 - ii. Osmality of fluids
 - iii. Control body weight
 - iv. Drive to eat and drink
 - v. Role of emotions
- iv. Pituitary Gland/ Hypophysis
 - a. Small gland, 1cm in diameter
 - b. 0.5 – 1g
 - c. Lies in the *Sella Turcica* (Bony cavitation at the base of the brain and it is connected to hypothalamus by Pituitary stalk.)
 - d. 2 lobes:
 - i. Anterior pituitary lobe (Adenohypophysis)
 1. The origin of this gland, from the pharyngeal epithelium that's why it will always explain the epithelial nature of its cells.
 2. Secretion is controlled by hormone *Hypothalamic releasing & inhibitory hormone*. (from the hypothalamus) *MOST important hypothalamic releasing & inhibitory hormone
 - a. TRH - Thyrotropin Releasing hormone, release of TSH
 - b. CRH – Corticotropin Releasing hormone, release of Adenocorticotropin hormone (ACTH)
 - c. GHRH – Growth hormone releasing hormone – Release of Growth hormone
 - d. GHIH - Growth hormone inhibiting hormone - inhibit release of Growth hormone
 - e. GnRH - Gonadotropin releasing hormone – release LH and FSH
 - f. PIH- prolactin inhibiting hormone – inhibit Prolactin

3. Anterior pituitary gland hormones
 - a. Growth Hormone –aka *Somatotropin*, responsible for growth of all soft tissues in the body that are capable of growing. Function to similar to amino acid, decrease blood glucose. Too much = Gigantism,
 - i. Increase at Adult, Acromegaly (Hand 2x size, Feet, from size 7 to size 14, nose, bosses of the forehead, lower jaw protruded, thickening of vertebra causing kyphosis)
 - ii. Dwarfism, same height as 4-5 years old
 - b. Prolactin – Production of milk
 - i. Tenderness in the breast: Peak of prolactin
 - ii. Excessive prolactin in males Erectile Dysfunction/ Impotence
 - iii. Excessive prolactin in females Amenorrhea.
 - c. FSH : From GnRH (Stimulate Gamete formation)
 - i. Female: Follicular cells > estrogen (Secondary sex characteristics)
Female sex drive (Libido)
 - ii. Male: production of sperm cells.
 - d. LH – from GnRH
 - i. Female: Triggers ovulation, stimulates sex hormone production, and production of progesterone.
 - ii. Male: Production of testosterone.
 - e. MSH – Melanocyte stimulating hormone
 - i. Increases skin pigmentation when present in excess
 - f. TSH – from TRH: Stimulate release of thyroid hormone in thyroid gland.
 - g. ACTH: Stimulate the release of aldosterone and cortisol.
- ii. Posterior pituitary lobe/ Neurohypophysis
 1. Originates from the neural tissues → explains the large number of glial cells in the area.
 - a. Glial cells, *Pituicytes*, does not release hormones
 - i. Support for terminal nerve fiber and nerve endings.
Which glial cells are seen in the posterior pituitary?
 - ii. Production of hormone for post. Pit. Gland comes from hypothalamus:
 1. Supraoptic nuclei
 2. Paraventricular nuclei
 - b. Hormone: Oxytocin (Exitocin)
 - i. Milk ejection via sucking reflex
 - ii. Contractions of uterus
 - c. ADH/ Vasopressin: Reabsorption of water

- v. Thyroid gland
 - a. Inferior to larynx
 - i. Thyroxine T4 –Tetraiodothyronine: 93% & Triiodothyronine T3 7% (Form of iodine)
 - 1. Maturation/ development and growth of CNS, regulate oxygen use, BMR, and cellular metabolism.
Insufficient amount at child results to Mental Retardation
Insufficient amount causes lethargy, Graves, myxedema
 - ii. Calcitonin
 - 1. Decrease blood Ca^{++} level
 - 2. Decrease osteoclastic activity/ bone matrix
- vi. Parathyroid gland
 - a. ParaTHORMone
 - 1. Posterior surface of the lateral lobe of the thyroid gland
 - 2. Increase blood Ca^{++} level
 - 3. Increase osteoclastic activity
- vii. Adrenal Gland
 - a. Superior to each kidneys
 - i. Adrenal Cortex
 - 1. 3 zones:
 - a. Zona Glomerulosa – Aldosterone, water and electrolytes/ Na
 - b. Zona Fasciculata – Cortisol, metabolism and resistance to stress
 - c. Zona Reticularis – Androgen, secondary sex characteristics of males
 - i. Stimulate growth of axillary and pubic hair
 - ii. If more in females they develop male characteristics, clitoris develops similar to penis. (Androgenital syndrome)
 - ii. Adrenal Medulla
 - 1. Epinephrine/ Norepinephrine: Fight/ flight response
- viii. Pineal Gland
 - a. Attached to the third roof ventricle of the brain
 - b. Hormone: Melatonin
 - i. Body biological clock
 - ii. Sleepiness
 - iii. Increased = sleepiness more during the dark
 - iv. Decreased = more awake
- ix. Thymus Gland
 - a. Behind the sternum between the lungs
 - i. Thymosin
 - 1. Maturation of the T-Cells
 - 2. Retard aging

- x. Pancreas
 - a. BOTH exocrine and endocrine system
 - GABIDS
 - i. Alpha cells – Glucagon, Increased blood glucose levels
 - ii. Beta cells – Insulin, Decreases blood glucose levels
 - iii. Delta cells – Somatostatin, balance/ controls the number of insulin and glucagon
 - Type I DM, IDDM. Ketone prone. No insuli. Skinny because if there is no insulin, blood glucose is everywhere. No source of energy. Fat will be used instead/ triglycerides (natural fat form inside the body). If triglycerides are digested ketones will be released in the blood will result to atherosclerosis.*
 - Type II NM, NIDDM. Deficient or kulang sa insulin. Adjusted with diet and exercises.*
 - Ketone – resistant.*
- xi. Ovaries and Testes
 - a. Gonads: produce gametes (female: oocytes, male: sperm)
 - i. Estrogen – Female sex characteristics/ sexual drive of females
 - ii. Progesterone
 - iii. Testosterone – maturation of male sex organ
 - iv. Relaxin – increase flexibility of the pubic symphysis (During labor)

BLOOD PHYSIOLOGY

i. Blood functions: BLOOD PH

Balance of acid/ base	Delivers
Levels Temperature	Immunity
Oxygen Transport	Nutrients
Osmosis	Oxygen transport
Delivers nutrient	Guards
Protection	Osmosis
Hormonal transport	Acid-base balance
	Nourishment

ii. Buffer system:

pH	7.34 -7.45
CO ₂ =	35 – 45mmHg
HCO ₃ =	22- 26 meq/L

Vomiting will result to? Gag- sound of vomiting, “aaaalkk-”

- a. RAC
- b. RAL
- c. MAL
- d. MAC

Diarrhea – HCO₃ (Base) is released, acid is left. ASSidosis

Hyperventilation – Throws away CO₂ (acidic), results to RAL

iii. Blood (Total circulating blood volume is about 8%)

- a. Plasma 55%
- b. Formed elements – 45% (RBC + WBC + Platelets = HCT))

i. RBC

- Male : **5.2 – 6.5m/mm³**
- Female : **4.5 – 5.5m/mm³**

ii. Hematocrit (HCT) – percentage of blood, viscosity

- Male: **42-52%**
- Female: **37-47%**
- Dehydration = Increase in HCT %, Decrease blood volume
- Bloated = Decrease in HCT % Increase blood volume

iii. Erythrocyte Sedimentation Rate (ESR)(Determines how much inflammation is in the body)

- Hallmark: Inflammation
- Male: **<15mm/hr**
- Female: **<25mm/hr**
- E.g. Arthritic conditions

iv. Hgb

- 65% of blood is iron (Fe)
- Male:**16mg/dL**
- Female: **14mg/dL**

v. WBC

- **5-10T/mm³**
- Decreased: Poor protection
- Increased: Infection

vi. Platelets

- **150T-400T/mm³**
- Decreased: Internal bleeding (Thrombocytopenia S/Sx):
 - Epistaxis
 - Petechiae
 - Echymosis

vii. Bleeding time:

- **1-6 minutes**

viii. Clotting time:

- **6-10 minutes**

ix. Red Flag Values:

	Hgb	HCT	WBC	Platelets
(n) Values	14-16mg/dL	37-52%	5T-10T	150T-400T
Red Flag	8mg/dL	25%	<5T	<50T

iv. Plasma:-

a. 55% of blood

- i. 90% H₂O
- ii. 10% Electrolytes

b. Terminology:

- i. Serum (-) clotting factor
- ii. Plasma (+) clotting factor *"Plas" clotting factor*

v. Proteins:

a. Albumin – most Abundant protein in the plasma

b. Fibrinogen – Clotting factor

c. Globulin – Immunoglobulin (Ig) G A M E D

- i. G – Crosses the placenta
- ii. A – Body fluids. E.g. sweat "pAAAwis, sAAAliva/ IAAAway, gAAAtas"
- iii. M – Food Antigen "mmmmmmm food"
- iv. E – Allergic reaction. "Allergeee" IgE attaches to Basophil> explodes> releases Histamine
Histamine> causes vasodilation of blood vessels
Histamine> causes bronchial constriction
* Antihistamine – slow acting, fast relief of signs
- v. D – activation B-cells, produced by bone marrow

Types of hypersensitivity reaction

I	Anaphylactic
II	Cytotoxic "Cytwotoxic" E.g. Incompatible blood transfusion
III	Immune complex
IV	Cell mediated delayed E.g. Contact dermatitis/ Eczema

- vi. Types of immunity NA NP AA AP
- a. na natural active: Chicken pox, naturally created antibodies against chicken pox
 - b. np natural passive: IgG mother passing IgG through placenta
 - c. aa artificial active: Vaccines (mild form of virus to create antibodies)
 - d. ap artificial passive: Tetanus Toxoid, given antibody
 - i. RHOGAM, given 72 hours. Protection of fetus. – Rh Incompatibility
Mother Rh⁻ Father Rh⁺ = Fetus Rh⁺ (only allowed the first time to be normal/ healthy birth)
Succeeding births with Rh⁻, mothers body has created antibodies for Rh⁺.
Further children, will die or live c complications.
Rhesus Dse, severe anemia, anti rh⁺ in the mother will destroy the fetus.
Erythroblastosis Fetalis – Fatal
- vii. RBC – aka Erythrocytes
- a. Life span: 120 days
 - b. Formation of RBC, Erythropoiesis
 - c. Deprivation of oxygen causes the hormone, erythropoietin to produce RBC.
 - d. From the kidneys 90%, liver 10%
 - e. Blood formation- Hematopoiesis
 - i. Infants In the middle of gestation the liver is the primary production of RBC's
 - ii. Adult – bone marrow
 - f. RBC formation
 - i. Proerythroblast
 - ii. Basophil erythroblast – First generation cell
 - iii. Reticulocyte
 - iv. Erythrocyte- anuclear (Mature RBC)
 - 1. Biconcave
 - 2. Strong membrane
 - 3. Designed to be small to fit through the smallest arteries.
 - 4. Nutrients required to created RBC's:
 - i. B 12 vitamin, vit b12 deficiency anemia
 - ii. I Iron, for Hgb iron deficiency, MC substance deficiency in anemia.
 - iii. F Folic acid, folic acid deficiency
- viii. Cell Terminology:
- a. Amount:
 - i. Increase in amount suffix **–cytosis**
 - ii. Decrease in amount suffix **–penia**
E.g. Thrombocytosis, Thrombocytopenia, Leukocytosis, leukocytopenia, polycythemia, anemia
 - b. Cell size – Cytic, e.g. microcytic (small), normocytic (normal), macro/ megaloblastic (big)
 - c. Cell color – Chromic e.g. hypochromic (decrease in color), normochromic

- ix. Oxygen related conditions:
- Anemia – Decrease in O₂ carrying capacity of RBC
 - Quantitative anemia:
 - Increase in destruction of RBC
 - Decrease in production of RBC
 - Qualitative anemia:
 - Abnormal maturation
 - Hypoxemia – Decrease of O₂ in blood
 - Hypoxia – Decrease O₂ tissue
- x. Common types of anemia: *Anemia: Decrease in oxygen carrying capacity of the blood cell.*
- IDA (Iron Deficiency Anemia)
 - Microcytic, hypochromic
 - MC in females d/t menstrual cycle
 - S/Sx
 - Headache
 - Irritability
 - Plummer – Vinson syndrome*: Form of esophageal webs. Chronic IDA. Complete depletion of iron. Difficulty with swallowing.
 - Post- Hemorrhagic Anemia
 - Normocytic, normochromic
Vicious cycle, bleeding causes decrease venous return > compensatory increase heart rate > increase blood flow > more bleeding (positive feedback)
 - Blood loss (amount in %)

%	Sx
20-30%	Decrease BP, dizziness
30- 40%	Decrease BP, Diaphoresis, Increase ADH
40-50%	Hypovolemic Shock, potential death
 - Vitamin B12 Deficiency Anemia
Vitamin B12 is best absorbed in what part of the intestine? Ileum
 - Affects CNS
 - Intrinsic factor – Protects Vitamin B12 to prevent Digestion > brush borders > ileum
 - (Pernicious Anemia: (-) intrinsic factor, Vitamin B12 is digested Macrocytic/ megaloblastic anemia, normochromic, Affects CNS)*
 - S/Sx
 - Decreased vibratory sense
 - (+) Rombergs Test
 - Folic Acid anemia*
 - Digested Macrocytic/ megaloblastic anemia, normochromic
 - MC in pregnant females
 - Aplastic anemia
 - 2° to Aplasia
 - Bone marrows replaced with fat
 - Idiopathic
 - Dx: Bone marrow Aspiration

- e. Hemolytic anemia
 - i. Increased destruction of RBC before its 120th day
- f. Thalassemia
 - i. *Problem with Hemoglobin*
 - ii. No problem with RBC production, but with reduced amount
Which disease is the structure of hemoglobin's that are produced normal but their amount reduced? Thalassemia
- g. Sickle Cell Anemia
 - i. (+) Hemoglobin S (Abnormal type of Hgb)
 - ii. Decrease O₂ -> RBC will form a sickle-shape
 - iii. Pain crisis, pain felt anywhere in the body or in any major organ of the body.
- h. Polycythemia
 - i. Primary polycythemia (idiopathic) Polycythemia Vera (Associated with acquired myeloproliferative disorders)
 Other types:
 - ii. Secondary polycythemia: Decreased O₂ > increase erythropoietin production > increase red blood cell count
- xi. WBC
 - a. *Only True Cell in the blood*
 - b. Lifespan
 - i. 4-5 hours (Circulation)
 - ii. 4-6 days (Tissue)

Granulocytes B E N – G	Count	Agranulocytes	Count
Neutrophils 4-6 hours (1-3 days)	50-70%	Lymphocytes	20-40%
Eosinophils	1-4%	Monocytes	2-8%
Basophils (Contain Histamine)	0-1%		

*Increase Neutrophils = Bacterial infection, **first line of defense***

Increase Lymphocytes = Viral infection / Chronic inflammation

Increase Eosinophils = Parasitic infection/ Allergic reactions/ Acute inflammation

**WBC's are able to leave the circulation.*

- xii. Process of phagocytosis: "Si NED may CP"
 - a. Neutrophils undergo margination
 - b. Emmigration – getting out
 - c. Diapedesis – *amoeboid like movement*, cells extend their cytoplasm to move.
 - d. Chemotaxis – attract neutrophils
 - e. Phagocytosis
- xiii. WBC related conditions:
 - a. Immunity Virus (Lymphocytes)
 - i. Cellular: T-Cell (origin Thymus gland, at the age of 8 years old, becomes smaller)
 - i. Killer – first to be attracted
 - ii. Helper – recruitment of B cells
 - iii. Suppressor – suppresses activity
 - ii. Humoral: B-Cell (origin from bone marrow)
 - i. When cell death > becomes plasma cells memory cells > becomes antibody.

- b. HIV
 - i. Opportunistic Virus (attacks when Immune system is low)
 - ii. Attacks helper T-cells
 - iii. No B cells, no memory cells, no antibodies
- c. GBS
 - i. Attacks suppressor T cells.
 - ii. Continuous activity
- d. Multiple Sclerosis
 - i. Mimics appearance of the myelin sheath
 - ii. Demyelination, the body kills the myelin sheath.
- e. Leukocytosis – increase in WBC, (+) infection
- f. Leukopenia – Felty's syndrome, Blood d/o common in rheumatic conditions
 - S – splenomegaly
 - L - leukopenia
 - A - anemia
 - A - arthritis
 - N - neutropenia
 - T – Thrombocytopenia
- g. Leukemia
 - i. Cancer of WBC
 - ii. Types:
 - i. Myelogenous – started in the bone marrow
 - 1. Fragile bones
 - ii. Lymphogenous – started in the lymph organs (spleen, liver, etc.)
 - What is the largest lymphoid organ? The Spleen
- xiv. Platelets – Thrombocytes
 - a. Function: to create clot. Platelet plug and clotting
 - b. Life span: 8-10 days.
 - c. Platelet plug process: (Stop bleeding, "Hemostasis")
 - 1. Vasospasm
 - 2. Platelet attraction/ cascade
 - 3. Platelet plug
 - 4. Desolution (Release of Plasmin, to dissolve clot)
 - d. Clotting factor (12) There is no clotting factor VI

Clotting Factors			
I	Fibrinogen	VIII	Anti Hemophilic Factor A
II	Prothrombin	IX	AHF B (Christmas Factor)
III	Thromboplastin	X	Stuart
IV-	Calcium	XI	AHF C
V	Labile	XII	AHF D (Hagemen)
VII	Stable	XIII	Fibrin stabilizing factor

- e. Clotting Process:
Blood vessel damage > thromboplastin > awakes prothrombin activator > calcium is added > resulting to prothrombin > thrombin > fibrinogen > fibrin stabilizing factor (CLOT) to dissolve (Plasmin). Sequence 3, 2 activator, 4, 2, thrombin, 1, 13
- xv. Platelet related conditions:
 - a. Thrombocytosis (Similar to polycythemia vera)
 - b. Thrombocytopenia
 - c. Hemophilia
 - i. S/Sx: Hemarthrosis (MC in knee joint)
 - ii. MC muscle affecting in muscle bleeding is Psoas
 - iii. Pain is felt around the buttocks instead of the front.
 - iv. Types:
 - 1. A – Absence of clotting factor 8, *Classic hemophilia*
 - 2. B Christmas – absence of clotting factor 9
 - 3. C – absence of clotting factor 11
 - 4. D – absence of clotting factor 12

CV PHYSIOLOGY

- i. Heart
 - a. Cone shaped muscle (tortora) Inverted pyramid (snell)
 - b. The heart is innervated by the *spinal segments from C3 to T4*
 - c. Relatively small, roughly same size as a closed fist
 - d. The heart rests on the diaphragm, and is located **mediastinum** (A mass of tissue extending from the sternum to the vertebral column between the two lungs)
- ii. Orientation of the heart
 - a. 2/3 of the mass of the heart lies left on the bodies midline
 - b. Orientation of the **apex** of the heart, (Anterior, Inferior, towards the left)
 - c. Orientation of the **base** of the heart, (Posterior, Superior, towards the right)
- iii. Pericardium
 - a. A fibrous connective sac that encloses the heart.
 - b. Types:
 - Fibrous pericardium: Outermost layer
 - Prevents over stretching the heart
 - **Anchors** the heart to the mediastinum.
 - Serous pericardium: Innermost layer
 - Serves as a double layer of the heart
 - 2 layers:
 - Visceral SP : AKA *Epicardium*
 - Parietal SP: outermost layer of the SP
- iv. Surfaces of the heart :
 - a. Anterior surface: "Sternocostal surface"
 - i. Right atrium and ventricle
What forms the most anterior portion of the heart?
 - b. Inferior surface: "Diaphragmatic surface"
 - i. Right and left ventricle
 - c. Posterior surface: "Base surface"
 - i. Left atrium and right atrium *What forms the most posterior portion of the heart?*
- v. Chambers of the heart (4)
 - a. Right and left atrium
 - b. Right and left ventricle
- vi. Heart muscles
 - a. Papillary Muscle – pulls on and tightens the chordate tendinae, preventing the valve cusps from everting.
 - b. Pectinate ms - Anterior surface of the atrium is rough
 - c. Trabeculae Carneae - Cardiac ridges fiber in the ventricles
 - d. Chordae Tendineae – cord like structure, where the cusps of the valves are attached
 - e. Papillary ms – cone shaped structure of trabeculae carneae
Where is the trabeculae carneae located? Ventricles

- vii. Valves of the heart
 - a. AV valves (Found between atrium and ventricle)
 - i. Tricuspid (R)
 - ii. Mitral (L)
 - b. SL valves, prevents back flow of blood from the arteries in the inferior heart chambers.
 - i. Aortic (L)
 - ii. Pulmonic (R)

Aortic valve is seen on the right or the left?

Pulmonary artery comes from what chamber of the heart? ® Ventricle
- viii. Blood pathway
 - a. Ascending Pathway

Ascending Aorta > ® brachiocephalic > ® subclavian and common carotid a.

Subclavian a. > vertebral and axillary a.

Vertebral a. > Basilar a. > posterior cerebral a.

Axillary a. > Brachial a. > radial a. and ulnar a.

There is no left brachiocephalic, (L) subclavian and common carotid artery directly arise from aorta.

Common carotid a. > internal and external carotid artery

External carotid a. > terminates at the TMJ and supplies the superficial structures of the scalp

Internal carotid a. > Anterior cerebral artery and Middle cerebral artery
 - b. Descending pathway

Descending aorta > Thoracic aorta > Abdominal Aorta > Common iliac a. > external and internal common iliac a.

Internal > lumbosacral plexus

External > Femoral a. > popliteal a. > Anterior tibial a.
- ix. Heart Sounds
 - a. S1 – “Lub” Longer Louder Lower in pitch. Closure of **AV valves**
 - b. S2 – “Dub” Shorter. Closure of **SL Valves**

(Sounds not audible to the ear):

 - c. S3 – Rapid filling of the ventricles (Samsung S3 phones are fast)
 - a. CHF (3 letters C – H – F, S3!), Ventricular Gallop
 - d. S4 – Atrial systole: MI or Hypertension, Atrial Gallop
- x. Conducting System
 - a. SA node – “Sinus node” Primary pacemaker of the heart
 - i. Below superior vena cava in the ® atrium.

What is the other name of the SA node?
 - b. AV node – “Junctional node”
 - i. Between atrium and ventricle

What is the MC site of heart block?

- c. AV bundle of His
 - i. Location on the interventricular septum
Foramen Ovale opening of the interatrial septum of the fetal heart, closes after birth and becomes the Fossa Ovalis
- d. Purkinje fibers (Largest pacemaker of the heart.)
 - i. Surrounds the whole ventricles
What is the largest pacemaker of the heart?
- xi. Coronary Arteries *What supplies blood to the heart?*
 - a. From aorta > R and (L) coronary artery
 - b. R **coronary artery** supplies R atrium and ventricle, (L) ventricle (minor portion), Interventricular septum, SA node, AV node, Bundle of His
55-60% population, the SA node is supplied by R C a.
 - c. (L) **coronary artery** (MC obstructed) (L) atrium, (L) ventricle, R ventricle (minor portion), Interventricular septum, SA node, Bundle of his
40-45% of population SA node is supplied by (L) C a.
What is 1st to be damaged in (L) C a.?
- (L) coronary a. > LADCA and circumflex
- d. LADCA > Supplies Anterior, Superior, Lateral surface of the heart
- e. Circumflex > Supplies Posterior surface of the heart
In MI, Muscles of the heart runs out of blood supply, d/t obstruction of coronary arteries.
- xii. Cardiac Action potential (5)
 - a. Charge of the cell is -88mV
 - b. Ions Na^+ , Ca^{2+} , K^+

Phase 0: Depolarization	Phase 1: Initial/ Partial Repolarization	Phase 2: Plateau	Phase 3: Repolarization	Phase 4: Resting membrane potential
Inward current of Na^+	Decrease Na^+ influx Outward current of K^+	Inward current of Ca^{2+}	Decrease of Ca^{2+} Influx Still outward current of K^+	Return to -88mV

On phase 0 of Cardiac Action potential which Ion enters?

Ca will always maintain the cardiac action muscle potential

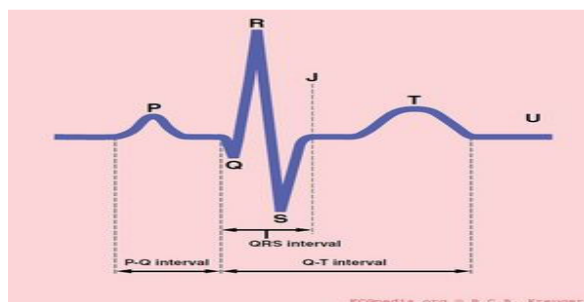
- xiii. Cardiac Cycle
 - a. Rhythmic pumping action of the heart
 - b. Systole – Ventricular contraction
 - c. Diastole – Ventricular relaxation
Cardiac Cycle starts at diastole
- xiv. Diastole:
 - a. Period of rapid filling of the ventricles
 - i. On the 1st third of diastole 75% of blood from atrium to ventricles passively
 - ii. Middle third of diastole, continuous blood flow
 - iii. Last third of diastole 25% of blood from atrium to ventricles via atrial contraction (*Atrial Kick*)

In this period are the AV valves are open or closed?

When is the most amount of blood transferred to the ventricles from atrium in Period of rapid filling?

- xv. Systole:
- a. Isovolumetric Contraction
 - i. Ventricular contraction will cause the tricuspid valve to close and builds pressure.
 - ii. Both AV and SL valves are closed.
What is the only period where both AV and SL valves are closed?
 - b. Ejection Fraction
 - i. Pressure must exceed 8mmHg (Pulmonic artery P_o), in the R ventricle to open Pulmonic valves.
 - ii. Pressure must exceed 80mmHg (Aorta) in the (L) ventricle to open Aortic valves.
 - iii. 1st 70% of blood is given to pulmonary a. and aorta.
 - iv. The last 2/3 of ejection, 30% Ventricles > Aorta & pulmonary a.
Pressures which the ventricles must overcome over the aorta/ pulmonary a. is called?
Afterload
 - c. Isovolumetric Relaxation
 - i. SL valves will close to prevent backflow.
 - ii. AV valves will open to restart the cycle.
- xvi. Hemodynamics
- a. Systolic: highest arterial pressure 120mmHg
 - b. Diastolic: Lowest arterial pressure 80mmHg
 - c. Pulse pressure (PP): Difference between Systolic BP & Diastolic BP
(SBP-DBP) 120-80 = 40mmHg
 - d. End Diastolic volume: Amount of blood left in the ventricles after diastole. **120mL**
What is the amount of blood after ventricular relaxation?
 - e. Preload: Initial stretching of the heart.
 - f. End Systolic Volume: The amount of blood left after systole. **50mL**
What is the amount of blood after ventricular contraction?
 - g. Mean Arterial Pressure: Arterial pressure with respect to time.
DBP + 1/3 PP (80mmHg + 13=93mmHg)
 - h. Stroke Volume (SV): Amount of blood pumped by the ventricles per contraction. **70mL**
SV = EDV-ESV
 - i. Cardiac output: Amount of blood pumped by the ventricles per minute. **Avg. 4-6L**
CO = SV x HR
CO/HR =SV

xvii. ECG



(Terminology: Interval: Longer Segment: Shorter)

- a. P- wave : Atrial depolarization
- b. QRS complex: Ventricular depolarization
- c. Premature Ventricular Contractions(PVC) Skipped heart beat
- d. T wave: Ventricular repolarization
- e. Segments/ Interval:
- f. P-Q Interval/ P-R Interval
 - i. From the beginning of P-wave to beginning of QRS complex.
What is the interval between the beginnings of Atrial Depolarization to Ventricular Depolarization?
Prolonged P-R interval: 1^o heart block
- g. Q-T Interval
 - i. From the beginning of the QRS complex to the end of the T-wave
What is the Interval between the beginnings of Ventricular depolarization to end of Ventricular repolarization?
- h. P-R segment
 - i. End of the P-wave to the beginning of QRS-complex
What is the Segment between the end of Atrial depolarization to the beginning of ventricular depolarization?
- i. S-T segment
 - i. End of QRS complex to end of T-wave
What is the segment between end of ventricular depolarization to the end of ventricular repolarization?
Elevated ST segment: Infarction
Depressed ST segment: Ischemia

Heart rate: Calculate the number of QRS complex in a six second ECG strip multiplied by 10.

xviii. Valves

Auscultation of valves	Location
A – 2 nd ® ICS	3 rd (L) ICS
P – 2 nd (L) ICS	3 rd (L) CC
M – 5 th (L) ICS	4 th (L) CC
T – 4 th (L) ICS	4 th ® ICS

xix. Heart Conditions

a. S/Sx

- i. Chest pain or discomfort/ Angina
 1. (+) Levine sign
 2. Referred pain to Jaw, shoulder, upper trapz, MC on (L) arm, always follows ulnar nerve distribution
 - a. Heart is innervated by spinal segment C3 – T4, somatic areas of these segments will be affected.
 3. Types of Angina
 - a. Chronic Stable Angina
 - i. Very predictable, precipitated by exertion.
 - ii. Responds to rest and nitrates (sublingual, max of 3 tables, c 5 min intervals, effects seen c/in 1 minute)
 - b. Unstable / Crescendo/ Preinfarction/ Progressive Angina
 - i. No response to rest and nitrates.
 - ii. C/I to exercise
 - c. Nocturnal Angina
 - i. Exertion (Dreams)
 - d. Prinzmetal Angina (Angina Inversa/ variant angina)
 - i. MC in women, post-menopausal
 - ii. Vasospasm of coronary artery s occlusion
- ii. Palpitations
 1. "Arrhythmia/ dysrhythmia"
 2. Excessive heart beat
 3. Benign cause; caffeine, anxiety
 4. Severe cause; coronary artery diseases
 5. Mitral valve prolapsed
- iii. Dyspnea (SOB)
- iv. Fainting "syncope" (Decrease oxygen in the brain.)
- v. Cyanosis (Bluish Discoloration of lips, toes, nail beds d/t decreased hemoglobin.)
- vi. Fatigue

xx. Laboratory findings:

a. Cardiac enzymes

- i. CK-MB - creatine kinase myocardial bond. Peak 12-24hours
- ii. SGOT- Serum Glutamic Oxalo-acetic Transminase- peak 24-48 hours
- iii. CPK - Creatinine Phosphokinase – peak 24hours
- iv. LDH – Lactate Dehydrogenase – peak 3-6 days

Which cardiac enzyme will first rise? CKMB

True myocardial infarction- S-T segment elevation + all cardiac enzymes elevated.

- xxi. Diagnostic Tools
 - a. Chest X-ray
 - i. Assess the size and shape of the heart
 - b. Myocardial Perfusion imaging
 - i. Thallium/ nuclear stress test, injected at the peak of exercise
 - ii. Thallium 201 – isotopes
 - c. Echocardiogram – use of US to see movement of valves, walls of the heart.
 - d. Cardiac Catheterization – inserted in the femoral artery in the inguinal area.
 - i. Dye injected through cinefluoroscopy
 - e. Central line/ Swan-ganz catheter
 - i. Subclavian artery/ jugular vein
 - ii. Determine
 - i. Central venous P_o
 - ii. Pulmonary a. P_o
 - iii. Pulmonary capillary wedge P_o
- xxii. Medical and Surgical Intervention
 - a. PTCA – Percutaneous Transluminal Coronary Angioplasty – catheter with balloon tip catheter
 - b. IV stents – Prevents recollapse
 - c. CABG – coronary artery bypass graft
 - i. Great Saphenous vein
 - ii. Int. Mammary artery
 - iii. Int. Thoracic artery
 - iv. Radial artery of nondominant UE
 - d. Heart transplant (Given immunosuppressive drugs to prevent rejection of drugs.)
 - i. Heterotopics – New + old heart, (2) hearts
 - ii. Orthotopics - New heart replaces old heart
- xxiii. Cardiac Conditions:
 - a. Pericarditis (Pericardial friction rub d/t decrease in pericardial fluid.)
 - i. Inflammation of pericardium, Inflammation d/t infection (HIV, sorethroat)
 - ii. Mimics chest pain
 - iii. Aggravating factors
 - 1. Trunk movements aggravate condition especially, Lateral/ side to side movement.
 - 2. Swallowing
 - iv. Relieving factors
 - 1. Kneeling on all fours (Quadruped, Cardiac workload is decreased in this position)
 - 2. Holding breath
 - b. Cardiac Tamponade (Excess pericardial fluid)
 - i. Not painful.
 - ii. Pt can suffer cardiac arrest d/t too much compression of the heart.

c. CHF (congestive heart failure/ cardiac decompensation)

i. Inability of the ventricles to contract effectively.

ii. Types:

1. Right –sided heart failure

2. Left – sided heart failure

Right – Sided Heart Failure	Left – Sided Heart Failure (Left, Lung, PuLmonary)
Backward heart failure Congestion of peripheries and organs (systemic) S/Sx: Bipedal edema Ascites Hepatomegaly Distended Jugular Veins Cyanosis CoR Pulmonale Venous stasis Spenomegaly	Forward heart failure Pulmonary symptoms. S/Sx: Cough SOB Orthopnea/ dyspnea S3 Gallop PND Fatigue/ Muscle weakness Tachycardia Diaphoresis Decreased urine output

d. Myocardial Infarction

i. aka Coronary Occlusion

Where is the most frequent location for a myocardial infarction to occur? Left Ventricle

ii. Decrease in blood supply > infarction (Cell death of myocardium)

iii. True MI has ST segment elevation and all cardiac enzymes elevated

iv. S/Sx

1. Chest pain

2. Cyanosis

3. Dyspnea

4. Fatigue

e. Heart valve Conditions

i. Stenosis/ Narrowing: Inability of the valve to fully open.

ii. Insufficiency/ Regurgitation: Inability of the valve to fully close.

iii. Prolapse: Cusps of valves bulges d/t decrease in strength of cusp.

1. MC in Mitral Valve (MVP/ Barlow's / Click Murmur/ Floppy valve Syndrome)

2. Etiology: (L) ventricle > right pressure 6x greater, wall of (L) ventricle > ® ventricle 3x, congenital anomalies.

3. Triad:

1. Palpitation

2. Dyspnea

3. Fatigue

xxiv. Congenital Anomalies

- a. Atrial Septal Defect (ASD)
 - i. Defect in interatrial septum
 - ii. Blood passes through Left to Right.
 - iii. Acyanosis
- b. Ventricular Septal Defect (VSD)
 - i. Defect in interventricular septum.
 - ii. Blood passes through Left to Right. (Pressure differences Right>Left)
 - iii. Acyanosis
- c. PDA (Patent Ductus Arteriosus)
 - i. Lungs are not developed as a fetus, oxygen is received via placenta from mother. Ductus Arteriosus connects pulmonary artery and descending aorta.
 - ii. Remnant of Ductus Arteriosus is Ligamentum Arteriosus
- d. Coarctation of Aorta
 - i. Constriction of proximal and distal aorta
 - ii. Narrowing -> turbulent blood flow
- e. Tetralogy of Fallot (True blue baby)
 - i. Pulmonary Artery Stenosis
 - ii. (Overriding) Aorta – towards the right
 - iii. (R) Ventricular hypertrophy/ aka Cor Pulmonale
 - iv. Interventricular Septal Defect (VSD)

xxv. Cardiac Rehabilitation

IER	Braddom (Old)	Braddom (New)
Phase I – In patient	Phase I Acute	Phase I In patient
Phase II – Out patient	Phase II Convalescent	Phase II Transitional
Phase III – Community	Phase III Training	Phase III Out patient
	Phase IV Maintenance	Phase IV Maintenance

*Descriptions remain the same, only name changes.

IER combines training and maintenance called Community Exercise Program

- a. Acute/ Inpatient Phase:
 - i. Pt is confined
 - ii. Goal: Prevent deconditioning, family education, and instruction only.
 - iii. Entry level level: 2-3 Mets
 - iv. Discharge level: 3-5 Mets
- b. Convalescent/ Outpatient phase
 - i. Period of recovery is 6 weeks
 - ii. Goal: Promote strong scar formation
 - iii. Entry level: 5 mets
 - iv. Discharge level: 9 mets (ascending stairs, playing competitive basketball)
- c. Training phase
 - i. Most difficult/vigorous phase with use of treadmill
- d. Maintenance phase
 - i. Most important phase
 - ii. Lifelong routine

- xxvi. Exercises intensity:
 - a. Karvonen's Formula
$$\text{THR} = (\text{MHR} - \text{RHR}) 60 - 80\% + \text{RHR}$$
$$\text{MHR} = 220 - \text{age}$$
- xxvii. Criteria for terminating an exercise program:
 - a. Unstable Angina
 - b. Resting BP 200/100mmHg
 - c. Acute systemic illness/ Fever
 - d. 2nd-3rd degree heart block
 - e. Recent Embolism
 - f. Uncontrolled Arrhythmias/ Dysrhythmias/ Palpitations
 - g. Uncontrolled Diabetes Mellitus
 - h. ST segment displacement $\geq$ 2mm
 - i. Increase diastolic BP
 - j. Active Pericarditis
- xxviii. Type's of Heart Block:
 - a. 1^o heart block – Prolonged PR Interval
 - b. 2^o heart block – Progressive lengthening of PR Interval
 - c. 3^o heart block – Complete heart block, no more impulses arriving at bundle of His and Purkinje Fibers

PULMO PHYSIOLOGY

- i. Pulmonary system
 - a. Function
 - i. Ventilation – act of moving air in and out of the lungs
 - ii. Perfusion – pulmonary blood flow
 - iii. Respiration – Transfer of gases between body cells and the environment.
 - 1. Internal respiration – between capillaries and tissues
 - 2. External respiration – between capillaries and alveoli
- ii. Upper respiratory tract
 - a. Nose
 - i. Largest mucosal surface area
 - ii. Vibrissae- for filtration
 - iii. (3) Cartilages of the nose *SALN (ose)*
 - 1. Septal
 - 2. Alar
 - 3. Lateral
 - b. Pharynx – Common area for respiratory and GI system.
 - i. aka Throat, passage way for food and air.
 - ii. 3 parts:
 - 1. Nasopharynx – filters and warms the air\
 - 2. Oropharynx – Conduits for air
 - 3. Laryngopharynx – conduits for air.
 - c. Larynx
 - i. Aka voicebox
 - ii. For sound production
 - iii. Ensures air passes through trachea
 - iv. 9 cartilages
 - 1. Unpaired (*U Try Epic Crying*)
 - a. Thyroid
 - b. Epiglottis
 - c. Cricoid
 - 2. Paired (*Pare, Aren't Corns Cute?*)
 - a. Arytenoid
 - b. Corniculate
 - c. Cuneiform
- iii. Lower Respiratory Tract (Tracheobronchial tree, 23 generations.
How many generations is coughing effective? 7 generations

- a. Trachea
 - i. aka windpipe
 - ii. tubular structure
 - iii. 16-20 cartilaginous half ring (Ant)
 - 1. T4-T5 angle of Louis/ carina (level of bifurcation)
 - 2. T2 suprasternal notch
 - 3. Last tracheal ring is the Carina

Tracheostomy: common site of intubation 2nd-3rd ring , If emergency in the larynx, cricothyroid membrane.
- b. Trachea to terminal bronchioles (Conduction zone)
- c. Respiratory bronchioles to capillaries (Respiratory zone)

What is the functional unit of the pulmonary system? Acinus
- iv. Primary Bronchi/ Mainstem bronchi (2)
 - a. Left and right

*MC site of aspiration of **large** objects? ® Wide, Short, more Vertical*
- v. Secondary Bronchi/ Lobar Bronchi (5)
 - a. MC site of **small** aspirated objects
 - b. (2) Left: superior and inferior
 - c. (3) Right: Superior, Middle and Inferior
- vi. Tertiary Bronchi/ Segmental Bronchi (18)
 - a. (10) Right
 - b. (8) Left
- vii. Terminal bronchioles (aka) transition zone
- viii. Respiratory Bronchioles
- ix. Alveoli (300m each lung)
- x. Lungs
 - a. Fissures
 - i. Right lung
 - 1. Horizontal
 - 2. Oblique
 - ii. Left lung: Oblique only
 - b. Coverings
 - i. Visceral Pleura(covers the lungs)
 - 1. Stretch sensitive
 - ii. Parietal Pleura(covers the inner ribs)
 - 1. Pain sensitive

c. Lobes & Segments:

Lobe	Right (10)	Left (8)
Upper	Apical Anterior Posterior	Apical-Posterior Anterior Lingula Sup & Inf
Middle	Lateral Medial	None
Lower	Superior Anterior basal Posterior basal Lateral basal Medial basal	Superior Anterior basal Posterior basal Lateral basal Medial basal

xi. Innervation/ Blood supply

a. Innervation

- i. Pulmonary plexus
 1. Sympathetic nerves
 - a. Thoracic-Lumbar
 2. Vagus nerve

b. Blood supply

- i. Bronchial artery supply:
 1. Plurae
 2. Airway
 3. Connective tissue
 - a. Elastin + Collagen (Elasticity of lungs (compliance))
- ii. Pulmonary Artery supplies alveoli

xii. Muscles for respiration

a. Relaxed inspiration

- i. Diaphragm (dome shaped)
 1. Piston action
 2. Inhalation, it moves downward and outward
 3. Exhalation, it move upward and inward
- ii. External Intercostals
 1. Fiber runs downward and outward

What is the origin of the external intercostals? Lower surface of the rib above

b. Forced inspiration (SUPAS)

- i. SCM
- ii. Upper trapezius
- iii. Pectoralis major and minor
- iv. Anterior, Middle, Posterior Scalenes
- v. Serratus Anterior and Posterior-Superior

c. Relaxed expiration

- i. There are no muscles involved in relaxed expiration
- ii. Elastic recoil of the lungs

- d. Forced expiration (e.g. coughing) ASI
 - i. Abdominals
 - ii. Serratus Posterior-Inferior
 - iii. Internal intercostals (fiber runs downward and inward)
- xiii. Mechanism to increase thoracic diameter

Normal AP:Lateral ratio: 1:2

 - a. Piston action by the diaphragm
 - i. Descent of the central tendon of the diaphragm , increase in vertical dimension
 - b. Pump handle mechanism
 - i. Increase AP diameter of the chest
 - ii. Sternum, acts on 1st-6th ribs

Simultaneous motions

 - c. Bucket- Handle mechanism
 - i. Increase in lateral diameter of the chest
 - ii. Ribs (ribs 7-10)
 - iii. Upward
 - d. Caliper motion
 - i. False ribs (8-12)
 - ii. Flare out to the side)
- xiv. Mechanics of breathing
 - a. Compliance = elasticity of airways and lung tissue
 - i. Ability of a structure to expand
 - b. Boyle's Law
 - i. Pressure is inversely proportional to the volume
 - 1. Increase pressure, decrease volume (Expiration)
 - 2. Decrease pressure, increase volume (Inspiration)
- xv. Control of respiration
 - a. Automatic control
 - i. Brainstem (Pons and medulla)
 - 1. Medulla Sets the inherent rhythmicity of breathing (Automatic Respiratory Center)
 - a. If the medulla acts alone, the breathing is weak and irregular
 - 2. Pons sets the rhythm and rate of breathing
 - a. If the pons acts alone, breathing is stronger, regular, and more effective
 - b. DRG (Dorsal respiratory group)
 - i. Dorsal Medulla (Nucleus Solitarius)
 - ii. For inspiration
 - c. VRG (Ventral Respiratory group)
 - i. Ventro-lateral medulla (Nucleus ambiguus, Retroambiguus)
 - ii. For expiration
 - iii. Mnemonic: VEX
 - d. Inspiratory ramp signal: on:off ratio 2:3

- e. Pons: (Pneumotaxic center)
 - i. Location: upper pons (nucleus propius)
 - ii. Function: “switching off of inspiratory ramp signal”
 - f. Apneustic center
 - i. Lower pons
 - ii. Function: prevents switching off of the inspiratory ramp.
mnemonic: PULA
 - g. Chemoreceptors
 - i. Central chemoreceptor (Ventral medulla)
 - 1. Stimulus: increase in H^+ ions
 - ii. Peripheral chemoreceptor
 - 1. Carotid- aortic bodies
 - 2. Stimulus: decrease in order of
 - a. O_2
 - b. $PaCO_2$
 - c. Acidosis
 - iii. Response to either is increase in Ventilation
 - iv. pH is direction proportional to CO_2 but inversely proportional to HCO_3^-
- xvi. Lung volumes and Capacities

TLC (6000mL) IRV+ERV+TV+RV	VC(4500mL) IRV+ERV+TV	IC (3500mL) IRV + TV	IRV(3000mL)
	OR		TV (500mL)
	IC+ERV	FRC (2500mL)	ERV (1000mL)
	RV (1500mL)	ERV+RV	RV (1500mL)

- a. Primary lung volumes
 - i. Tidal Volume
 - 1. Amount of air that goes in and out of the lung (at rest)
 - ii. Inspiratory reserve volume
 - 1. Amount of air that is inhaled after normal inspiration
 - iii. Expiratory reserve volume
 - 1. Amount of air that is exhaled after normal expiration
 - iv. Residual volume
 - 1. Amount of air that is left in the lungs after a maximum expiration
- b. Capacities
 - i. Functional residual volume
 - 1. Amount of air that is left in the lungs after a normal expiration
 - ii. Inspiratory capacity
 - 1. Amount of air that is max inhaled after normal expiration
 - iii. Total lung capacity
 - 1. Maximum amount of air that the lungs can hold. Amount of air in the lungs after a maximum inspiration
- c. Dead space
 - i. Normal volume of dead air space: 150mL

Pulmonary Assessment

xvii. Breathing Patterns

Type	Rate	Depth	Rhythm
Eupnea	(n)	(n)	Regular
Bradypnea	Slow	(n)/ shallow	Regular
Tachypnea	Fast	Shallow	Regular
Hyperpnea	(n)	Deep	Regular
Hyperventilation	Fast	deep	Regular
Dyspnea	Rapid	Shallow	Regular
Cheyne-stokes	Slow	↑ + ↓ + Apnea	Regular
Biot's	Slow	Shallow + Apnea	Irregular

Hyperventilation is associated with Metabolic Acidosis

xviii. 1st part of assessment

- a. History of patient and family
- b. Chest examination

xix. 2nd part of assessment

- a. ABG Analysis
- b. PFT
- c. X-ray
- d. Graded exam technique

xx. Common chest deformities

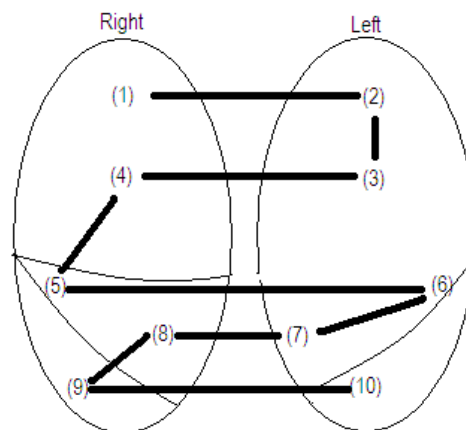
- a. Barrel chest AP:L 2:2 – associated with emphysema
- b. Pectus Carinatum
- c. Pectus Excavatum

xxi. Chest Mobility/ Chest expansion tests

Note for any asymmetry when the patient breathes in and out

- a. Upper lobes
 - i. Thumbs at the sterna notch, fingers placed above the clavicle
- b. Middle lobes
 - i. Thumbs at the xiphoid process, fingers placed on the lateral ribs
- c. Lower lobes
 - i. Thumbs at the lower thoracic spine (back), fingers placed on the lateral ribs.

xxii. Auscultation sites



xxiii. Breath Sounds

a. Normal

- i. Vesicular – soft pitch sound
- ii. Bronchial – hollow, loud, tubular, high pitched
- iii. Broncho-Vesicular – Softer than bronchial

b. Adventitious

- i. Crackles/ Rales – fine discontinued sounds
 - 1. Ex. Popping. Fizzing soda, hair rubbing
- ii. Wheezes – continuous high or low pitch sounds
 - 1. Associated with asthma
- iii. Stridor – snoring

c. Significance for auscultation

(n) breath sounds	(n) air filled lungs
Decrease	Hyperflation
Increase	Hypoinflation/ atelectasis
Crackles	Presence of secretions
Wheezes	Constricted bronchi
Balloon rubbing	Pleuritis/ pleurisy

xxiv. Voice transmission

a. Egophony

- i. “EEE” (N)
- ii. Consolidation/ pneumonia/ pleural effusion “AAA” – ab(n)

b. Bronchophony

- i. “99”
- ii. Ab(n) – louder “99”

c. Petriloquy (whispered) “1 , 2 , 3”

Palpation

xxv. Fremitus

- a. (n) vibration – (n) air filled
- b. Increased vibration = secretions
- c. Decreased vibration = hyperinflation

xxvi. Percussion

- a. (n)resonant
- b. Dull/ hyporesonant– increased secretions or atelectasis hyperinflation
- c. Hyper resonant – increase air, emphysema

xxvii. ABG findings

Condition	Mnemonics	S/Sx	Condition
Respiratory Alkalosis	DENTS	Dizziness, Early tetany, Numbness, Tingling, Syncope	Associated with hypervent
Respiratory acidosis	HARDy DiSC	Early – Headache, Anxiety, Restlessness, Dyspnea Late – Disorientation (confusion) Somnolence, Coma	Associated with hypoventilation
Metabolic Alkalosis	We Men ↑ Ms	Weakness, Mental dullness, ↑ DTRs, Ms twitching	MC with vomiting
Metabolic acidosis,	NaLoCo	Nausea, Lethargy, Coma	MC with diarrhea, assoc with Kussmaul's

xxviii. Coughing ability

a. Purpose of double cough (only good up to 7 generations!)

- i. 1st cough – to remove secretions
- ii. 2nd cough – to clear the air

b. Tracheal tickle – for those unresponsive

- i. Used to elicit reflex cough
- ii. Circular rub on trachea

How much secretions are produced in a day? 100mL/day

xxix. Sputum color

Color	Findings
Red/ scarlet	Blood
Rust	Pneumonia
Yellow	Infection
Green	Pus
Pink	Frothy Pulmo edema
Purple	Neoplasm, CA
Flecked	Carbon Particles
Clear	(n)

xxx. Other PE findings

	Emphysema	Chronic bronchitis
Chest shape	Barrel chested	(n)
Appearance	Use of accessory ms >tachypnea	Clubbing of finger Peripheral edema
Cor Pulmonale	None, except at the late stages	Prominent

xxx. Asthma

- a. Hypersensitivity to bronchial secretions due to various stimuli, resulting to wide spread bronchoconstriction
- b. 50% < 10y/o M:F 2:1, >30y/o M:F 1:1
- c. S/Sx
 - i. Triad
 - 1. Wheezes
 - 2. Cough
 - 3. Dyspnea
 - ii. Tachypneam use of accessory muscles
 - iii. (+) barrel chested
Kurschmann's spirals – thick stringy mucus
- d. Trigger Factors
 - i. Extrinsic Factors
 - 1. Allergens (dust, polles
 - 2. Food (chocolate, nuts, seafoods etc)
 - 3. Animal fur
 - 4. Drugs (Aspirin)
 - 5. Changes in climate
 - 6. Pollution
 - ii. Intrinsic Factors
 - 1. URT infection emotions
 - 2. Psychological stress
 - 3. Exercise
 - iii. Status asthmaticus – most serious type of asthma

xxxii. COPD vs CRPD

- a. V/Q ratio
 - i. COPD: <0.8, responsive to bronchodilators
 - ii. CRPD: >0.8
 - 1. Due to alteration of
 - a. Lung parenchyma
 - b. Chestwall
 - c. Neuromuscular apparatus
- In a child with >0.8 V/Q ration is this normal? Yes*

xxxiii. Emphysema & Chronic Bronchitis

Emphysema	Chronic bronchitis
Dysfuntion of air spaces distol to terminal bronchioles	3 mos productive cough for 2 consecutive years
Destruction of alveolar speta	d/t pollution, occupation
d/t smoking	heavy smoker > 40 cigarettes a day
increase proteolytic enzyme	production of sputum 100mL/ day
alpha 1 antitrypsin (destroyed by smoking) inhibits proteolytic enzyme	age bracket: >50 y/o +/-
age bracket: >60 years old	Aka Blue Boater
aka pink puffer	

xxxiv. Continued emphysema vs chronic bronchitis

	Emphysema	Chronic bronchitis
Dyspnea	Severe, hallmark	Mild
Cough	Less prominent After dyspnea	More prominent Before dyspnea, hallmark
Sputum	Scanty, mucoid	Copious, purulent
Bronchial infection	Less frequent	More frequent
Body built	Aesthenic, weight loss	Overweight
x-ray	Hyperinflated lungs, flat diaphragm, small heart	Increase broncho vesicular markings, "Dirty lungs" (n) lung size ® ventricular hypertrophy (cardiomegaly)

xxxv. Bronchiectasis

- a. MC affected areas are the terminal bronchioles
- b. Ab(b) permanent dilatation of bronchi. Bronchioles
- c. S/Sx
 - i. *Hemoptysis, hallmark*
 - ii. Fever
 - iii. Productive cough
 - iv. Recurrent infection
 - v. Dyspnea

xxxvi. Cystic Fibrosis

- a. Widespread abnormalities of exocrine glands
- b. Triad
 - i. Mucus glands
 - ii. Exocrine cells of pancreas
 - iii. Sweat glands
- c. S/Sx
 - i. Productive cough
 - ii. Bronchial infection
 - iii. Weight loss d/t malabsorption
 - iv. Salty sweat, increase in NaCL content (sweat test)
- d. X-ray: honeycomb lungs

xxxvii. CRPD

- a. Interstitial pulmonary fibrosis aka *Hammans-rich dse*
 - i. Idiopathic
 - ii. Associated with smoking
 - iii. Genetic predisposition
 - iv. Collagen disease
 - v. *Hallmark: progressive dyspnea*
 - vi. Death after 5-6 years old

- b. Pneumonia (intraalveolar infection)
 - i. MC to streptococcal virus
 - ii. 3 types
 - 1. Bacterial
 - 2. Viral
 - 3. Aspiration
 - iii. S/Sx:
 - 1. Fever and chills
 - 2. Productive cough
 - 3. Dyspnea
 - iv. Common pneumonia HS
 - 1. H. Influenza
 - 2. S. Pneumoniae
 - v. Hospital acquired
 - 1. Heregenosa
 - 2. Pseudomonas
- c. PTB d/t mycobacterium TB
 - i. 2-10 week, incubation
 - ii. 1st 2 weeks most infectious period
 - iii. Keep Pt in negative pressure room
 - iv. Hallmark: hemoptysis
 - v. S/Sx:
 - 1. Dyspnea
 - 2. Fever
 - 3. Enlargement of lymph nodes
 - vi. *TB in children is called primary complex*
- d. Atelectasis
 - i. 2 types
 - 1. Primary Atelectasis
 - a. Compressive atelectasis, e.g. pleural effusion
 - 2. Secondary Atelectasis
 - a. Obstructive atelectasis
- e. Pleuritis/ Pleural effusion
 - i. Inflammation of pleura
 - ii. Excess accumulation of pleural fluid
 - iii. s/Sx
 - 1. pleural rub
 - 2. dyspnea
 - 3. sudden acute pain

- f. Pulmonary Edema
 - i. Findings of water entering the alveoli
 - ii. d/t unequal capillary pressure
 - iii. associated with (L) sided CHF
 - iv. Hx of MI, mitral valve prolapsed/ stenosis
 - v. S/Sx:
 - 1. Pink/ frothy sputum (not productive)
 - 2. (+) crackles
 - 3. Dyspnea
 - 4. Non productive cough
 - 5. Sharp/ dull chest pain
- g. Pulmonary embolism
 - i. Lodging of large or small particles in the venous pulmonary circulation
 - 1. MC cause DVT
 - a. C – clotting disorder
 - b. O – oral contraceptives
 - c. V – venous stasis
 - d. A – air embolism
 - 2. S/Sx
 - a. Dyspnea
 - b. Cough
 - c. Sudden acute pain
 - d. Doorstop breathing (same c pulmonary edema)
 - e. FATAL
- h. Pneumothorax
 - i. Gas/ air in the intrapleural space
 - ii. Failure to cover a chest tube
 - iii. Trauma
 - iv. Treatment: P-tube inserted at 2nd-3^d ICS, if air. If effusion 8th-9th ICS.
 - v. S/Sx
 - 1. Dyspnea, cough, sudden chest pain
- i. SARS (Severe Acute Respiratory Syndrome)
 - i. CORONA virus, transmitted within 10 days
 - ii. S/Sx
 - 1. Lethargy, Sore throat, Dry cough, Dyspnea
- j. Bronchogenic Carcinoma
 - i. Aka. Lung cancer
 - 1. 3 types
 - a. Oat cells
 - b. Small cells
 - c. Squamous cells
 - 2. S/Sx: cough
 - 3. Dyspnea
 - 4. Hoarseness

xxxviii. Postural drainage

- a. Rationale: to prevent accumulation of secretions, and to remove secretions that have already been accumulated
- b. Duration: 20 – 30 – 45 minutes
 - i. Never exceed >45 minutes
- c. Trendelenburg position (head down position) reverse trendelenburg (position head up)
- d. C/i: recent head injury/ surgery, this treatment will cause increase in ICP
- e. Types of manual percussion used in PD, only perform 3-5 mins
 - i. Vibration
 - ii. Percussion (MC)
 - iii. Shaking
- f. Postural drainage
 - i. Initial position 5-10 minutes
 - ii. Percussion 3-5 mins
 - iii. Position: 5 mins, post
 - iv. If there is productive cough perform up to 4x a day

Segment	Position	Percussion
Upper Lobes		
Anterior apical	Sitting, leaning backward	Below clavicle
Posterior apical	Sitting, leaning forward	Above scapula
Anterior	Supine, flat	Over nipple area (male) Above the breast (female)
Posterior	¼ turn from prone on @/(L) side, reverse T-position	Scapula
Middle Lobes		
Middle lobe + lingula	¼ turn from supine, T –posn 15-30° or 12-16" of pillow	Below nipple
Lower Lobes		
Anterior	Supine T posn (30-45°)	Anterior lower ribs
Posterior	Prone T posn (30-45°)	Posterior lower ribs
Lateral	SL T posn (30-45°)	Lateral lower ribs
Superior	Prone bed flat	Below inferior angle
Posterior basal segment	Prone bed flat	Near lower thoracic spine

GIT PHYSIOLOGY

i. Organs of the digestive system

Alimentary tract	Accessory organs
Mouth	Teeth
Pharynx	Tongue
Esophagus	Salivary glands
Stomach	Liver
Intestines	Pancreas

ii. Layers of the alimentary tract:

- a. Mucosa- Innermost layer, absorption
- b. Submucosa – Meissner's/ Submucosal plexus
- c. Muscularis- Myenteric/ Auerbach's plexus
- d. Serosa - Outermost layer

iii. Enteric System- (Neural Control) Composed of 2 plexuses:

- a. Myenteric/ Auerbach's Plexus –Found in the Muscularis layer
 - i. Controls movement in the GIT
 - ii. Excitatory : Promotes contraction of the intestinal wall
 - iii. Inhibitory : Sphincters
- b. Meissner's/ Submucosal layer
 - i. Gastrointestinal secretions

iv. Hormones in GI motility :

- a. Cholecystokinin
 - i. Increased contractility of bile
 - ii. Inhibits stomach motility
- b. Secretin
 - i. Inhibitory effect in GI motility
- c. Gastric Inhibitory peptide
 - i. Decrease motor activity in the stomach (Still inhibitory, 2nd definition)

v. 2 types of movement in GI tract:

- a. Peristalsis – Propulsion forward.
- b. Mixing movement – (Segmental) Does not require a strong contraction, utilizes small contraction to increase mixing.

vi. GI blood supply:

- a. Splanchnic Circulation (Goes through hepatic circulation before the systemic circulation)
 - i. Gut, spleen, and pancreas then liver.
 - ii. Liver sinusoids

- vii. Ingestion of food *What are the two stages in the ingestion of food?*
- a. Mastication (chewing)
 - i. Teeth (anterior – cutting, posterior – grinding)
 - ii. Muscles (TIME) MCLO **M**edial: **C**losing **L**ateral: **O**pening, (Innervated by CN V)
 - iii. Chewing reflex – Once food enters the mouth, it will cause relaxation of the ms of mastication, and as the jaw drops the stretch reflex is created closing the mouth.
 - b. Swallowing (Deglutition)
 - i. Voluntary stage
 1. “Voluntarily” squeezes or rolls the bolus posteriorly towards the pharynx by pressure of the tongue.
 - ii. Pharyngeal stage
 1. Stimulates the swallowing receptor area.
 2. Trachea is closed, the esophagus is opened and fast peristaltic wave from the pharynx forces the bolus towards the upper esophagus (Less than 2 seconds)
 - iii. Esophageal stage
 1. Primary peristalsis
 - a. Continuation of the peristaltic wave that begins in the pharynx
 2. Secondary Peristalsis
 - a. Results from distention of the esophagus by retained food
 - b. *Gastroesophageal sphincter (Achalasia, if this sphincter has problems with relaxing)
- viii. Digestion in the mouth:
- a. Chemical digestion by saliva
 - i. Serous secretion for digestion of starches: Amylase (Star Amy)
 - ii. Mucous secretion for lubrication
- ix. Digestion in the esophagus
- a. No actual digestion
 - b. Propulsion via peristalsis
 - c. Mucous secretion for lubrication
- x. Motor function of the stomach :
- a. Storage for large quantities of food *Where are the foods stored in the stomach? The Fundus*
 - b. Mixing of food with gastric secretions (Bolus will turn into Chyme)
 - c. Slow emptying of food
 - d. *Body and antrum, Anatomical Terms
 - e. *orad and caudad, Physiological Terms
- xi. Storage function:
- a. Vagovagal reflex (Occurs when food is stored in the stomach)
 - i. Reduce the tone of the muscular wall (Food storage capacity ~1.5 liters)
- xii. Mixing and propulsion in the stomach:
- a. Gastric glands
 - b. Peristaltic constrictor waves/ mixing waves
 - c. Pyloric muscle can also contract
 - d. Chyme
 - e. Hunger contraction (Hunger pangs, occurs if no food is ingested within 12-24 hours (Guyton))

xiii. **Digestion in the Stomach**

- a. Food is stored in the fundus
- b. LES prevent reflux of stomach contents
- c. Mechanical digestion via segmental contraction
- d. No absorption
- e. Chemical digestion by Oxyntic cell secretion

Cell	Secretion	Function
Mucous neck cells	Mucus	Lubrication and protection
Peptic chief cells	Pepsinogen	+HCl to produce a proteolytic enzyme Pepsin for protein digestion
Parietal cells	HCl	
	Intrinsic factor *Pernicious anemia (-) IF	B12 absorption

xiv. **Emptying of the stomach**

- a. Intense antral(below) peristaltic contraction – aka the Pyloric pump
- b. Gastrin
- c. Duodenal factors (What are the factors that will inhibit emptying of the stomach?)
 - i. Too much chyme
 - ii. The chyme is excessively acidic

xv. **Small intestine**

- a. Where the **greatest amount** of digestion and absorption take place.

xvi. **Movement of Small intestine**

- a. Mixing contraction (Segmentation) (Simultaneous contraction)
- b. Peristalsis (Successive contraction)
- c. Propulsive movement
 - i. Gastroenteric reflex
 - ii. Gastrin, Insulin, Serotonin
 - iii. Secretin and glucagon
 - iv. Segmentation movement
 - v. *Ileocecal valve (ileocecal sphincter) will relax to propulse chyme towards large intestine

xvii. **Digestion and absorption in the Small Intestine**

- a. Dominating chemical digestion via
 - i. Pancreatic secretions
 - ii. Biliary secretions
 - iii. Intestinal secretions

Secretion	Function
Mucus	Lubrication and protection
Peptidase	Split Peptides into individual Amino Acids (PAA)
Maltase	Split Maltose > Glucose + Glucose
Lactase	Split Lactose > Glucose + Galactose
Sucrase	Split Sucrose > Glucose + Fructose
Lipase	Split Fats into > Glycerol + Free fatty

- xviii. Absorption via transport processes:
- Water: diffusion through tight junction
 - Carbohydrates: Na-Glucose cotransport; Fructose via facilitated diffusion
 - Lipids: diffusion with bile salts
 - Proteins: Na-Glucose co-transport or endocytosis
 - Na: diffusion down a electrical gradient
 - Cl: diffusion via solvent drag
 - HCO_3^- indirectly by secretion of H^+
- xix. Biliary Secretions by the liver and gall bladder
- For fat digestion and absorption
 - Emulsify large particles for more efficient lipase action
 - Base of absorption
 - Bilirubin, means of excretion of waste
 - Liver constantly produce bile and its concentrated in the gall bladder
 - Biliary secretions include HCO_3^- to neutralize the acidic chyme (Consider Diarrhea in Alkalosis)
- xx. **Chemical digestion by the Pancreas**
- Acini cells* secrete pancreatic digestive enzyme into the hepatopancreatic duct to empty into the small intestine (aids in digestion)
 - Enzymes are not activated until they are mixed with acidic chyme.

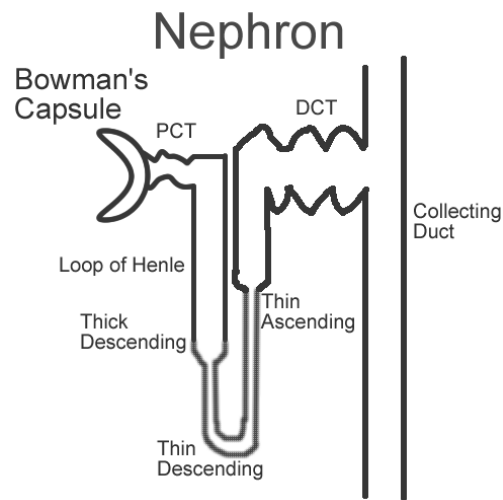
Secretion	Function
Trypsin	Split proteins into peptides
Chymotrypsin	
Carboxypolypeptidase	Split proteins into AA
Pancreatic Amylase	Carbohydrate breakdown
Pancreatic Lipase	Fat Digestion
Cholesterol esterase	
Phospholipase	

- xxi. Movement in the colon:
- Mixing movements (Haustrations)
 - Large circular constriction, and contraction 3 longitudinal muscle strips (Teniae Coli)
 - Propulsive movement (Mass movement)
 - Slow analward movement of haustral connections (Cecum and Ascending)
 - Mass movement (Transverse to Sigmoid)
 - *Gastrocolic and duodenocolic reflexes
- xxii. Absorption in the large intestine:
- Absorption of electrolytes occurs but backflow is prevented by tight junctions
 - Water flows as result of osmotic gradient. What is mostly absorbed in the large intestines?
 - Secretion of mucus lubrication
- xxiii. Defecation Reflex (Pooping reflex)
- Integrated in the sacral region of the spinal cord
 - Stimulated by distention of the rectal wall
 - Weak contraction of rectal wall
 - Relaxation of anal sphincters (May be inhibited by voluntary constriction of external anal sphincter)

RENAL PHYSIOLOGY

- i. Components of the renal system:
 - a. Kidneys (2)
 - b. Ureter (2)
 - c. Bladder (1)
 - d. Urethra (1)
- ii. Function of the kidney:
 - a. Elimination of waste products as urine is formed
 - i. Urea (Metabolism of amino acids (lowers blood glucose))
 - ii. Uric acid (Metabolism of nucleic acids)
 - iii. Bilirubin (Waste-product of hemoglobin)
 - iv. Creatinine (Metabolism of muscle creatinine)
 - b. Controls the plasma volume in the body
 - c. Acid-base balance
 - d. Regulation of BP (Renin Angiotension-Aldosterone Pathway)
 - i. RAAS activates if Dehydrated/ Hemorrhage (Decreased blood volume and decrease BP)
 1. Juxtaglomerular cells activate and releases Renin
Important substance regulating blood pressure that is produced by the kidneys?
 2. Renin > converts Angiotensinogen (in the liver) > increase Angiotensin I > reacts with ACE(Angiotensin Converting Enzyme) in the lungs converts into Angiotensin II (potent vasoconstrictor)> increase Angiotensin II > stimulates Adrenal Cortex to release Aldosterone (for reabsorption of sodium and water) > Increase in blood: increase blood pressure
 - e. Regulates Glucogeneos
 - f. Hormone production:
 - i. Calcitriol, active form of Vitamin D
What is the other name of Calcitriol? 1,25-Dihydroxylcholecalciferol
 - ii. Erythropoietin, stimulates production of new red blood cells in the bone marrow.
- iii. Anatomy of the Kidney
 - a. Bean shaped
 - b. Size: similar that of a clenched fist
 - c. Weight of the adult kidney: 150g
 - d. Location: Retroperitoneal, level of T12 & L3, @ kidney is lower than the left (d/t the liver)
 - i. External anatomy
 1. Layers of tissue
 - a. Renal capsule > Deep layer, acts as a barrier to trauma
 - b. Adipose tissue > Middle layer, shock absorption.
 - c. Renal fascia > Outer layer/ anchors the kidney to abdominal wall
 - ii. Internal anatomy
 1. Renal cortex – Superficial
 2. Renal medulla (Renal pyramids) – Deep

- iv. Nephron (1 kidney = 1 million nephrons)
- Functional unit of the kidney *What is the functional unit of the kidney?*
 - Responsible for the urine formation



- Two main structures
 - Renal corpuscle *What consists of the Renal Corpuscle?*
 - Glomerulus + Bowman's Capsule
 - Renal tubules
 - Loop of Henle, Ascending/ Descending Thick/Thin Convolved tubule
- Two types of nephron *Where does the luminal fluid flow last?*

Cortical Nephron	Juxamedullary Nephron
85% of total nephrons	15% of total nephrons
Lie superficially	Lie near the medulla
Short Loop of Henle	Long Loop of Henle
(-) Vasa Recta	(+) Vasa Recta
Cortical collecting tubule	Medullary collecting tubule

- v. Urine formation
- Glomerular filtration
 - Filtration membrane (Epithelial cells + Podocytes)
 - Will permit only water and small solutes
 - (-) permit proteins, platelets, blood cells
 - Net filtration rate (NFR) : 10mmHg (pressure that must be overcome to be filtered)
 - Glomerular filtration rate – amount of filtrate formed in all renal corpuscles in both kidneys in each minute (GFR): Male: 125mL/ min Female 105ml/min
What is the amount of filtrate in all Renal Corpuscle in both kidneys in each minute?
 - Tubular Reabsorption, 65% of fluid becomes reabsorbed
 - Proximal convoluted tubule
 - "Workhorse" of the nephron
 - Concentrated luminal fluid.
 - Thin Descending LOH
 - Permit reabsorption of water
 - (-) permit sodium, urea, chloride
 - Concentrated luminal fluid

- iii. Thin Ascending LOH
 - 1. Will permit reabsorption Na & Cl
 - 2. (-) permit water
 - 3. Diluted luminal fluid
 - iv. Thick Ascending LOH
 - 1. Will permit Na, Cl, K and other ions
 - 2. (-) permit water
 - 3. Diluted luminal fluid
 - v. Distal convoluted tubule,
 - 1. (-) permit water
 - 2. Diluted luminal fluid

ADH – activates when dyhydrated, attaches to distal convoluted tubule to reabsorb water. ADH AkoDistalHa!

The most dilute luminal fluid in the presence of ADH is found in the? Ascending LOH
- c. Tubular secretion - Occurs all throughout urine formation
- vi. Ureter (25 – 35cm long, 6mm diameter)
 - a. Long slender tube, made of smooth muscle (Propulsion of urine towards the urinary bladder)
- vii. Urinary bladder
 - a. Made of smooth muscle, known as the Detrusor muscle
 - b. Can contain 500ml of urine, maximum of 700-800mL or urine will cause dribbling.
 - c. Innervated by Sacral plexus **S2-S3**
 - d. Micturition reflex – “Voiding” urge to pee. Activates within 200-300ml of urine, This can be avoided, Although if the reflex returns, the stimulation will be stronger (Guyton)
- viii. Urethra
 - a. Male: 20cm long Female: 3 – 4cm long Why are female are more prone to UTI?
 - b. Two types
 - i. Internal urethral sphincter (Involuntary), keeps urethra closed.
 - ii. External urethral sphincter (Voluntary) Which urethral sphincter can a person control?
- ix. Types of incontinence (My **Stress** is **Overflowing**! I have the **Urge** to Pee)
 - a. Overflow incontinence (Atonic bladder)
 - i. Inability of the bladder to contract or empty completely
 - ii. Causes: Lower SCI, fecal impaction
 - b. Stress incontinence
 - i. Normal bladder, Weakness of supporting structure (Pelvic floor muscles)
 - ii. There is an increase in abdominal pressure
 - iii. Causes: laughing, sneezing, lifting, exercises (Any increase in abdominal pressure)
 - c. Urge incontinence (Overactive bladder, Neurogenic bladder)
 - i. Involuntary contraction of the bladder
 - ii. Hyperactive micturition reflex
 - iii. Causes: bladder infection, bladder tumor, alcohol abuse, drugs.

BASIC KINESIOLOGY

- i. Kinesiology
 - a. Kinesiology – Study of motion/ movement
 - b. Biomechanics – Application of mechanics to the human body
 - i. Statics – Deals with bodies at rest
 - ii. Dynamics – Deals with bodies that either accelerate or decelerate
 - c. Kinetics – Forces that either produce or arrest motion
 - d. Kinematics – Science of motions of bodies in space
 - i. Osteokinematics – Movement of bone
 - ii. Arthrokinematics – Minute movements occurring in a joint.
- ii. Cardinal Planes – imaginary planes that lie perpendicular to each other
 - a. Frontal/ Coronal/ XY plane – Z axis, divides the body into anterior and posterior, parallel to coronal/ frontal suture.
 - i. Abduction and adduction of shoulder and hip
 - ii. Flexion and extension of the thumb
 - iii. Radial and ulnar deviation
 - iv. Trunk lateral flexion
 - b. Sagittal/ Vertical/ YZ plane – X – axis, divides the body into left and right. Parallel to the sagittal suture.
 - i. Flexion and extension of shoulder and hip
 - ii. Abduction and adduction of thumb
 - iii. Dorsiflexion and plantarflexion
 - c. Transverse/ Horizontal/ XZ plane – Y axis, divides the body into superior and inferior
 - i. Internal and external rotation of the shoulder and hip
 - ii. Inversion and eversion
 - iii. Pronation and supination

Z frontal ay **X**Sag kay Transverse**Y**
- iii. Anatomical position –

Position		
Head	In midline	S – standing erect H – head is in midline A - arms slightly abducted F – forearm supinated feet apart T – Toes pointing forward
Shoulder	Slightly abducted	
Forearm	Supinated	
Palms	Facing forward	
Standing	Erect	
Toes	pointing forward	

In fundamental position, the palms are facing towards the hip.

- iv. Classification of joint structures and function
 - a. Synarthroses
 - i. Immovable
 - ii. Very slight movement
 - iii. Fibrous
 - iv. Function:
 - 1. Stability
 - 2. Shock absorption
 - 3. Force transmission
 - v. E.g. Sutures, gomphosis
 - b. Amphiarthroses
 - i. Slightly moveable
 - ii. Cartilaginous
 - iii. Function:
 - 1. Stability and mobility
 - iv. E.g. Pubis symphysis, IV joint, Sternocostal joint
 - c. Diarthroses
 - i. Freely movable
 - ii. Synovial capsule
 - iii. Subdivision
 - 1. Nonaxial – planar movement (brunnstrom)
 - 2. Uniaxial – one plane, one axis, one degree of freedom
 - a. E.g. hinge joint, pivot joint/Trochoid joint
 - 3. Biaxial - two planes, two axis, two degrees of freedom
 - a. E.g. Condylod joint, Ellipsoid joint, Saddle joint
 - 4. Triaxial – Multiaxial, three degrees of freedom
 - a. Ball and socket
 - iv. Function:
 - 1. For mobility
- v. Arthrokinematics
 - a. Rolling or Rocking – no same point meets the same point. Multiple points on an object contact multiple surfaces on another object.
 - b. Sliding or Gliding – one point meets multiple surfaces.
 - c. Spinning – one point meets the same point
- vi. Levers, a simple machine that works using a rigid bar that rotates at a certain axis.
 - a. Pivot /Fulcrum/ Axis (A)
 - b. Weight/ Resistance (W)
 - c. Force/ Effort (F)
 - d. Mechanical Advantage- Force arm/ weight arm

Level	Center	MA	Function	E.g.
Class I	Axis	1	Balance, stability	AO joint, unilateral stance
Class II	Weight	>1	Power	Double/ bilateral stance
Class III	Force	<1	Speed	MC in the body

vii. Kinetics

- a. COG slightly anterior to S2, (4-5cm/ 2 inches) 55% of the body's height
- b. Specific COG
 - i. Head – Sphenoid sinus
 - ii. Neck – Basioccipital
 - iii. Trunk – anterior to T11 (HAT)
 - iv. Upper extremities – above the elbow joint
 - v. Arm – medial head of triceps
 - vi. Forearm – Pronator teres
 - vii. Hand – Third MCP
 - viii. Lower Extremities – above the knee joint
 - ix. Thigh – Adductor brevis or magnus
 - x. Leg: Popliteus
 - xi. Foot: 2nd MTT
- c. LOG
 - i. T – Odontoid process, opistropheus, odontoid peg
 - ii. A - AO joint
 - iii. T - Cervical
 - iv. A - Thoracic
 - v. T - Lumbar
 - vi. P -hip
 - vii. A- knee
 - viii. A – Ankle
- d. Types of equilibrium (August 1997)
 - i. Stable equilibrium – When the COG is displaced, the body will return to its former position.
 - ii. Unstable equilibrium – When the COG is displaced, the body will seek a new position
 - iii. Neutral equilibrium – the COG is neither displaced or moved E.g. patient in a wheelchair, COG remains constant with no displacement although moving.
- e. Forces acting on the body
 - i. Gravity
 - ii. Muscles
 - iii. External forces
 - iv. Friction

viii. Kinematics

- a. Kinematic chain – combination of several joint uniting successive segments.
 - i. OKC – Open kinematic chain
 - ii. CKC – Closed kinematic chain

	OKC	CKC
Distal	Moving	Fixed
Proximal	Fixed	Moving

Open packed	Closed packed
Can be distracted a few mm	Ligament farthest apart
Ligaments are slacked	Maximally congruent
	Ligaments are taut
	Difficult to distract

- b. Resting position – position of the joint with the least congruency.

Joint	OPP
TMJ	Slightly Open/ ajar
Tongue	Resting on the hard palate
Shoulder	55° abduction 30° hor. Add
HU	70° flexion 10° Sup
HR	Full extension
PRU	70° flexion 35° Sup
DRU	10° supination
Wrist	Slight ulnar deviation
Hip	30° flexion, 30° abduction, slight ER
Knee	25° flexion
Ankle	10° Plantarflexion

- ix. Muscle contraction types

- Concentric – Shortening
- Eccentric – Lengthening
- Isometric – No change in joint angle.
- Isotonic – Same tone
- Isokinetic – Same rate of movement/ speed

- x. Muscle types

- Agonist/ prime mover – principal muscle that performs the motion
- Antagonist – muscle that opposes the agonist, passively lengthens to allow joint motion.
- Synergists – acts as a stabilizer, prevents/ rules out unwanted/ unnecessary motions. Supports
- Fixator (snell)

- xi. Insufficiency, occurs in two jointed muscles

- Active insufficiency – maximal shortening of muscle
- Passive insufficiency – maximal lengthening of muscle

- xii. End feels – sensation felt by therapist when performing passive ROM to the patient.

Normal Endfeel	
Soft	Soft tissue approximation
Hard	Bone to bone contact
Firm	Capsular/ ligamentous stretch
Pathologic	
Empty	Restriction d/t pain

Shunt	Spurt
Insertion is far from the joint	Insertion is near the joint
Stabilizing	Mobilizing

- xiii. Miscellaneous
 - a. Types of Bones
 - i. Long bones – length > than its width.
 - 1. E.g. femur, humerus, tibia, ulna etc.
 - ii. Short bones – length is almost = to its width
 - 1. E.g. carpal, tarsals
 - iii. Sesamoid – sesame seed shape
 - 1. Pisiform, patella
 - iv. Irregular bones – complex shape
 - 1. Hip bone, vertebra
 - v. Flat bones – thin and sometimes curved bones
 - 1. Scapula, skull, sternum, ribs

AK HEAD, NECK, TMJ

- i. Topic outline
 - a. Anatomy and kinesiology
 - i. Head
 - ii. TMJ
 - iii. Neck
- ii. Skull
 - a. Division of the bones of the skull: Cranium or cranial bones and facial bones
 - b. Layers of skull bones
 - c. Joints of the skull
 - i. Fibrous synarthrodial sutures
 - d. Connective tissues
 - i. **Sutural ligament** – CT connecting the bones together.
 - ii. The type of joint that will articulate the joints of the skull?
 - iii. Structural: Fibrous, Functional: synarthrodial
 - e. Between the bones
- iii. Three layers of the skull bones
 - a. External table (Compact bone)
 - b. Diploe (Trabecular or spongy bone)
 - c. Internal table (Compact bone)
- iv. Cranial bones (8) *What are the paired cranial bones?*
 - a. 1 frontal
 - b. 2 parietal bones
 - c. 1 occipital bone
 - d. 2 temporal bones
 - e. 1 sphenoid bone
 - f. 1 Ethmoid bone

The thinnest/softest part of the lateral part of the skull? Pterion

What artery and vein lie on top of this area? Anterior division middle meningeal artery
- v. Neonatal skull
 - a. Fontanelles
 - i. Anterior(frontal) 18 months
 1. Anterolateral sphenoid
 - ii. Posterior(occipital) end of the 1st year (12 months)
 1. Posterolateral (mastoid)
 - iii. At what age of the age will the fontanelles close
 - iv. Function:
 1. Allow the growth of the brain, allows the fetal skin to change in size
 2. If early close – microcephaly> brings out mental retardation
 3. Serves as a landmark for withdrawal of blood

- v. Layers of SCALP
 1. Skin – hair bearing, area where sebaceous glands contained.
Where do sebaceous cysts occur?
 2. Connective tissue
 - a. Fibrofatty layer, connects the skin to next layer
 - b. Provide passageway for nerves and blood vessels
 3. Aponeurosis: Provides an insertion site of the occipito frontalis
 4. Loose areolar tissue: Allow the superficial 3 layers of the skull to move over the bone/ pericranium
 5. Pericranium: Periosteum of the skull bones
- vi. Facial bones (14)
 - a. Paired
 - i. Nasal
 - ii. Lacrimal
 - iii. Zygomatic bone
 - iv. Maxillae
 - v. Inferior nasal conchae
 - vi. palatine
 - b. Unpaired
 - i. Mandible
 - ii. Vomer
 - c. Joined together by sutures, serves as the attachment of facial muscles.
 - d. Only movable joint in the skull? TMJ
- vii. Muscles of face
 - a. Occipitofrontalis (occipital and frontal bellies) raise the eyebrows, surprise
 - b. Orbicularis oculi (palpebral and orbital parts)
 - i. Palpebral – soft closure & dilates the lacrimal sac
 - ii. Orbital – forceful closure
 - c. Corrugator supercilli: Draws the eyebrows together, frowning
 - d. Nasalis
 - i. Compressor nasi: Compresses the aperture of the nose
 - ii. Dilator nares: Widens the aperture of the nose
 - e. Procerus: Wrinkling of the nose
 - f. Orbicularis oris
 - i. Surrounds the mouth
 - ii. Compresses the lips together, kissing
 - g. Levator labii superioris alaeque nasi
 - h. Levator labii superioris: Sneering, elevate lip
 - i. Zygomaticus minor: Protrude the upperlip
 - j. Zygomaticus major: Draws the angle of the lip sideways, smiling
 - k. Levator anguli oris: Lift angle of the outh sideways
 - l. Risorius: Sideward drawing of lips, grimace
 - m. Depressor anguli oris: Depress the angle of the mouth
 - n. Depressor labii inferioris: Depress lips

- o. Mentalis: Protrude chin forward, pouting
 - p. Buccinator: Compress the cheek inward, blowing
 - q. Platysma: Draw the buccal part of the mouth downward
 - r. *Severity of CNVII is Bells palsy*
 - viii. TMJ
 - a. Located between the condyle of the mandible, and the mandibular fossa of the temporal bone
 - b. Condylod joint
 - i. Movement: protrusion, retrusion, depression, elevation, lateral excursion
 - c. Arthrokinematics of mouth opening, tmj depression
 - i. Early phase
 - 1. 35-50%
 - 2. Predominant movement, **rolling** towards the posterior side.
 - ii. Late phase
 - 1. 50-65%
 - 2. Translation **anterior** and **inferior**
 - d. Functional Normal mouth opening 40mm, 25mm of rotation, 15mm translator glide
 - ix. Muscles of mastication
 - a. Primary MCLO
 - i. Temporalis
 - ii. Masseter
 - iii. Medial Pterygoid
 - iv. Lateral Pterygoid – only muscle for opening
 - b. Secondary
 - i. Digastric
 - ii. Geniohyoid
 - iii. Suprahyoid
 - iv. Mylohyoid
 - v. Infrahyoid
 - c. Retrusion: Temporalis, only muscle that will retrude the mandible
 - d. Protrusion: Lateral and medial pterygoid, masseter
 - e. Lateral excursion
 - i. Ipsilateral
 - 1. Temporalis
 - 2. masseter
 - ii. Contralateral
 - 1. Medial and lateral pterygoid(inferior head)
- Specific part of the lateral pterygoid that will depress the mandible*

x. Openings of the skull

Opening of the skull	Bone of the skull	Structures transmitted
Anterior cranial fossa		
Perforations at the Cribriform plate	Ethmoid	Olfactory nerves
Middle cranial fossa		
Optic canal	Lesser wing of sphenoid	Optic nerve, ophthalmic artery
Superior orbital fissure	Between lesser and greater wings of sphenoid	Lacrimal, frontal, trochlear, oculomotor, nasociliary and abducent nerves, superior ophthalmic vein
Foramen Rotundum	Greater wing of sphenoid	Maxillary division of the trigeminal nerve
Foramen Ovale	Greater wing of sphenoid	Mandibular division of the trigeminal nerve, lesser petrosal nerve
Foramen Spinosum	Greater wing of sphenoid	Middle meningeal artery
Foramen lacerum	Between petrous part of temporal and sphenoid	Internal carotid artery
Posterior cranial fossa		
Foramen magnum	Occipital	Medulla oblongata, spinal part accessory nerve, and right and left vertebral artery
Hypoglossal canal	Occipital	Hypoglossal nerve
Jugular foramen	Between petrous part of temporal and condylar part of occipital	Glossopharyngeal, vagus, and accessory nerves, sigmoid sinus becomes internal jugular vein
Internal acoustic meatus	Petrous part of temporal bone	Vestibulocochlear and facial nerve

xi. Bones of the neck

a. 7 cervical bones

- i. Has 3 atypical (atlas, axis, C7)
- ii. Uncinate process (Joints of von Lushka, uncovertebral joint)
 1. C3-C7; Limit lateral flexion

b. Features

- i. Body – small and broad from side to side
- ii. Spine – small and bifid
- iii. Vertebral foramen – large and triangular
- iv. SAP – Posterior and superior
- v. IAP – anterior and inferior
- vi. Others- transverses processes possess a foramen transversarium (for passageway of vertebral artery)

c. C1

- i. Two large lateral masses joined by anterior and posterior masses
- ii. No body, pedicle, lamina or spinous process

d. C2

- i. Has a large, tall body that serves as a base for the upwardly projecting odontoid process or dens

e. C7 (Vertebral prominens)

- i. Spine is not bifid, is long and slender similar that of the thoracic vertebra.

xii. Ligaments

- a. Anterior atlanto-occipital membrane
 - i. Continuation of the ALL, which runs as a band down the anterior surface of the vertebral column. The membrane connects the anterior arch of the atlas to the anterior margin of the foramen magnum
 - ii. Limits extension
- b. Posterior atlanto-occipital membrane
 - i. Similar to the ligamentum flavum and connects the posterior arch of the atlas to the posterior margin of the foramen magnum.
 - ii. Limits flexion

C1-C2

- c. Apical ligament
 - i. Median placed structure connects the apex of the odontoid process to the anterior margin of the foramen magnum
- d. Alar ligaments
 - i. These lie one on each side of the apical ligament and connects the odontoid process of the medial sides of the occipital condyles
- e. Cruciate ligament:
 - i. This ligament consists of a transverse part and vertical part. The transverse part is attached on each side to the inner aspect of the lateral mass of the atlas and bind the odontoid process of that anterior arch of the atlas. The vertical part runs from the posterior surface of the body to the axis to the anterior margin of the foramen magnum.
- f. Membrane tectoria
 - i. This is an upward continuation of the PL. It is attached above the occipital bone just within the foramen magnum. It covers the posterior surface of the odontoid process and the apical and cruciate ligaments.

Below C2

- g. ALL
 - i. Runs continuous from anterior surface of the vertebral column from skull to sacrum. The ALL is wide and is strongly attached to the front and sides of the vertebral bodies and to the IVD
- h. PLL
 - i. Runs continuous band down the posterior surfaces of the vertebral column from the skull to the sacrum. PLL is weak and narrow and is attached to the posterior borders of the discs
- i. Ligamentum nuchae continuation:
 - i. Supraspinous ligament: Runs between the tips of adjacent spine
 - ii. Interspinous: This connects the adjacent spine
- j. Intertransverse ligaments: Run between adjacent transverse processes, limits lateral flexion
- k. Ligamentum flavum: Connects the laminae of the adjacent vertebrae. Continuation of tectorial membrane

xiii. Joints of the neck

- a. Intervertebral joints
- b. Zygapophyseal joints
 - i. Facets are oriented 45° to the transverse plane/ frontal plane
 - ii. Inferior facets face Anteriorly and inferiorly a / p
 - iii. Superior facets face posteriorly and superiorly a \ p
- c. Arthrokinematics

	C0-C1	C1-C2	C2-C7
Extension	Roll Posteriorly Glide Anterior	C1 tilts posteriorly on C2	Body of C3 tilts posteriorly on C4 IAF of C3 will slide posterior and inferiorly on SAF of C4
Flexion	Roll Anterior Glide Posterior	C1 tilts anterior on C2	Body of C3 tilts anteriorly on C4 IAF of C3 will slide anterior and superior on SAF of C4
Lateral	Roll Right Glide Left	Negligible	
Rotation	Negligible		Body of C3 rotates of C4 Rotation to the right: The (L) IAF of C3 will slide superior and anteriorly on IAF of C4, and the ® IAF C3 will slide inferiorly and posteriorly on SAF of C4

xiv. Muscles of the neck

- a. Platysma
 - i. Depresses mandible and angle of the mouth
 - ii. Depressure the buccal area of the mouth
- b. SCM
 - i. Bilateral: sagittal plan flexion
 - ii. Unilateral: ipsilateral flexion and contralateral rotation
- c. Suprahyoid muscles
 - i. Digastric (anterior/ posterior belly): Depress mandible, elevates hyoid
 - ii. Geniohyoid
 - 1. Elevates hyoid bone
 - 2. Depress mandible
 - iii. Stylohyoid: Elevates hyoid bone
 - iv. Mylohyoid
 - 1. Depresses mandible
 - 2. Elevates hyoid bone and floor of the mouth
- d. Infrahyoid muscles
 - i. Thyrohyoid: Depresses hyoid bone or elevates larynx
 - ii. Omohyoid: Depresses hyoid bone
 - iii. Sternohyoid: Depress hyoid bone
 - iv. Sternothyroid : Depress larynx

- e. Anterior (CL) and middle scalene
 - i. Elevates the first rib
 - ii. Laterally flex and rotate the neck
- f. Posterior scalene
 - i. Elevates second rib
 - ii. Laterally flexes and rotates neck

Muscle	Ipsilateral	Contralateral
Rectus capitis Anterior	C0-C1 flexion (head) Head stabilizer	
Rectus capitis Lateralis	Head stabilizer and for proprioceptive feedback	Head ipsilateral lateral flexion
Longus capitis	Head flexion	Head ipsilateral lateral flexion and ipsilateral rotation
Longus colli	Head flexion	Neck flexion
SCM	Neck flexion	Neck ipsilateral lateral flexion neck contralateral rotation Extension of head and neck
Scalenes ant/ med/ post	neck flexion	Neck ipsilateral lateral flexion and ipsilateral rotation
Ant scalene only	Neck Rotation	Neck Contralateral rotation

Muscle	Bilateral contraction	Unilateral contraction
Rectus capitis posterior	Head extension	Head ipsilateral rotation
Splenius capitis and cervicis head	neck extension	Head and neck ipsi rotation
Obliquus capitis superior	Head extension	
Obliquus capitis inferior		Head ipsilateral rotation

Triangles of the neck

Boundaries

- i. Anterior: Posterior border of the SCM
- ii. Posterior: Anterior border of trapezius
- iii. Inferior: middle third of the clavicle
- iv. Floor: Semispinalis capitis, splenius capitis, levator scapulae, and scalenus medius

Subdivision of posterior triangle

- i. Occipital triangle
 - a. Anterior – posterior border of SCM
 - b. Posterior - anterior border of trapezius
 - c. Inferior - inferior belly of omohyoid
- ii. Supraclavicular triangle/ subclavian triangle
 - a. Anterior – posterior border of SCM
 - b. Superior – inferior belly of omohyoid
 - c. Inferior – clavicle

- iii. Posterior triangle contents
 - a. Arteries: Subclavian artery (3rd part), superficial cervical artery, suprascapular artery, occipital artery
 - b. Veins: External jugular vein, and its tributaries, subclavian vein
 - c. Nerves: Brachial plexus, spinal apart of accessory nerve, branch of cervical plexus

Anterior Triangle

- i. Boundaries
 - a. Anterior – midline of the neck
 - b. Posterior anterior border of SCM
 - c. Superior lower margin of the mandible

Subdivision of anterior triangle

- i. Submental triangle/ suprahyoid triangle
 - a. Anterior – midline of the neck
 - b. Lateral anterior belly of digastrics
 - c. Inferior – body of the hyoid bone
 - d. Floor – mylohyoid muscle
 - e. Contents – submental lymph nodes, anterior jugular vein
- ii. Submandibular triangle
 - a. Anterior – anterior belly of digastrics
 - b. Posterior – post. Belly of digastric, stylohyoid muscle
 - c. Superior – lower border of body of mandible
 - d. Floor – mylohyoid and hyoglossus muscle
 - e. Contents: submandibular salivary gland, facial artery/ vein, submandibular lymph node, hypoglossal nerve, vessels of mylohyoid muscle, carotid sheath, stylopharyngeal muscle, glossopharyngeal nerve, parotid gland
- iii. Carotid Triangle
 - a. Superior – posterior belly of digastrics
 - b. Inferior – superior belly of the omohyoid
 - c. Posterior – anterior border of the SCM muscle
 - d. Floor – part of the thyrohyoid, hyoglossus, and constrictor muscle of pharynx
 - e. Content: carotid sheath
- iv. Muscular triangle
 - a. Anterior – midline of the neck
 - b. Superior – superior belly of the omohyoid
 - c. Inferior – anterior border of the SCM
 - d. Floor – sternohyoid and sternothyroid muscle
 - e. Contents – thyroid gland, larynx, trachea, esophagus

Main blood vessels of the neck

Arteries	Vein
Common carotid artery	External, anterior, internal jugular vein
- External carotid artery	
- Internal carotid artery	

AK BACK AND SPINE

- i. Back and Spine
 - a. Functions
 - i. S – supports the head
 - ii. I – important attachment of muscles and ligaments
 - iii. R – Ribs attachment
 - iv. E – encloses and protects the spinal cord
 - v. N – nag provide ng mobility and flexibility of the trunk
 - vi. A – ability to transfer weight the head and body to the lower extremities
- ii. Vertebral spine
 - i. Length of the spinal cord
 - 1. Male: 72cm Female: 62cm

- iii. Vertebrae

	Infant	Adult
Cervical	7	7
Thoracic	12	12
Lumbar	5	5
Sacral	5	1
Coccyx	4	1
Total	33	26

- iv. Curves of the spine
 - i. 1° curve = Kyphosis
 - ii. 2° curve = Lordosis
 - 1. Cervical: 2-3 months upon lifting head in prone
 - 2. Lumbar: 10-12 months upon standing and walking
- v. Parts of a vertebra
 - a. Body
 - b. Transverse process
 - c. Spinous process
 - d. Vertebral foramen
 - e. Pedicle, anterior : Connects transverse process to body
 - f. Lamina, posterior: Connects the spinous process to transverse process
 - g. Vertebral arch, zygapophyseal joint, facet joint: Area between pedicle and lamina
- vi. Typical vertebra
 - a. Cervical, C3-C6 Typical (2x)
 - b. Thoracic (+2) T5-T8 Typical
 - c. Lumbar (+4) L1-L4 Typical

	Cervical	Thoracic	Lumbar
Vertebral body	Small	Moderate sized, heart shaped	Large/ massive, kidney shaped
Spinous process	Bifid spinous process, short	Long and inclined downward/ almost vertical	Short, Thick, Flat, quadrilateral
Transverse process	Transverse foramen	Costal Facets/ pits	Conical shape
Spinal canal/ Vertebral foramen	Largest and triangular	Smallest and circular	Triangular
Facets	SAF: Posterior/ Superior IAF: Anterior/ Inferior	SAF: Posterior/ Lateral IAF: Anterior/ Medial	SAF: Medial IAF: Lateral
Angle	45°	60°	90°

Structures passing through the transverse foramen? Vertebral artery and vein. Sympathetic Nerves.

vii. Atypical Vertebra

a. Cervical

i. C1, ATLAS

1. No vertebral body
2. No spinous process

ii. C2, AXIS

1. Dens/ Odontoid process
2. Short transverse process
3. Large, bifid spinous process

iii. C7, VERTEBRAL PROMINENS: Long non-bifid spinous process

b. Thoracic

i. T1: Horizontal spinous process

ii. T2-T4

1. Almost horizontal spinous process
2. Costal facets at the superior edge of the vertebral body

iii. T9: (+) Tubercles

iv. T10

1. (+) Tubercles
2. Costal facets at the outer/lateral part of the vertebral body

v. T11-T12

1. (+) Tubercles
2. Costal facets at the outer part of the pedicles

c. Lumbar

i. L5: Stout Transverse process

viii. 3 Joint complex

- a. Adjacent vertebral bodies and IV discs (1)
- b. Paired Superior facets and paired inferior facets (between pedicle and lamina) (2)

ix. 5 Joint complex (Cervical uncovertebral joints aka Joints of Von Luschka)

- a. Vertebral body of superior vertebra and paired base of pedicles of inferior vertebra

- x. Joints
 - a. Atlanto-occipital joints (A-O joint): Yes joint
 - b. Atlanto-Axial joints (A – A joint): No Joint
- xi. Important landmarks
 - a. Cervical
 - i. Hyoid C3
 - ii. Thyroid C4
 - iii. Cricoid C6
 - 1. Junction of the larynx and trachea
 - 2. Junction of the pharynx and esophagus
 - 3. Middle cervical sympathetic ganglion
 - 4. Laryngeal Prominens (aka Adams Apple)
 - 5. First ring of trachea
 - 6. Suprasternal notch
 - iv. Back of neck – suboccipital region
 - b. Thoracic
 - i. T2 spine = superior angle
 - ii. T3 spine = scapular spine
 - iii. T7 spine = inferior angle
 - iv. T10 = Xiphoid process (Not dermatomal level, w/c is T6)
 - c. Lumbar
 - i. L4 – iliac crest
 - ii. L5 – tubercle of iliac crest
 - iii. S2 – SI and PSIS
- xii. IV Discs
 - a. % of vertebral height: 25% of the vertebral height, 1:4
 - i. Annulus Fibrosus (Outer)
 - 1. Contains collagen (elasticity) and protein (lamellae, angled at 30°)
 - 2. *Sharpey's fibers* – attached to outside part of A.F.
 - 3. Hoop's stretch (Stretching of annulus fibrosus)
 - ii. Nucleus Pulposus (Inner)
 - 1. Contains collagen, water, and protein (Proteoglycans, strong affinity with water)
 - 2. % of H₂O 80-90%, through aging this amount decreases.
 - b. Pressure
 - i. Supine 25%
 - ii. Sidelying 75%
 - iii. Standing upright 100%
 - iv. Sitting upright 140%
 - v. Standing + bending forward 160%
 - vi. Sitting + bending forward 185%
 - vii. Standing + bending forward +weights 220%
 - viii. Sitting + bending forward + weights 275%
- Common direction of herniation
 - ix. Lumbar – posterolateral

xiii. Biomechanics

Biomechanics	Vertebral body	Spinous process	IV Discs	Foramina	Facets
Flexion	Approximates	Separates	Flattens (Ant) Pushed (Post)	Opens	Opens, anterosuperior movement
Extension	Separates	Approximates	Flattens (Post) Pushed (Ant)	Closes	Closes, Posteroinferior
Lateral Flexion	Approximates(Ipsi) Separates(Contra)	-	Flattens (Ipsi) Pushed (Contra)	Closes (Ipsi) Opens (Contra)	Closes (Ipsi) Opens (Contra)
Rotation	Rotates (ipsi)	Rotates (Contra)	At its weakest	Closes (Ipsi) Opens (Contra)	Closes (Ipsi) Opens (Contra)
Distraction	Separates	-	-	-	-
Compression	Approximates	-	-	-	-
Shearing	Sliding ant, post, lat	-	-	-	-

Coupling motions: Lateral flexion is always coupled with rotation, vice versa.

xiv. ROM of spine

Motion	AO joint	AA joint	C3-C7	Thoracic	Lumbar
Flexion	10°	5°	45°	15°	40°
Extension	25°	10°	45°	15°	25
Lateral flexion	L & R 5°	L & R 10°	L & R 30°	L & R 15°	L & R 20°
Rotation	L & R 0°	L & R 45°	L & R 30°	L & R 40°	L & R 5°
	50% flex/ ext	50% rot			

xv. Ligaments

a. Intersegmental lig

i. Connects many vertebrae

1. ALL,

- Connects that anterior aspect of vertebral body of many vertebrae
- Taut: extension
- Lax: flexion

2. PLL- connects the posterior aspect of the vertebral body of the many vertebrae

- Taut: flexion
- Lax: extension

3. Supraspinous ligament – connects the tip of spinous process of many vertebrae (C7 to sacrum)

- Taut: flexion
- Lax: extension

b. Intrasegmental lig

i. Connects individual vertebra

1. Ligamentum flavum/flava – connects the laminae of adjacent vertebrae

- Taut: flexion
- Lax: Extension

2. Ligamentum nuchae – connects the tip of spinous process of adjacent vertebrae (Occiput to C7)
 - a. Continuation of supraspinous ligament, in the neck region
 - b. Taut: flexion
 - c. Lax: Extension
 3. Interspinous lig – connects the adjacent spinous processes
 - a. Taut: flexion
 - b. Lax: Extension
 4. Intertransverse lig – connects the adjacent transverse processes
 - a. Taut: lateral flexion contralateral
 - b. Lax: lateral flexion ipsilateral
 - c. Other ligaments
 - i. Alar ligaments – “wing” controls the skull and atlas rotation over the axis
 - ii. Transverse ligament – holds the dens against the atlas
 1. Jefferson’s fracture – fracture of C1
 2. Hangmans fracture – fracture of C2
 - iii. Cruciform Lig – vertical ligament over the transverse ligament, limits rotation of skull and atlas.
 - iv. Accessory Atlanto-Axial ligament – (AAA lig) Functions similar to alar ligaments.
- xvi. Muscles of the back and spine
- a. Superficial muscles (TRaLaLA)
 - i. Trapezius (Upper, middle, lower) , shawl muscle
 - ii. Rhomboids –(Minor, Major)
 - iii. Levator Scapulae
 - iv. Latissimus Dorsi – broadest muscle “ crutch walking muscle:
 - b. Intermediate
 - i. Serratus Posterior
 1. Superior – forced inspiration
 2. Inferior – forced expiration

- c. Deep
 - i. Splenius cervicis, capitis (Bondage muscle)
 - ii. Erector spinae
 - 1. I - Iliocostalis
 - 2. L - longissimus
 - 3. S – spinalis
 - iii. Transversospinalis (oblique muscles of the back)
 - 1. S - Semispinalis
 - 2. M - Multifidus
 - 3. R – Rotatores
 - iv. Interspinalis muscle/ adjacent c interspinous process
 - v. Intertransversarii / adjacent c transverse process
 - vi. Levatores Costarum
 - vii. Suboccipital region
- d. Muscles of anterior trunk
 - i. Pectoralis Major and minor
 - ii. Rectus abdominis
 - iii. Transversus abdominis
 - iv. Obliques
 - 1. External oblique
 - 2. Internal oblique

HNBS CONDITIONS

- i. Torticollis (aka wry neck)
 - a. Affected muscle: SCM
- ii. Whiplash injury (Traumatic)
 - a. MOI: hyperextension followed by hyperflexion
 - b. Rear end collision
 - c. Acceleration- Deceleration injury
 - d. Ligament affected: ALL
 - e. S/Sx: Neck pain, tenderness, LOM, headache, dizziness, straightened cervical lordosis, radicular signs within 2 weeks.
 - f. Special Test:
 - i. (-) Spurling's Test, sitting position, lateral flexion + application of compression. Sx ipsilateral side.
 - ii. (-) Lhermitte's: Long sitting, PT passively flexes the head electrical sensation down the spine.
- iii. Spondylosis
 - a. OA of the cervical spine
 - b. Decrease in joint space, causes friction in joint space, formation of osteophytes (bone spurs)
 - c. S/Sx: neck pain
 - d. (+) Spurling's Test, d/t decrease in foramen space
 - e. (+) Lhermitte's Test
- iv. Different abnormal postures

Normal cervical lordosis	20-40°
Normal thoracic kyphosis	20-40°
Normal lumbar lordosis	30-50°

 - a. Forward head posture
 - i. Increased extension of AO joint, and upper cervical vertebrae
 - ii. Increase flexion at lower cervical and upper thoracic
 - iii. Mandible retruded
 - b. Flat neck
 - i. Exaggerated military posture
 - ii. Increase in flexion of AO joint
 - iii. Decrease in cervical lordosis, <20°
 - iv. Mandible protrusion
 - c. Postural Kyphosis/ Round back (can be corrected)
 - i. Increase in thoracic kyphosis >40°
 - ii. Protracted scapulae
 - iii. Forward head posture

- d. Structural Kyphosis
 - i. Congenital Kyphosis
 - 1. Present at birth
 - 2. E.g. failure of formation of the anterior vertebra, failure of segmentation of anterior aspect of vertebrae,
 - ii. Osteoporosis
 - 1. Decrease in bone mass per unit volume, decreased bone density
 - 2. First affects Cancellous (spongy) bone > compact bone
 - 3. Types:
 - a. Primary
 - i. Type 1 - Post menopausal osteoporosis
 - 1. Estrogen affected
 - b. Secondary
 - i. Type 2 – Senile (aging)
 - ii. Type 3 – Endocrine d/o, malnutrition
 - iii. Gibbus Deformity
 - 1. Collapse of a single vertebra
 - 2. (+) Dowager's hump: E.g. fracture, Pott's disease
 - iv. Scheuermann's Disease
 - 1. Other Names:
 - a. JDD, Juvenile Discogenic Disease
 - b. Juvenile Kyphosis
 - c. Adolescent Kyphosis
 - d. Vertebral Epiphysitis
 - e. Osteochondrosis of the spine
 - 2. Anterior vertebral wedging of at least 5° of 3 consecutive vertebrae
 - 3. Age: 12-16 years old, F>M
- v. Flat Upper back
 - a. Decreased thoracic kyphosis < 20°
 - b. Depressed clavicles
 - c. Depressed scapulae
 - d. Flat neck posture
- vi. Lordotic posture
 - a. Increased lumbar lordosis >50°
 - b. Anterior pelvic tilt/ counter nutation
 - c. Increase in LS angle
 - d. Increase in hip flexion
 - e. Decreased thoracic kyphosis
 - (n) lumbosacral angle 30°

- vii. Flat back
 - a. Decrease lumbar lordosis
 - b. Increase posterior pelvic tilt, nutation
 - c. Decrease in LS angle, $<30^\circ$
 - d. Increase in hip extension
 - e. Increase thoracic kyphosis
- viii. Spondylolysis
 - a. Defect of the pars interarticularis
 - b. Scotty dog with collar
- ix. Spondylolisthesis
 - a. Anterior slippage of the superior vertebra over the inferior vertebra
 - b. Scotty dog decapitated
- x. Grading of Spondylolisthesis

Grade:	% of slippage
Grade I	$<25\%$
Grade II	25-50%
Grade III	50-75%
Grade IV	$>75\%$
Grade V	100% Spondyloptosis

- xi. Classification of Spondylolisthesis (Wiltse) DID TRAP

Type	Cause
Dysplastic	Congenital abnormality of the pars
Isthmic	1 st , MC type, lack of normal continuity of the pars interarticularis
Degenerative	Degeneration of the pars interarticularis
Traumatic	Fracture other than the isthmus, E.g. pedicle, lamina
Pathologic	Bony lesions

- xii. Isthmic types
 - a. Lytic
 - b. Elongation but intact parts
 - c. Acute fracture
- xiii. Retrolisthesis: Backward slippage of vertebra
- xiv. Ankylosing Spondylitis (ASE)
 - a. Other names
 - i. Marie strumpells
 - ii. Bechterev's disease
 - b. Best position to promote: extension, position to avoid: flexion
 - c. Schober's test, T7 – S2 (n) $>5\text{cm}$, $<5\text{cm}$ indicates stiffness
 - d. Best exercise: swimming, to promote extension of the spine
 - e. X-ray findings: Bamboo spine

- xv. Spinal Stenosis (STF)
 - a. Narrowing of the spinal canal/ vertebral foramen
 - b. (n) diameter of spinal canal: 17mm
 - c. Diameter of spinal cord: 10mm
 - d. Cause: ligamentum hypertrophy, Facet joint hypertrophy, Disc herniation
 - e. Best position to promote: flexion, avoid extension
 - f. Types
 - i. Relative stenosis spinal canal <12mm
 - ii. Absolute stenosis spinal canal <10mm
- xvi. Spina Bifida
 - a. Failure of closure of the neural arch
 - b. Types
 - i. Spina bifida occulta
 - 1. Tuft of hair
 - 2. Dimple
 - ii. Spina bifida cystica
 - 1. Meningocele: Meninges and CSF protrudes
 - 2. Meningomyelocele: Meninges and CSF + SC protrudes
- xvii. Tzietze's syndrome
 - a. Costal chondritis
 - b. Inflammation of the costal cartilages
 - c. 12 pairs of ribs
 - i. True ribs (1-7)
 - ii. False ribs (8-10)
 - iii. Floating ribs (11-12)
- xviii. Coccygodynia: Painful coccyx and lower part of sacrum
- xix. TMJ
 - a. Normal mouth opening 35-50mm
 - b. Mouth opening 1st 26mm, will translate anteriorly
 - c. Dislocation, common direction of dislocation = anterior
 - d. (+) click mouth opening, TMJ disc displacement with reduction click
 - e. (+) click mouth closing, TMJ displacement with displacement click only
 - f. (+) pain upon mouth opening, TMJ capsulitis
 - g. (+) pain : " ", (+) swelling = TMJ synovitis
- xx. LBP
 - a. Back pain sources
 - i. Viscerogenic
 - 1. Problem c visceral organs
 - 2. Cannot isolate pain location
 - 3. E.g. Kidney problems

- ii. Vasculogenic: Problem c blood vessels
 - iii. Psychogenic: Hysterical Pt's
 - iv. Neurogenic: Nerve root compression
 - v. Spondylogenic: Bony deformities
 - vi. Mechanical/ Postural LBP: Faulty posture/ biomechanics
- b. Disc Herniation
 - i. Centralization – pain on lower back only
 - ii. Peripheralization – Radiating down to the lower extremity
 - iii. McKenzie if there is disc herniation, try to check to see if there is peripheralization by prone on hands position
- c. Bias
 - i. Extension bias, place the patient into extension
 - 1. Supine without pillow
 - 2. Prone without pillow
 - ii. Flexion bias
 - 1. Supine with pillow under popliteal area
 - 2. Prone with pillow under the abdomen, anterior hips.
- xxi. Scoliosis:
 - a. Lateral curvature of the spine with vertebral rotation
 - b. Types
 - i. Structural/ Nonfunctional
 - 1. Irreversible curvature
 - 2. 3 types (NOI)
 - a. Neuromuscular – CP, SCI, PMD, Polio
 - b. Osteopathic – Hemivertebra, failure of segmentation of side.
 - c. Idiopathic – Unknown
 - i. Infantile 1st 3 years
 - ii. Juvenile 4-14 years old
 - iii. Adolescent 10-16 years old
 - ii. Nonstructural/ functional/ postural/ positional
 - 1. Reversible curvature, can be corrected
 - 2. Due to:
 - a. Faulty/ Habitual posture
 - b. Presence of muscle guarding
 - c. Leg length discrepancy
 - d. Hip dislocation

- c. Terms:
- i. Primary curve: the first curve of appear
 - ii. Secondary curve: the next curve to appear
 1. Types:
 - a. Compensated, if primary curve was compensated
 - b. Decompensated
 - iii. Major curve: largest curve
 - iv. Minor curve: Smallest curve
 - v. Apical vertebra: farthest from the midline, most rotated but the least tilted.
 - vi. End vertebral – most tilted
 1. Sup EV
 2. Inf EV
 - vii. MC Right (Dextro), Left (Levo)
- d. Methods of measuring
- i. Cobb's method (MC)
 1. SEV and IEV drawn a line and measure the angle intersecting.
 2. Draw a line perpendicular to each line.
 - ii. Risser Ferguson
 1. Draw line pass through each vertebra.
- e. Severity of scoliosis
- i. Mild: $<20^\circ$
 - ii. Moderate: $20-40^\circ$
 - iii. Severe: $>40^\circ$
 1. $>40-50^\circ$ associated with pain and DJD of the spine
 2. $>60-70^\circ$ cardiopulmonary changes, decreased life expectancy
- f. Curves of scoliosis

Convex	Concave
High shoulder Posterior rib hump Prominent Scapula Vertebral body rotates on this side	High Pelvis (+) anterior rib bump Spinous process rotates on this side

Grading

- g. Nash- moe (pedicle) method
- i. The vertebral body is divided into 6 segments
 - ii. No rotation 0
 - iii. Going to midline + 1
 - iv. 2/3 to midline++ 2
 - v. At midline +++ 3
 - vi. Beyond midline ++++ 4

0	0	0	0	0	0	0
				0		+1, +
			0			+2, ++
	0					+3, +++
0						+4, ++++

- h. Rods
 - i. Posterior approach
 - 1. Harrington rods, sublaminar wire, MC use
 - 2. Luge sublaminar wire
 - 3. Cotrel –Dubousete
 - a. Use pedicle screws
 - ii. Anterior approach
 - 1. Thoracotomy, most effective
- xxii. Orthoses
 - a. Apex above T6 Milwaukee, or @ T6 level
 - b. Apex below T6 Miami T7 & T9
 - c. Apex below T8 Boston, Wilmington, Yamamoto

ST: HEAD, SPINE, & PELVIS

i. Head

a. For increasing intracranial pressure

- i. Compression of neural structures
- ii. (n) ICP 4-15mmHg, Red flag >20mmHg, neurological impairment >40mmHg
- iii. S/Sx:
 1. Change in LOC, irritable, confused, lethargic, coma
 2. Change in VS: increase BP, Temp Decrease HR/PR irregular RR
 3. Pounding headache
 4. Vomiting (projectile), CN X irritation
 5. Unequal Pupils CN III

iv. Special Test

1. Romberg Test

- a. Pt standing straight feet together, eyes open first, and eye closing after.
- b. Held at least **20 seconds**,
- c. Eyes closed: (+) if there is postural sway, out of balance (Sensory ataxia)
- d. Eyes open: Cerebellar ataxia, not a positive test for Rombergs

Lobes	Function
Paleo/ Anterior/ spinocerebellum	Postural Reflexes/ Muscle tone
Archi/ flocculonodular/ vestibule	Balance
Neo/ Posterior/ cerebrocerebellum	Coordination

e. Sharpened Romberg test: Tandem position

2. Walk or stand tandem test

- a. Walk in tandem position, (+) postural sway, out of balance

b. For coordination, ability to execute tasks smoothly accurately and controlled.

i. PCML, cerebellum, basal ganglia(amplitude of movements), cortex

1. Finger to nose
2. Finger to thumb
3. Hand flip test
4. Pronation –supination
5. Finger drumming test
6. Hand thigh test
7. Past pointing (index fingers 6 inches apart), eyes open first, then eyes closed.
8. Heel to knee test
 - a. (+) test: Dymetria, tremor, difficulty
 - b. Indication: Incoordination 2° to damage cortex, PCML, cerebellum, basal ganglia

- c. For proprioception
 - i. Position sense without visual input
 - ii. (+) test: imbalance from posture, out of balance of the patient, difficulty upon performing the test.
 - iii. Indication: dysfunction of proprioceptive centers (PCML)
 - iv. Tests:
 - 1. Proprioceptive finger- nose test
 - 2. Proprioceptive movement test
 - a. Short sitting, legs dangling, eyes closed, hold the toes at the sides and move up or down and ask the patient the position of the toe.
 - 3. Past pointing
- d. For the Ear
 - i. 3 sections of the ear: outer, middle, inner
 - 1. Outer ear:
 - a. pinna/ auricle
 - b. Ear canal
 - c. Outer portion of the tympanic membrane/ ear drum
 - d. Function: Mechanical reception of sound from the external environment
 - 2. Middle ear:
 - a. Ossicles (maleus, incus, stapes)
 - b. Opening of Eustachian tube, (modifies pressure)
 - c. Function: Conduction of outer sound towards the inner ear.
 - 3. Inner ear:
 - a. Cochlea (organ of hearing)
 - b. Labyrinth (vestibular apparatus)
 - c. Semi circular canals (for balance)
 - d. Function: Sensory reception of sound / Sound interpretation
 - ii. Conduction of sound
 - 1. Air conduction: sound energy ossicles -> ear drum
 - 2. Bone conduction: bony prominences -> cochlea
 - iii. Deafness
 - 1. Types
 - a. Sensorineural: inner ear affected, this side of the ear will not be able to interpret the sound. Can hear with good ear.
 - b. Conduction: Middle ear affected. Intact inner ear, bone conduction will go directly to cochlea
 - c. Mixed: inner and middle ear affected.

iv. Special Tests:

1. Tuning fork: 512 Hz

a. Weber Test (Test for lateralization)

- i. Place tuning fork on vertex, as where the patient can hear the sound?
- ii. If the patient can hear on one side stronger, this is "Lateralization."
- iii. Follow up with an MRI

b. Rinne Test

- i. Compares air conduction vs bone conduction
- ii. Place tuning fork over mastoid, and time how long Pt will hear the sound and compare with time over external auditory meatus (1-2cm) perpendicular
- iii. (+) test Air conduction > bone conduction (normal)
- iv. (-) test bone conduction > air conduction (abnormal)

c. Schwabach test

- i. Activate tuning fork, place over mastoid process and time how long Pt will hear the sound. PT will place tuning fork over his mastoid and time and compare results.
- ii. Pt > PT: Conduction
- iii. PT > Pt: Sensorineural

e. For TMJ

i. Chvostek test

1. Tap the parotid gland.
2. (+) test: twitches over masseter muscle
3. Indication: tetany, hypocalcemia, or hyperkalemia

ii. Jaw reflex

1. Checks status cranial nerve V
2. (+) test hyperactive closure of the mouth

Cervical

f. For neurological symptoms

i. Foraminal compression test (Spurlings)

1. Pt is sitting, passive side flexion and apply downward compression
2. (+) Radicular pain, towards the side flexed. localized pain at the neck radiating down the shoulder and arm, dermatomal distribution. Paresthesia & tingling numbness.
3. Indication: Radiculitis/ pressure on the nerve root.
4. If pain is on the opposite side, d/t muscle spasm (Reverse Spurlings)

ii. Maximum cervical compression test

1. Extend, side flex, rotate and downward compression.
2. (+) radicular pain to decide where the head is rotated.
3. Indication: radiculitis/ pressure on the nerve root/ facet syndrome.
4. Test vertebral artery (major blood supply of the neck) from subclavian artery. By holding position for at least 20-30seconds (+) Dizziness

- iii. Distraction
 - 1. One hand cuffing the jaw, and the opposite on the
 - 2. (+) relief of pain
 - 3. Indication: relief of pressure of the nerve root.
- iv. Shoulder abduction (relief test)
 - 1. (+) Relief of pain (bakody sign)
 - 2. Relief of pressure on the nerve root (C4-C5)
- v. Scalene cramp test
 - 1. Chin towards the clavicular fossa
 - 2. (+) localized pain on the neck: Trigger points on the scalene muscles
 - 3. Between Anterior/ middle scalene the lower brachial plexus exits
- vi. Jackson compression test
 - 1. (+) test radicular pain
 - 2. (+) radiculitis, pain on nerve root
- vii. Valsalva
 - 1. Hold breath and bear down
 - 2. (+) pain
 - 3. Indication: Increased intrathecal pressure, space occupying lesions, Increased pressure in the spinal cord
- viii. ULTT (brachial plexus tension test/ Elvey's)
 - 1. Determine status of neural structures of the upper extremity
 - 2. Passively performed by the PT
 - a. 1st to move shoulder (always depressed) > forearm > wrist/hand > elbow

	ULTT1	ULTT2	ULTT3	ULTT4
Shoulder	Abducted 110°	Abducted 10°	Abducted 10° IR	Abduction 10-90°, ER
Elbow	Extended	Extended	Extended	Flexed
Forearm	Supinated	Supinated	Pronated	Supinated
Wrist	Extended	Extended	Flexed ulnar deviation	Extended Radial deviation
Fingers and thumb			Flexed	extended
Nerve bias	Median, AIN C5-C7	Median, Axillary, Musculocutaneous	RayDial	U(rat)lnar Bikel Sign (active)

- 3. Provocative: contralateral sideflexion
- 4. (+) radicular pain
- 5. Indication: Upper extremity neural structures dysfunction

g. Tests for TOS

- i. Brachial plexus, subclavian artery and vein
 - ii. Special tests
 - 1. Adsons test
 - a. UE extended, ipsilateral head rotation
 - b. (+)Diminished radial pulse
 - 2. Allen test
 - a. Shoulder 90° abduction , elbow flexion, rotated contralaterally
 - 3. Costoclavicular syndrome test (military brace)
 - a. Assume attention posture shoulder hyper extended
 - 4. Wright test (hyper abduction test)
 - a. Highest accuracy if there is TOS
 - b. “always right”
 - 5. Halstead maneuver
 - a. Contralateral head rotation
- (+) Diminished radial pulse
Indication: TOS
- 6. Roo’s test
 - a. 3 minutes maintain position
 - b. (+) inability to maintain the position
 - c. Indication: TOC

h. For UMNL

- i. Romberg’s
- ii. Lhermitte’s sign
 - 1. (+) sharp pain/ electric shock from the spine towards the extremities
 - 2. Indication: Dural/ meningeal irritation, increased thecal pressure, MS

i. Vascular signs

- i. Vertebral artery/ cervical quadrant test
 - 1. Extend, sideflex, and rotate to the same side
 - 2. (+) nausea, dizziness vertigo
 - 3. Indication: Compromised vascular sign for the brain.
- ii. Dekleyn-nieuwenhuys Test
 - 1. Vertebral artery test minus the sideflexion.
 - 2. Extension and rotation only
- iii. Dix-hallpike maneuver
- iv. Hautant’s Test
 - 1. (B) arms flex 90°, closed eyes> neutral neck and observe
 - 2. (+) arm wavering with eyes open, indication: non vascular, articular problem of the shoulder joint.
 - 3. (+) head and neck extension and rotation
 - a. (+) arm wavering with eyes closed, indication: vascular problem.

- v. Barre's test
 1. Standing (B) upper ex
 2. (+) simultaneous forearm pronation and falling of (B) hand
 3. Indication: compromised blood supply to the brainstem.
- vi. Underburg's test
 1. Component 2 of hautant's test + marching in place
 2. (+) out of balance
 3. Indication: compromised blood supply to the brainstem.
- vii. Naffziger's test (30 secs/ cough)
 1. Compress the jugular vein for 30 seconds, then ask the patient to cough
 2. (+) pain on neck area
 3. Indication: increased intrathecal pressure, nerve root lesion, and compromised blood supply of the brain.
- j. For cervical instability
 - i. Sharp purser test
 1. (+) clunk, PT feels a backward translation
 2. Indication: Reversal of subluxation
 - ii. Pettman's distraction test
 1. Test integrity of tectorial membrane
 2. (+) pain
 3. Indication: lax or dysfunctional tectorial membrane, or pathology
- ii. Thoracic
 - a. Slump Test
 - i. (+) radicular pain localized in the thoracic area and towards extremities
 - ii. Indication: Meningeal irritation/ nerve root involvement/ increased intrathecal pressure
- iii. Lumbar

	ST1	ST2	ST3(Sidelying)	ST4(Long sitting)
Cervical spine	Flexion	Flexion	Flexion	Flexion with rotation
Thoracic and lumbar spine	Flexion	Flexion	Flexion	Flexion
Hip	Hip flex 90°	Hip flex 90° with abduction	Flex 20°	Hip flex 90°
Knee	Extension	Extension	Flexion	Extension
Ankle	DF	DF	PF	DF
Nerve Bias	SC, cervical and lumbar nerve roots, sciatic	Obturator	Femoral	SC, cervical, and lumbar nerve roots, sciatic nerve

- a. For neurologic dysfunction
 - i. Sitting root test
 1. Slump test actively performed
 2. (+) pain localized in lumbar or radicular pain
 3. Indication: Radiculitis/ pain on nerve root

- ii. Prone knee bending(Nachlas test)
 1. Maintained for 45-60 seconds
 2. (+) felt posteriorly, nerve root impingement/ disc prolapsed
 3. (+) pain felt anterior, nerve stretch femoral nerve.

	PKB1 (Basic)	PKB2	PKE
Cervical spine	Rotation to test side	Rotation to test side	
Thoracic and lumbar spine	Neutral	Neutral	Neutral
Hip	Neutral	Ext/add	Ext/Abd/ER
Knee	Flex	Flex	Ext
Ankle	-	--	DF
Foot	-	-	Eversion
Nerve Bias	Femoral nerve L2-L4 nerve root	Lateral femoral cutaneous	Saphenous nerve

- b. SLR test
 - i. Aka laseuges
 - ii. (+) Unilateral 0-30° hip joint, 30-50° sciatic nerve, 50-70° hamstrings muscles, 70-90° SI joint
 - iii. (+) bilateral SLR <70° SI joint , >70° lumbar area
 - iv. Well leg raising test, tests good leg.

	SLR1	SLR2	SLR3	SLR3	SLR5
Hip	Flexion /adduction	F	F	F/iR	F
Knee	Extension	E	E	E	E
Ankle	Dorsiflexion	D	D	P	D
Foot	-	Ev	Inv	Inv	
Toes	-	E			
Nerve bias	Sciatic/ tibial nerve	Tibial	Sural	Common Peroneal	Nerve root (disc prolapsed)

1. Bragard sign/test
2. Bowstring test/ cramp test (Modification sciatic nerve tension Deyerle's sign)
- c. For neurologic dysfunction:
 - i. Brudzinski & Kernig's test
 1. Neck is flexed, pain appears, knee is then flexed and relief of pain
 2. Indication: meningeal and dural irritation, nerve root involvement
 - ii. Valsalva
- d. Joint dysfunction
 - i. Milgram's test
 1. Lift foot 5-10cm
 2. (+) inability to maintain the position for 30 seconds
 3. Indication: hip joint affectation/ increased intrathecal pressure
 - ii. Beevor's Sign
 1. (+) umbilicus (T10) goes out of midline
 2. Indication: abdominal dysfunction
 - iii. Trendelburg's
 1. Gluteus medius weakness

- iv. Pheasant's test
 1. Prone, PKB to buttocks
 2. (+) Trunk extension upon knee flexion in a prone position
 3. Indication: lumbar spinal segment instability

- v. Ely's Test
 1. PKB
 2. (+) trunk/ hip flexion upon knee flexion
 3. Indication: Tight Rectus femoris

e. For intermittent claudication

Neurogenic	Vascular (Arterial)
Nerve root pain	Cramping
Buttocks and posterior thigh	Calf
Spinal extension, flexion bias	Exertion
Relieved by trunk bending	Relieved rest even in standing

- i. Stoop test
 1. Walk 50m(165 feet)
 2. (+) test: relief of pain upon stooping
- ii. Bicycle test of van Gelderen
 1. (+) test: relief upon leaning forward
- iii. Treadmill test:
 1. 15 minutes
 2. 1-2mph, desired speed of Pt
 3. Note:
 - a. Ambulation time and when Sx appeared

f. For malingering

- i. Hoover's test
 1. (+) pressure felt from the remaining lower extremity
 2. Refusal
- ii. Burn's Test
 1. (+) out of balance/ refusal
 2. Indication: malingering

iv. Pelvis

a. For neurological involvement

- i. SLR
- ii. PKB 45-60 seconds

b. For SI involvement

- i. Gillet's test (Sacral fixation test/ ipsilateral posterior rotation test)
 1. Unilateral stance, knee to chest, Therapist palpates PSIS and sacrum
 2. (+) Upslip of PSIS or PSIS goes upward
 3. Indication: abnormal biomechanics of SI joint
- ii. Flamingo Test
 1. (+) Pain at the pubic symphysis
 2. Indication: Dysfunction at the SI joint / hip joint

- iii. Piedallu's sign
 - 1. (+) PSIS at involved side goes superiorly
 - 2. Indication: abnormal torsion movement of the SI joint
- iv. Gaenslen's test
 - 1. Sidelying, uninvolvement on the bottom with knee to chest. Suspected is above, is extended
 - 2. (+) pain at SI joint
 - 3. Indication: SI Joint pathology
- v. Supine to sit or long sitting test
 - 1. (+) upon assuming long sitting position the medial malleoli unequal
 - 2. Indication: Functional leg length discrepancy
- vi. Goldthwait's test
 - 1. Supine, PT's hand under interspinous spaces and hip flex
 - 2. (+) Pain is felt before the spaces move,
 - 3. Indication: SI Joint pathology
 - 4. (+) Pain is felt after the spaces move upon hip flexion
 - 5. Indication: lumbar spine
- vii. Yeoman's test
 - 1. Prone, flex knee to 90° hip abduct with extension
 - 2. (+) pain
 - 3. Indication: SI, hip, lumbar, tight rectus femoris
- viii. Patrick's test (FABERE Test)
 - 1. (+) pain on resistance upon downward push on knee
 - 2. Indication: can be ipsi/ contralateral, rectus fem tightness
- ix. Transverse anterior/ posterior stress test
 - 1. Anterior – ASIS, Posterior – PSIS
- c. For limb length
 - i. Leg length test
 - 1. True leg length ASIS to medial malleolus
 - a. Problem with anatomical
 - 2. Apparent leg length umbilicus to medial malleolus
 - a. Problem with functional/ biomechanics/ posture
 - ii. Forward bending test
 - iii. Plumblin test
 - Surface landmarks
 - 1. Through the earlobe
 - 2. Through the shoulder joint
 - 3. Midway of the trunk
 - 4. Through the greater trochanter
 - 5. Anterior to knee joint
 - 6. Anterior ankle joint
- d. For stress test of SI joint
 - i. Approximation test or sidelying compression test
 - 1. (+) pain, Indication: SI joint pathology

ANAKINES SHOULDER

- i. Shoulder Complex
 - a. Bones
 - b. The only bone that will connect the Upper extremities to the trunk: Clavicle
- ii. Clavicle
 - a. Orientation of clavicle, scapula and humerus
 - b. Convex medially 2/3 (vascular supply and blood vessels passes through), concave lateral 1/3 (common site of clavicle)
 - c. Slightly **superior to horizontal** plane, **posterior to frontal** plane.
- iii. Scapula T2-T7
 - i. Medial (5-6cm or 2-3 fingerbreadths)
 - ii. Lateral
 - iii. Superior border
 - iv. Rotated **anteriorly to the frontal plane 35°** (aka Scapular plane)
 - v. Glenoid fossa: tilted upward 5-10°
 - 1. Paralysis of upper trapezius, subluxation of humerus
- iv. Humerus
 - i. Anatomical neck
 - ii. Surgical neck – MC site of humeral fractures
 - iii. Greater Tubercle – palpable by Internal rotation
 - iv. Retroverted 20° posterior to frontal plane
 - v. Angle of inclination 125°
- v. Joints (3)
 - a. True joints
 - i. Sternoclavicular joint
 - 1. Type: Saddle/ sellar type of joint, 3° of freedom

OPP	arm by the side
CPP	full elevation
CP	pain @ extremes ROM, especially horizontal adduction and full elevation

2. Arthokinematics:

- a. Sternal end of clavicle
 - i. Vertical: convex
 - ii. Horizontal: concave

Movement	Roll	Slide	Ligament
Elevation	Superior	Inferior	Costoclavicular ligament
Depression	Inferior	Superior	Interclavicular ligament
Protraction	Anterior	Anterior	Posterior SC ligament
Retraction	Posterior	Posterior	Anterior SC ligament
Rotation	Spins on articular disc		Coracoclavicular ligament

3. Ligaments:

- a. SC ligaments Anterior/ posterior
- b. Interclavicular ligament, - prevent depression of the clavicle at the SC joint
- c. Costoclavicular ligament – prevent elevation of clavicle

4. Movement

Movement	Plane	Axis	ROM
Elevation-depression	Frontal	Z	Elevation 30-45 Depression 5-10
Protraction-retraction	Horizontal	Y	15°
Transverse, Upward rotation	Horizontal	Y	40°

ii. Acromioclavicular joint

1. Plane type of joint 3° of freedom,
2. Ligaments
 - a. Acromioclavicular ligament
 - b. Coracoclavicular ligament
 - i. Conoid (medial/center): taut in retraction and upward rotation
 - ii. Trapezoid(lateral): taut in Protraction and Downward rotation

3. Movement

Movement	Plane	ROM
Upward and downward rotation	Scapular plane	20-30°
Rotational adjustments	Horizontal plane: Internal and external tilting Sagittal plane: Anterior and Posterior tilting	10-30°

OPP	arm by the side
CPP	90 abduction
CP	pain @ extremes ROM, especially horizontal adduction and full elevation

4. AT what movement of the humerus will posterior tilting of the scapula occur.
Abduction and flexion

iii. Glenohumeral joint

1. Ball and socket joint 3° of freedom

OPP	55° abduction 30° horizontal Adduction
CPP	Full abduction and lateral rotation
CP	ER>ABD>IR

2. Ligaments

- a. Coracohumeral ligament – provides passive stabilization at the GH joint
 - i. Prevents the downward displacement of the head of the humerus, AP stabilization, prevent excessive adduction
- b. Glenohumeral ligament
 - i. Superior
 1. Prevent downward displacement, and AP movement of the humeral head.

- ii. Middle
 - 1. Becomes taut from 45-60° abduction and External rotation, excessive anterior displacement of humeral head
- iii. Inferior
 - 1. Taut @ 90° abduction
 - 2. Prevent inferior displacement of humeral head
 - 3. Parts:
 - a. Anterior band, prevent ER
 - b. Axillary pouch
 - c. Posterior band, prevent IR
- iv. Points of weakness
 - 1. Foramen of Weitbrecht between Superior and middle
 - 2. Foramen of Ruvier between Middle and inferior
- c. Bursae Prevents friction and allows lubrication
 - i. Subacromial bursa
 - ii. Subdeltoid bursae
 - iii. Subcoracoid bursa
- d. Osteokinematics
 - i. Active abduction
 - 1. If GH is in full IR: 60° of abduction
 - a. The greater tubercle will be exposed and hit the coracoacromial arch.
 - 2. If GH is in 90° of ER: 90° abduction
 - ii. Passive abduction
 - 1. Can reach up to 120°

Movement	Planes of motion	axis	ROM (GH only)	ROM (ST only)	ROM (GH + ST)
Flexion-extension	Sagittal	X	120°	60	180
Abduction	Frontal	Z	120	60	0-180
ER-IR	Forizontal	Y	-	-	ER: 0-90 IR: 0-70

Movement (OKC)	Roll	Slide
Abduction	Superior	Inferior
Adduction	Inferior	Superior
Flexion 140°	Anterior	Posterior
Flexion 140-180°	Posterior	Anterior
Extension	Posterior	Anterior
External rotation	Posterior	Anterior
Internal rotation	Anterior	Posterior

b. False joint

i. Scapulothoracic joint – motions occurring in SC and AC joint are composite of what?

1. Movable base for the humerus
2. Increases the ROM
3. Provides stability for GH joint
4. Provide shock absorption on an outstretched hand
5. Permits elevation of the body in closed chain activities: Depression
6. Movement: protraction, retraction, elevation depression, upward and downward rotation.

	SC joint	AC joint
Elevation	30-45°	Elevation: 20°
Depression	Depression 5-10°	
Protraction	15° protraction	
Retraction	15 ° retraction	
Rotation	40°	20°

vi. Scapulohumeral rhythm

- a. After 30° of abduction, there is 2:1 ratio between glenohumeral joint and scapulothoracic joint. 0-30° setting phase.>30°

vii. Six kinematic principles associated with full abduction of the should

Principle 1	Based on a generalized ration 2:1 scapulohumeral rhythm, active shoulder abduction of about 180° occurs as a result of simultaneous 120° of GH joint abduction and 60° of ST UR
Principle 2	The 60° of UR of the scapula during full shoulder abduction is the result of a simultaneous elevation at the SC joint with UR at AC joint
Principle 3	The clavicle retracts at the SC joint during shoulder abduction
Principle 4	The scapula posteriorly tilts and ER during full shoulder abduction
Principle 5	The clavicle posteriorly rotates around its own axis during shoulder abduction
Principle 6	The GH joint ER during shoulder abduction

viii. Companion motions of the shoulder girdle and GH joints

GH motion	ST motion	SC motion	AC motion
Flexion	UR, posterior tilting medial tilting followed by lateral tilting in higher ranges of flexion	Elevation Posterior rotation Protraction	Upward rotation Horizontal and Sagittal plane rotational adjustments
Extension	Downward rotation, anterior tilting, Reverse of all triplanar motion occurring inflexion	Depression Anterior rotation Retraction	Downward rotation Horizontal and Sagittal plane Rotational adjustments
Abduction	UR, posterior tilting, medial tilting initially followed by lateral tilting in higher ranges of abduction	Elevation Posterior rotation Retraction	Upward rotation Horizontal and Sagittal plane Rotational adjustments
Adduction	Downward rotation, anterior tilting, lateral tilting initially followed by medial tilting when returning to anatomic position	Depression Anterior rotation Protraction	Downward rotation Horizontal and Sagittal plane rotational adjustments
Lateral rotation and horizontal abduction	Retraction	-	-
Medial rotation and horizontal adduction	Protraction	-	-

ix. Phases of abduction

a. 0-90

- i. SC joint 25° elevation
- ii. AC joint: 5° upward rotation
- iii. GH joint: 60°
- iv. Terminated by: hitting of greater tubercle with the glenoid

b. 90-180

- i. SC joint: 5° elevation; 35° posterior rotation of the clavicle
- ii. AC joint: 25° upward rotation
- iii. GH joint: 60°
- iv. Terminated by: resistance of adductor muscles E.g. Latissimus Dorsi

x. Muscles

- a. Responsible for movement and stabilization of the scapula during GH joint motion
- b. Originate on the thorax and on the scapula
 - i. Serratus anterior
 - ii. Trapezius
 - iii. Rhomboids major and minor
 - iv. Pectoralis minor
 - v. Levator Scapulae

Muscle	Origin	Insertion	Innervation	Action	Paralysis
Serratus anterior aka saw muscle Lowest 5 digitations: strongest portion of the muscle	First 8-9 ribs	Medial border of scapula	Long Thoracic Nerve	Draws the scapula forward around the thoracic wall rotates scapula	Medial winging of scapula, inability to elevate arm >100°
Trapezius - Upper - Middle - Lower	Occipital bone, ligamentum nuchae, spine of 7 th vertebra, and spines of all thoracic vertebra	Upper fibers into lateral third of the clavicle, middle and lower fiber inserts into acromion and spine of the scapula	CN XI	All 3: upward rotation Upper: elevation Middle: retraction Lower: depression	Inability to achieve full scapular UR Elevation is limited to 120° and shifts the orientation of the glenoid fossa inferiorly
Rhomboids Minor	Ligamentum nuchae and spines of C7-T1	Medial border of the scapulae	Dorsal scapular nerve	Retractions	Scapula is placed in protracted position on the thorax
Rhomboids Major	Spine of 2 nd thoracic to T5	Lower Medial border of the scapula		Retraction	
Pectoralis minor	3 rd , 4 th , 5 th ribs	Coracoid process	Medial pectoral nerve	Scapular depressor, if scapula is fixed elevate the 3 rd , 4 th , and 5 th ribs	Reduce strength during scapular depression, and downward rotation of the scapula against resistance
Pectoralis major			Lateral		
Levator scapulae	Transverse process first 4 cervical vertebrae	Upper medial border of the scapula	C3 ,C4 dorsal scapular nerve	Elevation of the scapula, upward rotation	Reduced ability to elevate and downwardly rotate the scapula
Glenohumeral stabilizing muscles of the shoulder					
Supraspinatus “workhorse of abduction” max contract 90-180°. active all throughout ROM	Supraspinous fossa	Greater tuberosity	Suprascapular nerve	Abduction of arm and stabilizes shoulder joint	Decrease strength and endurance in shoulder abduction

Infraspinatus	Infraspinous fossa	Greater tuberosity	Suprascapular nerve	Laterally rotates arm and stabilizes the shoulder joint	Complaints of pain and discomfort
Teres minor	Upper 2/3 of lateral border of the scapula	Greater tuberosity	Axillary nerve		
Subscapularis	Subscapular fossa	Lesser tuberosity	Upper and lower subscapular nerve	Medially rotates arm and stabilizes the shoulder joint If in IR – adductor If in ER – abductor If overhead – assist in extension	
Biceps Brachii	Supraglenoid tubercle and coracoids process of the scapula	Radial tuberosity and bicipital aponeurosis and into deep fascia of the forearm	Musculocutaneous nerve	Supinator of forearm and flexes at elbow joint weak flexor of the shoulder	
Triceps Brachii	Infraglenoid tubercle	Olecranon process	Radial nerve	Extensor	
Large muscle movers of the shoulder					
Deltoid Anterior Middle (greatest activity 90-120°) posterior	Lateral third of the clavicle, acromion, and spine of the scapula	Deltoid tuberosity	Axillary nerve	Abduct shoulder Ant: flexes Post: Extends	In paralysis of the supraspinatus, the deltoid can abduct the arm in full range but with less strength
Teres Major	Lower 1/3 of the lateral border of scapula	Medial lip of the bicipital groove	Lower subscapular nerve	Medially rotates and adducts the arm, stabilizes the shoulder joint	
Pectoralis Major Clavicular Sternal Costal	Sterna: manubrium body of the sternum Costal: cartilages of the first 6 ribs Clavicular: clavicle	Lateral lip of the bicipital groove	Medial and lateral pectoral nerves	Adducts arm and rotates it medially clavicular fibers also flex arm	
Coracobrachialis	Coracoid process	Medial aspect of the shaft of humerus	Musculocutaneous nerve	Shoulder flexion and also weak adductor	

- xi. Passive and dynamic stabilization of the glenohumeral joint
 - a. Passive stabilizers:
 - i. Coracohumeral ligament
 - ii. Superior GH ligament
 - iii. Negative atmospheric pressure within the capsule
 - b. Dynamic stabilization
 - i. Rotator cuff muscles
 - ii. Biceps brachii
 - iii. Deltoid
- xii. Antagonists and synergists of scapular upward rotation
 - a. Synergists
 - i. Upper trapezius
 - ii. Lower trapezius
 - iii. Serratus anterior
 - b. Antagonists
 - i. Rhomboids
 - ii. Pectoralis minor
 - iii. Levator scapulae
- xiii. Scapular retraction
 - a. Synergists
 - i. Rhomboids
 - ii. Trapezius
 - b. Antagonists
 - i. Pectoralis minor
 - ii. Serratus anterior
- xiv. Scapular protraction
 - a. Synergists
 - i. Pectoralis minor
 - ii. Serratus anterior
 - b. Antagnosists
 - i. Rhomboids
 - ii. Trapezius
- xv. Scapular elevation
 - a. Synergists
 - i. Upper trapezius
 - ii. Rhomboid
 - iii. Levator scapulae
 - b. Antagonists
 - i. Pectoralis minor
 - ii. Lower trapezius
 - iii. Lower serratus anterior

- xvi. Scapular depression
- a. Synergists
 - i. Pectoralis minor
 - ii. Lower trapezius
 - iii. Lower serratus anterior
 - b. Antagonist
 - i. Upper trapezius
 - ii. Rhomboids
 - iii. Levator scapulae

Flexors	Extensors	Abductors	Adductors	ER	IR
Pectoralis major (clavicular head) Coraco brachialis Biceps brachii	Latissimus dorsi Teres major Long head of triceps brachii Pectoralis major(costosternal head)	Deltoid Supraspinatus Biceps brachii (long head)	Pectoralis major Latissimus dorsi Teres major Triceps brachii (long head)	Infraspinatus Teres minor Posterior deltoid	Subscapularis Teres major Pectoralis major Latissimus dorsi Anterior deltoid

- xvii. Force couple
- a. Force couple is define in mechanical terms as two forces whose points of application occur on opposite sides of an axis and in opposite direction to produce rotation of the body
 - i. Scapular upward rotation
 1. Synergists
 - a. Upper trapezius
 - b. Lower trapezius
 - c. Serratus anterior
 - ii. Scapular downward rotation
 1. Synergists
 - a. Rhomboid
 - b. Pectoralis minor
 - c. Levator scapulae
 - iii. Shoulder elevation
 1. Deltoid
 2. Supraspinatus
 3. Infraspinatus
 4. Teres minor
 5. Subscapularis

AK ELBOW AND FA

i. Bones

Distal humerus	Radius	Ulna
Trochlea	Radial head	Olecranon process
Coronoid fossa	Fovea of radius	Coronoid process
Capitulum	Radial neck	Trochlear notch and longitudinal crest
Radial fossa	Radial tuberosity	Radial notch
Medial and lateral epicondyles	Ulnar notch of radius	Supinator crest
Medial and lateral supracondylar ridge	Styloid process	Tuberosity of the ulna
Olecranon fossa		Ulnar head
		Styloid process

ii. The elbow joint

- Hinge joint, Sagittal plane, mediolateral axis through the epicondyles
- Carrying Angle – (N) amount of cubitus valgus
 - Female: 10-15°
 - Male: 5-10°
 - < 5° = cubitus varus, aka gunstock deformity
 - >15° = cubitus valgus
- Elbow flexed neutral forearm position to palpate lateral epicondyle
- Medial epicondyle is more prominent than lateral
- Capitulumtrochlear groove

iii. Elbow – ligaments

- MCL – Limits:
 - excessive valgus stress that would force the forearm laterally
 - Anterior fibers also limit extension, posterior fibers limit flexion
 - Prevents subluxation at humeroulnar articulations
- LCL – Limits:
 - Excessive varus stress that would force the FA medially
 - Prevents subluxation of humeroulnar articulation by securing humerus to ulna
 - Stabilizes radial head and humeroradial articulation

iv. Elbow - Osteokinematics

Motion	Plane	Axis	ROM
Flexion-Extension	Sagittal	X	0-150°

	UH joint	RH
OPP	70° elbow flexion 10° supination	Full extension Fullsupination
CPP	Extension with supination	Elbow flexed 90 5 supination
CP	Flexion>extension	Flexion>extension

- v. Forearm joints
 - a. Proximal radioulnar joint
 - b. Distal radioulnar joint
- vi. Forearm ligament
 - a. Annular ligament
 - i. Maintains integrity of PRU articulation
 - ii. Prevents dislocation
 - iii. Prevents excessive radial distraction
 - b. Oblique cord
 - i. Fibers run perpendicular to interosseous membrane to offer major stabilization to proximal RU joint
 - ii. Taut in full supination

* Oblique cord, Annular ligament, Brachioradialis to prevent distraction force placed on the hand
 - c. Quadrate ligament
 - i. Limits spinning of radial head
 - ii. Maintains radial head up against ulnar radial notch
 - iii. Reinforces capsule
 - d. Dorsal and palmar RU ligament
 - i. Stabilizes distal RU connection
 - ii. Reinforces joint capsule
 - e. Interosseous membrane
 - i. Provides stability for both superior and inferior RU joints
 - ii. Taut when forearm is in neutral
 - iii. Slack when forearm is Supinated or pronated
 - iv. Serves as attachment for muscles
 - v. Functions
 - 1. Absorbs shock
 - 2. Transmits force along the kinetic chain to protect the forearm bones from injury

vii. Forearm

Motion	Plane	Axis	ROM
Pronation -supination	Horizontal	Y	0-80°

- viii. CKC
 - a. Shoulder internal rotation > forearm supination
 - b. Shoulder external rotation > forearm pronation

ix. Muscles of the elbow and forearm joints

- a. Flexors
 - i. Biceps brachii
 - 1. Multi jointed muscle
 - 2. Selectively recruited when the task requires elbow flexion with forearm supination such as carrying a loaded tray or bringing spoon to the mouth.
 - 3. Greatest moment arm: 90-110° flexion

- ii. Brachialis
 - 1. Only one jointed elbow flexor
 - 2. “workhorse” of elbow flexors
 - 3. Largest elbow flexor
 - iii. Brachioradialis
 - 1. Longest elbow flexor
 - 2. Aka supinator longus
 - 3. Peak force 100-120° of elbow flexion
 - 4. Only elbow flexor whose belly is located in the forearm
 - 5. Acts as supinator toward neutral position in a pronated forearm Vice versa.
- b. Extensors
 - i. Triceps
 - 1. 2 jointed muscle
 - 2. Lateral head: strongest
 - 3. Greatest extension force 70-90° of elbow flexion
 - ii. Anconeus
 - 1. Accessory muscle for elbow extension
 - 2. It initiates most low-load elbow extension motions
- c. Pronators
 - i. Pronator Teres/ Quadratus
 - 1. Pronator quadrates performs forearm pronation regardless of elbow position
- d. Supinators
 - i. Supinator
 - 1. The only muscle whose sole action is supination
 - 2. Always recruited for forearm supination, regardless of speed and load
 - 3. Acts alone during slow supination when the arm is extended
- x. The elbow complex
 - a. Biceps brachii vs supinator
 - i. Biceps brachii is 4x stronger supinator in elbow flexed and 2x in extension vs supinator muscle.

AK WRIST & HAND

- xi. Bones and joints
 - a. Distal radius and ulna
 - i. Ulnar tilt 25° , reason why ulnar deviation > radial deviation
 - ii. Palmar tilt 10° , reason why wrist flexion > wrist extension
 - b. Carpal bones
 - i. Proximal row: scaphoid, lunate, triquetrum, pisiform
 - ii. Distal row: trapezium, trapezoid, capitate, hamate
 - c. Lister's tubercle/ dorsal tubercle, separates the tendon EPL and ECRB
- xii. Carpal bones
 - a. Capitate: os magnum
 - b. Scaphoid: navicular
 - c. Trapezium: greater multangular
 - d. Trapezoid: lesser multangular
 - e. Triquetrum: triangular
- xiii. Wrist ligaments
 - a. Extrinsic ligaments
 - i. Dorsal radiocarpal
 - ii. Radial collateral
 - iii. Palmar radiocarpal
 - iv. Radioscapholunate
 - v. Radiolunate
 - vi. Radioscaphocapitate
 - vii. Triangular fibrocartilage complex (TFCC)
 - 1. Functions
 - a. Primary stabilizer of the distal RU joint
 - b. Reinforces the ulnar side of the wrist forms part of the concavity of the RU joint
 - c. Helps transfer part of compression forces that naturally cross the hand to the forearm.
 - viii. Articular disc
 - ix. RU joint capsular ligament
 - x. Palmar ulnocarpal (ulnotriquetral and ulnolunate)
 - xi. Ulnar collateral ligament
 - xii. Meniscus homologue

- b. Intrinsic ligaments
 - i. Short (distal row)
 - 1. Dorsal
 - 2. Palmar
 - 3. Interosseous
 - ii. Intermediate
 - 1. Lunotriquetral
 - 2. Scapholunate
 - 3. Scapho trapezial
 - iii. Long
 - 1. Palmar intercarpal (inverted V)
 - iv. Lateral leg (capitates to scaphoid)
 - v. Medial leg (capitates to triquetrum)
 - vi. Dorsal inter carpal (trapezium-scaphoid-lunate-triquetrum)
- c. Wrist Kinematics
 - i. RC is greater in flexion about 50° of flexion, MC 35° flexion
 - 1. Dorsal wrist ligaments taut
 - ii. MC is greater in extension, 50° of extension, RC 35° flexion
 - 1. Volar wrist ligaments taut
 - iii. Radial deviation
 - 1. Roll: lateral/ Radially
 - 2. Glide: medially/ ulnarly
 - 3. Ulnar collateral ligaments
 - 4. At radiocarpal: ulnar collateral ligament + palmar radiocarpal
 - 5. AT midcarpal: palmar inter carpal, medial leg becomes taut
 - iv. Ulnar deviation
 - 1. Roll: medially/ ulnarly
 - 2. Glide: lateral/ radially
 - 3. At radiocarpal: Radial collateral ligaments + palmar ulnocarpal
 - 4. At midcarpal: Palmar intercarpal, lateral leg becomes
- xiv. Metacarpals(5)
 - a. 2nd MCP joint abduction
 - i. Roll: lateral
 - ii. Glide lateral
 - b. 3rd MCP acts as axis.
 - c. 4th and 5th MCP joint abduction
 - i. Roll: medial
 - ii. Glide: medial
 - d. Deep transverse ligament – keep the position of the MC in place, affectation results to splay deformity

- xv. Phalanges
 - a. Shallow bases
 - b. Volar/palmar plate – limits extension, promotes flexion
 - i. Strengthened by collateral ligaments
 - ii. Cruciate/check reins ligaments
- xvi. Pollex
 - a. Trapezium articulates with 1st CMC
 - b. Saddle/sellar type of joint
 - c. Abduction and adduction occurs in the Sagittal plane, x axis
 - d. Flexion and extension occurs in the frontal plane, z axis
 - e. Opposition: 1st abduct> then flexion
 - f. Ulnar collateral ligament and anterior Oblique ligament abduction, flexion and opposition
 - g. Intermetacarpal ligament and posterior oblique, abduction and opposition
 - h. Radial collateral ligament – taut in all motion of thumb, except for extension.
 - i. Abduction/ adduction (convex moving)
 - i. Rolls: anterior
 - ii. Glides: posterior
 - j. Flexion/ extension (concave moving)
 - i. Rolls: laterally
 - ii. Glides: laterally
 - k. Opposition
 - i. Phase 1:
 - 1. Roll: anterior
 - 2. Glide posterior
 - ii. Phase 2:
 - 1. Roll: medially
 - 2. Glide: medially
- xvii. Arches of the hand
 - a. (2) transverse (1)longitudinal
 - i. Proximal transverse arch (carpal)
 - 1. Keystone: capitate
 - ii. Distal transverse arch (head of MC)
 - 1. Keystone: middle finger
 - iii. Longitudinal arch
 - 1. Between 2nd and 3rd finger (keystone)
- xviii. Muscles of the hand and wrist
 - a. Extrinsic
 - i. Anterior compartment
 - 1. Superficial
 - a. Flexor carpi radialis
 - b. Palmaris Longus
 - c. Flexor carpi ulnaris (innervated by ulnar nerve)
 - i. Humeral head
 - ii. Radial head

2. Intermediate
 - a. Flexor Digitorum Superficialis
 - i. Humeroulnar head
 - ii. Radial head
3. Deep
 - a. Flexor pollicis longus
 - b. Flexor digitorum profundus
 - i. Only muscle capable of flexing of DIP
 - ii. Dually innervated
 - iii. Unresisted closure of the hand, only active muscle is FDP
- ii. Lateral compartment (innervated by the radial nerve)
 1. Brachioradialis
 2. Extensor carpi radialis longus
- iii. Posterior compartment
 1. Superficial
 - a. ECRB
 - b. ED
 - c. EDM
 - d. ECU
 - e. The strength of contraction of the wrist extensors is in direct proportion to the effort of the grip – the harder the grip, the stronger the contraction of the wrist extensors.
 - f. If the wrist is allowed to flex during finger flexion, the grip is markedly weakened.
 2. Deep
 - a. APL
 - b. EPB
 - c. EPL
 - d. EIP
- b. Intrinsic muscles
 - i. Small muscle
 1. Palmaris brevis
 - a. Corrugates skin to improve grip of palm
 2. Lumbrical
 - a. Lumbricals are electrically silent in MCP flexion unless the IP joints are extended
 - b. Lumbricals do not participate in grip and rarely contract synchronously with FDP

3. Interossei
 - a. Found to have high levels of activity in pinching, grasping, and power grip
 - b. Serves to rotate the finger to fit the surface of the object, strengthen the grip, stabilize the proximal phalanges against the metacarpal head, and stabilize the extensor tendons on the dorsum of the MCP joint through its attachment of the extensor hood.
4. Adductor pollicis
 1. Abductor pollicis brevis
 2. Flexor pollicis brevis
 3. Opponens pollicis
- ii. Thenar
 1. Abductor pollicis brevis
 2. Flexor pollicis brevis
 3. Opponens pollicis
- iii. Hypothenar
 1. Opponens digiti minimi
 2. Flexor digiti minimi
 3. Abductor digiti minimi – inserts into pisiform
- xix. Special topics
 - a. Carpal tunnel
 - i. Roof : transverse carpal ligament:
 1. Attachment medially pisiform and hook of hamate
 2. Laterally scaphoid and trapezium
 - ii. Contents:
 1. (4) FDP
 2. (4) FDS
 3. (1) median nerve
 4. (1) FPL
 - b. Extensor tunnel
 - i. Attachment of extensor retinaculum
 1. Medially: pisiform and hook of hamate
 2. Laterally:
 3. Distal end of radius
 - ii. Contents
 1. APL, EPB
 2. ECRL, ECRB
 3. EPL
 4. EDC, EIP
 5. EDM (little finger)
 6. ECU
 - c. Anatomic snuffbox
 - i. Fovea radialis
 1. Anterior/ radial border: APL/ EPB
 2. Posterior/ ulnar border: EPL
 3. Floor: scaphoid trapezium
 4. Content: radial artery

- d. No man's land
 - i. The area where the finger flexor tendons are tightly enclosed within the tenosynovium
 - ii. Located in the palm between the distal palmar crease and the creases of the PIP in the palm
 - iii. Generally, primary repair of the tendons in this region is contraindicated
 - iv. Increase in chance of infection, no coating of flexor tendon
 - v. zones
 - 1. Distal phalanx
 - 2. Refer above no mans land
 - 3. Lumbricals origin
 - 4. Carpal tunnel
 - 5. Proximal to carpal tunnel
- e. Extensor assembly
 - i. Tendinous system
 - 1. Tendons of extrinsics extensors
 - 2. Almost all intrinsic muscles, except:
 - a. PB, OP, ODM, ADM, FDM
 - ii. Retinacular system: Fascia and ligaments (to retain and stabilize tendons and skin)
 - iii. Function
 - 1. Extends the digits in different positions of finger flexion
 - 2. Provides shortcut to allow full flexion
 - iv. Remember:
 - v. Motor input into the lateral bands for each finger:
 - 1. Extrinsic extensors
 - 2. Lumbricals
 - 3. 2 interossei
 - vi. Central band
 - 1. Serves as the back bone of the extensor mechanism
 - 2. Transmits extensor force from the ED across the IP joint
 - 3. Boutonniere deformity
 - vii. Lateral band
 - 1. Transmit extensor force from the ED, lumbricals, and interossei across the PIP and DIP joint
 - 2. Mallet finger lesion
 - viii. Dorsal hood transverse fibers
 - 1. Stabilize the ED tendon over the dorsal aspect of the MCP joint from a sling around the proximal end of the proximal phalanx, thereby assisting the ED in extending the MCP joint
 - ix. Dorsal hood oblique fibers
 - 1. Transfer force from the lumbricals and interossei to the lateral bands of the extensor mechanism, thereby assisting with extension of the PIP and DIP joints
 - x. Oblique retinacular ligament: Helps coordinate movement between the PIP and DIP of the fingers.

f. Flexor Pulleys

i. Annular pulleys:

1. Attached to the shafts of the proximal and distal phalanges and to the sides of the volar plates of the MCP, PIP, and DIP joints at the junction with the extensor hood and the retinaculum.
2. Prevents bow stringing effect of long flexor tendons during flexion.

ii. Cruciate pulleys

1. Attach on the shafts of the phalanges and cross to from distal attachments on the volar plates of the PIP and DIP joints.

iii. These pulleys prevent bowstringing of the long flexor tendons during flexor activities

iv. Severance of a pulley causes loss of finger motion.

g. Power and precision grips

Grip	Description	Finger function	Thumb function	Example of uses
Precision				
Lateral pinch grip	Aka. Key grip . The least precise. Small object placed between the index finger and thumb.	Index Finger is abducted at the MCP joint and partially flexed at the MCP and IP joint. Muscles include FDP, FDS, and first dorsal interossei	Thumb is abducted and IP joint is flexed. Muscles include FPL, FPB, and adductor pollicis	Placing a key in the ignition, grasping papers
Three-Prong chuck	Aka Pad to pad . Originally palmar prehension grip. Pads of distal 2 nd and 3 rd digits contact pad of thumb for the purpose of picking up or grasping objects	MCP and PIP joint flexion. Possible DIP flexion. The FDS performs the task if the DIP is in extension, but if the DIP is partially flexed, the FDP is also activated. Interossei work to provide MCP abduction and adduction.	Thumb is positioned in opposition, flexed at MCP and IP joints and adduction at CMC joint. Muscles include adductor pollicis working with FPB to stabilize the first metacarpal while FPL flexes the thumb joints	Grasping a coin, using a writing utensil such as pen or pencils
Tip to Tip grip	Aka. Tip prehension grip . Thumb faces one of the fingers. Most often used is index finger. Purpose of picking up or manipulating small objects	The finger used is in flexion at all 3 joints. Some lateral movement of the MCP towards the ulnar side occurs to allow the finger to meet the thumb with the fingertip facing a radial direction. FDP plays a greater role.	Thumb is in opposition with flexion at MCP and IP. Thumb is also adducted so the tip is aligned with opposing finger's tip. FPL plays an important role in this grip	Picking up a small object such as a pin, bead, or strand of hair.
Power				
Hook Grip	Finger digits provide this grip. Use primarily to support or carry an object	Fingers and palm are primary contributors. Primary function from FDS and FDP secondarily with IP joint flexion especially at the PIP.	Thumb does not contribute to this grip. Held in extension away from the hand.	Carrying a suitcase or briefcase. Weight lifters may use a hook grip to lift a barbell off the floor

Cylinder Grip	Grasp objects that are cylindrical in shape and large enough that the thumb does not touch the fingers. The fingers and palm are on one side of the object and thumb is on the opposite side.	Flexion of MC and IP with activity of FDP. The FDS assists when increased forces are required. Interossei muscles flex MCP joints and move in an ulnar direction and counteracted by Radiocollateral ligaments. Fingers lie adjacent and parallel to each other with contact with adjacent fingers	Thumb position varies. Often it flexes and adducts acting as a vice to secure the object to fingers and palm. Rotation to provide opposition at CMC joint but depends on the width of the grip.	Holding a beverage container or a telephone receiver incorporates the cylinder grip.
Spherical Grip	Grasp round objects. Fingers are spread apart more than in the cylinder grip. Interossei work more in this grip. The grasp is adjusted to the size and shape of the spherical object.	MCP abducted and partially flexed. Abductor and Adductor interossei co-contract to stabilize MCP joints. Extrinsic flexor muscles work to flex the fingers. ED contracts to counteract the flexor force and control release of object.	Thumb position varies according to size and weight of the object. Thumb opposed the other fingers and is flexed at MCP and may be flexed at IP joint. Adductor pollicis and other thenar muscles are active.	Holding a ball or an apple
Fist grip	Aka. Digital palmar prehension pattern . Palm and fingers flex around the object and the thumb wraps around from the other direction to enclose the object. Objects grasped are usually not large since the thumb closes over the object and often incontact with the digits.	All three joints of the digits are flexed. Digits are usually parallel to one another. Firm grasping is provided by the FDS and FDP.	Thumb is held in adduction against the object or in opposition to fingers. Thumb is used to provide a counterforce against the finger flexors to maintain the object in the hand.	Used for powerful grasps of objects. Grasping a broom handle, baseball bat, hammer, rake.

UE ORTHO CONDITIONS

Shoulder

i. Shoulder Dislocation

- a. OPP: 55° abduction 30° horizontal adduction
- b. CPP: Abduction + External rotation
- c. MOI
 - i. Anterior: ABER
 - ii. MC: Horizontal adduction & IR
 - iii. FOOSH
 - 1. Scaphoid fracture
 - 2. Lunate dislocation
 - 3. Colle's fracture
 - 4. Smiths fracture
 - 5. Elbow dislocation
 - 6. Supracondylar humeral fracture
 - 7. Shoulder dislocation (posterior)
 - 8. Clavicular fracture (Last to be injured)
 - a. Clavicle ossification at 21 years old
- d. MC direction: Anterior
- e. Nerve involved: Axillary, with Deltoid atrophy
- f. Common direction of dislocation:

Superior	SC & AC	Posterior	Hip
Anterior	GH	Anterior	Knee
Posterior	Elbow	Lateral	Patella
Anterior	Lunate	Posterior	Ankle

g. Lesions

- i. Bankart lesion – anterior shoulder dislocation, labral tear of anterior/ inferior
- ii. Hill Sach lesion – anterior shoulder dislocation, labral tear of posterior/ lateral
 - 1. Avulsion of the greater tuberosity
 - 2. Proximal humeral fracture
- iii. Benett's lesion – posterior shoulder dislocation, labral tear of posterior

What is the resting position of the wrist joint?

Ulnar Deviation + Neutral position of the wrist

- h. Medial to lateral, palpation greater tuberosity, bicipital tendon, lesser tuberosity
- i. Management:
 - i. Kocher's maneuver: Anterior dislocation – Horizontal adduction + IR
 - ii. PNF pattern: Contraindicated: D2 Flexion Best: D2Extension
 - iii. Surgical Intervention
 - 1. Bankart Surgery MC performed on glenoid labrum
 - a. Pt must wear a sling for 4 weeks

2. Magnuson: Subscapularis tendon transfer from lesser tuberosity to greater tuberosity (Avoid ER)
 3. Putti-Platt
 - a. Subscapularis tendon transfer from lesser tuberosity to glenoid labrum
 - b. CP: First motion lost is ER>ABD>IR
- ii. Frozen Shoulder
 - a. Adhesive Capsulitis/ Diabetic Periarthritis
MC disease associated with F.S.? Diabetes Mellitus
Situs Inversus
 - i. Reversal of organs from right to left
 - b. Stages: (259 – 412 – 526)
 - i. Freezing/ painful phase
 1. 2.5 to 9 months
 2. Pain even at rest
 - ii. Frozen/ stiffness phase
 1. 4-12 months
 2. Pain on movement, increase stiffness, decrease pain
 - iii. Thawing/ resolution phase
 1. 5 to 26 months
 2. (-) pain, (-) synovitis (+) LOM
 - c. Special test: For quick assessment of ROM, Apley's scratch.
- iii. Rheumatoid Arthritis: MC in GH joint
- iv. Osteoarthritis: MC in AC joint
- v. AC joint injury
 - a. Separation of AC joint, Step deformity (Step off sign)
 - b. Rockwood classification (AC joint injury)
 - i. AC sprained + CC intact
 - ii. AC torn + CC sprained
 - iii. AC + CC torn, CC space widening from 25-100%
 - iv. " " Clavicle d/p posteriorly
 - v. " " CC space widened >100% Deltoid + trapezius detach from distal clavicle
 - vi. " " Clavicle displaced inferiorly

- vi. Impingement syndrome
 - a. Tendinitis – Tendon inflammation.
 - b. Tendinosis – Tendon degeneration
 - c. Tenosynovitis – Inflammation of synovial membrane covering the tendon
 - d. Tenovaginitis – Thickening of tendon sheath
 - What is the most common cause of anterior shoulder pain?*
 - Subacromial Tendinitis, bursitis*
 - e. MC impingement Subacromial tendinitis
 - f. 2nd MC biceps tendinitis
 - i. Speed's test (Most reliable for long head bicipital tendinitis) Ludington's ((+) proximal Muscle tear)
 - ii. Yergasons (Primarily for transverse humeral ligament, short head of biceps tendinitis)
 - iii. Heuter's Sign: (+) Intact biceps, (-) distal tendon tear
 - iv. What part of the tendon is MC torn? Proximal portion
- vii. Supraspinatus tendinitis/ rotator cuff/ swimmer's
 - a. MC backstroke swimming, MC for anterior D/L of GH joint
 - b. Peritendinitis Calcarea (seen through x ray)
 - c. Types
 - i. 1^o impingement (> 40 years old)
 - 1. Poor posture
 - 2. Osteophyte formation
 - 3. Acromion types
 - a. Flat
 - b. Curve (Population)
 - c. Hook (impingement)
 - ii. 2^o impingement
 - 1. Irritation of the articular side
 - 2. Neer's classification
 - a. Edema & hemorrhage
 - b. Bursitis, tendinitis, fibrosis
 - c. Bony spurs & tendon rupture
 - d. Multiple tendon rupture
 - iii. Special test:
 - 1. Hawkin's Kennedy Test
 - 2. Empty Can – most optimal resisting full can motion
 - 3. Neer's Impingement (IR + elevation)
 - 4. Drop arm (supraspinatus tear)
 - 5. Yocum's (supraspinatus tendinitis from bottom up, Top to bottom Dugas test)

- viii. CRPS (complex regional pain syndrome, RSD, shoulder hand syndrome, Causalgia, algoneurodystrophy, post traumatic osteo, post traumatic vasospasm, post-infarctionsclerodactyl)
- a. Stages

I. Acute	II. Dystrophic	III. Atrophic	IV. Psychological
Localized pain Warm, and patchy skin Spotty osteoporosis Increased hair nail growth Burning pain	Diffused pain Cold, clammy dry scaly skin Diffused osteoporosis Beginning atrophy Hyperhydrosis (Excessive sweating)	Variable pain (Severe pain) Cool skin Generalized osteoporosis Pathologic fracture Severe irreversible deformity Atrophy	Psychological d/t Atrophic stage depression suicidal tendencies

- ix. TOS

- a. Brachial plexus, subclavian artery
- MC lower brachial plexus, C8-T1
 - D/t
 - Cervical rib
 - Clavicle and first rib
 - Superior Thoracic outlet
 - Scalene triangle (between anterior and middle scalene)
- b. Special tests
- Costoclavicular (military brace)
 - Roo's - 3 minutes
 - Adson's (hold breath)
 - Halstead
 - Allen's, All (+) diminished radial pulse. MC used since it is more superficial.

- x. Elbow

- a. Epicondylitis
- Lateral (Tennis Elbow)
 - MC affect muscle ECRB, if common Mobile Wad of Three
 - Special Test
 - Method 1 – Cozens(Active)
 - Method 2 – Mills (passive)
 - Method 3 (Active)
 - Medial (Golfer's elbow, throwers elbow,[Pedia: Medial Apophysitis, Little league's elbow])
 - MC Pronator Teres and FCR
- b. Pulled elbow
- aka Nurse maids elbow
 - Radial head dislocation from annular ligament
 - LOM: supination
 - MOI: Elbow extension and pronation

- c. Pushed Elbow: MOI: FOOSH Posterior elbow dislocation
- d. Olecranon bursitis
 - i. Acute: Student's Elbow
 - ii. Chronic: Miner's/ Draftsman's elbow
- e. Boxer's elbow/ hyperextension overload syndrome
 - i. MOI: Hyperextension + Valgus
- f. Volkmann's Ischemic Contracture
 - i. Aka Anterior forearm Compartment syndrome
 - ii. SC humeral fracture d/t FOOSH injury
 - iii. MC lateral>medial fracture
 - iv. Brachial artery vasospasm
 - v. (n) forearm pressure is 5-10mmHG, <15mmHg is safe. 30mmHg starting of ischemic contracture. 40mmHg Surgery, Fasciotomy
- g. Cubital Tunnel Syndrome
 - i. Ulnar nerve affectation
 - ii. MC site of ulnar nerve compression, by **FCU**
- h. Panner's (capitulum)
 - i. Osteochondritis/osis/ avascular necrosis of the capitulum , males>females
 - ii. Avascular Necrosis:

Scaphoid	Preiser's
Lunate	Keinbocks
Distal Lunate	Burns
Spine/ V. Endplate	Scheuermann's Dse
Vertebral Body	Calves
Femoral Head	LCPD/ children Chandler's/ Adult
Navicular	Kohlers
2 nd Metatarsal	Freiberg's
Talus	Diaz Disease
Calcaneus	Sever's Disease

xi. Wrist

- a. Function
 - i. W & H – 90%
 - ii. Thumb – 40-50%
 - iii. Index – 20%
 - iv. Middle – 20%
 - v. Ring – 10%

- b. Flexor Tunnel**
 - i. 4 Tendons of FDP
 - ii. 4 Tendons of FDS
 - iii. 1 Median Nerve
 - iv. 1 FPL
 - v. FCR possible 11th in carpal tunnel
- c. Extensor tunnel**
 - i. APL & EPB
 - ii. ECRL & ECRB
 - iii. EPL
 - iv. ED & EI
 - v. EDM
 - vi. ECU
- d. Flexor zone**
 - i. Insertion of FDP
 - 1. Sweater-finger sign
 - 2. Jersey finger
 - ii. No man's land
 - 1. FDS (Flexor Sublimis)
 - 2. Fibrous Lacertous
 - iii. Neck of MCP
 - iv. Carpal tunnel: Release
 - v. Distal forearm
- e. Extensor Zone**
 - i. DIP
 - ii. Midphalanx
 - iii. PIP
 - iv. Proximal phalanx
 - v. MCP
 - vi. Metacarpal
 - vii. Carpal bones
- f. Scaphoid fracture**
 - i. MOI: FOOSH
 - ii. Avascular necrosis – Preiser's
 - iii. Malunion proximal 3rd
 - iv. Watson's test: Anatomic snuff box tenderness

- g. Lunate D/L
 - i. MOI: FOOSH
 - ii. Avascular necrosis (Kienbocks)
 - 1. Stages
 - a. **S** -sclerosis
 - b. **F**- fragmentation
 - c. **C** – collapse
 - d. **A** – arthritis
 - iii. Special test: Murphy's sign
 - 1. Make a fist, and check height of MCP, if equal (+)
 - iv. Terry-Thomas Sign (check scaphoid and lunate gap, >3-5mm gap), scapholunate dislocation
- h. Madelung (congenital, malformed wrists)
 - i. Radius deformity (radial shortening)
 - ii. LOM pattern
 - 1. **E** – wrist extension
 - 2. **S** – Supination
 - 3. **P** – Pronation
 - iii. Wrist flexion is not affected
- i. Dupuytren's (palmar fascia)
 - i. Shortening of palmar fascia
 - ii. MC in (B) hands, if unilateral MC ®
 - iii. Affection of 4th and 5th digits
 - iv. Stages:
 - 1. Nodule formation (loose)
 - 2. Nodule formation + skin (tight)
 - 3. 1 or 2 finger contracture
 - 4. Fixed flexion contracture
 - v. Peyronies – contracture of penis, raised lesion
 - 1. D/T syphilis
 - vi. Surgery to correct Dupuytren's, McCash
- j. Mallet finger/ deformity
 - i. Rupture of lateral slip/ band EDC
 - ii. Signs: DIP flexion
- k. Bouttoniere
 - i. Central slip of EDC
 - ii. Sign: DIP hyperextension + PIP flexion + MCP extension
- l. Swan neck/ FDSwan
 - i. Rupture of FDS
 - ii. Signs: DIP flexion, PIP extension
- m. Skiers (gamekeeper's thumb) UCL of the thumb
 - i. UCL tear + adductor pollicis tear (Stener's Lesion)

- n. Jersey Finger
 - i. Rupture of FDP
 - ii. MC in 3rd digit
 - iii. Sweater-finger sign
 - o. EPL tendon rupture
 - i. MC tendon rupture in RA
 - p. De Quervains Tenosynovitis
 - i. MC affected APL, EPB other MC FPL, 2nd most common affected EPL
 - 1. FPL tendon rupture d/t RA (Mannerfelt syndrome)
 - ii. Special test: Finkelstein's Test (passive)
 - q. Trigger Finger
 - i. Mechanical Locking
 - ii. Pulley Most affected : A1 (MCP)
 - iii. MC finger affected 3rd and 4th digit
 - 1. Annular pulleys
 - a. A1 A2 MCP
 - b. A3 PIP
 - c. A4-A5 DIP
- xii. Congenital Deformities

Deformity	Presentation
Lobster clawhand	Congenital absence of 2 nd to 4 th CMC
Clasped finger	Congenital trigger finger
Guttering	Ulnar neuropathy/ lesion/ interosseus atrophy
Vaughn Jackson	4 th and 5 th EDC rupture
Camptodactyly	Bent Little finger
Clinodactyly	Little finger curve
Syndactyly	Webbed finger -Simple skin -Complex bony union
Polydactyly	Excessive digits
Macroductyly	Long fingers
Adactylia	Absence of finger
Acheria	Absence of wrist and hand
Amelia	Absence of limb
Meromelia	Absence of part of a limb E.g. Radius/ ulna Radius is Missing, Radialmeromelia
Phocomelia	Sealed limbs
Floating thumb/ Ponce Flottant/ Vagrant Thumb	No bone in the thumb

ST: UPPER EXTREMITY

Special Test	Position	Action	(+) Result	Indication
Anterior instability				
Anterior Drawer Test	Supine, arm in Abduction 80-120° ER 30° Flexion 20° Stabilize scapula	Give anterior force to proximal humerus	(+) Apprehension (+) pain (+) click labral tear	Anterior instability
Apprehension test/ crank test	Supine Abduction 90° Elbow flexion 90°	Slowly ER the arm	(+) apprehension	Anterior instability
Fulcrum Test	Supine Abduction 90° Elbow flexion 90°	Slowly ER the arm, place a fist at the proximal humerus	(+) apprehension	Anterior instability
Relocation/ Fowler's test/ Jobes	Supine Abduction 90° Elbow flexion 90°	Slowly ER the arm. Apply posterior force to the proximal humerus	(+) relief from apprehension	Anterior instability
Dugas	Sitting involved hand on C/L shoulder	Elbow depressed	(+) apprehension	Anterior instability
Posterior Instability				
Posterior Drawer test	Supine Elbow flexion 80-120°	Horizontal adduction and IR	(+) apprehension	Posterior Instability
Norwood's test	Supine Elbow flexion 80-120°	Downward force on elbow	(+) apprehension (+) increase posterior slippage	Posterior Instability
Inferior Instability				
- Best position to test inferior instability? 20-50° abduction with neutral rotation				
Subluxation grading	-	Grade 1 <1cm Grade 2 1-2 cm Grade 3 >2cm	-	-
Sulcus sign	Arms at the side	Pull down on distal humerus	(+) sulcus	Shoulder subluxation
Feagin Test	90° abduction or placed on therapists hand	Downward and lateral force on proximal humerus	(+) sulcus	Shoulder subluxation
AC joint				
AC shear Test	Therapists posterior hand is on spine of scapula, anterior hand clavicle	Compress	(+) excessive movement (+) pain	AC joint pathology

Scapulothoracic				
Lateral Winging “sliding door paralysis”	-	Ask patient to retract scapula	-	Trapezius C3-C4 CN XI
Medial Winging “Open book paralysis”	-	Ask patient to perform wall push ups	-	Serratus anterior Long thoracic, C5, C6, C7
Thoracic Outlet Syndrome				
Roo's	90° shoulder abduction 90° elbow flexion	Open close hands repeatedly for 3 minutes	(+) unable to finish 3 minutes	TOS
Adsons	Ipsilateral neck rotation and extended UE	Ask patient to hold breath and therapist palpates over radial artery	(+) diminished radial pulse	TOS
Costoclavicular (Military Brace)	Extend (B) UE	Palpate over radial artery	(+) diminished radial pulse	TOS
Halstead	Contralateral neck rotation and ipsilateral extended UE	Palpate over radial artery	(+) diminished radial pulse	TOS
Subscapularis				
Lift Off Sign (active)	Place dorsum of the hand on the back	Ask patient to lift arm	(+) if unable to do	Subscapularis lesion
Spring back test (passive)	Therapist Places dorsum of the hand on the back	Passively lift arm and ask to hold	(+) inability to hold position	Subscapularis pathology
Shoulder				
Horn Blowers Sign/ Signe de Clairon	Abduct arm 90° elbow flexed to 90°	Resist ER	(+) cannot resist	Teres Minor lesion
Elbow				
Carrying angle male 5-10° female 10-15° >20° cubitus valgus, <5° cubitus varus				
Cozen/ method 1	-	Resist wrist extension	(+) pain at lateral epicondyle	Lateral Epicondylitis
Mills/ method 2 (passive)	-	Passive elbow extension + wrist flexion	(+) pain at lateral epicondyle	Lateral Epicondylitis
Maudley's/ method 3	-	Resist 3 rd finger extension	(+) pain at lateral epicondyle	Lateral Epicondylitis
Medial epicondylitis Test (passive)	-	Passive elbow extension, wrist extension.	(+) pain at medial epicondyle	Medial Epicondylitis
Valgus Stress Test/ abduction force test	-	-	-	MCL
Varus Stress Test/ Adduction force test	-	-	-	LCL

Median Nerve				
Pronator Teres test	-	Resist pronation	(+) Paresthesia and tingling sensation on median nerve distribution below the pronators teres	PTSyndrome
Pinch grip test "ok" sign, Kiloh-Nevin	-	Ask patient to make an OK sign	(+) tip to tip (+) pulp to pulp	Intact median n. Lesion
Phalens Test	-	Perform for 1 minute	(+) paresthesia on median nerve distribution	Carpal tunnel syndrome
Reverse Phalens Test, prayers test	-	Perform for 1 minute	(+) paresthesia on median nerve distribution	Carpal tunnel syndrome
Tinel's sign for the wrist	-	Tap over the flexor retinaculum	(+) paresthesia on median nerve distribution	Carpal tunnel syndrome
Carpal compression test	-	Compress over median nerve for 30seconds	(+) paresthesia on median nerve distribution	Carpal tunnel syndrome
Flick sign	-	Flick hands	(+) disappearance of paresthesia over median nerve distribution	Carpal Tunnel Syndrome
Ulnar Nerve				
Tinel's for elbow	-	Tap over cubital tunnel	(+) paresthesia on medial forearm/ ulnar nerve distribution	Ulnar nerve compression
Elbow flexion test	Elbow flex and depress	Wait 3 minutes	(+) pain and paresthesia over	Ulnar nerve neuropathy
Froment's Sign	Paper in between index and thumb	Ask Pt to grasp paper, therapist tries to pull paper	(+) Thumb flexion/ compensation (aka Jeannes sign, hyperabduction compensation)	Ulnar nerve lesion
Radial Nerve				
Long Horn Sign	-	Active wrist extension	(+) unable to perform	Radial nerve injury
Others				
Watson	-	Palpate over anatomic snuffbox	(+) tendernesss	Scaphoid fracture
Murphy's	-	Make a fist	(+) unequal MCP heads	Lunate dislocation
Finkelstein	-	Make a fist, passive ulnar deviation	(+) pain	DQT
Bunnel Littler	-	(-) PIP flexion > flex MCP	=(+) PIP flexion	Intrinsic muscle tight
		(-) PIP flexion > flex MCP	=(-) PIP flexion	MCP capsule lesion

Tight Retinacular ligament	-	(-) DIP flexion > flex PIP (-) DIP flexion > flex PIP	=(+) DIP flex =(-) DIP flex	Tight retinacular ligament PIP capsule lesion
Allens Test	Palpate ulnar and radial artery,	Occlude (B) arteries ask patient to close hands	Check fill time for (B) arteries (+) delay	Patency of ulnar and radial arteries
Digital Artery blood flow	-	Check nailbeds	(+) blood flow does not return, 1-3 seconds	arterial insufficiency
Wrinkle/ shrivel	-	5-10 min water soak	(+) wrinkling (-) wrinkling	Intact nerve Dennervation
Ninhydrin sweat test	-	Wash hands, wait 30 mins. Blot hand on bond paper, apply ninhydrin .	(+) purple discoloration (-) purple discoloration	Intact nerve Dennervated

ANAKINES HIP

- i. Lower Extremity
 - a. Bones - 206
 - b. Muscles-620
 - c. UE-64, carpals (8)
 - d. LE-62, Tarsals (7)
 - e. LE Function
 - i. Ambulation
 - ii. Weight-bearing
 - iii. Posture
- ii. Bones
 - a. Pelvis (Pelvic Girdle)
 - i. Innominate bones (2)
 - 1. Ilium (means wing/ ear) Largest, Superior.
 - a. Iliac crest = L4, highest point in the pelvis (Homologous: Acromion)
 - b. Iliac tubercle = L5
 - c. Iliac spines
 - i. ASIS – Level of S2, (Origin of Sartorius, Inguinal Ligament/ paupart's/ aponeurosis of external oblique)
 - 1. Measurements
 - a. LLD, Q-Angle
 - b. If (B) Fulcrum for hip abduction/adduction
 - ii. AIIS – origin of Rectus femoris straight head
 - 1. Reflected head (ilium above acetabulum)
 - iii. PSIS – S2, Dimple of Venus
 - 1. Measurement
 - a. Schober's Test (AS)
 - iv. PIIS
 - d. *Duvurney's Fracture: Fracture of the iliac wing
 - 2. Ischium (means fish), Posterior.
 - a. Ischial tuberosity, (sit bone) Functions: Weight bearing when sitting
 - i. Origin of hamstrings
 - ii. BF (injures IT) ST SM Add Magnus
 - b. Ischiogluteal bursa
 - i. Prolonged sitting causes bursitis, known as:
 - 1. Tailor's, Weaver's, Boatman's bottom

3. Pubis (means crotch), Antero-Inferior
 - a. Common site of adductor origin
 - b. Superior Ramus, origin of pectineus
 - c. Pubic Tubercle/ Body origin of adductor longus
 - i. Osteitis Pubis
 1. Affects Add. Longus
 - a. Spasticity
 - b. Rectus abdominis
 - c. “moth eaten” appearance
 - d. Inferior Ramus
 - i. **M**agnus
 - ii. **G**racilis
 - iii. **B**revis
 - e. Obturator foramen
 - i. Lumbar and sacral nerves
 - ii. Blood Vessels
4. Acetabulum
 - a. Triradiate ligament connects ilium, ischium, pubis (Fused after birth)
 - b. Covered by fibrous cartilage, glenoid labrum (Cotyloid ligament)
 - c. Orientation
 - i. LIA (femoral head (SAM), covered by hyaline + fibrous cartilage)
- ii. Sacrum
 1. Sacral promontory
 2. Sacral body
- iii. Coccyx
 1. Aka tail bone
 2. Painful coccyx, “Coccygodynia”
 - a. MOI: Falling in a seated position
- iv. Pelvic Inlet (Pelvic Brim) – carries abdominal organs
 1. Male: Heart Female: Oval
 2. Male: smaller Female: larger
 3. Borders:
 - a. Ant: Symphysis pubis
 - b. Post: Sacral promontory
 - c. Lat: Iliopectineal line
- v. Pelvic outlet – carry’s reproductive organs
 1. Ant: Pubic arch
 2. Post: Coccyx
 3. Lat: Ischial tuberosity

b. Pelvic Tilt

i. ASIS=PSIS

1. ASIS<PSIS = APT
2. ASIS>PSIS = PPT

ii. Nutation/ Counternutation

Nutation	Movement	Counter-nutation
Sacral flexion	AKA	Sacral Extension
Smaller	Pelvic Inlet	Larger
Larger	Pelvic Outlet	smaller
Posterior	Pelvic Tilt	Anterior
Anterior + Inferior	Sacral promontory	Posterior + Superior
Posterior + Superior	Coccyx	Anterior + Inferior
Approximate	Pelvic Wing	Distract
Distract	Ischial Tuberosity	Approximate
Sacroiliac Sacrotuberous Sacrospinous	Ligaments Restrict	

iii. During labor – Litotomy position – nutation + posterior pelvic tilt

iv. LACEB

v. KPNF

c. Level Structures

i. ASIS=PSIS >pelvic Tilt

ii. Iliac crest (L) = (R) > pelvic obliquity/ list

iii. Pubic Tubercle =Greater trochanter True LLD, Coxa vara, D/L

d. Femur

i. 25% of the general height

1. Neck shaft angle

- a. Adult 125°
- b. Child 160°
- c. Increase coxa valga
- d. Decrease coxa vara

2. Femoral Torsion

- a. Ante/ Retro (n) angle 8-15°
 - i. Increase anteversion (intoe, patellar squinting, SPITT, knocked knees)
 - ii. Decrease Retroversion (out toe, frog eyes/ grasshopper, SSETT, bowleg)
- b. Special test for IR/ER: Craig's/ Ryder's

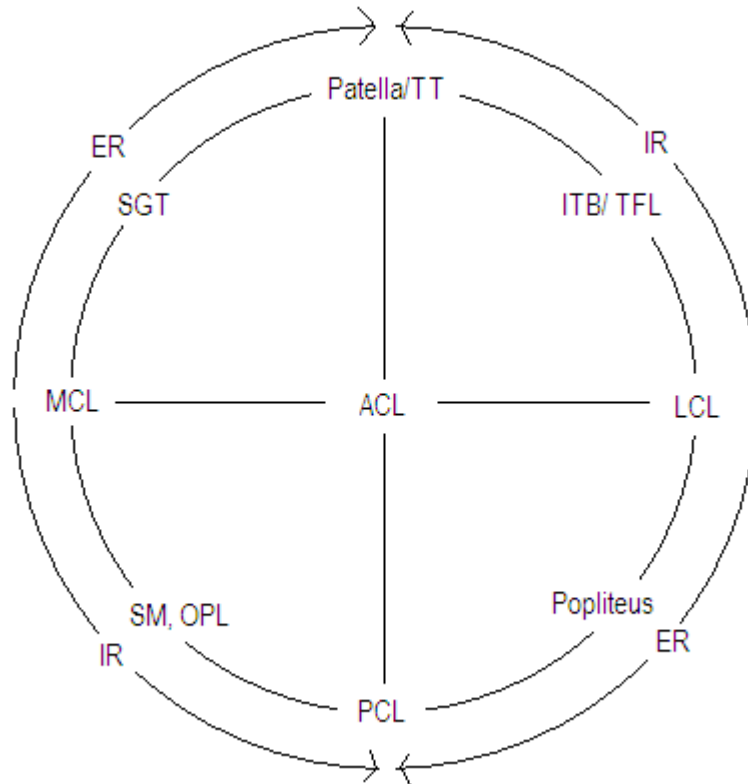
3. Center edge angle
 - a. Aka Angle of Wiberg
 - b. (n) 20-30°
 - c. Increase more stable
 - d. Decrease less stable
 - e. Aging increases center edge angle
4. Q angle
 - a. Landmarks: ASIS> Midpatella> Tibial tuberosity
 - b. (n) males: 13° females: 18°
 - c. Increase: genu valgum, increased anteversion, patella baja
 - d. Decrease: genu varum, increase retroversion, patella alta
- ii. Femoral head (covered by hyaline and fibrous)
 1. Blood supply:
 - a. Obturator Artery/ ligamentum teres 20%
 - b. Medial Circumflex Artery/ Artery of Trueta 80%
- iii. Femoral Neck
- iv. Trochanter
 1. Greater
 - a. G Med/Min
 - b. Red Carpet Muscles
 - i. P - piriformis
 - ii. O – obturator internus
 - iii. O – obturator externus
 - iv. S - Superior gemellus
 - v. I – Inferior gemellus
 - vi. Q uadratus Femoris -> quadrate tubercle
 2. Lesser
 - a. Iliopsoas
- v. Femoral Shaft
 1. MC type of fracture, Oblique 2nd MC Spiral
 - a. Fixation
 - i. Elderly – external fixation
 - ii. Young – internal fixation
- vi. Condyle
 1. Medial condyle larger, longer
 - a. MC Osteochondritis Dissecans,
 - b. Lateral aspect (LAMFC)
 - c. MC for OA
 2. Lateral condyle, more anterior
 - a. MC for RA
- vii. Intercondylar groove

- e. Tibia
 - i. Tibial Plateau
 - 1. Medial C-shaped, Lateral O-shaped, MCLO
 - 2. Tibial Notch
 - ii. Tibial Tuberosity
 - 1. Attachment for Patellar tendon
 - a. Osgood Schlatter Disease
 - i. M>F, adolescents
 - ii. Bilateral, NO US (epiphyseal plate)
- f. Fibula
 - i. Head
 - 1. Location for Common Peroneal nerve
 - ii. Neck
 - iii. Shaft
- g. Patella
 - i. Sesamoid bone, largest (smallest: Fabella, back of the knee)
 - ii. Diamond shaped.
 - iii. Superior pole: Quads tendon
 - iv. Inferior pole: Patellar tendon
 - v. Increases leverage power by 35% of quads
 - vi. Patellar tilt
 - 1. Occurs flexion and extension
 - 2. Increase ROM
 - 3. Prevents Jamming
 - 4. Patellar tracking, abnormal movement towards lateral
 - vii. Fracture:
 - 1. Bipartite – into 2, occurs at inferior pole
 - 2. Tripartite – into 3, similar to peace sign
 - 3. Management: Kirschner Wire
 - viii. Displacement
 - 1. Alta – High, Camel Sign (prominence of infrapatellar bursa)
 - 2. Baja – Low, decrease quads leverage, prone to stiff knee (arthrogenic knee)
 - 3. Parva – Irregularly Shaped
 - 4. Tarda - small, chondritis
- iii. Joints and ligaments
 - a. Pelvis
 - i. SI Joint
 - 1. SI ligaments
 - a. Weakens by Relaxin,, and ankylosing Spondylitis
 - 2. Movement: Nutation/ Counter-Nutation

- ii. Sciatic notch, divided by sacrospinous ligament
 - 1. Greater
 - a. Piriformis passes through and divides (if sciatic nerve passes through piriformis = “fat wallet syndrome”)
 - i. Superior GSF
 - 1. Superior gluteal nerve/ artery
 - 2. Sciatic nerve
 - ii. Inferior GSF
 - 1. Inferior gluteal nerve/ artery
 - 2. Lesser, closed by sacrotuberous ligament
- b. Acetabulo Femoral joint/ Hip joint
 - i. OPP: FABER 30 30 slight
 - ii. CPP: EXABIR, most stable position of the hip Extension, most limited IR
 - iii. CP = IR>F>AB (IRFAB) if there is RA, IR>F>E>Contractures (IRFEC)
 - iv. Ligaments
 - 1. Iliofemoral ligament
 - a. Aka Y ligament of bigelow
 - b. Strongest ligament of the body
 - c. Restricts: EXER small amount AD
 - d. This is used if there is weakness of iliopsoas, jackknife
 - i. Psoatic gait, weak iliopsoas (swing phase)
 - ii. Difference, Backward lurch (Stance, GMax weakness)
 - iii. Psoatic limp – LCPD (FABER swing)
 - 2. Ischiofemoral ligament
 - a. Restrict IR, EXAB small amount of AD
 - 3. Pubofemoral ligament
 - a. Restrings AB, EXER
 - v. Factors – hip stability
 - 1. Architecture/ stability
 - 2. Ligaments – Y lig
 - 3. Atmospheric pressure is negative, creates a suction
 - vi. Dislocation – posterior
 - 1. MOI Dashboard injury, FADIR + PCL
- c. TibioFemoral joint/ knee joint
 - i. Modified hinge joint
 - 1. 2 joints
 - a. Medial, more weight bearing OA
 - b. Lateral, more synovium, RA
 - ii. (n) position of the knee, Genu valgus 6°
 - iii. OPP: 25° flexion
 - iv. CPP: full extension + ER
 - v. CP flexion> extension

- vi. Screw Home Mechanism
 - 1. Last 20° of extension **FIR -TER**
 - a. Tibia ER on femur (OKC)
 - b. Femur IR on Tibia (CKC)
 - 2. Muscles:
 - a. Unlocks the knee: popliteus (deepest ms of back of knee) **FER-TIR**
 - i. OKC – Tibial IR
 - ii. CKC – Femoral ER
 - b. Locking: none
 - 3. Cannot be controlled, it is a mechanical event.
- vii. Ligaments
 - 1. Cruciates (LAMP)
 - a. ACL
 - i. SPL from medial tibial plateau to lateral femoral condyle
 - ii. Slack 30-60°
 - iii. Taut – Extension/ IR
 - iv. Weakest ligament of knee
 - v. Prevent femur backward, tibia forward
 - vi. MOI hyperextension, Rotation
 - vii. O'Donoghue Triad
 - 1. ACLExt c Rot
 - 2. MCL Valgus force
 - 3. Medial Meniscus Foot Flat/ grind
 - viii. Special test
 - 1. Lachman's
 - 2. Anterior drawer test (Hamstrings)
 - b. PCL
 - i. SAM from lateral tibial plateau to medial femoral condyle
 - ii. Strongest ligament in the knee
 - iii. Slack: Ext, Taut: flexion
 - iv. Prevents forward femur, tibia, backward
 - v. MOI: Dashboard, hyperflexion
 - vi. Special test
 - 1. Posterior Sag sign
 - 2. Godfrey Test
 - 3. Gravity
- 2. Collaterals
 - a. MCL
 - i. Slack in 90° flexion
 - ii. Taut: in extension + ER
 - iii. Prevents Valgus force/ abduction
 - iv. Ossification of MCL Pellegrini Steina Disease
 - v. Frog swim' Swimmers knee(MCL)

- b. LCL
 - i. Slack in 90° flexion
 - ii. Taut in extension + ER
 - iii. Prevents Varus force/ adduction
 - c. Genu valgus
 - i. Dislocation laterally
 - ii. Causes
 - 1. Weight/ GRF
 - 3. Coronary/ Meniscal ligament
 - a. Medial meniscus to the tibia
 - d. Patellofemoral joint
 - i. Functional plane
 - 1. Gliding rule: same
 - 2. Function
 - a. Leverage
 - b. Protection
 - 3. Loading
 - a. W -walking = 0.3x BW
 - b. A – ascending = 2.5x BW
 - c. D – Descending = 3.5x BW
 - d. S – squats = 7x BW
 - e. Rotatory Instability



Muscle groups

Hip

Flexors

- iv. Iliopsoas (sit ups muscle)
 - a. Iliacus
 - i. Femoral nerve(L2, L3, L4)
 - b. Psoas major
 - i. Lumbar plexus (L1, L2, L3)
 - c. Insertion: Lesser Trochanter
 - d. Action: hip flexion/ abdominal flexion
 - e. Conditions
 - i. Weakness: increase lordosis during sit ups
 - ii. Site of hemorrhage
 - iii. Iliopsoas bursitis (no rectus femoris)
 - f. Most active: Heel off to toe off, acceleration
 - g. Special Test:
 - i. Thomas Test
 - h. Poland's Disease
 - i. Palmaris longus
 - ii. Piramidalis
 - iii. Pecs major sternal head
 - iv. Plantaris
 - v. Psoas Minor
- v. Rectus Femoris (SLR ms)
 - a. Part of Quads
 - b. 2 joint muscle
 - c. Hip flexion Knee extension
 - d. 2 heads:
 - i. Straight head AIIIS
 - ii. Reflected head: ilium above acetabulum
 - e. Innervated by femoral nerve
 - f. Active insufficiency: hip flexion and knee extension
 - g. Passive: Hip extension and knee flexion
 - h. Special Test:
 - i. Ely's Test (most sensitive)
 - ii. Thomas Test (Kendall)
- vi. Sartorius (Tailor's muscle)
 - a. Longest muscle of the body
 - b. ASIS to proximal medial tibial condyle, (Pes anserine area)
 - c. Innervated by femoral nerve
 - d. Action:
 - i. FABER of hip
 - ii. FIR of knee

- vii. TFL-> ITB
 - a. Innervated by Superior gluteal nerve (L4, L5, S1)
 - b. Action
 - i. Hip FABIR
 - ii. Knee EXER
 - c. Special Test:
 - i. Ober Test

Extensors

- viii. Gluteus Maximus
 - a. Innervated: Inferior Gluteal nerve
 - b. 3 fibers
 - i. Upper - abduction
 - ii. Middle - Extension
 - iii. Lower- adduction
 - c. Action
 - i. Hip extension (active)
 - 1. Strengthen by hip ext c ER
 - ii. Extension: ER
 - iii. Flexion: IR
 - d. Protected by large fat pad
 - e. Most active in initial contact
 - f. Covers the red carpet muscles
- ix. Hamstrings
 - a. Innervated by sciatic nerve
 - b. Parts (4)
 - i. Semimembranosus
 - 1. Oblique popliteal ligament
 - ii. Semitendinosus
 - 1. Palpable
 - iii. Biceps femoris
 - 1. Long headTibial branch of sciatic nerve
 - 2. Short head, common peroneal nerve
 - iv. Adductor magnus (hamstring part)
 - 1. Obturator/ sciatic nerve
 - c. Action Hip extension
 - i. Knee rotation by BF + ST (ER), SM(IR)
 - d. Gait deformity –
 - i. terminal knee extension impact/ abrupt knee extension (Deceleration)
 - e. Most active in gait: Deceleration

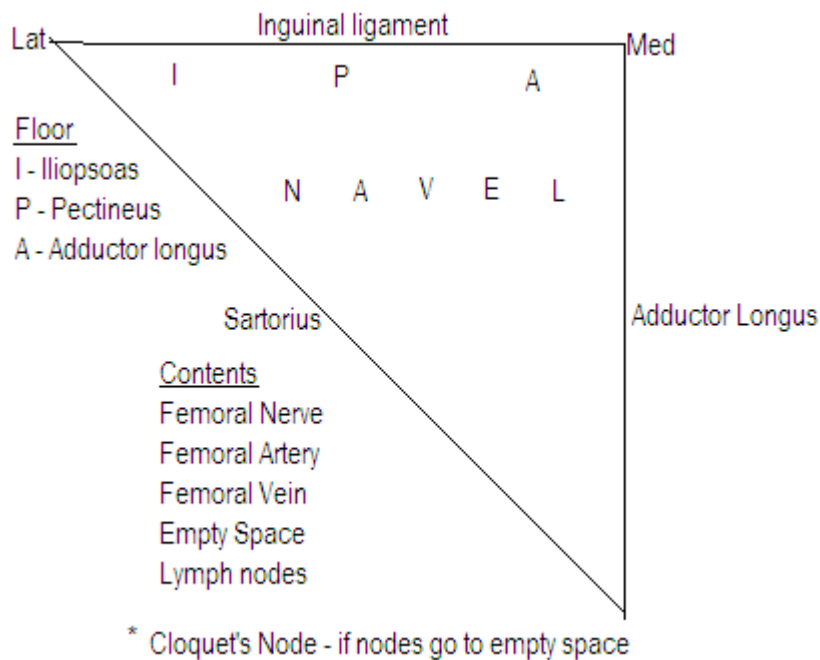
Abductors

- x. G. Medius (main abductors)
- xi. G. Minimus (main IR)
 - a. Abductors and IR
 - b. Innervated: Superior Gluteal nerve (L4, L5, S1)
 - c. Gait: Waddling/Chorus Girl/ Mae West/ Cheerleader/ Trendelenburg
 - d. MC cause of trendelenburg gait is OA
 - i. Uncompensated (pelvis list/ obliquity drops to opposite side of lesion)
 - ii. Compensated (Trunk lean to ipsilateral side)
 - iii. (The foot that is in midstance is the weak side)

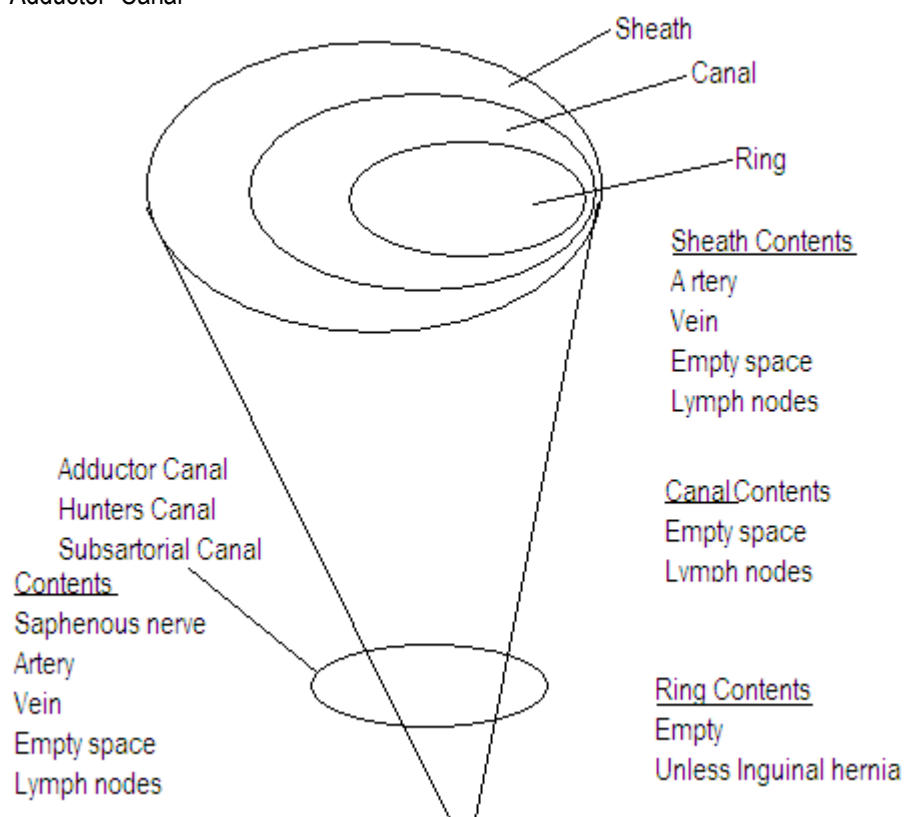
Adductors

- xii. Pectineus
 - a. Not innervated by obturator nerve, femoral nerve
 - b. Action flexion and adduction, irregardless of position
- xiii. Magnus –
 - a. Biggest, inserts in adductor tubercle
 - b. Hamstring
 - c. Adductors
- xiv. Gracilis
 - a. Weakest adductor
 - b. Tendon graft muscle
 - i. Gracilis
 - ii. PLantaris
 - iii. Palmaris longus
 - iv. PAcs major
- xv. Brevis - thickest
- xvi. Longus - Strongest adductor
 - a. All
 - i. Innervated by obturator nerve except pectineus
 - ii. ER for hip flexion
 - iii. IR for hip extension
 - iv. Active in foot flat to midstance to heel off
- xvii. Red Carpet muscles
 - a. P – Piriformis
 - i. ER in extension
 - ii. IR in flexion (similar to G. Max)
 - b. O – Obturator internus
 - c. O – Obturator externus
 - d. S – superior gemellus
 - e. I – inferior gemellus
 - f. Q – Quadratus femoris: inserts Quadrate tubercle
 - g. All: Inserts into greater trochanter

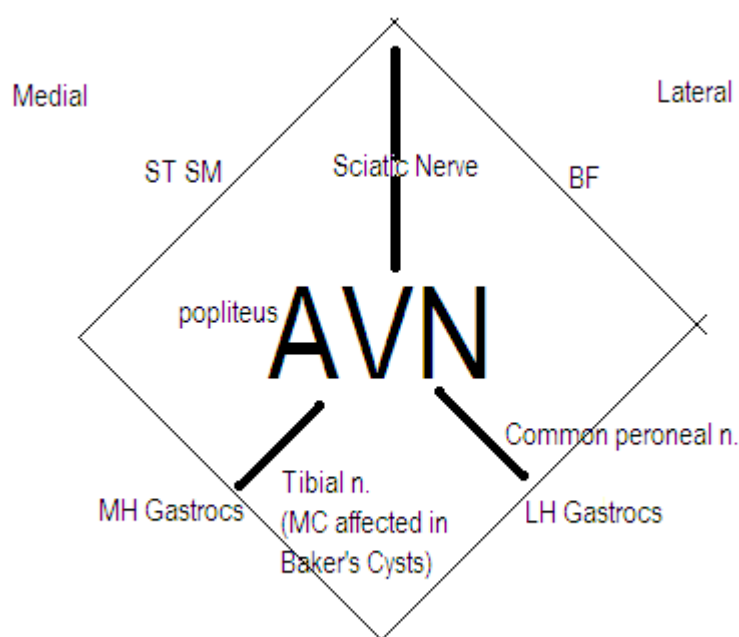
- xviii. Quadriceps
- Action: knee extensors
 - If no quads > genu recurvatum to stand
 - Affects the PCL
 - Innervated by femoral nerve L2, L3, L4
 - Supported PCL
 - Parts (4)
 - Rectus femoris
 - Vastus lateralis -Strongest
 - Vastus intermedius –deepest
 - Articularis Genu (plica/ capsule) (counterpart anconeus)
 - Vastus Medialis
 - VMO (workhorse)
 - Controls patellar tracking
 - 1st to atrophy, last to recover
 - VML
 - Condition
 - Weakness
 - Difficulty stairs
 - Ascending
 - Descending (Most difficult)
 - Ramps, sitting. Standing
 - ACL injury
 - Co-contraction hamstrings before quads
- xix. Landmarks
- Femoral Triangle



b. Adductor Canal

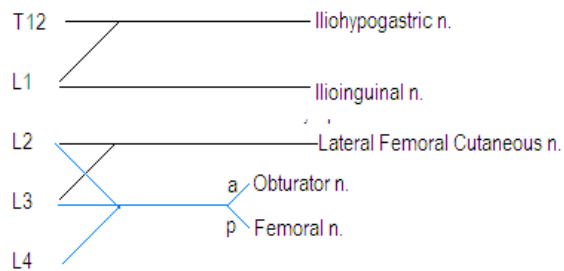


c. Popliteal fossa

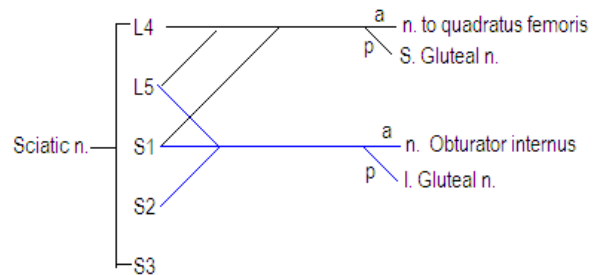


- xx. Blood and nerve supply
- Supplied by Descending/ thoracic/ Abdominal Aorta > common iliac artery > external & internal iliac artery
 - External iliac artery> (pass throughinguinal ligament) > femoral artery (gives a brach called profunda femoris/ deep femoral artery)> (pass through popliteal artery) > popliteal artery > posterior & anterior tibial artery> dorsalis pedis
 - Venous puncture at venous puncture
- xxi. Lumbar and sacral plexus

Lumbar Plexus



Sacral Plexus



ANAKINES LEG, ANKLE, & FOOT

Leg

- i. Bones
 - a. Number of bones in the foot: 26
 - b. Tibia (Tibia)
 - i. Tibial Tuberosity
 1. Osgood Schlatter Disease (avascular necrosis, traction apophysitis/ inflammation)/ Jumper's knee
 - a. Male > Female
 - b. Young 10-14 years old
 - c. Bilateral affection
 - d. Self-Limiting, best treatment is rest
 - e. E.g. basketball/ volleyball
 - f. Precautions: Ultrasound
 2. Housemaids knee – prepatellar bursitis
 - ii. Tibial plateau
 - iii. Tibial Condyles
 1. Medial
 2. Condyle
 - iv. Tibial Crest (pressure sensitive)
 1. Site of MC Malunion fracture (Distal 3rd OF THE TIBIA)
 - v. Medial Malleolus
 1. Shin splints
 - a. Traction apophysitis of calcaneus by tibialis posterior (Braddom 2012)
 - c. Fibula (Splint)
 - i. Head (pressure sensitive area)
 1. Common peroneal (most commonly entrapped since it's the most superficial nerve in the LE, median nerve in UE) aka Cross Leg Palsy
 - a. Fracture of fibular head
 - b. PTB socket
 - c. Bed Ridden with LE Externally rotated
 - d. Ice massage on lateral knee
 - e. Conditions
 - i. Weakness (Gait: heel strike, Deformity: Foot drop, Gait Deviation: Foot slap (Steppage), compensation: excessive Hip/knee flexion during swing phase)
 - ii. Neck
 - iii. Shaft
 - iv. Lateral Malleolus

d. Medial vs lateral malleolu

Medial Malleolus		Lateral malleolus
Shorter	Size	Longer
Inversion>eversion	Motion	
0-35°	ROM	0-15°
Anterior	Position (n) Fick Angle 5-10°	Posterior

(n) values	
NSA	125°
Femoral Anteversion	15°
Sacral	30°
Pelvic inclination	50-60°
TF shaft angle	6° valgum
TTorsion	20°
Hallux Valgus	<15°
Fick	5-10°
Q-angle	13-18°

ii. Joints

i. Superior TibioFibular Joint (Forgotten joint)

1. Gliding joint
2. Dorsiflexion of ankle causes upward glide
3. PlantarFlexion of ankle causes of downward glide

ii. Inferior TibioFibular Joint

1. Syndesmosis

iii. Muscle

a. Leg muscles

i. Anterior (4) (aka Pretibials)

1. Tom – Tibial Anterior (primary dorsiflexors)
2. Dick – Extensor Digitorum Longus
3. Harry – Extensor Hallucis Longus
4. Potter – Peronius Tertius
5. All
 - a. Action: Dorsiflexors/ ankle extesion
 - b. Peak activity: Heel-strike
 - c. Test: walk on heels
 - d. Innervation: Deep Peroneal nerve/ Anterior Tibial Nerve

ii. Lateral

1. Peroneus Longus
2. Peroneus Brevis
3. All
 - a. Action: Eversion
 - b. Innervation: Superficial Peroneal nerve(Musculocutaneous nerve of the leg)

iii. Posterior

1. Superficial

- a. Gastrocs-Soleus (Primary plantarflexor) (triceps surae, 2 heads gastrocs 1 head soleus)
 - i. Gastroc
 1. Lateral Head
 2. Medial head
 - ii. Soleus
 1. From tibia
- iii. MMT (as many tip toes possible)
 1. Gastroc-soleus knees extended (Active insufficiency of gastrocs)
 2. Soleus c flexed knee

Gastroc	Soleus
Type II	Type I
Fast glycolytic	Anti-Gravity muscle
Jumping	
Running	

- b. Plantaris/ Freshman's nerve/ Fisherman's nerve

2. Deep

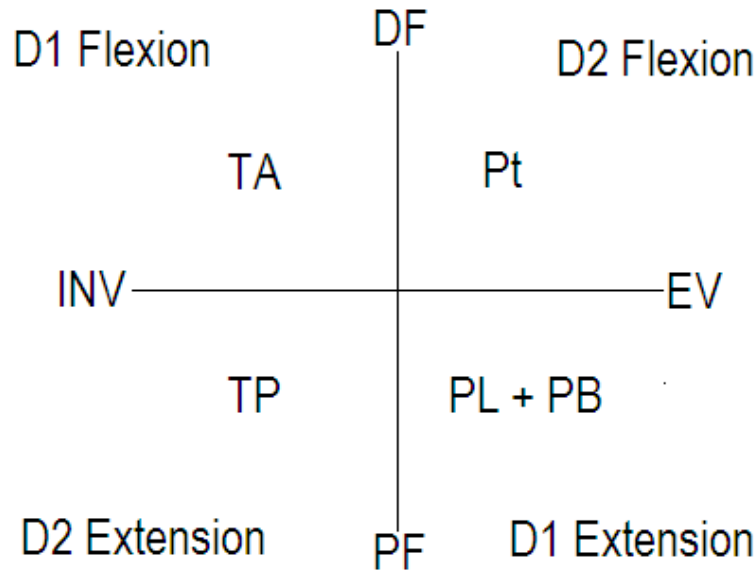
- a. Tom – Tibialis Posterior (inverter par Excellence, primary inverter)
- b. Dick – Flexor Digitorum Longus
- c. Harry – Flexor Hallucis Longus
- d. Potter – Popliteus (key)
 - i. Lateral femoral condyle into lateral tibia

3. All

- a. Planter flexors/ ankle flexion, except popliteus
- b. Peak activity: push off (Gait Dev: Apropulsive gait pattern)
- c. Test: Walk on toes
- d. Innervation: Tibial nerve/ Posterior Tibial nerve
- e. Trivias:
 - i. Deepest muscle of the calf: Tibialis posterior
 - ii. Deepest muscle of the knee: Popliteus

iv. Cutaneous Supply

1. Posterolateral – Sural nerve, Tibial
2. Posteromedial – Saphenous, Femoral
3. Anterior leg and dorsum of the foot, Superficial peroneal nerve (Musculocutaneous)
4. First dorsal webspace – Deep peroneal nerve
5. Sole of the foot – Tibial nerve



- v. MC deformity of LE – clubfoot
 - 1. In order:
 - a. A-Adduction
 - b. I- Inversion
 - c. P – plantarflexion
- vi. MMT direction of resistance

Ankle Joint

- iv. Bones
 - a. Talus
 - i. Talocrural joint/ Ankle joint (Ankle mortise with Talus)
 - 1. Type: Pure Hinge, DF: 0-20, PF 0-50
 - 2. OPP: 10° PF
 - 3. CPP: Full dorsiflexion
 - 4. CP: PF>DF
 - 5. Trivia
 - a. Most congruent joint in the body
 - b. MC injured joint in the sports (Mx: PRICE)
 - ii. Ligaments
 - 1. LCL
 - a. MC sprained ligaments
 - i. Fewer ligament
 - ii. Separate
 - iii. Inversion>eversion
 - b. ATFL - Anterior talofibular ligament **MC sprained**
 - c. CFL - Calcaneofibular ligament **2nd MC sprained**
 - d. PTFL posterior talofibular ligament

2. MCL –deltoid ligament **Least commonly sprain in the ankle**

Bony damage will occur first before damage to MCL

- a. ATTL – anterior talotibial ligament
- b. PTTL – posterior talotibial ligament
- c. CTL – calcaneotibial ligament
- d. TN – tibionavicular ligament

Ligament	MOI	Special Test
Anterior talofibular ligament	PF +INV	Ant. Drawer
Calcaneofibular ligament	DF + INV	Talar Tilt
Deltoid ligament	EVE	Kleiger Test

Cotyloid ligament (Acetabular labrum)

v. Foot

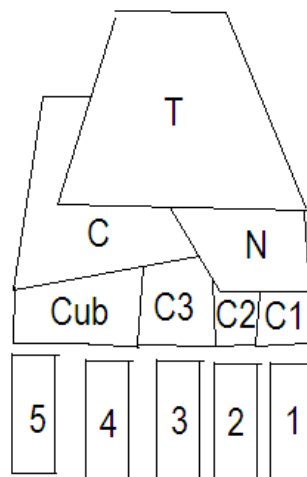
- a. Tarsals 7
- b. MT 5
- c. Phalanges 14
- d. Total: 26

i. Ossification process

- 1. Come - CApitate
- 2. Home - hamate
- 3. To - Triguetrum
- 4. Luneta -lunate
- 5. To -trapezium/ trapezoid
- 6. See - Scaphoid
- 7. Philippines – Pisiform

Talus	Calcaneus
Astragalus (-) muscle attachment TB of the foot	Larger First to ossify 1 st Fractured (lover's heel) "Os calcis" Sustentaculum tali – medial side (support head of talus)

Concertina Collapse if fracture D/t Pott's disease



e. Metatarsals

i. Trivias

1. Longest : 2nd , Stoutest – 1st

ii. Conditions

1. March Fx – 2nd MTT Shaft “ MarSHAFT:
2. Jones Fx – 5th MTT base (P. brevis & Tertius Insertion)
3. Freiberg – AVN of head of 2nd MTT
4. Iselin – AVN of 5th MTT

iii. Configuration

1. Forward projection of metatarsal head $2 > 3 > 1 > 4 > 5$
2. $2 > 1 > 3 > 4 > 5$ Toes (Morton's/ Grecian/ Atavistic Foot) (23%)
3. $3 > 4 > 2 > 5 > 1$ Fingers
4. $1 > 2 > 3 > 4 > 5$ Egyptian foot; Most common type (69%)
5. $1 = 2 > 3 > 4 > 5$ Square foot

f. Arches

i. Medial longitudinal arch

1. Bones:
 - a. Talus (Keystone: head of talus)
 - b. Navicular (important)
 - c. Calcaneus
 - d. 1 , 2, 3 cuneiforms
 - e. 1, 2 , 3 Metatarsal
2. Ligament : Spring ligament (plantar calcaneonavicular ligament)
 - a. Proximal attachment: sustentaculum tali
3. Tendon: Tibialis posterior tendon
4. Special Test
 - a. Navicular drop test: Feiss line
 - i. Med. Malleolus > MTP
 - ii. Navicular should be within the line
 - b. Flat foot deformity/ Pes Planus
 - i. Pronated
 - ii. Pes Valgus
 - c. High Arch foot
 - i. Pes cavus (MC in Charcot Marie Tooth [Peroneal muscular atrophy, eversion is lost 1st, (+) inverted champagne bottle])
 - ii. Supinated, Pes varus, Inverted

ii. Lateral longitudinal arch

1. Bones
 - a. Calcaneus
 - b. Cuboid (key stone)
 - c. 4th-5th MTT
2. Ligament: Long plantar ligament
3. Tendon: Peroneus Longus tendon

iii. Transverse arch

1. Bones:

- a. 3 cuneiforms
- b. Cuboid
- c. 1-5th metatarsals (bases) (2nd cuneiform keystone)

2. Support: Intrinsic muscles of the foot

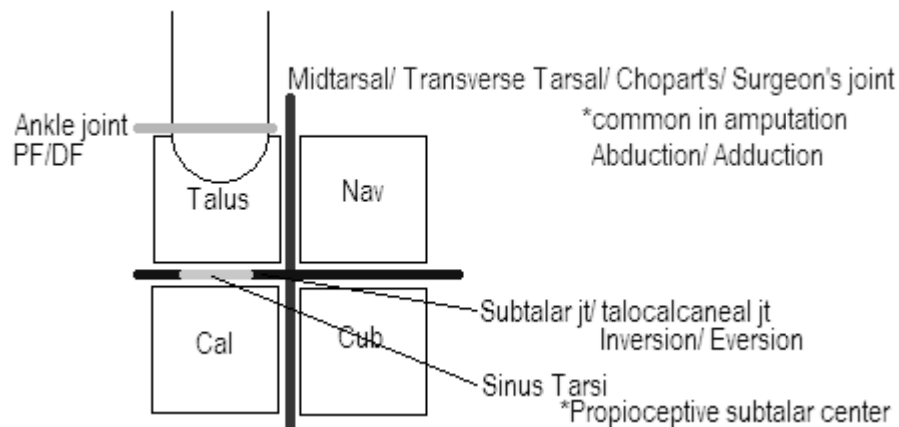
3. Conditions

- a. Splay foot
 - i. Muscle weakness (intrinsic muscle)
 - ii. Collapse of transverse arch

vi. Joints

- a. Talocrural
- b. Subtalar
- c. Midtarsal

	OKC	Mnemonic	CKC
Supination	INV, ADD, PF	SINADP	INV, ABD, DF
Pronation	EV, ABD, DF	PEVABD	EV, ADD, PF



- i. Interosseous talocalcaneal ligament in Sinus Tarsi (connections to cerebellum)

d. Syme's ankle disarticulation

e. Chopart

- i. TALO NA
- ii. CCPA

vii. Muscles of the foot

a. Dorsal compartment

- i. EDB Only muscle on the dorsum of the foot, with no counter part in the hand

b. Plantar compartment

i. Layer 1

1. Abductor Hallucis
2. Flexor Digitorum Brevis
3. Abductor Digiti Minimi

ii. Layer 2 (Dick and Harry (Master knot of Henry))

1. Lumbricals
2. Quadratus plantae

iii. Layer 3

1. Flexor Hallucis Brevis
2. Abductor Hallucis
3. Flexor Digiti minimi brevis

iv. Layer 4

1. Interossei (7) (TP and PL tendons)
 - a. 4 dorsal
 - b. 3 plantar

Hand	Foot
Interossei	Interossei
Lumbrical	Lumbricals
FDS	FDB
FDP	FPL
EDC	EDL

v. Originate in calcaneus

1. EDB
2. A
3. F
4. A
5. Q

c. Nerve supply

i. Tibial

1. Medial plantar nerve (4)
 - a. Flexor hallucis brevis
 - b. Abductor hallucis
 - c. Flexor Digitorum brevis
 - d. 1st lumbrical
2. Lateral plantar nerve

- d. Blood supply
 - i. Femoral artery> profunda femoris> posterior thigh (hamstrings)
 - ii. Femoral artery> popliteal artery> anterior and posterior tibial artery
 - iii. Posterior tibial (gastrocs)
 - iv. Anterior tibial (DF)> dorsalis pedis
 - 1. Palpation of DP artery
 - a. Lat to TA
 - b. Lat to EHL
 - c. Between 1st and 2nd MTT bones
 - d. Between Dick and Harry (EDL and EHL)
- e. Landmarks of the foot
 - i. Tarsal tunnel (medial side of the foot)
 - 1. Flexor retinaculum of ankle (lancinate ligament)
 - a. Tom (TP)
 - b. Dick(FDL)
 - c. And (PTA)
 - d. Very(PTV)
 - e. Nervous (PTN) (tarsal tunnel syndrome, sole numbness)
 - f. Harry (FHL)
 - 2. Extensor Retinaculum
 - a. Tom (TA)
 - b. Hanks (EHL)
 - c. And (Ant. TA)
 - d. Very (vein)
 - e. Nervous (DeepPN)
 - f. Dick (EDL)
 - g. Tracy (PTert)

LE ORTHO CONDITIONS

Pelvis

- i. Osteitis Pubis
 - a. Affects pubic body/ tubercle/ symphysis pubis
 - b. Decrease bone density
 - c. Ca⁺⁺ by adductor longus, Rectus Abdominis
 - d. (+) moth eaten appearance
 - e. Mx:
 - i. Roods
 - ii. Botox
 - iii. Pelvic Rhythmic Exercises E.g. Kegel's/ Pelvic Elevator/ Pelvic Clock
- ii. Innominate syndrome
 - a. Anterior innominate, AILSUP
 - i. Supine – LE becomes longer
 - ii. Sitting: - LE becomes shorter
 - b. Posterior innominate, PISSUP
 - i. Supine – LE becomes shorter
 - ii. Sitting – LE becomes longer

Hip condition

iii.

	LCPD	SCFE
Etio	Idiopathic	Idiopathic
	AVN of femoral head M>F M: 3-12 F: 5-15 Coxa Plana (+) crescent sign	Downward Slippage of head of femur M>F M/F: 7-15 (+) Coxa Vara = decrease NSA
Gait	(+) Psoatic Limp	(+) Trendelenburg gait
Body Build	Short and Thin	Obese and Tall
LOM	ABIR	FABIR
POI	EXABIR	FABIR
Ortho	Scottish Rite Toronto Trilateral brace Petrie cast Salter stirrup	Von Rosen Ilfeld Pavlik -MC Frejka Pillow

Stages of LCPD

- a. Necrosis
- b. Fragmentation – flattening
- c. Revascularization
- d. Remodelling
- e. Healed Stage

LCPD in adults – Chandlers (Alcohol, Steroid)

iv. Dislocation

- a. MOI dashboard injury (usually accompanied with PCL injury)
- b. MC dislocation: posterior
- c. Direction: Hip FADIR
- d. Mx: Arthroplasty/ Partial/ Total (MC partial/ head)
 - i. Material: Titanium Alloy (most durable), molybdenum, ceramic, plastic
 - ii. Approach:
 1. Posterolateral (G med weakness)
 2. Anterolateral
 3. Lateral (G med weakness, for sure)
 4. Posterior (G max)
 - iii. Fixation:
 1. Cemented: PMMA (elderly, promotes early weight bearing)
 - a. C/i: Ultrasound (Ogvie's Syndrome)
 2. Non-cemented (6 months before ambulation)
 - a. Pins, Fixtures, plates, rods, screw
- e. Driving
 - i. ® = 4-6 months
 - ii. (L) = 1 week

v. Coxa Valga vs Coxa Vara

	Coxa Valga	Coxa Vara
NSA	Increase	Decreased
Length	Longer	Shorter
I/L compensation	Hip/ knee flexion Subtalar pronation Genu Recurvatum	Ankle PF Subtalar Supination
C/L compensation	Ankle PF Subtalar supination	Hip/ knee flexion Subtalar pronation Genurecurvatum

Anteversion	Retroversion
squinting	Frog eyes
Intoeing	Out-toeing
SPITT	SSETT
SSET	SPITT

- vi. Coxa Saltans / Snapping hip
 - a. Runner's Knee (amateur runners), wearing incorrect footwear, overflexing knee (>30° knee flexion), uneven terrain
 - b. Cause:
 - i. Tight ITB/ TFL
 - ii. G. MAX overriding greater trochanter
 - iii. Iliopsoas bursitis
 - c. Special Test:
 - i. Ober's Test
 - ii. Noble compression
 - d. Mx:
 - i. Stretching (D2 extension of lower extremity + trunk lean to opposite side)
 - ii. Ice (Slow)
 - iii. Tendon release(Surgical Mx)

vii. Sitting Position

Position	Hip	Knee	Ankle	Condition
W	IR	ETT	Out-toeing	Pellegrini Steina Dse
J	IR	ITT	In-Toeing	
I	ER	ITT	In-Toeing	

*W – MC

viii. Bursitis

- a. Hip
 - i. Ischiogluteal bursitis (Boatmans, Tailors, weavers)
 - ii. Trochanteric bursitis
 - 1. Greater Trochanter
 - a. GMED/MIN too strong
 - b. ITB/TFL tight
 - 2. Presentation of pain: Pain on the lateral portion
 - 3. Pain presentation
 - 4. Lateral: Trochanteric bursitis
 - 5. Anterolateral = meralgia Parasthetica
 - 6. Posterolateral = fat wallet syndrome
 - 7. Lateral: Bald Trochanter
 - a. Covered by GMED/GMin
 - b. Weakness reveals the trochanter

ix. Strain (muscle stretch)

- a. MC Quads (Rectus Femoris)
- b. 2nd MC Hamstrings (Biceps femoris)
- c. Adductors: (Adductor Longus)
- d. Predisposition: 2 jointed

- x. Contusion (blunt force trauma)
 - a. MC quads (Charley Horse)
 - b. 2nd MC Hams
 - c. Adductor: Magnus/ Longus
 - i. If always contused > Myositis ossificans
 - ii. (if bones: HO)
- xi. HO & MO
 - a. Decrease ROM, Hard endfeel,
 - b. Mx:
 - i. Early stages: gentle stretching, ROM, ultrasound (pulsed)
 - ii. Late stages: Bed Rest, Dissodium Ethydrionate
*Alkaline Phosphatase increase if there is HO & MO
 - c. HO
 - i. SCI- HIP> KNEE> SHOULDER> ELBOW
 - ii. CVA – Shoulder
 - iii. TBI – Shoulder
 - iv. Burns – Posterior Elbow
 - d. MO:
 - i. LE: Quads> Adductors (Rider's Bone)
 - ii. UE: Brachialis> Brachioradialis> Biceps Brachii
 - iii. Whole body: Quads
- xii. Knee

Osgood Schlatter/ Jumpers knee	Patellar tendinitis	Sinding-larsen-johansson	Osteochondritis Dissecans
Tibial tuberosity bilateral Patellar Tendon Idiopathic Self-Limiting M>F adolescence US C/i Mx: Rest/ Strengthening Calcaneal Apophysitis/ severs/ haglunds dse	Insertional tendinopathy Similar symptoms, but unilateral C/i: US Mx: Taping, Excercises for quads Descending/ downhill running difficulty	Inferior pole of patella "Softening" – osteomalacia M>F 2-8 y/o C/i: US	Separation "chipping" Lateral aspect of medial femoral condyle "LAMFC" D/t OA of the knee C/i: US

Patella Alta	Patella Baja	Arthrogenic	Genu Valgus (n) 6°	Genu Varus	Genu Recurvatum
Excessively high Decreased Q-angle "Camel Sign" Patellar tendon rupture	Excessive low Patellar Capsule Increased Q angle	Ankylosis	3-5 y/o With coxa vara Increased anteversion Increase Q angle MCL	OA and RA LCL	PCL affection Weak quads

xiii. PFPS

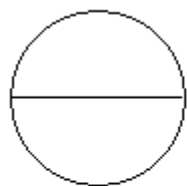
a. Chondro malacia patella (old name)

- i. Collection of diseases
- ii. Prolonged sitting
- iii. $>90^\circ$ knee flexion
- iv. S/Sx
 - 1. Patellar malalignment/ maltracking
 - 2. Weakness of VMO
 - 3. Tightness of ITB
 - 4. (+) theatre sign/ cinema sign/ movie gowers sign
 - a. Knee is subjected to $>90^\circ$ flexion

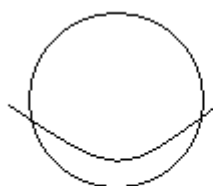
v. Mx:

- 1. Patellar taping (lateral to medial Up/ down or horizontal)
- 2. Splints with lateral buttress
- 3. Strengthen VMO/ quads
- 4. SLR
- 5. Stretching ITB (D2 extension)
- 6. Surgical MX:
 - a. Lateral: retinacular release
- 7. Special test
 - a. Clarkes
 - b. Waldrons
 - c. McConnel's sign
- 8. Miserable Alignment Syndrome
 - a. Broad pelvis
 - b. Increased anteversion
 - c. Increased Q angle
 - i. Bayonet Sign
 - d. Related 2-5-7
 - e. Patella alta
 - f. Genu valgum
 - g. Subtalar pronation
 - h. External tibial torsion
- 9. Stages of PFPS
 - a. Swelling (acute)
 - b. Fissuring
 - c. Deformed surface/ ulceration
 - i. Worn out of infrapatellar surface
 - d. Fem. Cartilage involvement
 - i. Bed of patella

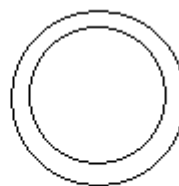
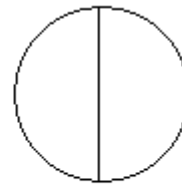
- xiv. Bursitis
 - a. Pes anserine bursitis
 - i. Overactive Sartorius
 - ii. Figure of 4
 - b. Housemaids knee/ prepatellar bursitis
 - i. Always kneeling
 - c. Vicar's/ clergyman's/ nun's knee/ parsans knee
 - i. Superior infrapatellar bursa
 - ii. Tall kneeling
 - d. Baker's cyst/ popliteal bursa
 - i. PF weakness
 - ii. Affects Tibial nerve
- xv. Ligaments (sprain)
 - a. ACL
 - i. Knee hyperextension/ knee flexion + rotation
 - ii. 6-10months
 - iii. Most difficult: 6" lateral *step ups*/ downs
 - 1. Stages of sprain
 - a. I strained
 - b. II partial tear (Start surgery)
 - c. III complete
 - d. Functional instability (O' Donoghue)
 - iv. Strengthen hams>quads
 - b. MCL
 - i. Pellegrini Steini Disease – ossification of MCL
 - ii. Laxity , Valgus force
 - iii. Grading
 - 1. I 0-5mm
 - 2. II 6-10mm
 - 3. III 11-15mm
 - 4. Complete rupture >15mm
 - c. LCL
 - i. MOI: dashboard injury
 - d. Patellar fracture



Transverse (MC)



Sleeve

Marginal
(Degradation/
Quads Ca)

Vertical (Vastus lat >VMO)

xvi. Leg

- a. Blounts (Tibia Vara)
 - i. Excessive Genu varum
 - ii. Children 2-10 years old
 - iii. Male > female
 - iv. Intentional Fracturing & resection
- b. Anterior compartment syndrome
 - i. Stress fracture of distal 1/3 of Tibia
 - ii. Decreased blood supply
 - iii. Inflammation of anterior compartment
 - 1. Compression TA's and Pt, DPN
 - iv. Gait deformity: Foot slap
 - v. Surgical Mx: Compartment P° Release (up to 800mmHg and fasciotomy)
 - vi. 5 ps
 - 1. Pain
 - 2. paresthesia
 - 3. Pallor
 - 4. Paralysis
 - 5. Pulselessness
 - 6. Polar –cold
 - vii. C/i: superficial heat/ US
- c. Stress fracture
 - i. MC site of stress fracture
 - 1. Tibia > Fibula > Metatarsals > tarsals > femur
- d. Achilles Tendon Rupture

Achilles Tendinitis	Achilles Rupture
Overuse	Partial/ full tear
Difficulty in uphill/ descending stairs	Overuse
Early Mx: PRICE, Relaxation exercise (Roods, inhibitory techniques), Taping (Distal > proximal)	Vehicular accidents
Late Mx: Eccentric calf strengthening	Plasty TQ Z & H
	Position in PF

xvii. Ankle

- a. Sprain
 - i. MC sprain LCL, ATFL
 - ii. Reason:
 - 1. Inv > Eve (2:1)
 - 2. Few/ Spread ligament
 - 3. Lateral Malleolus is longer
 - iii. ATFL MOI: PF Inversion
 - 1. ST: anterior drawer test
 - iv. CF MOI: Dorsiflexion inversion
 - 1. ST: Talar Tilt test

- v. Posterior drawer
 - 1. ST: Ant or Poster Drawer test
 - vi. Deltoid ligaments (avulsion fracture of medial malleolus)
 - 1. ST: Kleiger's Test
 - vii. Early Mx: PRICES
 - viii. Late Mx: Proprioceptive exercises
- b. AVN
 - i. Kohlers: Navicular
 - ii. 2nd MTT: Freiburgs
- c. Calcaneal Apophysitis (Haglunds/ Severs, common with Osgood Schlatter)
 - i. Overweight, Unilateral stance, Flats
 - ii. Medial calcaneus, degeneration
 - iii. Mx: lose weight,, foot othotics (UCBL,)
- d. Tarsal coalition
 - i. Congenital defect
 - ii. Fusion of calcaneus + navicular/ talus + calcaneus
- e. Triple arthrodesis (Relief of pain from OA)
 - i. Severe Ankle osteoarthritis
 - ii. TALO NA
 - iii. TALOCA
 - iv. Ca-Cub
 - v. Surgical: intentional, 50% balance left (TALO CA)
- f. Plantar Fasciitis
 - i. Inflammation of PF
 - ii. Overweight, overuse
 - iii. Windlass Effect, First step phenomenon (1st step in the morning)
 - iv. Mx: Stretch + big toe extension + Eversion
 - 1. Strengthen TP> support MLA
- g. Dancer's Tendinitis
 - i. MC in ballet dancers
 - ii. Flexor hallucis longus
 - iii. Hyper extension of toes
 - iv. Mx: maintain toes in neutral (Toe Insert)
 - v. Gait: apropulsive gait
- h. Tarsal tunnel syndrome
 - i. Tibial Nerve affectionation
 - 1. TP
 - 2. D - FDL
 - 3. N, tibial
 - 4. H – FHL
 - ii. Surgical Mx: Fascial release

i. Gout vs Pseudo Gout

Gout	Pseudo Gout
Monosodium Urate excess 1 st MTP, Podagra Knee, gonagra Wrists: Cheiragra Ankle: Ankle Gout Cochicine Dorsal UE	Calcium Pyrophosphate excess Calcific Bursitis Calcaneal Spurs Posterior

j. Talipes Equinovarus

- i. AIP, congenital clubfoot
- ii. Adduction, inversion, plantar flexion
- iii. Splint: Denis Browne splint

k. Pes planus vs Cavus

Pes Planus	Pes Cavus	Pes Valgus	Pes Varus
Flat foot Collapse of MLA	High arch foot, hollow foot, Pes Arcuatus, Inverted champagne bottle	PEVABDA Over activity of peronals Weak: evertors Tibial n effect	SINADPA Invertors Weak: peroneals Superficial peroneal n

l. Hallux valgus, Rigidus, vs turf toe

Hallux valgus	Hallux Rigidus	Turf Toe
(n) <15° Cause: tight toe box/ narrow shoes Medial deviation of 1 st Metatarsal Correlated 1. Metatarsus abductus 2. SPITT Surgical Mx: 1 st say resection: Schmitz	Ankylosis (congenital defect) (-) push off (apropulsive gait) (-) forefoot creases	Hyper extension injury of 1 st MTP (Traumatic) E.g. Sprinters,

m. Metatarsalgia vs Mortons neurma

i. Metatarsalgia

1. Pain in metatarsal head, more on medial side 1st, 2nd, 3rd
2. D/t shear. Tight shoes
3. Mx: wide toe box, open shoes
 - a. Metatarsal bar modification (head to base weight transfer)

ii. Mortons neruroma

1. Palmar Interdigital nerve
2. Electric like pain
3. MC 3rd and 4th
4. Special test
 - a. Morton's Test

- n. Charcot neuropathy/ diabetic neuropathy
 - i. Decrease in sensation
 - 1. Neurotraumatic – trauma to nerves
 - 2. Neurovascular – decrease blood supply to nerves (vasa nervosum) viscous blood d/t increase in viscosity
 - ii. Wagners ulcer grading
 - 1. I - Superficial
 - 2. II – Deep, involvement of bones, muscles, and tendone
 - 3. III – Deep, + osteomyelitis (Start amputation)
 - 4. IV – Gangrene of forefoot
 - 5. V – Gangrene of the entire foot

ST: LOWER EXTREMITY

i. Hip

a. Hip assessment

Hip Assessment	Landmarks	Orientation	(+) CHD
Nelaton's line	ASIS to Ischial tuberosity	Diagonal	Femoral head above the diagonal
Bryant's triangle	1 st : ASIS 2 nd : Isc. Tuberosity	® triangle	Difference measurement on femoral head
Helgenreiner's	ASIS	Horizontal	FH moves upward and locates outward quadrant
Perkin's line	Acetabular fossa	Vertical	Acetabular fossa goes up/ FH outward quadrant
Shenton'line	Femoral head	Curve	Curve is distorted of broken curves

b. Hip pathology

i. Patrick's Test (FABERE test, Figure of 4, Jansen's test)

1. Supine c knee is FABER (hip/ knee)
2. Push the medial knee
3. (+) inability to lower the knee
 - a. Midrange: Hip Joint
 - b. End range: SI joint
4. Gives stress to the hip joint

ii. Trendelenburg sign

1. Indication: G. Medius weakness
2. Standing in one leg
3. Note:dropping hip on opposite leg
4. (+) dropping hip
5. *MC cause: OA

Trendelenburg sign	Trendelenburg gait
(-) compensation	(+) compensation, by lateral trunk bending upon single limb stance

iii. Quadrant Test (Scouring)

1. Supine with hip and knee flexed
2. Examiner will give axial load
3. (+) pain
4. Indication OA of the hip

- iv. Craig's test (Ryder's method)
 1. Femoral Anteversion
 2. **Prone**
 3. **Palpation** (g. troch)
 4. **Parallel** to bed
 5. IR and ER of the hip
 6. (+)excessive movement

c. Hip muscle pathology

i. Thomas test "iliothomas"

1. Supine with 1 hip and knee flexed, knee to chest.
2. Examiner will push the flexed leg
3. Note: if the straight leg rises off the table
4. Indication: Hip flexion contracture
5. Modified:
 - a. 1 jointed/ 2 jointed hip flexion contracture
 - b. Supine with legs dangling (knee to chest of other limb)

	Flexion of the knee	Elevation of the pelvis
1 jointed	<70°	(+)
2 jointed	>70°	(-)

ii. Staheli's test (infants)

1. Hip flexion contracture
2. Prone with legs dangling
3. Examiner will slowly extend one leg
4. (+) pelvis rises

iii. Rectus femoris contracture test

1. (B) are passive

Kendall's	Supine	Knee to chest	(+) knee extension
Ely's Test	Prone	Heel to butt	Hip flexion

Rectus FemELY's

iv. Hamstrings contracture Test

1. 3 methods
 - a. Method 1: 90-90 SLR (active)
 - i. Supine
 - ii. Hips and knee 90° flexion
 - iii. Let subject extend one knee
 - iv. (+) politeal angle <120°
 - v. (n) extensor lag is 25°, for children extensor lag starts at 6 y/o = 155°
 - vi. * most reliable test
 - b. Method 2: Sit and reach (active)
 - i. Long sitting and reach toes
 - ii. (+) unable to reach
 - iii. *Most unreliable test – trunk tends to lateral bend

- c. Method 3: Tripod Sign (passive)
 - i. Sitting with legs dangling
 - ii. Examiner should passively extend one knee
 - iii. (+) backward trunk lean

d. Tightness

Test	Ober's Test	Phelp's	Noble Compression	Take of Shoe Test (TOST)
Indication	ITB tightness	Gracilis Tightness	ITB Friction syndrome	1° & 2° Biceps femoris strain
Position	Sidelying	Prone	Supine	Standing
Examiner	Extends and abducts top leg, and let subject lower leg	Subject will abduct, examiner will flex the knee, and abduct again	Knee flexion, palpate lateral femoral condyle > extend knee	Subject will actively remove shoes on the affected side by means of the opposite foot.
(+)	Inability to lower leg	Increased abduction	Pain on lateral femoral condyle	ER of the hip and flexion of the knee

e. Piriformis Test

- i. Sidelying
- ii. With hips and knee flexed
- iii. Examiner will push the upper leg
- iv. (+) pain accompanied by:
 - 1. IR + hip extension (Freiberg's sign)
 - 2. ER + abduction (Sign of Pace & Nagel)
- v. Indication: Piriformis tightness

f. Other tests:

- i. Leg Length Discrepancy
 - 1. True
 - a. Structural E.g. Fracture, CHD
 - b. Landmarks: ASIS > Medial Malleolus
 - 2. Apparent/ Functional
 - a. Compensation E.g. Scoliosis, Pes planus
 - b. Landmarks: Umbilicus > medial malleolus
 - c. *xiphoid process can be used as well.

g. CHD

Ortolani's	Barlows
Tested: Few weeks from birth	Tested: 3 months to 6 months
Correction Test, The hip is already dislocated	Dislocatable
"Correctiolani"	Supine
Supine	Inward + Outward pressure
Inward pressure	*one time test
(+) relocation click	(+) Reduction click

- i. Gallezi's/Alli's –
 - a. Hook lying
 - b. Note: Height on (B) knees
 - c. (+) lower leg/ knee
 - d. Femoral shortening – lateral view
 - e. Tibial shortening – anterior view
- ii. Telescoping Test (Piston test/ Dupuytren's Test)
 - a. Hips and knees flexed 90°
 - b. Examiner will perform piston action(downward first and upward)
 - c. (+) excessive translation
 - d. *note: administer on (B) sides

Mc-Murray	Bounce Home	Apley's (compression)	Apley's(Distraction)
Supine Supine with hip/ knee flexed 90° + IR = Lateral Meniscus + ER = Medial Meniscus (+) Pain + clicking	Supine Hip/ knee fully flexed Slowly extend knee *note: end feel (+) Springy/rubbery Block	Prone with 90° knee flexed Compress and rotate Meniscal lesion ER = lateral meniscus IR= medial meniscus	Same posn Distract + rotate Ligamentous lesion

h. Ligamentous Lesions

MCL	LCL
Valgus Stress Test	Varus Stress test
Supine/ sitting with knees extended Examiner will give valgus force to the knee/ abduction force (+) pain/ apprehension/ excessive movement	Same posn Examiner gives varus force/ Adduction force

i. Cruciates

i. PCL

Post. Drawer Test	Godfrey's Test/ Gravity Test	Posterior Sag Sign
Supine with hips and knee flexed 90° Examiner will draw the tibia backward (+) excessive movement	Supine with hip and knee 90° Examiner will hold both flexed legs (+) Tibia will sag backwards	Supine with hips 45° and knee 90° flexed (+) tibia will sag backwards

ii. ACL

Anterior drawer's Test	Lachman's Test
Supine hip and knee flexed 90° Examiner will draw the tibia anterior (+) excessive movement of tibia anterior	Supine hip and knee 20-30° flexion Examiner will draw the tibia anterior (+) excessive movement of tibia anterior

iii. Lachman's Test modifications

Lachman modification		Fulcrum
I.	–	Examiners Thigh
II.	Stable	Examiners Knee
III.	Drop leg	Between examiners thigh
IV.	–	Examiner's thorax
V.	–	Bed (Eye of examiner parallel to the knee)
VI.	Prone	Prone
VII.	Active (No touch)	Forearm
VIII.	Maximal Quadriceps test	Arm/ Hand

Lachmans displacement stages

- 1: 3-6 mm
- 2: 6-9mm
- 3: 10-16mm
- 4: 16-20mm

iv. Rotatory instability

1. AI: Opposite, PI: Same
2. Special tests
 - a. Slocums > Anterior
 - b. Houghstons > Posterior

v. Patella

Patellar Tap	Brush Test/ Stroke Test/ Wipe Test/ Bulge Test
Ballotable patella Supine, knee extended or flexed Examiner will tap the patella (+) Floating of patella (Dancing patella) Indication infrapatellar effusion	Spine stroke on medial side of the patella 2-3x (+) examiner will feel fluid on the medial aspect of the knee Indication: swelling

Patellofemoral pain syndrome

Clarke's Sign	McConnell's	Waldron's
Aka patellar grind test Contract the quads & resist the patella. (+) pain and unable to hold contraction	Sitting position & femur in ER Contract quads isometrically in 120° > 90° > 60° > 30° > 0° of knee flexion for 10 seconds each. * note where the pain is elicited > let the subjects knee extend fully, and push the patella medially then perform the 2 nd procedure: (+) if there is decreased pain upon returning position	Standing Examiner Palpate the patella and subject perform squats (+) pain and crepitation

j. Other tests:

i. Wilson's Test

1. Indication: osteoarthritis dissecans
2. Sit c legs dangling
3. Actively knee extension with tibia in IR
4. (+) pain on media condyle upon rotating in IR
5. To confirm:
 - a. ER the tibia, if there is decreased pain
 - b. (+) osteochondritis dissecans

- ii. Patellar dislocation/ Fairbank's apprehension test
 1. MC dislocation of the patella > laterally
 2. Supine, make sure quads are in a relaxed position
 3. Examiner will push patella laterally
 4. (+) contraction of the quads in order for the patella to be mid line

ii. Ankle & Foot

Special	Indication	Position/ Procedure	(+)
Anterior Drawer Test (ankle)	ATFL	Supine, 20° plantarflexion and draw talus anteriorly	Pain + excessive movement
Talar Tilt Test	CFL	Supine, foot in relaxed and examiner tilts ankle abduction and adduction	Pain + excessive movement
Kleiger's Test/ External rotation Stress Test	Deltoid Ligaments	Sit with legs dangling and examiner ER talus	Pain on medial side of ankle

iii. Other Tests

	Indication	(+)
Squeeze test (leg)	Syndesmosis Sprain	Pain
Homan's sign (Passive DF)	DVT	Pain
Hoppas test	Calcaneal fracture	Pain, (-) PF
Thompson's (Simmonds/ Achilles rupture test)	Achilles tendon rupture	(-) PF
Morton's Test (MTT heads)	Morton's neuroma,	(+) pain/ paresthesia

iv. Other tests for ankle and foot

a. Feiss Line

- i. Landmarks: medial malleolus > 1st MTP
- ii. stages
 1. 1° mild 1/3 navicular drop
 2. 2° 2/3 navicular drop
 3. 3° Navicular rests on the floor

b. Duchenne's test

- i. Foot is in relaxed position
- ii. Examiner will push the first MTP/ Big toe
- iii. (+) Dorsiflexion of the remaining toes
- iv. Indication: superficial peroneal nerve lesions

c. Tinel's sign

- i. Ant: Anterior tibial nerve/ superficial peroneal nerve
- ii. Post: Posterior tibial nerve/ tibial nerve

d. Buerger's test

- i. Patency of the blood supply of the lower ex, subjects leg is elevated 2-3 minutes
- ii. Examiners observes for any discoloration
- iii. (+) blanching/ pale in color
 1. To confirm:
 - a. Dependent position
 - i. Sitting with legs dangling
 - ii. Examiners observes 1-2 minutes
 - iii. (+) >2 minutes of discoloration

GAIT ANALYSIS/ PATHOLOGY

i. Basic terminology

NYU	RLA
Heel Strike	Initial contact
Foot flat	Loading response
Mid stance	midstance
Heel off	Terminal stance
Toe off	Preswing
Acceleration	Initial swing
Midswing	Midswing
Deceleration	Terminal swing

ii.

Stance	HS	FF	Mst	HO	TO
Hip	+20 flexion	15 flexion	N	10-20 flex	10-20 flex
Knee	N	15 flexion	5 flexion	N	30 flex
Ankle	N	5PF	5DF	N	20PF
MTP	25 extension	0	0	21 extension	55 extension

Swing	Acc	Msw	Dec
Hip	20 flexion	30 flexion	30 flexion
Knee	60 flexion	30 flexion	0
Ankle	10PF	N	N

HS HIP AY FLEXED +20

FOOT FLAT AY 15, N SA

MISTANCE PUSH OFF

EXTEND 10- 20, KNEE

N FLEX 15 5 N FLEX30

ANKLE N 5PF 5DF N

20 PF SA TOE OFF MTP EX25

0 0 EXTEND 21 AT 55

SWING MO ANG HIP FLEX ITO NG 20 30 30, ANG KNEE NAMAN

FLEX NG 60 30 0 (2x)

ANKLE 10 PF N N

iii. (n) Values

	Stride length	Step length	Cadence	Step Width	Fick Angle	Pelvic Lateral Displacement
Male	146cm	73cm	110 steps/ min	5-10cm or 2-4 in	7°	1 ¾ in or 4.5cm
Female	126cm	63cm	116 steps/min			
Average	70-82cm		90-120 steps/min			

Walk this way

126cm sa girls, 146cm sa boys

70-82 mean ng stride length sa all

Cadence 116 sa girls. 110 naman sa boys

90-120 steps per minute sa all

Trunk rot ay 7 degrees

Toe out ay 7 degrees

Pelvic tilt rot 8 degrees

5cm moves COG

Up down side to side it moves

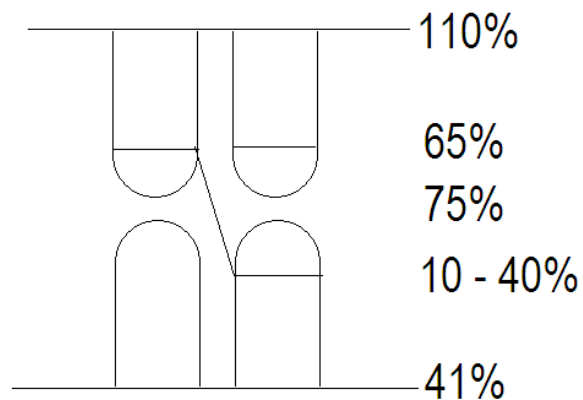
Normal VO2 12, 3mph 3 METS

iv. Determinants of gait

v. Ambulation level, SCI

- a. SO - T2
- b. TA - T3 - T11
- c. HA - T12 - L2
- d. CA - L3

vi. Energy Expenditure in amputees



vii.

	H	K	A
HS-FF	E GMAX	EQUADS	E DF
FF-Mst	CGMAX	C QUADS	E PF
Mst-HO	-	-	E PF
HO-TO	E IP	E QUADS	C PF
Acc-Msw	CIP	C HAMS	C DF
Msw-Dec	E H	CQUADS	C DF

viii. Peak activity

- Tibialis anterior and quads – peak at heel strike
- Hamstrings – “main decelerator” of hip flexion and the rapidly swinging leg towards knee extension (JC)
- Gluteus medius and minimus peaks at midstance
- gastrocsoleus – peaks at toe off

ix. Determinants of gait

Vertical	Transverse pelvic rotation (swing)	Reduces downward displacement of COM
	Muscle activity of knee	
	Muscle activity of ankle and foot	
	Pelvic tilting (stance)	Reduces upward displacement of COM
	Knee flexion (stance)	
Side to side	Adducted femur; tibiofemoral angle	Reduces side to side displacement of COM

- Vertical displacement
 - Displaced highest at midstance
 - Displaced lowest at double support
- Horizontal displacement
 - Displaced highest at unilateral leg stance (2cm)
 - Displaced lowest at double support
- 5cm moves COG; LOG?

x.

- In stance phase there are 2 periods of DLS: initial contact and terminal stance
- 1 period of SLS: midstance
- Running-period of DLS does not exist

xi. Questions

- A person with painful, arthritic hip has the following gait abnormality:
 - Internal rotation of the leg at the hip
 - Vaulting on the extended knee
 - Trunk extension at heel strike
 - Lateral trunk shift over the affected joint during stance

If to compensate, answer is (ii.)

If excessive stride length: hip flexor contracture

- b. In gait analysis of your patient, you note that the ankle is maintained in plantarflexion during foot flat through midstance. This is done by the patient to compensate for
 - i. Excessive plantarflexion and uncontrolled tibial advance
 - ii. Excessive dorsiflexion and uncontrolled tibial advance
 - iii. Decreased contralateral step
 - iv. No roll of
 - v. Lack of ability to oppose plantar flexion
- xii. Gait Pathology
 - a. Foot slap/ foot flat
 - i. Heel stike
 - ii. Weak: pretibial
 - iii. Nerve affectation: deep peroneal nerve
 - b. IC/ forefoot
 - i. Weak: pretibials
 - ii. Heel pain and PF contracture (pes Equinus deformity)
 - c. Bouncing gait
 - i. Mid/ terminal stance
 - ii. Lack of DF
 - iii. Tight plantar flexors
 - d. Heel contact in heel off
 - i. Weak/flaccidity Plantarflexors
 - ii. Pes calcaneus deformity
 - e. Pes cavus deformity
 - i. Weightbearing
 - ii. Other name:club foot deformity
 - iii. Spint: denis browne splint
 - iv. Orthosis
 1. Scaphoid
 2. Navicular pad
 3. Cookie
 4. MLAS
 5. UCBL
 6. Thomas heel (medial)
 7. Reverse Thomas heel (opposite side, lateral)
 - f. Pes planus
 - i. Weightbearing lateral
 - ii. Weak: tibialis posterior
 - iii. Orthosis:
 1. Reverse Thomas heel

- g. Foot drop deformity
 - i. Swing phase
 - ii. Weak dorsiflexors
 - iii. Spastic plantarflexion
 - iv. Compensate: excessive hip and knee flexion
- h. Vaulting
 - i. Midstance
 - ii. Inability to lift the involved LE (Swing)
- i. Excessive toe
 - i. Tight: ER
 - ii. Lacking: IR
- j. Rapid ex of knee after IC (extensor thrust)
 - i. Weak hamstrings
 - ii. Spastic quads
- k. Genu recurvatum
 - i. Weak: Quads
 - ii. Stance phase
- l. Flex knee – stance, lack knee ex – decal
 - i. Contracture of hamstrings
- m. Gmax gait
 - i. Backward trunk lean
- n. Forward trunk lead/ circumducting gait
 - i. Weak hip flexors
- o. Excessive length
 - i. Hip flexion contracture
- xiii. Gait
 - a. Antalgic gait
 - i. Time in stance phase
 - ii. Shorter step length on involved LE
 - iii. Ipsilateral trunk lean with hip pain, contralateral trunk lean with knee and foot pain
 - iv. Common causes: OA, Fx, tendinitis
 - v. Most commonly involved joint in OA

OA		
Knee	HIP	C5- C6
1 st MTP	DID	L4-L5
1 st CMC	PIP	Ac joint

Most common involved joint in RA

MCP	Shoulder
Wrist	Ankle
PIP	Cervical
Knee	Hip
MTP	Elbow
	TMJ

- b. Plantar flexor gait
 - i. Ankle PF during swing
 - ii. Toe drag
 - iii. Weak? dorsiflexors
 - iv. Hip hiking (happens?) in swing phase
 - v. Circumduction or excessive hip knee flexion
 - vi. Vaulting of the stance limb
- c. Equinus gait
 - i. Seen with talipes equinovarus (club foot)
 - ii. WB> decrease (limp is present)
 - iii. Position of tibia and foot
 - iv. Compensates: external rotation of femur
- d. Psoatic gait
 - i. Classical manifestations: hip FADER during swing
- e. Quadriceps gait
 - i. Knee is in hyperextension (genu recurvatum)
 - ii. Compensates through trunk flexion
- f. Trendelenburg and waddling gait
 - i. Weak hip abductors
 - ii. Bilateral weakness – “wobbling” gait “chorus girl swing”
- g. Parkinsonian gait
 - i. Main motor impairment include muscle rigidity, resting tremor, bradykinesia, postural instability
 - ii. Difficulty in limb advancement
 - iii. Pelvic and trunk motion are diminished
 - iv. Abnormally stopped posture, limitation in trunk rotation
 - v. Festinating gait – increased speed with shortening of stride length
 - vi. Good
 - 1. Resting tremor
 - 2. Rigidity
 - vii. Bad
 - 1. Postural instability
 - 2. Bradykinesia/ akinesia
 - viii. Brace: Jewett
 - ix. Types of brace/ limiting motions:

FE	Chairback	FE	Taylor
FEL	Knight	FEL	Knight Taylor
EL	W"EL"liam	FLR	Cow horn
		F	Jewett
		Ex	Cruciform
		All	Plastic body jacket

MEE	WFE
Herniation	Spinal Stenosis
Osteoporosis	Spondylolisthesis
AS	DJD
PTCFx	

- h. Ataxic gait
 - i. Widebase gait
 - ii. PI; high guard
 - iii. Characterized by decrease in balance due to affection of cerebellum
- i. Tabetic gait
 - i. Taps heel twice to feel grounds
 - ii. Proprioception affection of dorsal column
- j. Scissoring gait
 - i. Spastic adductors
 - ii. CP spastic diplegia
- k. A patient with low back and leg pain has been seen 4 times in outpatient PT. Presently, the patient reports that the pain ceased. During gait evaluation, the patient's tibia advances excessively during leading response to midstance. Additional assessment must include
 - i. Achilles tendon reflex testing
 - ii. Sensation testing of the foot dorsum
 - iii. The sign of the buttock test
 - iv. Tibialis anterior strength testing

EXERCISE PHYSIO

i. Energy systems

Energy System	Mole of ATP/min	Time to fatigue	Role
Immediate: ATP system or Phosphagen system (ATP and PCr)	4	5-10 secs	Immediate burst of energy. Muscles will store ATP.
Short-term: Glycogen-lactic acid system	2.5	1-2 minutes	Short duration
Long-term: aerobic system	1	Beyond 2-3 minutes	Endurance

These systems act concurrently, not one after the other. Within the first 5 minutes of the exercises these 3 will work together.

a. Phosphagen-system

ATP stored in the muscle, as the energy is used it will be depleted. (using what is existing)

b. Glycogen-lactic acid

Using muscles will use up glucose. When glucose is broken down (glycolysis), it is oxidized. In the process, there is creation to pyruvate leading to the production of ATP. With the presence of 2 mols of Hydrogen and pyruvate, together will form lactic acid. Lactic acid will not accumulate since it is being used. As the exercises progress >2 mins the system will fail.

- Glucose either is used for physical energy or becomes stored through glycogen.
- 2nd in line to Amino acids are utilized to produce ATP (protein) .
- Lactate is not necessarily harmful, it will be used to resynthesized ATP. Harmful, if there is unbalance of production of Lactate and ATP. More supply than demand = lactic accumulation, causing muscle fatigue.

c. Aerobic system use of oxygen

If exercise > 2 minutes, body will produce more ATP in the process.

Capacity to produce ATP is indefinite as long as there is oxygen. Failure of the ms system, (limit of muscles).

Muscle endurance vs cardiovascular endurance

ii. ATP production

- a. Muscle glycogen > glucose (from blood) > 2 ATP (ADP + P) + pyruvate + 2H = Lactate > gluconeogenesis
- b. Gluconeogenesis used 2 ways
 - i. Glucose > if used during Anaerobic glycolysis, does not require oxygen. (Repeat cycle)
 - ii. Glycogen > (storage)
- c. As exercise progresses, oxygen uptake will decrease, lactic acid production will increase (Inbalance) and will now engage in aerobic system to take in more oxygen > gluconeogenesis in aerobic system > similar cycle as first.

- iii. Blood Lactate threshold
 - a. Onset of blood lactate accumulation (OBLA) or lactate inflexion point (LIP)
 - b. LT and VO_{2max} or workload
 - i. Sedentary individual 40-60% of VO_{2max}
 - ii. Endurance-trained 70% VO_{2max} (VO_2 max - max O_2 Consumption)
 - c. LT at steady state exercise (SSE)
 - i. Maximum intensity SSE can be maintained
 - ii. Prescribe intensity as % of LT
- iv. Factors related to lactate threshold
 - a. Low tissue oxygen
 - b. Reliance on glycolysis
 - c. Activation of fast-twitch muscle fibers
 - d. Reduced lactate removal after 2 minutes.
- v. Oxygen Deficit and EPOC
 - a. Oxygen deficit
 - i. Difference between total oxygen used during exercise and additional oxygen that would have been used in steady state aerobic metabolism
 - b. Excess post-exercise oxygen consumption (EPOC)
 - i. Total oxygen during recovery
- vi. Energy Expenditure: TDEE (total daily energy expenditure)
- vii. Factors affecting TDEE:
 - a. BMR (basal metabolic rate) / RMR (resting metabolic rate) kilocalories/mass/ hour - 60-75%
 - b. TEF (Thermic effect of feeding/ food consumption (DIT, Dietary Induced Thermogenesis)) - 10%
 - c. Thermic effect of physical activity - 15-30%
- viii. Other factors:
 - a. Environment, hot or cold.
 - b. Circulatory dynamics
 - c. Male/ female 10:5%
- ix. Resting Daily Energy Expenditure
 - Factors
 - a. BMR
 - b. Height
 - c. Body Mass
 - d. Age
- x. Basic Training Principles
 - a. Individuality
 - i. Consider specific need/ abilities of individual
 - b. Specificity –SAID
 - i. Stress physiological system critical for specific sport
 - c. FITT
 - i. Frequency, Intensity, Time, Type
 - d. Progressive Overload
 - i. Increase training stimulus as body adapts.

- xi. Long, slow distance
 - a. Low-intensity exercise
-57% VO_{2max} or 70% HR_{max}
 - b. Duration > than expected in competition
 - c. Based on idea that training improvements are based on volume of training
- xii. High – intensity Continuous Exercises
 - a. May be the best method for increasing VO_{2max} and lactate threshold
 - b. High intensity exercises
 - i. 80-90% HR_{max}
 - ii. At or slightly above lactate threshold
 - c. Duration of 25-50min
 - i. Depending on individual fitness level\
- xiii. Factors affecting maximal Aerobic power

Intrinsic	Extrinsic
Genetic	Activity levels
Gender	Time of day
Body composition	Sleep deprivation
Muscle mass	Dietary intake
Age	Nutritional status
Pathologies	Environment

- xiv. Adaptation to aerobic training
 - a. Increase oxidative enzymes
 - b. Increase size and number of mitochondria
 - c. Increase slow contractile and regulatory proteins
 - d. Increase capillary density
 - e. Increase blood volume, cardiac output and O_2 diffusion
 - f. Decrease glycolytic enzymes
 - g. Decrease Fast-fiber area
- xv. Influence of gender, initial fitness level, and genetics
 - a. Men and women respond similarly to training programs
 - b. Training improvements is always greater in individuals with lower initial fitness
 - c. Genetics plays an important role in how an individual responds to training
- xvi. Anaerobic power
 - a. Depends on ATP-PC energy reserves and maximal rate at which energy can be produced by aTP-PCR system
 - b. Maximal effort
 - c. Cyclists and speed skaters highest
 - d. Power = force x distance/ time
- xvii. Adaptations to anaerobic training
 - a. Increase wet mass of muscle
 - b. Increase muscle fiber cross sectional area
 - c. Increase protein and RNA content
 - d. Increase capacity to generate force

PVD

i. Peripheral Vascular Anatomy

- a. Circulatory system: For transport and nourishment
- b. Sympathetic Nerves: Major neurologic control of blood vessel tone.
- c. Norepinephrine: Principal neurotransmitter that causes vasoconstriction
- d. Function: Transportation/ filter nutrients, gases, and waste
- e. Blood Vessels – the framework of peripheral vascular system

ii. 3 components:

- a. Arterial
- b. Venous
- c. Lymphatics

iii. Tunics – Layers of the blood vessels

- a. Tunica intima – innermost layer
 - i. Endothelium – *What is the only layer present in all blood vessels?*
**Site of plaque deposits of cholesterol/ fibrous tissue will calcify and will protrude causing decreased blood supply (Atherosclerotic plaque), once broken d/t increased blood pressure and exposed to blood flow it will activate platelets to the site of plaque, and also activates fibrin causing clot formation (Thrombosis) if dislodged (Embolus).*
 - ii. Connective Tissue
 - iii. Basement Membrane
- b. Tunica media – middle layer; smooth ms (Seen in arteries)
 - i. Ability to vasoconstrict/ vasodilate
- c. Tunica Adventitia/ externa – outermost; collagen (Seen in veins, T. adVEIntitia)
 - i. Composed of elastic and collagenous fibers.

iv. Artery vs. Veins

Artery	Veins
- Prominent T. Media - Vasoconstriction - Vasodilation - Propels blood away from the heart - Efferent vessel - Oxygenated	- (+) valves: condensation of T. Intima - Propels blood towards the heart - Prominent T. Adventitia. - T. adventitia

v. Arterial System:

- a. Carries oxygenated blood away from the heart
- b. Muscular, elastic, Tubular extension of the heart
- c. Arteries proximal to the heart has increased pressure (elastic) Distal to the heart decreased pressure (Muscular) (Sullivan)
- d. Arteries (Resistance vessel)>Arterioles>Capillaries aka exchange vessel (Site of gas exchange)
- e. *Best site of gas exchange in the tissue? Interstitial space*
- f. *Aorta – aka Wind Kessel Vessel*

- vi. Venous System:
 - a. Carries De-Oxygenated blood towards the heart
 - b. (-) Tunica Media, (+) Valves
 - c. Carries 60-64% percent of blood
 - d. Veins aka Capacitance Vessel.
 - e. Can resist pressure up to 400mmHg
 - f. Venous return is dependent on contraction of muscle to return blood to the heart.
 - g. 3 types:
 - i. Superficial- underneath the skin, 2 principal veins in lower extremity:
 - 1. Greater Saphenous (Runs medial aspect of leg and thigh to join the femoral vein)
 - 2. Lesser Saphenous (Runs towards the Lateral malleolus and posterior leg to join with the Popliteal vein) LSP
 - ii. Deep Veins – Large veins that already follows the structure of the arteries.
 - 1. ((3) Anterior/ Posterior tibial and peroneal veins)
 - iii. Perforating/ Communicating vein – Connects the superficial and deep veins.
- vii. Lymphatic system:
 - a. Cannot be found in cornea and lymphatic (hymepas)
 - b. Cervical, axillary, inguinal

Calcification of smooth ms of aorta: Monkborg's Disease

Peripheral Vascular Diseases

- viii. Arterial Disorders: Athero-Occlusion of endothelium Arterio-Hardening of tunica media
 - a. ASO (Arteriosclerosis Obliterans), Chronic occlusive arterial disease
 - i. MC arterial D/o
 - ii. Risk factor: Smoking
 - iii. Early stages: Intermittent claudication (Leg pain)
 - iv. Late stages: Gangrene
 - v. Pulse: Decreased/ Absent
 - vi. Pale upon elevation, dusky red on dependency
 - vii. Large to medium sized arteries (Seen lower extremities)
 - viii. Femoral a.: MC affected artery in ASO
 - ix. Femoral a./ Anterior Tibial a. MC in DM
 - x. Abdominal Aorta/ common iliac a./ femoral : 2nd MC in Non-DM

Maraming ASO sa Pilipinas na medium to large sized!
 - b. TAO (Thromboangitis Obliterans)
 - i. Aka Buerger's Dse
 - ii. MC (2nd) Arterial disorder
 - iii. Distal segment to proximal (Ascending progression)
 - iv. Chronic inflammation + occlusion of small arteries. (Seen Upper extremities)
 - v. Young males, smoker
 - vi. Medium – small sized arteries

TAO: Small Young Male persons like to eat Buerger's after Smoking

- c. Vasospastic disorders
 - i. Precipitated by cold and emotional stress
 - ii. Sx for >2 years
 - iii. Bilateral involvement
 - iv. Minimal or absent gangrene of finger tips
 - v. Absence of other diseases associated with vasospastic attacks
 - vi. Abnormal vasoconstriction of arteries affecting small arteries, MC in female.
 - 1. Raynaud's disease
 - a. Cold hypersensitivity
 - 2. Raynaud's syndrome
 - a. Cold hypersensitivity + emotional stress
 - 3. Raynaud Phenomenon
 - a. With underlying pathology E.g. SLE
 - vii. Classic triad of Raynaud's / French Flag Sign
 - 1. Digital or occasionally proximal hand pallor with numbness
 - 2. Cyanosis
 - 3. Reactive hyperemia and rubor of skin
- ix. Venous disorders
 - a. Varicose Veins (Affects the valves of the veins) d/t prolonged standing/ obesity/ pregnancy/ cross legs.
 - i. Distended, swollen superficial veins
 - ii. May lead to varicose ulcers.
 - iii. S/Sx:
 - Aching heavy leg
 - Appearance of spider vein
 - b. Superficial vein thrombosis
 - i. Clot formation + inflammation affecting superficial veins (Saphenous vein)
 - ii. S/Sx:
 - pain along the course of saphenous vein
 - c. Deep Vein Thrombosis/ Thrombophlebitis (2° to immobilization)
 - i. Clot formation + inflammation affecting deep veins
 - ii. Virchow's triad (HIV)
 - Hypercoaguability
 - Intimal wall damage
 - Venous stasis (blood is stagnant during immobilization)
 - iii. Prev. measure: Early mobilization (BEST prevention)
 - iv. Special Test: Homan's test and Cuff's Test: 40mmHg (+) Pain
 - v. Most serious complication: Pulmonary Embolism

Pt is undergoing anticoagulant therapy (Warfarin (oral/ aka Coumadin) & Heparin):
Immobilization

- d. Chronic venous insufficiency (CVI)
 - i. MC venous disorder, **MC PVD**
 - ii. Increased pressure in deep venous system
 - iii. Stages:
 - Edema & pigmentation
 - Edema & pigmentation + varicosities + trophic skin changes (dermatitis)
 - Ulceration; MC manifestation
 Where can you find edema + pigmentation in PVD?

x. Lymphatic Diseases

- a. Lymphadenopathy
 - i. Enlargement of nodes with/ without tenderness
 - ii. Disease of the lymph nodes
- b. Lymphedema
 - i. Increase in girth of an extremity or other body part due to disorder of transcapillary fluid exchange resulting from impaired function of the lymphatic channels
 - ii. Overaccumulation of fluid in interstitial tissues and fibrotic changes in soft tissue
 - i. In lymphatic system obstruction
 - iii. Fluid in the interstitial space:
 - i. Promotes reabsorption of fluid
 - 1. Plasma protein in the blood
 - 2. Interstitial pressure
 - ii. Retain fluid in interstitium
 - 1. Capillary blood pressure
 - 2. Plasma protein in interstitial fluid

1° Lymphedema	2° Lymphedema
Hereditary No cause (idiopathic) 3 types: Lymphedema Praecox > 35 & below Lymphedema Tarda > 35 & above Tarda – MaTanda! Milroy's disease (Congenital)	With cause: Infection (3 rd world) Malignancy (Western) Surgery Males: prostate CA Females: Breast CA and lymphoma

Arterial Insufficiency	S/Sx	Venous Insufficiency
Weak pulses	Pulse	Strong pulses
Cool to touch	Temperature	Warm
Regularly shaped on lateral malleolus	Ulceration	Irregular shaped wounds on medial malleolus
Pain and pale upon elevation, Resting pain	Upon elevation	Relief of pain upon elevation
Cyanosis/ paleness, gangrene	Color	Pigmentation
	Others	Edema

- c. Other Lymphatic notes:
 - i. Stemmer's Sign: Unable to pick-up skin at the edge of the 2nd toe of the affected extremity
 - ii. Intrabdominal pressure should be at least (40mmHg) if venous pressure is 40mmHg
 - iii. Takayasu's Dse – pulseless Dse

xi. Tests:

- a. Claudication Time: measuring the time and the distance where the pain is felt, treadmill.
 - i. For Arterial Insufficiency
 - ii. Exercise to the point of pain only. Never beyond pain
 - iii. Common prescription of exercise: Walking daily 2x/day for 10-15 min (Siegelman)
- b. Ankle Brachial Index (ABI)
 - i. For Arterial Diseases
 - ii. Take pressure of lower and upper extremity using BP Apparatus using Doppler US.
 - iii. LE: Place cuff around calf area, and place probe on Dorsalis Pedis/ Post. Tibial artery.
 - iv. UE: Place cuff around arm, and place probe on Radial artery.
 - v. Start machine sound starts, If (-) sound, stop BP cuff check pressure.
 - vi. ABI = LE pulse/ UE pulse
 - vii. Significance of ABI:

>1.2	Falsely elevated
1.19-0.95	Normal
0.94-0.75	Mild artery disease, Intermittent claudication
0.74-0.50	Moderate arterial disease, rest pain
0.50	Severe arterial disease

ORTHOSES

- i. An external device applied to body segment/s to:
 - a. Immobilize an injury
 - b. Assist weak or paralyzed muscles
 - c. Prevent undesired motion
- ii. Part's of a shoe
 - a. Sole
 - i. Inner sole – directly in contact with foot
 - ii. Filler: in between inner and outer sole
 - iii. Outer sole – directly in contact with the ground
 1. Ball – widest part of the sole, corresponds to the metatarsal heads
 2. Shank – Narrow part of the sole
 - a. Steel shank – longitudinal arch support
 - iv. Heel
 1. Pitch: inclination from vertical
 - a. S: Spring – 1/8"
 - b. O: Oxford – 6/8 - 8/8" 1inch
 - c. M: Military – 10/8" 1 ¼ inch
 - d. C: Cuban – 12/8" 1 ½ inch
 2. Breast
 - b. Upper
 - i. Vamp: anterior section
 - ii. Quarter: posterior section
 - iii. Lace stay : Opening: eyelets, Aglets: plastic
 - iv. Toe box
 - v. Throat
 - vi. Tongue
 - vii. Toe spring – facilitates toe off
- iii. Orthotic Principle
 - a. Heel stress/ pain
 - i. Internal modification: heel cushion
 - ii. External modification: SACH (Solid Ankle Cushion Heel), between sole and the heel.
 - b. Metatarsal head pain/ metatarsalgia
 - i. Internal modification:
 1. Metatarsal pad/ Internal Metatarsal bar, 2nd-4th MT head. (Transfer weight to MT shaft)
 2. U-shaped pad – 3rd MT head only
 3. Sesamoid pad – applied proximal to 1st MT head
 4. Morton's Extension – applied distal to 1st MT head

- ii. External modification: -
 - 1. Metatarsal bar – placed at the ball of the foot, 1st-5th MT heads
 - 2. Rocker Bar – shifts roll over point proximal to MT heads
 - a. Facilitates push off
 - c. Control of Valgus/ Varus
 - i. Valgus – PEVABDA
 - ii. Varus – SINADPA
 - iii. Wedge
 - 1. Sole (Lavarus Medieval)
 - a. Lateral: corrects varus
 - b. Medial: corrects valgus
 - 2. Heel
 - a. Lateral: corrects varus
 - b. Medial: corrects valgus
 - iv. Flare
 - 1. Sole
 - a. Lateral: corrects varus
 - b. Medial: corrects valgus
 - 2. Heel
 - a. Lateral: corrects varus
 - b. Medial: corrects valgus
 - v. Rearfoot = “Corrective”
 - vi. Forefoot = “Supportive”
 - 1. E.g. RF Varus, FF valgus
 - 2. RF- Valgus posting, FF valgus posting
 - d. Toe pain
 - i. Internal: Toe crest, placed at the sulcus of the foot.
 - e. MLA weakness
 - i. Internal: Scaphoid pad, Navicular pad, Cookie placed at medial internal sole.
 - f. LLD
 - i. <1” Heel lift
 - ii. >1” Sole lift
- iv. AFO (ankle foot orthosis)
- a. Components
 - i. C – calf bands
 - ii. A – ankle joint/ mechanism
 - iii. T – two lateral uprights
 - iv. S – Shoe/ Foot attachment
 - b. Types of shoe foot attachments (StiCaShoe)
 - i. Stirrup – MC, attached to the shank. Mechanical ankle joint level with anatomic ankle joint
 - ii. Calipers – attached to the heel of the shoe. Mechanical ankle joint is distal to the anatomic ankle joint.
 - iii. Shoe inserts – attached to the stirrup. For valgus and varus control

- c. Joint control
 - i. Axis
 - 1. Single – DF/ PF
 - ii. Free motion : no restrain c ankle motions
 - iii. Limit stops:
 - 1. PF stop/ Posterior stop, limit PF
 - 2. DF stop/ Anterior stop, limit DF
 - iv. Limited stops: limits (B) PF and DF
- d. Assists
 - i. DF assist
 - ii. DF – PF assist, assists (B) DF + PF
 - iii. Springwire DF assist
 - iv. VAPC clasp type, Veteran's Administration Prosthetic Center, DF assist
- e. BiCAAL, Bi-channel Adjustable Ankle Lock
 - i. Full adjustable ankle lock
- f. Corrects valgus/ varus deformity
 - i. T strap
 - 1. Medial: corrects valgus
 - 2. Lateral: corrects varus
- g. Buckled Insert
 - i. Corrects valgus deformity
- v. Plastic Designs
 - a. Posterior leaf spring (PF)
 - i. Prevent's PF
 - b. Modified PLS
 - i. Prevents PF + DF
 - c. Spiral
 - i. Has full turn on leg
 - ii. Medial condyle > medial malleolus
 - iii. Prevents DF-PF, Inv-Eve, Rotation
 - d. Hemi-spiral
 - i. Has Half turn on leg Medial condyle > lateral malleolus
 - e. Solid – ankle
 - i. Used for predetermined positions
 - f. AFO – with flange
 - i. For maximum ankle valgus-varus control
- vi. KAFO
 - a. Components
 - i. AFO + 2 thigh bands, 2 lateral bands, knee joint mechanism
 - ii. Axis
 - 1. Single: Flexion + Extension MC used
 - 2. Polycentric: Flexion + Extension + Slight transverse rotation. Seen in Knee orthosis s uprights

- iii. Types of knee joints
 - 1. Free motion: No restrictions of movements
 - 2. Offset knee joint
 - a. Knee joint is posterior to the knee
 - 3. Drop ring lock
 - a. MC used, locks knee to extension
 - 4. Pawl lock
 - a. Easier to use, d/t bail control
 - 5. Adjustable locks
 - a. Fan lock – for Pt's with knee flexion contracture
 - b. Serrated lock – 6 intervals lock
- iv. Controls for the knee
 - 1. PatelloFemoral Disorder
 - a. Palumbo – infrapatellar + suprapatellar strap
 - b. Patello-Femoral Orthosis – infrapatellar strap
 - 2. For control of knee is frontal and Sagittal plane
 - a. Swedish knee cage – genu recurvatum/ knee hyperextension
 - b. Three-way knee – for genu recurvatum (more cosmetic than SKC)
 - c. Typical knee orthosis: (+) 2 lateral uprights & bands
 - d. Miami
 - i. Thermoplastic type with polycentric axis
 - e. C.A.R.S. – U.B.C.
 - i. Has more extensive knee controls (valgum/ varum)
 - ii. Canadian Arthritis and Rheumatoid Society – University of British Columbia
 - f. Supracondylar knee orthoses
 - i. For genu recurvatum
 - 3. For rotation control
 - a. Lenox-Hill Derotation orthosis
 - b. Lerman multi-ligamentous derotation orthosis
 - 4. Scott craig orthosis – KAFO
 - a. Designed to help SCI pt's to stand and walk without additional hip orthosis
 - b. Key Design: $10^{\circ}DF$
 - c. Hip hyper extension: jack knife, support of iliofemoral ligement

vii. HKAFO

- a. Basic Components
 - i. KAFO, Pelvic band/ Silesian belt
 - 1. Pelvic band: Controls rotation
 - 2. Silesian belt: control rotation and abduction/ adduction

- b. Hip joints
 - i. Drop ring lock (MC),
 - ii. Pawl lock
 - iii. Two-position hip lock (0-90°) 0 standing, 90 sitting
 - viii. Fracture Orthosis
 - a. The fracture site but allowing standing and ambulation possible
 - b. Femoral rothosis
 - c. Tibial rothosis
 - d. UCBL Insert– molded on the midfoot and hindfoot
 - i. Immobilization: the intertarsal of tarsometatarsal joints
 - ix. Special purpose orthosis
 - a. PTB orthoses
 - i. Weight bearing on patellar tendon
 - ii. Eliminates weight bearing on tibia & ankle
 - b. Ischial wt bearing orthosis
 - i. Weight bearing on Ischial tuberosity
 - ii. Eliminates WB: on femur and knee
 - c. Patten-bottom orthosis
 - i. WB: Patellar tendon and Ischial tuberosity
 - ii. Eliminates weight bearing on whole lower limb
 - iii. Uprights terminate on the floor
 - iv. Toes suspended on midair
 - v. Sole/shoe elevation on opposite leg
 - x. Orthoses of children
 - a. For angular and rotational deformities
 - i. Denis browne splint
 - 1. For congenital clubfoot, pronated foot, abnormal tibial torsion
 - 2. Components
 - a. Spreader bar (for valgus/varus correction)
 - b. Foot plates (rotational)
 - ii. A-Frame splint
 - 1. Distal segment – same with denis browne splint
 - 2. Prox: A frame
 - 3. Pads:
 - a. Thigh pad
 - b. Calf pad,
 - c. Pressure pad
 - 4. For patient with more proximal tibial torsion
 - iii. Torsion shaft splint
 - 1. For Pt's with mild scissoring, mild spasticity, abnormal toe-in/ toe-out
 - iv. Hip rotation control
 - 1. Controls hip rotation during ambulation
 - v. External rotation control strap
 - vi. IR control strap

- b. CHD, positioners no immobilizers
 - i. Von rosen -ABIR
 - ii. Ilfeld -ABIR
 - iii. Pavlik harness – FABER
 - c. LCPD
 - i. Trilateral-ABIR
 - ii. Toronto-ABIR
 - iii. Scottish-Rite- FAB, lightest
 - d. Severe paralytic d/o
 - i. Standing frame – designed for standing and for use of swing through gait
 - ii. Parapodium – for young paraplegics to help them stand without crutches
 - iii. Reciprocating orthosis – used for Pts with reciprocal walker and using reciprocating gait pattern
 - xi. Spinal orthotics
- Cervical
- a. Collar
 - b. SOMI
 - c. Poster appliance
- CervicoThoracic
- d. Custom (Cuirass and Minerva)
 - e. HALO

Cervical	Collars	F	E	L	R
	SOMI	x	X		
	Poster Appliance	X	X	X	X
CT	Custom (Cuirass and Minerva)	X	x	x	X
	Halo	X	X	X	X

LumboSacral WiCK TiK CoP

Orthoses		F	E	L	R
LS	Williams		x	X	
	Chairback	X	x		
	Knight	x	x	X	
TLC	Taylor	x	X		
	Knight taylor	x	x	X	
	Cowhorn	X		x	X
	Plastic body jacket	x	x	x	x
	Jewtt (ant. Hyper extension)	X			
	Cruciform	X			

- xii. Scoliosis
 - a. Apex above T6 – Milwaukee(T6)
 - b. Apex below T6 – Miami(T7, T8, T6-T8)
 - c. Apex below T8 – Boston, Wilmington, Yamamoto

- xiii. Upper limb orthosis
 - a. Static
 - i. Function
 - 1. S - stretching
 - 2. I - immobilization
 - 3. S – support
 - ii. Conditions
 - 1. Nerve injuries, Post-op conditions, Burns, Arthritis, Fracture, and soft tissue injuries
 - b. Dynamic
 - i. Function
 - 1. S - stretching
 - 2. I - immobilization
 - 3. S - support
 - 4. C - corrective
 - 5. A - assistance
 - 6. I – improvement of function
 - ii. Conditions
 - 1. Nerve injuries, weak/ paralyzed muscles, post op conditions, post hip replacement

Conditions

- c. Swan-neck/ boutonniere deformity
 - i. Silver or plastic ring
- d. Mallet finger
 - i. Mallet slipover static splint
- e. Trigger Finger
 - i. MC in 4th, PIP slip
- f. Dequevains Tenosynovitis
 - i. Thumb posterior splint with wrist strap/ thumb spica
- g. Carpal tunnel syndrome
 - i. Cock up splint
- h. Axillary/ shoulder burn
 - i. Airplane splint, Axillary splint
 - ii. 90° shoulder abduction, ER, slight flexion
 - iii. Worn 24 hours
- i. Elbow burn
 - i. Posterior elbow splint
 - ii. Extend and Supinated
- j. Hand burn
 - i. Claw hand – intrinsic minus deformity
 - ii. Resting hand splint(intrinsic plus position
 - iii. Wrist 15-20 extended, MCP flexion 70°, IP's extended & thumb slightly abducted

- xiv. Dynamic
 - a. Humeral fracture orthosis
 - i. For Pt's with humeral fracture and shoulder subluxation
 - ii. Allows arm swing
 - b. BFO (balance forearm orthosis)
 - i. SCI c5-c6 SCI
 - ii. For feeding purposes

PROSTHESIS

- i. A device applied to replace the function of a missing part
- ii. Prosthetic Principle/ Partial foot amputations
 - a. Transmetatarsal & Tarsometatarsal (Lisfranc)
 - i. Anterior weight bearing areas are lost
 - ii. Shoe fillers/ rocker bar(shifts roll over point to the heel)
 - b. Midtarsal (Choparts)
 - i. Spares Calcaneus & Talus
 - c. Pirogoff
 - i. Vertical resectioning of calcaneus
 - d. Boyd (Talectomy)
 - i. Surgical removal of the talus
 - * Partial foot amputations = Low Heels
 - e. Syme's (Ankle Disarticulation)
 - i. 0.6cm transection above the malleoli
 - f. Transtibial (BKA)
 - i. MC type of amputation
 - ii. Types
 - 1. Short <20%
 - 2. Standard (**20-50%**) Siegelman
 - 3. Long >50%
 - iii. Ideal shape of slump: Cylindrical
 - iv. MC contracture: Knee flexion
 - g. Knee Disarticulation
 - i. 3 types
 - 1. T – Tenoplastic / Kirk & Callendar (TEKiCa)
 - a. Section femur is covered by quads
 - b. Levels of adductor tubercle
 - 2. O - Osteplastic/ Gritti –Stokes
 - a. Arthrodesis of patella to section femur
 - b. Level of femoral condyles
 - 3. T –True Disarticulation
 - a. Simplest and least traumatic
 - b. Preserves the femoral epiphysis
 - h. Transfemoral (AKA)
 - i. 3 Types
 - 1. Short <35%
 - 2. Standard 35-60%
 - 3. Long >60%
 - ii. Ideal shape of stump: Conical
 - iii. MC contraction: Hip FAB

- i. Very Short AKA (Hip disarticulation)
 - i. Above the lesser trochanter
 - ii. Prosthesis Rx: Canadian Hip Disarticulation Prosthesis
 - j. Hemipelvectomy/ Transpelvectomy
 - i. Lower half of the pelvis
 - ii. Prosthesis Rx: CHDP
 - k. Hemicorporectomy
 - i. Removal of pelvis, sacrum L4-L5
 - ii. Prosthesis Rx: Sitting bucket articulated to CHDP
 - l. Bilateral AKA
 - i. Primary means of ambulation – WC with posterior displaced wheels 1.5-2"
 - ii. Prosthetic Rx: Stubby Prosthesis
 - 1. Quadrilateral socket with posteriorly protruding rocker bar (Prevents Pt from falling backwards)
 - 2. Reduces height about 12" (Low COG, increase balance)
 - m. Bilateral BKA
 - i. BK prosthesis should be 1" shorter than the pre-amputated leg (For foot clearance)
 - n. Combination Single BKA + Single AKA
 - i. BKA is the better leg
 - ii. AK prosthesis should be ¼ to ½ inch shorter than the BK prosthesis (For foot clearance)
 - iii. Stair climbing: BKA (ascending) AKA (Descending)
- iii. Prosthesis
- a. BK vs AK

	BK	AK
Components	FOO t ankle assembly SH Ank SO cket SU sension	FOO t and anke assembly SH Ank K nee joint SO cket SU sension

- b. BK Prosthesis
 - i. Foot Ankle Assembly
 - 1. Nonarticulated components
 - a. FAA is directly attached to the shank
 - b. (-) mechanical ankle joint
 - c. Quiet and lighter
 - d. Keel (Wt. supporting structure inside the prosthetic foot)
 - i. SACH (Solid ankle cushion heel) Wooden Keel with cushion heel
 - 1. Has slight Medio-lateral motion to accommodate uneven terrain.

- ii. SAFE (Stationary Attachment Flexible Endoskeleton)
 1. Rigid bolt block (SA)
 2. Flexible keel (Flexible Endoskeleton)
 3. Has Medio-Lateral motion, heavier than SACH.
- iii. STEN(Stored Energy)
 1. Has 3 pieces wooden keel and 2 pieces rubber plugs
 2. Heavier than SACH
- iv. Seattle
 1. **Delrin Keel** that acts as a cantilever
 2. Heavier than SACH
 3. Mod: Seattle – Lite Foot

*Safe st's in seattle are heavy
- v. Carbon copy II
 1. Has 2 carbon composite plates
 2. Has medio-lateral motions
 3. Light wt.
- vi. Quantum
 1. Keel is in a **spring module**
 2. (-) Spring cushion
 3. Medio-lateral motion
 4. Light wt.
- vii. Flex-walk
 1. Has composite graphite
 2. Has medio-lateral motion
 3. Light wt., carbon color
- viii. Flex-foot
 1. (+) Absorbs energy
 2. Distal Segment is same with flex walk
 3. Proximal segment extends upward to form as the shank. Carbon color
 4. Light wt.

*Car Quo Walks Foot kaya Light Wt.

* Such SAFE Carbon Qud walk M-L motion

2. Articulated Components

- a. Presence of mechanical ankle joint
- b. Heavy but can be adjusted
- c. Single Axis
 - i. MC type since, inherent stability
 - ii. DF + PF
- d. Multi Axis/ Polycentrics
 - i. DF + PF/ Inv + Eve / Slight Rotation

- ii. Shank
 - 1. Exoskeletal (Crustacean)
 - a. MC type of shank
 - 2. Endoskeletal (Central Support/ Pylon)
- iii. Socket
 - 1. PTB socket
 - a. MC prescribed
 - b. (+) Presence of soft insert
 - i. Provides additional support areas
 - ii. Reduces edema
 - iii. Improves circulation
 - 2. Hard socket
 - a. (-) soft insert
 - i. For Pt's with good tissue covering and heavy perspiration
 - 3. ISNY socket
 - a. Icelandic Swedish New York PTB socket
 - b. Thermoplastic materials
 - c. Thin to dissipate heat, enhances comfort since it is made up of thermoplastic, can be reshaped again.
 - 4. Pressure Tolerant Areas (bulges/ build ups)
 - a. Patellar tendon
 - b. Pretibial Ms
 - c. Postero-distal aspect of residual limb
 - d. Popliteal fossa
 - e. Distal end, medial flare, lateral of Tibia
 - f. Tibial and fibular shafts
 - 5. Pressure Sensitive areas (Reliefs/ channels)
 - a. Anterior crest of tibia
 - b. Fibular head and neck
 - c. Peroneal/ Fibular nerve
 - d. Anterior Tibia
 - e. Hamstrings tendon
- iv. Suspension (Calf supra supra Thigh)
 - 1. Calf suspension
 - a. MC prescribed
 - b. Provides slight medio-lateral stability
 - 2. Supracondylar suspension
 - a. High medial and lateral walls, encompasses the femoral condyles)
 - b. Provides better medio-lateral stability
 - 3. Supracondylar-suprapatellar suspension
 - a. High medial-lateral- anterior walls
 - b. Encompasses the femoral condyles and the patella
 - c. Provides greater medio-lateral stability

4. Thigh corset
 - a. Medio-lateral stability and additional suspension
 - b. Quads atrophy, heat.
- c. Biomechanics (FAA)
 - i. Inset
 1. MC used
 2. Prosthetic foot and shank is located medially in relation to the socket.
 - ii. Outset
 1. Prosthetic foot and shank is located laterally in relation to the socket.
 - iii. Posterior wall is higher than anterior wall
 1. Special device: rotator unit
 - a. Absorbs rotational forces acting on the stump
- d. AK Prosthesis
 - i. Foot ankle assembly
 1. Single axis MC, for inherent stability
 - ii. Shank
 1. Exoskeletal (Crustacean)
 - iii. Knee joint/ mechanism
 1. Axis
 - a. Single
 - i. MC
 - ii. Knee Flexion - extension
 - b. Polycentric
 - i. Knee flexion- extension, slight rotation
 2. Friction
 - a. 3 types
 - i. Constant
 1. MC used
 2. During Swing phase, friction remains constant all throughout
 - ii. Variable
 1. Increased resistance during acceleration and decelerations
 2. Decreased resistance during midswing
 - iii. Fluid control
 1. Cadence –dependent motion
 2. 2 types
 - a. Pneumatic - air
 - b. Hydraulic – oil
 3. Extension Aid/ Extension stop
 - a. Extension aid
 - i. Aids during swing phase
 - b. Extension stop
 - i. Situated on prosthetic limb, to prevent terminal impact

4. Knee locks
 - a. 2 types
 - i. Wt. activated
 - ii. Manual lock
- iv. Socket
 1. Quadrilateral sockets
 - a. MC used
 - b. Walls
 - i. Posterior wall, horizontal shelf for Ischial tuberosity wt. bearing
 - ii. Anterior wall, higher than posterior wall to prevent dislodging of Ischial tuberosity from the horizontal shelf) d/t translatory and angular pelvic motions.
 - iii. Lateral wall, Should be pre-adducted to give leverage to the gluteus medius
 - iv. Medial wall, should accommodate the adductor muscles to prevent adductor roll
 - v. ALAT anterolateral walls should be 2.5-3" higher than the postero-medial walls
 2. Ischial containment socket
 - a. Ischial tuberosity is within the socket
 - b. Wt. bearing: ascending ischio-pubic ramus
- v. Suspension (SuPaSiPer)
 1. Suction suspension
 - a. MC prescribed
 - b. 4G's criteria
 - i. Good tissue covering
 - ii. Good limb length and shape
 - iii. Good single limb balance
 - iv. Good hand strength and dexterity
 2. Partial suction with Axillary suspension
 3. Silesian bandage
 - a. Has medio-lateral control
 - b. Pt's active in sports
 4. Pelvic belt
 - a. Has (B) medio-lateral and rotational control
- vi. Biomechanics
 1. Inset
 2. Initial socket flexion = 5°
 3. Initial socket adduction = 7°
 4. Special device
- e. Speed of ambulation
 - i. (N) speed = 3mph
 - ii. Single BK = 2-2.5 mph
 - iii. Single AK = 1.5mph

- f. Metabolic requirements
 - i. Wheelchair ambulation = 9%
 - ii. Crutch walking = 60%
 - iii. Single BKA = 10-40%
 - iv. Double BKA = 41%
 - v. Single AKA = 65%
 - vi. Double AKA = 110%
 - vii. Single AKA + Single BKA = 75%
- g. Prosthetic fitting in children
 - i. Above below elbow *3-6 months*
 - ii. Below above knee joint : *8-10 months*
 - iii. Actively controlled Terminal Device - *2 years old*
 - iv. Actively controlled elbow joint – *2-3 years old*
 - v. Functional hand - *3 years old*
 - vi. Actively controlled knee joint *3-4 years old*
- h. Gait deviations
 - i. BK prosthesis (Seen in stance phase)
 - 1. Excessive knee flexion at heelstrike ($>15-20^\circ$)
 - a. Excessive dorsiflexion of prosthetic foot
 - b. Stiff heel cushion
 - c. Excessive anterior tilt of socket in relation to prosthetic foot
 - d. Excessive anterior displacement of socket in relation to prosthetic foot
 - 2. Insufficient knee flexion ($<15^\circ$)
 - a. Excessive PF of prosthetic foot
 - b. Too soft heel cushion
 - c. Excessive posterior tilt of socket in relation to prosthetic foot
 - d. Excessive posterior displacement of socket in relation to prosthetic foot
 - 3. Forward bending
 - a. Pain on the end of stump/ residual limb
 - 4. Early lateral thrust
 - a. Excessive inset
 - b. Abducted socket
 - 5. Early knee flexion at push off
 - a. Excessive DF of prosthetic foot
 - b. Too soft DF bumper/ anterior stop
 - c. Excessive anterior tilt of socket in relation to the foot
 - d. Excessive anterior displacement of socket in relation of the foot
 - 6. Delayed knee flexion
 - a. Excessive plantarflexion of prosthetic foot
 - b. Too stiff DF bumper/ anterior stop
 - c. Excessive posterior tilt of socket in relation to the foot
 - d. Excessive posterior displacement of socket in relation of the foot

ii. AK prosthesis (Seen in stance and swing phase)

Stance phase

1. Foot rotation at heel strike
 - a. Too stiff heel cushion/ PF bumper
2. Foot slap
 - a. PF bumper too soft/ heel cushion
3. Lateral trunk bending
 - a. Weak gluteus medius
 - b. Abducted socket
 - c. Short prosthesis
 - d. Hip abduction contracture
 - e. Pain on distal limb/ insecurity
4. Wide walking base/ abducted gait
 - a. Pain on crotch area: increased medial wall
 - b. Contracted hip abductors
 - c. Prosthesis is too long
 - d. Shank is in valgus position
 - e. High medial wall
5. Uneven step length
 - a. Insecurity or fear
 - b. Hip flexion contracture
 - c. Insufficient knee friction (longer steps taken by the prosthetic leg)
6. Swing-phase whips
 - a. Medial = external rotation
 - b. Lateral = internal rotation
7. Exaggerated lordosis
 - a. Contracted back extensors
 - b. Hip flexion contracture
 - c. Weak abdominals and hip extensors
 - d. Insufficient support from the anterior wall
 - e. Improperly shaped posterior wall

Swing Phase

8. Vaulting
 - a. Prosthesis is too long
 - b. Decrease knee flexion
 - c. Too small socket
 - d. Excessive PF of prosthetic foot
 - e. Insecurity / fear
 - f. Inadequate suspension (piston action)
 - g. Manual lock
 - h. Too tight extension aid
 - i. Insufficient knee friction\
9. Circumduction
 - a. Prosthesis is too long

10. Hip hiking
 - a. Prosthesis is too long
 11. Excessive heel rise
 - a. Forceful hip flexion
 - b. Absence of extension aid
 - c. Insufficient knee friction
 12. Insufficient heel rise
 - a. Manual lock
 - b. Extension aid too tight/ stiff
 - c. Excessive knee friction
 13. Terminal knee impact
 - a. Criteria Visible and Audible
 - b. Tight extension aid
 - c. Extension stop is too old/ Resilient
 - d. Knee friction is insufficient
 - e. Pt's fear of buckling
- iv. UE prosthesis
- a. Components
 - i. TeD – Terminal Device
 - ii. Inter – Interposing joints
 1. Wrist, elbow ,shoulder
 - iii. SO - socket
 - iv. SU - suspension
 - v. CO – Control cable
 - b. Terminal device
 - i. Voluntary Opening Hook – MC and Most functional (2 cycles)
 - ii. Voluntary Closing Hook – Most physiological (4 cycle action)
 - iii. Cosmetic Hand – Rigid/ Semi rigid, non functional
 - iv. Myoelectric hand- muscle rectified by electrodes into hand function
 - v. Motor driven hand – computer generated hand function
 - c. Interposing Joints
 - i. Wrist
 1. Fixed
 - a. Uses only 1 terminal device
 2. Quick change
 - a. Uses many terminal devices
 - ii. Elbow
 1. Mechanical – prepositioned 5-125° elbow flexion
 2. Electrical – has more control and versatility
 - iii. Shoulder
 1. Prepositioned to different ADL's

- d. Socket
 - i. Standard socket – two-walled socket. (outer & inner wall)
 - 1. MC
 - 2. Inner wall encapsulates the stump
 - 3. Outer wall shaped like the real arm or forearm
 - ii. 3 walled socket – used for infants
 - 1. Outer wall – shaped like the outer arm of the infant
 - 2. Middle wall - preshaped for a year or two, to reduce cost
 - 3. Inner wall - encapsulates the stump
 - iii. Single walled socket
 - 1. Very short below elbow amputation
 - iv. Muenster type socket
 - 1. Very short Below elbow amputation
 - 2. To gain power
 - v. Splint-type socket with step up hinge
 - 1. Pt's with LOM of elbow flexion
 - 2. For every 1° of elbow flexion = 2° at the socket
- e. Suspension/ Harness
 - i. Figure of 8 harness: MC prescribed
 - ii. O-ring harness: adjustable
 - iii. Figure of 9 harness
 - iv. Modified shoulder saddle harness – used by patient who performs heavy lifting
- f. Control cable/ system
 - i. Bowden control
 - 1. MC
 - 2. Single cable sliding inside a single housing
 - ii. Fair-lead control
 - 1. Single cable sliding inside dual housing
- v. Upper extremity prosthetic training
 - a. Don and doff prosthesis
 - b. Open and close the terminal device
 - c. Hold and release the object
 - d. Transfer objects
 - e. Perform ADL's
 - f. Maximum weight UE prosthesis is 5lbs

ASSISTIVE DEVICES

- i. Most stable to least stable
 - a. Parallel bars
 - b. Walker
 - c. Crutches
 - i. Axillary
 - ii. Non-Axillary
 - d. Cane
 - i. Quad cane
 - ii. Bilateral cane
 - iii. Unilateral cane
- ii. Parallel bars
 - a. Used for initial training or ambulation
 - b. Height: level of greater trochanter, 20-30° elbow flexion
 - c. Width: 2 inches from each greater trochanter
- iii. Walker
 - a. Provides anterior and lateral stability
 - b. Transmits up to 85% of BW
 - c. Widens BOS
 - d. Used by elderly patients
 - e. Parts
 - i. Handgrip/ Handpiece
 - ii. Uprights (4)
 - iii. Lateral bar/ brace
 - iv. Anterior bar/ brace
 - v. Extension piece
 - vi. Adjustable pin/ notch
 - vii. Rubber suction tips
 1. Prone to wear and tear
 2. Diameter 1-1.5" diameter
 3. Height: 2-3"
 - f. Measurement of walker
 - i. 10-12" anterior to the Pt
 - ii. Hand grips: level of greater trochanter, 20-30° elbow flexion
 - g. Types of walker
 - i. Stander (collapsible)
 - ii. Stair climbing walker
 1. Addition 2 posterior uprights with hand grips
 - iii. Hemiwalker
 1. Used by CVA patients
 2. One handgrip located at the center of the walker

- iv. Reciprocal walker
 - 1. Pt's with RGO
 - 2. Using RG pattern
- v. Rollator walker
 - 1. Anterior rollating walker
 - a. Promotes trunk extension
 - b. Accommodate a flexed posture (stoop posture)
 - 2. Posterior rollating walker
 - a. Promotes trunk extension
 - b. Corrects stoop posture
 - c. E.g. CP spastic diplegia
 - d. Goal: Velocity/ speed, posterior rollating walker with 4 wheels
- h. Gait pattern – walker
 - i. (B) legs are good legs when using a walker
 - ii. FWB / 100% WB: walker>left foot>right foot (half a step forward than the first step.)
 - 1. Do not parallel (B) feet, it will diminish balance
 - iii. PWB/ 1-99%WB: walker> bad leg > good leg (half a step forward than the first step)
 - 1. E.g. 2lbs of weight is still considered PWB
 - 2. E.g. Arthroplasty
 - iv. NWB/ 0%WB: walker> good leg (no bad leg)
 - 1. E.g. Single BKA/ AKA, arthroplasty
 - 2. No stair climbing
- iv. Crutches
 - a. Provides lateral stability
 - b. Widens BOS
 - c. Always use (2)
 - d. Types
 - i. Axillary crutches, transmits 80% BW
 - 1. Parts
 - a. Axillary pad
 - b. Double uprights
 - c. Handgrip/ handpiece
 - d. Extension piece
 - e. Adjustable pin/ notch
 - f. Rubber tip
 - i. Diameter 1-1.5" diameter
 - ii. Height: 2-3"
 - ii. Orthocrutch / type of Axillary crutch
 - 1. Single upright

- iii. Non-Axillary crutches/ Lofstrand/ Forearm/ Canadian transmits 40-50% BW
 - 1. Parts
 - a. Forearm cuff
 - b. Single upright
 - c. Hand piece/hand grip
 - d. Extension piece
 - e. Adjustable pin
 - f. Rubber tip
- e. Measurement – Axillary crutches
 - i. In standing
 - ii. Axillary pad: 2-3 fingers, 2" below the axilla
 - 1. If Pt's continues to lean on axillary pad > impingement of radial nerve, (wrist drop)
 - iii. Hand grip – 20-30° elbow flexion, level of greater trochanter
 - 1. Platform attachment – for patients with arthritic hand, fracture of wrist/ distal forearm. Elbow will be at 90° flexion
 - iv. Rubber tip: 2 inches lateral, 6 inches anterior from 5th toe
 - v. Standing technique:
 - vi. Tripod stance: 4" lateral and 4" anterior from the 5th toe (COG is between the crutches and feet), cannot be used in crowded areas
 - vii. Pt's ht. subtract 16 inches
 - viii. In Supine: Axillary pad= over the anterior Axillary fold
 - 1. Rubber tip = 6-8" from the lateral heel, or at the lateral heel, add 2 inches (without shoes)
 - ix. In sitting: ® shoulder abducted to 90° elbow flexed, (L) shoulder abducted to 90° elbow extended
 - 1. Axillary pad is placed on the tip of flexed elbow (olecranon process)
 - 2. Rubber tip is placed on middle or 3rd finger
- f. Measurement – forearm crutches
 - i. FA cuff: Opening: anterior and medial
 - ii. FA cuff is at the proximal forearm/ or 1-1.5" below the olecranon process
 - iii. Handrip: level of greater trochanter 20-30° flexion
 - iv. Rubber tip : 2 " lateral and 6" anterior to 5th toe
- g. Crutch walking muscles
 - i. Shoulder depressors
 - 1. Latissimus Dorsi
 - 2. Pectoralis minor
 - 3. Lower-trapezius
 - ii. Shoulder adductors
 - 1. Pectoralis major
 - 2. Latissimus dorsi
 - iii. Flexors/ extensors/ abductors of the shoulder
 - 1. Deltoids

- iv. Elbow extensor
 - 1. Triceps
 - a. Weakness, use Canadian crutch (triceps cuff)
 - 2. Anconeus
- v. Wrist extensor
 - 1. ECRL
 - 2. ECU
 - 3. ECRB
- vi. Finger and thumb flexors
 - 1. FDS
 - 2. FDP
 - 3. FPL/FPB
- h. Gait patterns – crutches
 - i. 4 point gait: ® crutch > (L) foot > (L) crutch ® foot (affected foot is always last), safest, slowest, most difficult to teach
 - ii. 3 point gait: (B) crutches > bad leg > good leg
 - iii. 2 point gait: ® crutch + (L) foot > (L) crutch + ® foot, for speed
 - iv. Swing gaits:
 - 1. Swing-to gait: (B) crutches > swing (B) legs level with crutches, best for paraplegics
 - 2. Swing-through gait: (B) crutches > swing (B) legs beyond crutches
 - a. Most difficult to perform
 - 3. Drag-to gait: crutches forward > drag (B) feet on the floor
 - a. d/t fear
 - 4. Simultaneous gait: ® crutch + ® foot > (L) crutch + (L) foot
 - a. Problem with coordination
- i. Stair climbing
 - i. Good leg up, bad leg down
 - ii. Ascending
 - 1. Good leg > bad leg > crutches
 - 2. Or Good leg > bad leg + crutches
 - iii. Handrail
 - 1. On same side of good side, place crutches on bad leg side of arms
 - iv. Descending
 - 1. Crutches > bad leg > good leg
 - 2. Crutches + bad leg > good leg
 - v. Pt position
 - 1. Guard belt/ safety belt (4-6 inches), placed on waist
 - 2. Level surface: posterolateral near the weak side, hands placed on guarding belt, other hand standby assist on the shoulder
 - 3. Ascending: posterolateral near the weakside
 - 4. Descending: Anterolateral near the weakside
 - 5. If a patient has the tendency to fall > hug the patient grab the safety belt and ask the patient to sit down

- v. Cane
 - a. Transmits 10-20% of body weight
 - b. Cane is always placed on the opposite side
 - c. Parts:
 - i. Handgrip
 - ii. Single upright
 - iii. Extension piece
 - iv. Adjustable pin/ notch
 - v. Rubber tip
 - d. Measurement – cane
 - i. 6 inches lateral from the 5th toe
 - e. Different types of cane
 - i. Standard- Adjustable aluminum cane
 - 1. (J shaped, crook)
 - ii. Adjustable – aluminum off-set cane
 - 1. Gives better support
 - iii. Quad cane
 - 1. Narrow part is closes to the body
 - 2. Large-based
 - 3. Small-based (stair climbing)
 - f. Gait pattern
 - i. Method 1
 - 1. Cane> bad leg > good leg
 - ii. Method 2
 - 1. Cane + bad leg>goodleg
 - iii. IF knee is involved
 - 1. Place the cane on the ipsilateral side (according to magee)
 - iv. When turning, always turn on the good side, making small circles and do not pivot

WHEELCHAIR

- i. Parts of a wheelchair
 - a. Backrest- connected to the back tube
 - i. Height: Seat of the wheel to base of Pt's axilla and subtract 4 inches
 - ii. Width: 16-18inches
 - iii. High back rest – quadriplegic patient
 - iv. Low back rest – paraplegic
 - b. Hand grip
 - c. Back post
 - d. Tipping level
 - e. Arm rest
 - i. 90° of elbow flexion
 - ii. Removable: for transfer: sliding board transfer of paraplegic patients.
 - iii. Types
 1. Full length – full support to forearm
 2. Desk length
 - f. Clothing guard
 - g. Seat – attached to seat rail
 - i. Depth: from the back to the popliteal area, subtract 2 inches
 1. Too narrow: sacral sitting/ pressure ulcer
 2. Too wide: Pt's will slide off
 - ii. Width: form the widest part of the buttocks add 2 inches (1 inch from each side)
 1. Too narrow – pressure sore
 2. Too wide – scoliosis
 - iii. Seat height
 1. Seat to down to the heel/ foot rest add 2 inches (PBM: 90° hip and knee flexion)
 - a. Too high = too much pressure on Ischial tuberosity
 - b. In sitting, perform sitting pushup every 20 minutes
 - c. Paraplegics – wrap around the back to prevent anterior falling
 - h. Wheel/ tire
 - i. Pneumatic - air
 - ii. Rubber (width 1 inch)
 - iii. Diameter 22" or **24"** (ideal)
 - i. Wheel lever/ lock brakes – always lock the wheel chair
 - j. Hand rims – propulsion
 - i. Diameter: 18" – 20"
 - ii. Projections
 1. Oblique (C5) shoulder flexors
 2. Vertical (C6) Wrist extension c tenodesis effect
 3. Horizontal (C7) Elbow extension
 4. Friction (C7) Elbow extension

- k. Caster wheels – guides the turning of the wheelchair
 - i. (n) diameter: 5"- 8" (ideal)
 - ii. Types
 - 1. Small-sized – 3 inches (COG is more on the rear wheel, for easy turning)
 - 2. Large-sized – 12 inches (COG is more on the caster wheels), for rough surfaces)
- l. Foot plate/ foot rest
 - i. Heel loop/ toe loops – lower extremity extensor spasticity (Foot dislodges from the foot plate)
 - ii.
- m. Leg rest
 - i. Fixed
 - ii. Removable/ detachable/ swing-away (transfer)
 - iii. Elevating leg rest (for lower extremity edema)
- n. Cross brace – tie the urinary catheter- to draining bag
- ii. Types of wheel chair
 - a. 12-9 moderate speed
 - b. 3-9 faster speed
 - c. Outdoor wheel chair – large wheels behind, small wheels in the front
 - i. power
 - d. Indoor wheel chair – large wheels in the front, small wheels in the back
 - i. Mobility
 - e. Standard wheel
 - i. Collapsible/ for easy storage
 - f. Hemi-wheelchair
 - i. Low seat
 - ii. One arm drive (2 hand rims on one side only, larger:ipsi smaller: Contralateral)
 - g. Sports wheel chair
 - i. Low seat, low back rest, rigid frame, high chamber (slanted wheels)
 - ii. Wheelie: paraplegic, requires rigid frame
 - h. Reclining wheelchair
 - i. For patients with postural instability and for pressure relief
 - ii. Anti-tipper (tipping lever) to prevent tipping backward
 - i. Tilt in space
 - i. Extensor spasticity
 - ii. For pressure relief
 - j. Amputee wheel chair
 - i. Bilat AKA on wc, COG is displaced psoterior
 - ii. Posteriorly displaced 1.5-2 increase
- k. Motorizes
 - i. Joystick
 - ii. Push button
 - iii. Sip and puff (S4) – patients should have controls memorized

- iii. Measurements
 - a. Stairs
 - i. Ratio for every 1 inch rise there should be 12 inch length
 - ii. Avg step height is 7 inches
 - iii. Resting area: 60" x 60"
 - iv. 1:12 or 8.3%
 - b. Grab bars from the floor 33-36"
 - i. Diameter 30mm or 3"
 - c. Door width minimum 32" but ideal is 36"
 - d. Corridors or hallway: minimum is 36 inches, ideal is 42"
 - e. Toilet seat 17-19"
 - f. Distance of bed from room wall
 - i. 90° turn at least 32"
 - ii. 180° turn at least 60"
 - g. Zigzag technique: one hand is stabilized and the other hand is moving vice versa to get up the ramp

ORAD & ETHICS

- i. Ethics, Basic principles
 - a. Autonomy
 - i. Freedom of an individual to choose and implements one's decisions
 - ii. Free from undue influence, coercion, duress, deceit, or constraint
 - iii. A form of personal liberty
 - iv. Self determination
 - b. Beneficence
 - i. A duty to promote the health and welfare of patients above the consideration, while attending and honoring personal autonomy
 - ii. Seeing only what will benefit the patient
 - c. Non maleficence
 - i. Doing no harm to patient
 - d. Justice
 - i. Giving what is due to an individual
 - e. Veracity
 - i. Truthfulness
 - ii. Ensuring accuracy and precision in giving information
 - 1. Patient: telling the truth so that appropriate care may be provided
 - 2. Therapist: disclose facts or information regarding treatment so that the patient can exercises autonomy
 - f. Confidentiality
 - i. Right of the patient over the information regarding his/ her body or condition
 - ii. Ensuring the privacy of the patient concerning his medical care, throughout care and even after provision of care.
 - g. Role of fidelity
 - i. Practicing with the constraints of one's profession
 - ii. Defined and bound by a professions legislated scope of practice. \
- ii. Ethical dilemma
 - a. If there is a conflict with different principles
 - b. How to solve?
 - i. Gather all the facts that can be known
 - ii. Decide which ethical principles are involved
 - iii. Clarify your professional duties in the situation
 - iv. Describe the general nature of the outcomes of consequences desired
- iii. PPTA standard of practice
 - a. Legal and ethical considerations
 - b. Client care
 - c. Education
 - d. Administration
 - e. Research

- iv. Republic act 5680
 - a. Enacted June 21, 1969
 - b. Contents:
 - i. Composition of board of PT/ OT
 - ii. Application to examination
 - iii. Board examination
 - 1. Requirements for application
 - 2. Scope of examination (APK(30), MES(25), PT APPS(45))
 - 3. Ratings to pass the board examination
 - c. Amendments
 - i. Separation of PT/OT
 - ii. Expansion of powers of the board
 - iii. Expansion of scope of practice
 - iv. Integration of all PT's nationwide
 - d. APTA , definition of PT a profession to develop and maintain and restore physical function.
- v. Republic act 7277
 - a. Magna Carta for Persons With Disabilities
 - i. Contents
 - 1. Education
 - 2. Telecommunication
 - 3. Accessibility law (BP 344)
- vi. Code of ethics of PPTA
 - a. Responsibilities
 - i. To the profession
 - ii. To patient/ client
 - iii. To colleagues
 - iv. To public
- vii. Professional behavior (WCPT)
 - a. Accountability
 - b. Altruism – others first before self
 - c. Compassion
 - d. Cultural competence
 - e. Ethical behavior
 - f. Integrity
 - g. Personal/ professional development
 - h. Social responsibility
 - i. Teamwork

- viii. PPTA
 - a. Founded December 8, 1964
 - i. President: Royson Mercado
 - ii. VP internal: Michael Gabilo
 - iii. VP external: Baldomero Ranjo II
 - iv. Secretary: Christian Rimando
 - v. Treasurer: Arthur Benedict King
 - vi. PRO: Vanessa Sardan
- ix. Notes
 - a. The National PT day/ World PT day
 - i. September 8
 - b. National disability and prevention and rehabilitation week
 - i. July 17-23
 - c. Founding president of PPTA
 - i. Jose Inoturan

ORAD

- x. Functions of a manager
 - a. Planning
 - b. Organizing
 - c. Controlling
 - d. Budgeting
 - e. Delegation
 - f. Conflict management (conflict between therapist/ therapist/ patient)
 - g. Decision making
 - i. Statement of goals
 - ii. Agenda building
 - iii. Determination of organization structure
 - iv. Continuous evaluation of decision
- xi. Personnel recruitment and selection
 - a. To fill positions with the best candidates from among persons who are qualified for those positions
- xii. Recruitment
 - a. Means attracting applicants from outside the organization
 - b. Direct vs. indirect
 - c. May be:
 - i. Passive recruitment
 - ii. Active recruitment
 - iii. Aggressive recruitment
- xiii. Performance appraisal
 - a. A means by which the supervisor tells the employee (or vice versa) how he is doing(
 - b. A form of feedback
 - c. Should be used as a motivator to continue learning and growing on the job

- xiv. Retention and development
 - a. The supervisor must be concerned with keeping enough competent employees for a long enough periods of time to assure reasonable degree of effectiveness and efficiency in carrying out the department's work and achieving its goal.
- xv. Managerial technique for employee development
 - a. Job rotation
 - b. Coaching (junior to senior therapist feedback)
 - c. Understudy
 - d. Vertical loading
- xvi. Policies and procedures
 - a. Policies – a set forth criteria for what is to be done
 - b. Procedure – How it will be implemented

Department planning, design, and constructions

- xvii. Early considerations
 - a. Existing services
 - b. Type and size of hospitals
 - c. Populations characteristics
 - d. Referral sources
 - e. Changes in delivery and practice
 - f. Services and activities
- xviii. First requirements
 - a. Accessibility
 - i. Convenient location if to serve both in-patients and out-patients
 - 1. Can be clustered with other departments
 - 2. Better if located on main floor
 - b. Functional areas
 - i. Hydrotherapy area
 - ii. Gymnasium
 - iii. Treatment cubicle area
 - iv. Support area/ non-treatment areas
 - v. Size of functional areas may be dependent on:
 - 1. Type of patient
 - 2. Anticipated utilization
 - 3. Cost restraints
 - 4. Architectural barriers
 - c. Staffing

- xix. Space requirements
 - a. Defining functional areas
 - b. Number of functional area elements
 - i. $E = TO/UA$
 - ii. E: elements x the suggested dimension in table 3.1 of Hickok = size of the functional area
 - iii. T: Estimate number of treatments per day
 - iv. O: occupancy time of each treatment of patient, expressed in minutes including time preparation
 - v. A: total available time per day that a cubicle could be used for treatment
 - vi. U: Utilization ratio. The proportion of daily available time during which a cubicle is likely to be occupied. The ratio is usually greater than .50 or less than 1.00
- xx. Available time and space
 - a. E.g. Gymnasium and Hubbard tank areas requires at least 10 feet of ceiling height for installation of electric hold apparatus and high mounted objects (over head pulleys)
- xxi. Gymnasium
 - a. The recommended realistic size should be one-third of the total space\
 - b. The presumption is that the gymnasium will be one of the high utilization areas.
- xxii. Patient waiting areas
 - a. Doctor (1973) suggested that 16 square feet for average ambulatory patient and 25 square feet for patient in a wheel chair
 - b. Patton (1965) suggested a formula for space required for waiting
- xxiii. Future expansion
 - a. If this was considered during the originally planning, future costs can be reduced significantly and there will be minimal disruption of the service at the time of expansion.
- xxiv. Questions
- xxv. Given an estimated total number of 15 patient per day, compute for the total number of cubicles you will put up, assuming that each treatment time will last for 1 hour and the clinic will be open for 6 hours a day; (50% utilization ratio)
 - a. 3
 - b. 4
 - c. 5
 - d. 6
 - e. 7

Anteswe: $E=TO/UA$
 $=15 (1)/ \frac{1}{2} (6)$
 $=5$

Leadership

- xxvi. The state where a person or a froup of persions is able to influence others or agree on a goal and work towards it

- xxvii. Style of leadership
 - a. Autocratic leadership
 - i. Also referred to as authoritarian, boss-centered, dictatorial leadership
 - ii. Characterized with close supervision
 - b. Bureaucratic leadership
 - c. Participative
 - d. Meaningful involvement of employees in decision making process and in the maintenance of cohesive group interaction
 - e. Laissez-faire leadership 'free rein
 - f. Paternalistic leadership
 - i. Father-son/ daughter relationship
- xxviii. New leadership approaches
 - a. Transactional leader
 - i. A leader who clarifies subordinates role and task requirements, initiates structures, provides rewards and displays consideration for subordinates.
 - b. Charismatic leader
 - i. A leader who is able to motivate
 - c. Transformational leader
 - i. A leader distinguished by a special ability to bring about innovation and change through different major strategic ways
 - d. Interactive leaders
 - i. Who is concerned with consensus building, is open and inclusive, and encourages participation
 - e. Servant leaders
 - i. A leader who works to fulfill subordinates needs and goals as well as to achieve the organization's larger mission
- xxix. Factors that influence leadership styles
 - a. Work assignment
 - b. Personality and ability of the employee
 - c. Attitude of employee towards the manager
 - d. Personality and ability of the manager
- xxx. Seven deadly sins by Marcus
 - a. Being a boss instead of a leader
 - b. Being indifferent towards discipline and recognition when they are warranted
 - c. Being too busy to train subordinates, and instead applying pressure to increase productivity
 - d. Using snap judgement in selection of employees rather than selection techniques and interviews
 - e. Letting the eager beaver's job grow like topsy while letting the goldbrick get away with substandard performance
- xxxi. Fair labor practice
 - a. Labor code of the Philippines
 - i. Presidential decree (442-1974) – labor code of the Philippines
 - ii. Republic act of 6175 (1989) – amendments of the labor code

- xxxii. Regular employee
 - a. Has rendered at least 1 year
- xxxiii. Probationary employment
 - a. Probationary period should not exceed six months from the date the employee started working.
One becomes a regular employee upon completion of his six-month period.
- xxxiv. Wages
 - a. Include overtime, night differential, rest day, holiday, and 3th month play
- xxxv. Computing overtime
 - a. On ordinary days
 - i. Number of hours in excess of 8 hours (125% s hourly rate)
 - b. On a rest day, special day or regular holiday
 - i. Number of hours in excess of 8 hours (130% x hourly rate)
- xxxvi. Computing pay for work done
 - a. A special day (130% x basic pay)
 - b. A special day which is also a scheduled rest day (150% x basic pay)
 - c. A regular holiday (200% x basic pay)
 - d. A regular holiday, which is also a scheduled rest day (260% x basic pay)

RHEUMA

- i. Rheumatoid arthritis
 - a. Autoimmune d/o
 - b. More common in Female, 30-45 y/o affecting small joints
 - c. If male, 60y/o, larger joints are affected E.g. shoulder, hip
 - d. *Type of arthritis: Erosive*
 - e. Inflammatory (Disabling), Chronic , Systemic
 - f. Theories of etiology
 - i. **(+) RF** (series of antibodies IgM, IgG is recognized as their antigen.)
 - 1. Seropositive (Associated with poor prognosis, increased extraarticular manifestation, severe disease.)
 - ii. **(+) HLA DR4**, if seronegative
 - iii. **T-Cell**, proliferation involving the synovium of the diarthrodial joints (Pannus formation, causes erosion)
 - g. S/Sx:
 - i. (+) pain, morning stiffness (Increased pain at rest)
 - h. Diagnostic criteria:= (7) (*1st 4 should be present for 6 consecutive weeks*)
 - i. Morning stiffness
 - 1. Inflammation >1 hour
 - 2. Non-inflammatory <30 mins (OA)
 - ii. Arthritis on 3 or more joints
 - iii. Hand arthritis (can be asymmetrical)
 - iv. Symmetrical
 - v. Rheumatoid/ subcutaneous nodules
 - vi. Erosion in x-ray (d/t Pannus formation)
 - vii. (+) RH Factor
 - i. X-ray
 - i. A - Abnormal alignment
 - ii. B – Bone involvement
 - iii. C – Cartilage involvement
 - iv. D – Deformities
 - v. E – Erosion
 - vi. S – Soft tissue Swelling
 - j. MC affected joints: (it will never involve the DIP)
 - i. MCP – MC affected
 - ii. Wrist
 - iii. PIP
 - 1. Swan-neck –
 - a. rupture of lateral slip/ band
 - b. volar capsule swelling of PIP
 - c. reflex muscle spasm of intrinsic

- d. rupture of EDC at the DIP
2. Boutonniere deformity
 - a. Rupture of Central slip
3. Ulnar drift
 - a. Causes Z deformity (wrist towards radial (partial dislocation of carpals) , MCP towards ulnar side)
4. Vaughn Jackson Deformity – 4th and 5th digits
 - a. Rupture of the 4th and 5th extensor tendons
5. Mannerfelt syndrome
 - a. Rupture of the FPL at the first CMC
- iv. K – Knee
 1. Knee flexion contracture
 2. subluxation
- v. M – MTP
 1. Hallux valgus
 2. Metatarsalgia
 3. Hammer toes (flex PIP joint)
 4. Apropulsive gait – absence of push off
 - a. Shoe modification high toe box, wide ball of the foot.
 - b. Rocker bottom – for push off
- vi. S – Shoulder
 1. Glenohumeral joint, limited IR, Adduction
 2. If OA, AC joint
 3. Subluxation
- vii. A – Ankle
 1. Pes planus/ flat foot
- viii. C – cervical spine
 1. C1-C2 (most mobile)
 - a. Laxity of transverse atlantal ligament (Cord compression)
 - b. Fracture of the Odontoid process (Hangmans/ tear drop fracture), it will move superiorly and penetrate medulla oblongata (respiratory center, body will be deprived of oxygen) [Asphyxia]
- ix. H – Hip
 1. Protrusion acetabuli
 2. LOM - IR
- x. E – Elbow
 1. Elbow flexion contracture
- xi. T – TMJ
 1. Loss or limitation of mouth opening
- k. Rheumatoid nodules
 - i. Extensor of forearm
 - ii. Popliteal bursa

- I. Extraarticular manifestations
 - i. Heart – Pericarditis
 - ii. Lungs – Caplan’s syndrome “capLUNGS”,
 1. Rheumatoid nodules found in the lungs
 - iii. Eyes - Sicca syndrome - dryness
 - iv. Nerves
 1. Median, MC – CTS
 2. Ulnar – Cubital Tunnel syndrome
 3. Post. Tibial n. – tarsal tunnel syndrome
 4. Radial – Posterior interosseous n. (PIN)
- ii. Sjogrens syndrome
 - a. Other names
 - i. Gougerot’s
 - ii. Mikulicz’s
 - iii. Autoimmune exocrinopathy
 - b. Triad
 - i. Xerophthalmia – dryness of eyes
 1. Schirmers, filter paper in the eye, 5mm in 5 mins
 2. Rose Bengal, dye in the eye shows area of dryness
 - ii. Xerostomia – dryness of mouth, bilateral parotiditis
 - iii. Arthritis
 - c. Manifestations
 - i. Dyspareunia – painful sexual intercourse, 2° to dryness of female genitalia
- iii. Felty’s Syndrome
 - a. Mnemonics

*“Tita Felty’s
Blood is SLLAANT
Penia – Penia, Felty’s”*
 - b. Splenomegaly
 - c. Lymadenopathy
 - d. Leukopenia
 - e. Anemia
 - f. Arthritis
 - g. Neutropenia
 - h. Thrombocytopenia
- iv. **Stage of Rheumatoid Arthritis**
 - a. Stage I – Early
 - i. (+) Juxtaarticular osteoporosis,
 - ii. (-) bone destruction, deformity, & extraarticular manifestations.
 - b. Stage II – Moderate
 - i. (+) juxta articular osteoporosis, with or without slight bone destruction,
 - ii. Rheumatoid nodules(signs of progression to severe stage)
 - iii. *Adjacent* muscle atrophy

- c. Stage III – Severe
 - i. (+) deformities > LOM,
 - ii. (+) extraarticular manifestations, Rheumatoid nodules, tenosynovitis,
 - iii. *Extensive muscle atrophy*
- d. Stage IV – Terminal
 - i. All of the above, (+) ankylosis (bony fusion)
- v. **Functional Classification in RA**
 - a. Class I:
 - i. Independent in *all activities*
 - b. Class II:
 - i. Independent in Self care and vocational
 - ii. Limited in avocational activities
 - c. Class III:
 - i. Independent in self care
 - ii. Limited in vocational and avocational activities
 - d. Class IV:
 - i. *Dependent* in all activities
- vi. Polymyalgia Rheumatica
 - a. Elderly Female 50y/o
 - b. Pain and stiffness in the muscles,
 - c. Common in shoulder girdle and neck muscles affected
 - d. Common affected knee joint
 - e. *Low grade fever*
 - f. Muscle pain > joint pain
 - g. Mnemonics: SECRET
 - Ang secret ni lola poly:*
 - i. S – Stiffness
 - ii. E – Elderly
 - iii. C – Caucasian
 - iv. R – Rheumatism (knee joint)
 - v. E – Elevation of ESR (inflammation) 60-100mm/hr
 - vi. T – Temporal Arteritis (Giant Cell Arteritis)
- vii. JRA/ JIA (Juvenile rheumatic/ Idiopathic)
 - Types:
 - a. Pauciarticular/ Oligoarticular –
 - i. <4 joints; for the first 6 months KAW +TMJ (knee ankle wrist)
 - ii. F>M 5 y/o
 - iii. If 1 more joint is added after 6 months it is called *Extended Pauciarticular JIA*
 - iv. 2 types
 - 1. (+) ANA (antinuclear antibody) (+) *iridocyclitis* (Most disabling involvement of the eye (unequal size of the pupils))
 - 2. (-) ANA

- b. Polyarticular - >5 joints affected
 - i. Aka polyarthritis
 - ii. F>M 8 y/o
 - iii. Affects the small joints of the hands
 - iv. 2 types
 - 1. (+) RF Adult like RA
 - 2. (-) RF
- c. Systemic/ still's disease
 - i. Most rare among the 3
 - ii. F=M
 - iii. Most Severe type
 - iv. **High grade fever, daily, od**
 - v. Still's Rash in the presence of fever (salmon –pink)
 - 1. Disappears with depression of fever
 - 2. Nonpruritic rash
- viii. SLE (Systemic Lupus Erythematosus)
 - a. Female
 - b. 15-45 y/o
 - c. Type of arthritis: Non-Erosive
 - d. African Americans/ Hispanics
 - e. Systemic
 - f. 11 criteria (4 or more / 11)
 - i. D – Discoid Rash (when heals, leaves a scar)
 - ii. O – Oral Ulcer
 - 1. Immunologic disease
 - iii. P – Photosensitivity
 - 1. Presence of sun exposed areas of the body
 - iv. A – Arthritis
 - 1. Jaccoud's Arthritis – non erosive but deforming
 - 2. MC swan neck deformity that is reversible (since the problem is not erosive but capsular laxity)
 - v. M – Malar
 - 1. Butterfly rash (Over malar eminence)
 - 2. Forehead, nasal fold is spared
 - vi. I – Immunologic
 - 1. Infection, MC cause of mortality
 - 2. CNS lupus
 - 3. Renal
 - 4. Cardiac Involvement
 - vii. N – Neurologic
 - 1. Psychosis, CVA manifestation. Mov't d/o
 - viii. R – Renal
 - 1. Lupus Nephritis

2. Undergo Dialysis

- ix. A – ANA (+)
 - 1. 99% with ANA, have SLE
- x. S – Serositis
 - 1. Heart - pericarditis
 - 2. Lungs - pleuritis
- xi. H – Hematologic
 - 1. Felty's syndrome
- g. Symmetric involvement of joints is small = large
- h. Associated wolf
- i. Alopecia
 - i. Discoid
 - ii. Systemic disease
 - iii. Side-effect of medications
- j. “Si Lupe Baluga(**black**), Luka Luka (**psychosis**), na-inlove(**pericarditis**), kay kalbo (**alopecia**) nagkikita pag gabi (**photosensitivity**), lips to lips (**oral ulcers**), lupaypay (**seizures**). Jaccoud si Kalbo gamit ang (L) hand, na anal si Lupe, napaihe (**proteinurea**), saksi ang buwan (**crescent**), wolf (**lupus**) hiningal (**pleuritis**)”
- ix. Drug induced SLE
 - a. Drugs causing:
 - i. Procainamide – anti arrhythmic drugs
 - ii. Hydralazine – arteriolar dilator
 - iii. Isoniazid - TB
 - iv. Methyldopa – Anti- depressant
 - v. Chlorpromazine – anti psychotic
 - b. No rashes
- x. Seronegative Spondyloarthropathies
 - a. (-) rheumatoid factor
 - b. Involvement of the spine
 - c. (+) HLA B27
 - d. F=M
 - e. *Arthritis type: Entesitis* (inflammation of the entheses[connection of tendon to ligament to bone])
 - f. Types:
 - i. Psoriatic
 - ii. AS
 - iii. Reiters
 - iv. IBD

xi. Psoriasis

- a. with joint involvement (Psoriatic Arthritis)
- b. Affecting skin
- c. With spine involvement (Asymmetric)
- d. Manifestation
 - i. Oligoarticular – 4 or less joints affected progresses to Polyarthritis
 - ii. Involvement of *DIP joint*, next is knee joint (WB)
 - iii. Auspitz sign – bleeding when removal of skin
 - iv. Koebner Phenomenon
 - v. Onycholysis – Nail lysis
 - vi. “Pencil in a Cup” deformity
 - vii. (+) Dactylitis (Sausage Digits)
 - viii. Mnemonics

*“Psoriatic
Koebner DIP his Sausage
In Auspitz sause in a cup”*

*“DAKO Pencil
Psoriasis ito!”*

xii. AS (ankylosing spondylitis)

- a. Male, 20-40 years old
- b. Systemic, symmetric, Inflammatory
- c. Other names
 - i. Marie Strumpell (Ascending)
 - ii. Von Bechterev (Descending)
- d. Manifestation
 - i. Kyphosis deformity
 - ii. MC affected peripheral joint is the Shoulder joint
 - iii. Morning back pain, SI joint affectation (Sacroilitis)
 - 1. Special test:
 - a. Patrick's, FABERE, Figure of 4
 - b. Gaenslen's Test
 - c. Schober's Test
 - d. Occiput to Wall test
 - e. Pelvic Rock/ Compression
- e. Calcification of IVD (annulus fibrosus) connecting vertebrae
 - i. Syndesmophytes – vertical running bony spurs
- f. Lung problems – chest expansion (CRLD)
- g. Extra articular:
 - i. Heart: AortitiS
 - ii. Eyes: Anterior UveitiS
- h. X-Ray finding: Bamboo Spine

- i. Management:
 - i. Best exercise: Swimming
- j. DDx
 - i. DISH (Diffuse Idiopathic Skeletal Hyperostpsos)/ Forestier's Dse
 - ii. >40 y/o
 - iii. Non systemic
 - iv. Calcification of ALL
 - v. X-Ray finding; Bamboo Spine
- xiii. Reiter's (Reactive Arthritis)**
 - a. Males STD 9:1, Chlamydia
 - b. Female Food Born 1:1, Salmonella
 - c. Asymmetric
 - d. Cause: Infection + HLA B27
 - e. Ankle joint affected
 - f. Triad
 - i. Conjunctivitis
 - ii. Urethritis
 - iii. Arthritis (Enthesis)
 - g. Mnemonics:
 - i. C - Conjunctivitis
 - ii. U - Urethritis
 - iii. K – Keratoderma Blennorrhagicum, seen in the palm/ soles of the foot
 - iv. A – Arthritis (Enthesitis)
 - v. C – Circinate Balinitis, inflammation of penis
- xiv. IBD (inflammatory bowel disease/ Enteropathic Arthritis)**
 - a. Ulcerative Colitis – large intestine (colon)
 - i. Increased upper extremity involvement
 - b. Crohn's Disease – most severe, more chronic
 - i. Increased lower extremity joint involvement
 - ii. May affect the whole GIT, from mouth to rectum (Skip Lesion).
 - c. MC affected joint: Knee joint (Enthesitis)
 - d. Manifestation:
 - i. Diarrhea
 - ii. Abdominal pain
 - iii. Vomiting
- xv. Progressive Systemic Sclerosis (PSS)**
 - a. Female>male
 - b. Increase production of collagen
 - c. Major criteria (1 major at least 2 minor)
 - i. Scleroderma proximal to MTP or MCP joint
 - d. Minor Criteria
 - i. Digital pitting scars
 - ii. Sclerodactyly
 - iii. Bibasilar Pulmonary fibrosis

- e. Stages:
 - i. Edematous stage
 - ii. Sclerotic Stage, Sclerodactyly
 - iii. Atrophic stage
 - 1. Very thin appearance in the skin
- f. Classification
 - i. Localized
 - 1. Morphea – thickening of the skin of the anterior trunk
 - 2. En Coup De Bande – Thickening skin of Extremities
 - 3. En Coup De Sabre – Thickening skin of the Face
 - 4. Tobacco Pouched lips – Thickening of skin of lips
 - ii. Systemic
 - 1. Organ affected:
 - a. Kidney (Most feared complication in scleroderma)
 - b. Heart
 - c. Lungs (Honey Comb lungs)
 - 2. Types:
 - a. Diffused Systemic Scleroderma
 - i. Elbow
 - Knee
 - b. Limited Systemic Scleroderma
 - i. Hands
 - ii. Feet
 - iii. Telangiectasia (dilation of small venules found in the face(Spider veins)
- g. Crest Syndrome
 - i. C – Calcinosis
 - ii. R – Raynauds Phenomenon
 - iii. E – Esophageal Dysmotility
 - iv. S – Sclerodactyly
 - v. T – Telangiectasia
- xvi. **PM- DM**
 - a. PM
 - i. Muscle weakness of proximal muscles, no evidence of rash
 - b. DM
 - i. Characteristic rash
 - ii. Can develop PM later on
 - c. Types
 - i. PM: Adult Idiopathic Poly Myositis
 - ii. DM: Adult idiopathic Dermato Myositis, MC in females
 - iii. PM-DM: associated with malignancy, MC in males age of 40 y/o
 - iv. PM – DM: associated with vasculitis/ seen in children commonly
 - v. PM – DM: Associated with collagen vascular disease
 - vi. PM – DM: Inclusion body myositis, MC in males weakness of proximal and distal muscles

- d. Rashes
 - i. Heliotrope rash – purplish discoloration around the eyes
 - ii. V sign rash – anterior neck and chest
 - iii. Shawl sign rash - arms
 - iv. Mechanic's hand rash – Hands
 - v. Gottron's Papules – rash on the joint (Cancerous)
- xvii. Gouty Arthritis (acute attack)
 - a. Males 30 and above 7.0mg/dl
 - b. Females 60 and above 6.0 mg/dl
 - c. (+) hyperurecemia
 - i. Too much intake of uric acid
 - d. MC in 1st MTP
 - e. Build up of monosodium urate in:
 - i. Joint
 - 1. Wrist – Chieragra
 - 2. Knee – Gonagra
 - 3. 1st MTP – Podagra
 - ii. Cartilage
 - 1. Tophi
 - iii. CNS
 - 1. Lesch Nyhan: severe spasticity
 - f. Pharma: Colchicines
 - g. Vs Pseudogout
 - i. Build up of calcium pyrophosphate Dihydrate (CPPD)
 - ii. Knee joint
 - iii. Associated with chondrocalcinosis
- xviii. Osteoarthritis
 - a. MC in female, 60 y/o and up
 - b. Hallmark: joint space narrowing, articular cartilage degeneration
 - c. Pain during activity/ weight bearing
 - d. (+) osteophyte formation
 - e. (+) gull wing sign
 - f. Types:
 - i. Primary
 - 1. Degenerative disc disease/ Degenerative joint disease
 - ii. Secondary
 - 1. Trauma
 - g. MC affected joints
 - i. UE/ pinching joints: DIP, PIP, 1st CMC, AC
 - ii. LE: knee, hip, 1st MTP
 - h. DIP: Heberden's, PIP: Bouchard's (signs of primary OA)
 - i. Most disabling complication: Affectation of 1st CMC
 - j. Hip: Malum Coxae Semilis >70 y/o (most disabling)
 - k. Knee: Genu Varum deformity

- I. 1st MTP: Hallux Valgus
- m. Facet: C5-C6
- n. Variant: Erosive inflammatory arthritis aka Crain's Disease
 - i. Females 50-55 years old
 - ii. Affected: PIP, DIP joints
 - iii. Similar to RA, but there is erosion at DIP joint.
- xix. FMS (Fibromyalgia Syndrome)
 - a. Non – inflammatory, rheumatic d/o, non – articular
 - b. Affects tissue
 - c. Tender points (localized pain, (trigger points: referred pain)
 - d. 10-45 y/o
 - e. 11/18 tender points, should be present above and below the waist.
 - i. Occiput
 - ii. Lower cervical
 - iii. Trapezius
 - iv. Supraspinatus
 - v. 2nd rib
 - vi. Lateral epicondyle
 - vii. Gluteals
 - viii. Greater trochanter
 - ix. Medial fat pad of the knee
 - f. Psychological effect
 - i. Raynaud's
 - ii. Sleep disturbance (Hardest to manage)
 - 1. Best management: aerobic exercises
 - 2. Promote relaxation: Biofeedback
 - iii. Morning stiffness
 - iv. Fatigue

FRAC, D/L, AMP & OP

i. Osteoporosis

- a. Female, Caucasian
- b. 60y y/o F= M
- c. Gold standard in diagnosing osteoporosis
 - i. DEXA – Dual energy X-ray Absorptiometry
- d. Definition: Progressive bone disorder
 - i. Decrease is total bone mass 2° imbalance in resorption activity
- e. Osteoblast – bone forming cell
- f. Osteoclast - bone resorption
 - i. Osteomalacia – (N) or increase in BMD

WHO

Classification	Standard deviation
Normal bone mass density	1.0
Osteopenia	1.0-2.0
Borderline osteoporosis	2.0 – 2.5
Osteoporosis	>2.5
Severe Osteoporosis	>2.5 with Fragility fractures

- g. Types:
 - i. Primary/ Involutional Osteoporosis
 - 1. Gonadal function: Hormonal (Parathormone)
 - 2. Senile Osteoporosis
 - ii. Secondary
 - 1. Osteoporosis with other disease
- h. Causes
 - i. Hormonal imbalance
 - 1. Male: testosterone
 - 2. Female estrogen
 - ii. Increase caffeine intake
 - iii. Drug induced
 - 1. Prednisone – corticosteroids (increased rate of resorption)
- i. Diagnosis
 - i. X-ray: Codfish vertebra (biconcavity of vertebra, expansion of IVD)
- j. Appearance
 - i. Kyphosis
 - ii. Protruding abdomen
 - iii. Dowager's Hump
- k. Management
 - i. Calcium intake:
 - 1. 1.200-1,500mg day

ii. Fracture

- a. Definition: Deformity/ A break or discontinuity of a bone
- b. General Appearance
 - i. Spiral – torsional force, fastest to heal
 - ii. Oblique – diagonal, angular force
 1. Forms an angle along the axis of the bone
 - iii. Transverse – forms a right angle along the axis of the bone, Horizontal, angular force
 - iv. Compression – axial loading, angular force
 1. MC: T8 – L3 (spine)
 - v. Avulsion – 2° strong pull of muscles
 - vi. Pathologic – underlying disease, body weight
 - vii. Comminuted – bone is broken down into several fragments
 - viii. Stress – repetitive, cumulative microtrauma
- c. Completeness
 - i. Complete – total discontinuity within a bone
 - ii. Incomplete – Crack or avulsed, does not destroy the totality of the bone
- d. Communication
 - i. Open/ Compound
 1. (+) wound, prone to infection
 2. MC infection (Osteomyelitis)
 - ii. Closed/ Simple
 1. (-) wound or breaking of skin
 2. (+) abrasions, contusions
- e. According to location
 - i. Proximal
 - ii. Distal
 - iii. Anterior
 - iv. Posterior
- f. According to position
 - i. Displaced - leave the anatomical position
 - ii. Undisplaced – does no leave the anatomical position
- g. Specific classification

Salter & Harris Classification (Epiphyseal Plate Fx)

Stage	Manifestations
Stage I	Epiphyseal plate Fx
Stage II	Epiphyseal plate + Triangular segment of metaphysic
Stage III	Epiphyseal plate + Epiphysis
Stage IV	Epiphyseal plate + metaphysis + Epiphysis
	Stoppage of bone growth
Stage V	Comminuted epiphyseal plate

Gustilo-Anderson Classification (Open Fx)

Stage	Manifestations
Stage I	Fx, wound <1cm
Stage II	Fx, wound >1cm but less than 10cm
Stage III	A – Adequate tissue covering peritoneal coverage
	B – Bony exposure
	C – Circulatory Problems

Garden classification (femoral neck fracture)

Stage	Manifestation
Stage I	Incomplete fracture
Stage II	Complete Fx without displacement
Stage III	Complete Fx with partial displacement
Stage IV	Complete Fx with complete displacement

Le Fort Classification

Stages	Manifestation
Stage I	Horizontal Fx Low maxillary fracture
Stage II	Posterolateral Fx of the face Nasal bridge and inferior lacrimal wall;
Stage III	Zygomatic arch, ethmoid bone, supraorbital fissure

- h. Eponyms, nomenclature of ones
- i. Humeral fracture
 - i. Supracondylar Fx:
 - 1. MOI: Hyperextension
 - 2. Malgaigne fracture
 - 3. Complication: Volkmann's ischemic contracture
 - ii. Transcondylar fracture of the elbow/ Posada's
- j. Radius & Ulna
 - i. Monteggia – fracture of ulna c radial head dislocation
 - ii. Gallezi –fracture of radius c D/L of DRU joint
 - 1. Other names
 - a. Dupuytren's fracture
 - b. Reverse Monteggia
 - c. Piedmont
 - d. Fracture of necessity
 - iii. Nightstick – fracture of ulna

- k. Distal radius & ulna
 - i. Colle's – dorsal displacement
 - 1. aka Pott's fracture
 - 2. MOI: FOOSH
 - 3. Silver-fork deformity
 - ii. Smith's – volar displacement
 - 1. MOI: FOOSH
 - 2. Garden-spade deformity
- l. Hand
 - i. Palmar rim/ Lennier's
 - ii. Bennett's – 1st CMC
 - iii. Rolando's – comminuted Bennett's
 - iv. Boxer's – neck of the 5th MCP
 - 1. Aka Fist Fighter's/ street fighters
- m. Spine
 - i. Jefferson's fracture- C1/ atlas
 - ii. Hangman's fracture – C2
 - iii. teardrop fracture- Odontoid process
 - iv. Clay Shovelers - C7 spinous process, excessive extension of cervical region
 - v. Chance Fx: seatbelt fracture, lap belt fracture
- n. Hip
 - i. Straddle fracture – double vertical fracture of the pelvis
 - 1. Pubic ramus
 - ii. Ischioacetabular fracture/ Walter's
 - iii. Malgaigne fracture – fracture and dislocation of femoral neck
 - iv. Duvigne's fracture – fracture of iliac wing (ALA)
 - v. Subtrochanteric – common in elderly, below the trochanters
 - vi. Intratrochanteric – common in elderly, along the trochanters
 - 1. 3-4cm does not interrupt blood supply
- o. Ankle and foot
 - i. March fracture - shaft of 2nd MTP
 - ii. Cotton's – Transmalleolar fracture
 - 1. MOI: excessive inversion or eversion
 - iii. Calcaneus – MC fracture of the tarsal
 - 1. MOI: forceful DF
 - iv. Shepherd's/ Talus – MOI forceful DF & Eversion
 - v. Nutcracker/ - anterior calcaneus and cuboid
 - vi. Jones's – base of 5th metatarsal
- iii. Fracture anomalies
 - a. Malunion (union in a faulty position)
 - i. Rotation
 - ii. Angulation
 - iii. Shortening

- b. Delayed union (failure of the fracture ends to unite at a given time)
 - i. UE – 10 weeks
 - 1. Hand: 5 weeks
 - ii. LE – 20 weeks
 - c. Non-union (failure of the fracture ends to unite)
 - i. MC cause: poor blood supply
 - ii. Intervention:
 - 1. Bone grafting
 - 2. acceptance
 - d. Factors:
 - 1. Infection
 - 2. Malnutrition (Vit C deficiency (essential for wound healing))
 - 3. Mobilization
 - 4. Impaired circulation
 - iv. Bone Healing
 - a. Inflammatory stage
 - i. 3-5 days , first 48 hours
 - ii. Hematoma formation (inflammatory cells)
 - iii. Prostaglandin mediation
 - iv. Initiation of fibroblast proliferation
 - v. Increase blood, faster healing
 - b. Reparative phase
 - i. Continuation of fibroblast activity
 - ii. Formation of soft callus
 - iii. weeks
 - c. Remodeling phase
 - i. Hard callus formation
 - ii. WB is allowed
 - iii. Months to years
 - iv. Dormant Quiescence
 - v. Principles of fracture healing
 - a. Reduction
 - i. Replacement of bones as near the anatomical position as possible, manipulation (MC)
 - b. Maintenance of reduction
 - i. traction
 - c. Preservation function
 - i. Rehabilitation
 - ii. MC site of traction: Tibia

- vi. Traction
 - a. Skin traction
 - i. Adhesive rubber strip: 5-6lbs
 - ii. Dunlop skin traction, UE
 - iii. For children
 - 1. Russel's
 - 2. Box
 - iv. Pearson attachment
 - a. Allows knee flexion
 - b. Skeletal traction
 - i. Commonly attach to tibia
- vii. Fixation
 - a. Internal fixation
 - i. Intermedullary nails
 - ii. Internal medullary rods
 - iii. Pins and screws
 - b. External fixation
 - i. Plates
 - ii. CAST
- viii. **Dislocation**
 - a. Dissociation of a joint
 - b. MC
 - i. S – SC and AC
 - ii. A – GH joint (ABER)
 - iii. P – Elbow joint (nurse/housemaids elbow)
 - iv. A – Lunate/ Wrist
 - v. P – Hip (CHD , AHD) FADIR
 - vi. A - Knee
 - vii. L – Patella (2° to vastus lateralis)
 - viii. P – ankle (excessive DF)
- ix. **Amputation**
 - a. Absence of removal of a limb
 - b. MC
 - i. UE: (L) intercalary radial limb
 - ii. LE: (L) fibula
 - c. MC amputation
 - i. UE: Trauma -> PVD
 - ii. LE: PVD -> Trauma
 - iii. Generally: PVD

d. Congenital:

- i. Amelia – absence of the entire limb
- ii. Hemimelia – absence of majority art of a limb
- iii. Acheria – absence of wrist of hand
- iv. Adactylia – absence MCP/MTP
- v. Aphalangia – Absence of fingers and digit
- vi. Apodia – absence of the foot
- vii. Phocomoelia – flipper like stump, absence of arm and forearm

e. Acquired

- i. Partial toe – excision of one or more toes
- ii. Ray Ressection – 3-5th digits
- iii. Syme's – ankle disarticulation
- iv. Lisfranc's – TMT disarticulation
- v. Boyd – spared calcaneus
- vi. Pirigoff – vertical dissection of calcaneus
- vii. Foot le fort – horizontal dissection of calcaneus
- viii. Chopart – midtarsal (talo Na CC Pa)
 1. Surgeons joint
- ix. Hemipelvectomy
 1. Lower part of the pelvis
- x. Hip Hindquarter
 1. Si joint, symphysis pubis
- xi. Hemicorporectomy
 1. Bilateral limbs, pelvis, L4-L5
- xii. Gritti-Stoke's disarticulation
 1. Knee through condyles

f. Levels of amputation

- i. Above elbow amputation –
 1. 0 % Shoulder disarticulation
 2. 30% - Humeral neck dissection
 3. 50 % - short above elbow
 4. 90 % - long above elbow
 5. 100 % - elbow disarticulation
- ii. Below Elbow Amputation
 1. 35% - Very short below elbow
 2. 55% - Short below elbow
 3. 90% - long below elbow
 4. 100% - Wrist disarticulation

iii. Knee

Above knee		Below knee
<33%	Short	<20%
33-66%	Standard/ medium	20-50%
>66%	Long	>50%

iv. Stump measurements

Level	Reference point
BKA	Medial tibial plateau
AKA	Medial inguinal line
AEA	Tip of Acromion
BEA	Medial Epicondyle

v. Energy expenditure

Wheelchair ambulation = 9%

Crutch walking = 60%

Single BKA = 10-40%

Double BKA = 41%

Single AKA = 65%

Double AKA = 110%

Single AKA + Single BKA = 75%

BURNS & ULCERS

i. Skin injury caused by thermal

- a. Heat
- b. Cold
- c. Chemical
- d. Electrical(iceberg type)
- e. Lightning
- f. Radiation

ii. Zones of burn injury

- a. Zone of Coagulation - irreversible damage cell death occurs (+) infection and if not treated will lead to a deeper burn injury
- b. Zone of Stasis – Cells may die within 24 – 48 hours if without specialized treatment.
- c. Zone of Hyperemia - Minimal cell injury, should recover.

i. Classification (Degree of burn injury)

Degree of burn	Class	Level	Color	Blister	Edema	Pain	Scar	Healing time
Superficial	1 st degree	Epidermis	Red/ hyperemic	None	Slight	Delayed on set	None	3-7 days
Superficial partial thickness	2 nd degree	Epidermis, Upper Dermis	Bright red	Intact	Moderate	Extremely Painful	Minimal	7-21 days
Deep partial thickness	2 nd degree	Epidermis, Most dermis	Waxy white/ Mixed red	broken	Marked	Little to None	Moderate Hypertrophic/ keloids	21-35 days
Full thickness	3 rd degree	Epidermis, Dermis, Subcutaneous tissue	White/ grey/ Black/ Charred	Wet	Severe (extensive)	Anesthetic	Severe, Eschar formation	Grafting

a. Notes

- i. Keloids- beyond the boundaries of the burn injury
- ii. Hyper trophic scars, within the boundaries of burn injury
- iii. If developing hypertrophic scars, Mx: Pressure garments (up to 2 years))
- iv. Full Thickness burn (3rd degree burn) E.g. Chemical

b. Subdermal burn (4th degree burn)

- i. Damage to the dermis, epidermis, subcutaneous tissue, extending to muscle and bone
- ii. E.g. Electrical burn, prolonged contact with flame.

c. Electrical burn

- i. Complications:
 1. Ventricular fibrillation
 2. SCI
 3. Renal failure
 4. Cataracts
- ii. "Tip of the Ice Berg" phenomenon

ii. Rule of nines

- a. A quick estimation of percentage of burn injury
- b. Used during emergency
- c. A more detailed version done inside the hospital : Lund and Browder chart: inside the hospital

Rule of Nine's (Adult & Children)

Area		Adult	Total	Children	Total
Head and neck	Anterior	4.5%	9%	9%	18%
	Posterior	4.5%		9%	
Anterior Trunk	Upper	9%	18%	9%	18%
	Lower	9%		9%	
Posterior Trunk	Upper	9%	18%	9%	18%
	Lower	9%		9%	
One UE	Anterior	4.5%	18% (2)	4.5%	18% (2)
	Posterior	4.5%		4.5%	
One LE	Anterior	9%	36% (2)	7%	28% (2)
	Posterior	9%		7%	
Genitalia/ Perineum		1%	1%	1%	1%
		Total	100%		101%

iii. Burn wound healing

- a. Epidermal healing
 - i. Retention of viable cells (healthy skin cells) that will allow re-epithelialization
- b. Dermal healing
 - i. Phases (3) (IPM)

Inflammatory	Proliferative	Maturation
Cardinal signs: Pallor, rubor, dolor Calor, tumor, loss of function	Fibroblast and granulation Tissue formation	Remodelling and Maturation of scars (2 years)

c. Complications:

- i. *Infection*, MC leading factor of mortality
- ii. Pulmonary complication: pneumonia (intraalveolar infection)
- iii. Metabolic Complications E.g. Negative Nitrogen balance
- iv. HO: MC: post elbow > Hip > shoulder
- v. MO: MC site UE: brachialis, LE: Quads

d. Severity of burn injury

	Minor	Moderate	Major/ Critical
2 nd degree burn	<15%	15-30%	>30%
3 rd degree burn	<2%	2-10%	>10%

i. Major burns

1. Inhalation injury
2. Electrical burn
3. Burns on face, hands, feet, & perineum
4. Burns with complications
5. Burns with fractures

- e. Splint
 - i. Immediately worn upon admission (24 hours)

f. Common deformities and orthosis

Area	Common deformity	Position to stress	Splint
Neck	Flexion	Hyperextension	Cervical/ plastic collar
Shoulder	Adduction & IR	Abduction 90°/ ER/ Slight flexion	Airplane splint/ Axillary splint
Elbow & FA	Flexion & pronation	Extension & supination	Posterior elbow splint
Hand	Intrinsic minus (claw hand)	Intrinsic plus: Wrist extension: 15-20° MCP flexion: 70° IP extension & slight Thumb abduction	Resting hand splint
Hip	Flexion, Adduction, IR	Extension, abduction Neutral rotation	Posterior hip splint
Knee	Flexion	Extension	Posterior knee splint
Ankle	Plantarflexion	Dorsiflexion	AFO

g. Grafts

- i. Delay ROM Exercise Even passive/ active post graft 3-5 days post graft.
- ii. Allograft/ Homograft – same species (cadaver skin) (temporary)
- iii. Autograft – patients self (permanent)
- iv. Xenograft/ Heterograft – from another species; usually a pig (temporary)
- v. Biosynthetic graft - artificial graft (meshed graft)
- vi. Splint-thickness graft – Epidermis and upper layer of dermis from donor's skin site
- vii. Full-thickness graft – has (B) epidermis and whole dermis from donor skin site
- viii. Cultured skin – Laboratory grown skin
- ix. Z-plasty – used to lengthen burn wound contracture

h. Common drugs given:

- i. Surface organisms: Silver Nitrate
- ii. Pseudomonas: Silver Sulfadiazine (silvadene)
- iii. Prior to skin grafting: Nitrofurazone/ Furasin
- iv. Penetrates thick eschar: Sulfamylon (Mafenide Acetate)
- v. Liquefies Thick eschar: (Travase/ elase)
- vi. Open technique: (+) use of drugs without dressing
- vii. Closed technique: (+) use of drugs with dressings

i. Management

- i. Whirlpool –
 1. NSS (normal saline solution) clean wound/ debridement
 - a. Contains NaCl .9g/100ml H₂O
 2. Povidone iodine, Sodium hypochloride (bleach), Hydrogen Peroxide (PNH)
 - a. For infected wound
 3. Dakin's Solution for (+) AIDS
 - a. Boric Acid + Bleach

iv. Ulcers

a. Arterial Vs Venous

	Arterial ulcer	Venous Ulcer
Granulation	Pale red	Bright Red
Location	Lateral Malleolus, Toes, Feet, anterior Tibia	Medial Malleolus
Edema	(+)/ (-)	(+)
Gangrene	(+)	(-)
Pain	Upon elevation	Little to no pain on dependency
Pulse	Diminished/ absent	Normal
Other	Pallor or elevation Dusky Rubor on Dependency (reactive hyperemia)	Cyanotic on Dependency Venous stasis
Position	Dependent	Elevated
Hemosiderin Staining	(-)	(+)

b. Pressure Ulcer/ Decubitus ulcer/ Pressure sores/ Ischemic Sore/ Bed Sore

- i. Definition: area of unrelieved pressure resulting to ischemia, usually over bony prominences
- ii. Provide relief
 1. Supine q 2 hours
 2. Sitting q 15-20 mins
- iii. Predominant sites
 1. Supine:
 - a. Occiput
 - i. MC in infants (d/t heaviness)
 - b. Spine of scapula
 - c. Posterior elbow
 - d. Sacrum
 - i. MC in elderly
 - e. Coccyx
 - f. Gluteals
 - g. Posterior heel
 2. Sidelying:
 - a. Ear
 - b. Lateral shoulder
 - c. Lateral epicondyle of elbow
 - d. Lateral trunk/ ribs
 - e. Greater Trochanter
 - f. Lateral & medial condyle
 - g. Lateral & medial malleoli
 - h. Lateral aspect of 5th MTT and toe

3. Sitting
 - a. Spine of scapula
 - b. Vertebral spines
 - c. Ischial tuberosity
 - d. Sacrum/ coccyx

4. Prone
 - a. Ear
 - b. Ant shoulder
 - c. Breast
 - d. ASIS
 - e. Genitalia
 - f. Patella
 - g. Anterior tibia
 - h. Dorsum of foot
 - i. Toes

c. Diabetic Ulcer

- i. Loss of protective sensation
- ii. D/t peripheral neuropathy
- iii. Gloves and stocking sensation
- iv. Plantar aspect of the foot
- v. Best gait pattern: NWB

Stages of ulcer (E Di Sugat Malalim Ba?)

Stages	Manifestation
Stage I	(E) Epidermis, non –blanchable erythema, Warm/ cool, Pain/ itching
Stage II	(D) Dermis, abrasion, blister (>1cm), shallow Crater
Stage III	(S) Subcutaneoustissue, deep crater
Stage IV	(M, B) Muscle & Bone. (+) undermining Tunneling, (+) Sinus Tracts

III & IV usually requires surgery.

v. Debridement

- a. Process of removal of necrotic tissue
- b. Types
 - i. Autolytic
 1. (+) Enzymes
 2. Allow's patients own skin to remove necrotic tissues
 - ii. Enzymatic
 1. (+) enzymes, liquefy the necrotic tissues
 2. PT will perform the process (enzymatic debridement)
 - iii. Mechanical
 1. (+) enzymes & dressings, PT will remove the necrotic tissues (mechanically)

- iv. Sharp
 - 1. Done by PT
 - 2. (-) anesthesia
 - 3. Leave viable cells and necrotic tissue removal
 - v. Surgical
 - 1. Done by Doctor
 - 2. Use of anesthesia
 - 3. Remove both viable cells and necrotic tissue
- vi. Dressings (2- 1 – 1)
 - a. Dry Wounds
 - i. Hydrogel – Best, facilitates AD
 - ii. Transparent Films (for stage I and II ulcers) facilitates AD
 - iii. Wet to Wet/ Continuous wet –
 - 1. (2) Gauze Impregnated
 - b. Minimal drainage
 - i. Hydrocolloids – for granulation tissue, facilitates AD
 - ii. Wet to dry – (1) impregnated (1) not impregnated gauze
 - c. Moderate to high/large/ copious drainage (CAsE IoCo Dry)
 - i. Calcium alginate – Best, facilitates AD
 - ii. Semipermeable dressing , facilitates AD
 - iii. Iodosorb
 - iv. Collagen dressing
 - v. Dry to dry – (2) not impregnated gauze
- vii. Trivias
 - a. Low voltage electrical burn – 500-1000 volts
 - b. High voltage electrical burn - >1000 volts
 - c. Home injuries – 110 or 220 volts
 - d. Minimum temperature to cause burn: 44°C/ 111°F
 - e. Short duration temperature to cause burn: 45°C
 - f. VAC> vacuum assisted closure
 - i. Uses 125mmHg
 - ii. Purpose: Controls edema, increase localized blood flow, removes infectious materials
 - g. HBO> Hyperbaric Oxygen
 - i. Duration 90-120 minutes per day (3-4x a day)
 - h. ES> Electrical Stimulation
 - i. HVPGC – High voltage pulsed galvanic current, destroys bacteria
 - ii. Electrode application
 - 1. Clean wound: Anode
 - 2. Infected wound: Cathode

CNS

- i. Nervous System
 - a. Primitive Structures
 - i. Ectoderm – Brain, Skin, Appendages
 - 1. Neuroblastoma/ Neurofibromatosis – Brain lesion
 - a. Manifestation: Café Au Lait Spots
 - ii. Mesoderm – Internal organs, Skeletal muscles, Ligaments, Bones
 - iii. Endoderm – Linings (Endothelium)
 - b. Divisions
 - i. Central nervous system
 - 1. Brain
 - a. Forebrain (Prosencephalon)
 - i. Telencephalon
 - 1. Cerebrum
 - 2. Basal Nuclei
 - ii. Diencephalon
 - 1. Thalamus
 - 2. Hypothalamus
 - 3. Epithalamus
 - 4. Subthalamus
 - b. Midbrain (Mesencephalon)
 - i. Tectum
 - ii. Cerebral peduncles
 - c. Hindbrain (Rhombencephalon)
 - i. Metencephalon
 - 1. Pons
 - 2. Cerebellum
 - ii. Myelencephalon
 - 1. Medulla Oblangata
 - 2. Spinal Cord
- ii. Meninges
 - a. Dura Mater (Pachymeninx) (E.g. condition: Pachymeningitis)
 - i. Outer membrane
 - ii. Function: Protection
 - iii. Space:
 - 1. Above: Epidural space
 - 2. Below: Subural space
 - b. Arachnoid Mater (Leptomeninges)
 - i. Function: Regulation
 - 1. Below: Subarachnoid space (drain for CSF)

- c. Pia Mater (leptomeninges)
 - i. Function: Nutrition
 - d. R/L Condition:
 - i. Nuchal rigidity – 1st symptom of meningitis, difficult c neck/ head movement
 - 1. + anterior horn cell affectation and paralysis (Polio)
- iii. Cerebrum
- a. Frontal
 - b. Parietal
 - c. Temporal
 - d. Occipital
 - e. Insular/ Island of Reil
 - f. Rhinencephalon
 - i. Olfactory bulb (receives the olfactory nerve through cribriform plate)
 - g. Limbic Lobe
 - i. Emotions
 - ii. Sexuality
 - iii. Mnemonics:
 - 1. P - Parahippocampus
 - 2. U – Uncus, CN I (Lesion: Misinterpretation of smell, olfactory hallucination)
 - 3. C – Cingulate Gyrus
 - 4. H – Hippocampus – Integration of memory
 - a. Short term memory is converted into long term memory
 - 5. A – Amygdala (Lesion: Hypersexuality, Kluver-Bucy)
 - a. Controls: libido

	Left Hemisphere	Right Hemisphere
Function	Verbal, Calculating, and Abstract thinking Interpretation of speech Sterognosis	Non-verbal, spatial, temporal And synthetic function
Lesion	Aphasia	Loss of visuo-spatial awareness and perceptual defecits Anosognosia Dyspraxia Hemineglect Cortical extinction

90% of cases the left hemisphere is dominant

- iv. Gyrus & Fissure
 - a. Median Longitudinal Fissure (separates ® and (L) hemisphere)
 - b. Lateral Sylvian fissure (Separates temporal lobe from all other lobes)
 - c. Central fissure of Rolando (separates Frontal & Parietal lobe)
 - d. Parieto-occipital fissure
- v. Saltatory conduction
 - a. Happens in myelinated axons
 - b. Conserves Energy
 - c. Fast Conduction

- d. Mnemonics
 - i. C - CNS
 - ii. O – Oligodendrocytes
 - iii. P - PNS
 - iv. S – Schwann cells
- vi. Fibers
 - a. A – large myelinated increase NCV
 - i. A – Delta, Fast pain/ acute pain
 - b. B – Medium myelinated, <15 m/s
 - c. C - Small unmyelinated; decrease NCV 2-3m
 - i. C – Slow pain/ chronic pain
- vii. Gray Matter
 - a. Composes the cerebral cortex/ surface of the brain
 - b. Surface area of the gray cerebral cortex is increased by the presence of gyri/ sulci
 - c. Cell bodies
 - d. *Nuclei* – group of bodies located outside the cerebral cortex but within the cerebrum E.G CN Nuclei
 - e. *Ganglia* – Group of cell bodies located outside the cerebral cortex & cerebrum E.g. DRG
- viii. White matter
 - a. Nerve fibers/ processes of neurons
 - b. White color d/t the myelin
 - c. *Fascicles/ Tracts* – processes of neurons having the same function tend to bundle together when located in the cerebrum
 - d. *Columns/ Funiculi* – “ ” located outside the cerebrum
 - e. Types of fibers
 - i. Commissural/ transverse fibers – connects ® and (L) hemisphere,
 - 1. E.g. Corpus Callosum, anterior commissure (Olfactory bulbs from (B) temporal lobes), Hippocampal fissure/ commissure of the fornix ((B) hippocampus)
 - 2. Lesion: MC structure affected with manifestation of: LOC > Coma
 - a. “Diffuse Axonal Injury” (TBI)
 - ii. Association fibers – connects structures on the same hemisphere
 - 1. E.g Arcuate Fasciculus (connects Broca’s and Wernicke’s)
 - 2. Short association/ U-fibers
 - a. Connect adjacent gyri E.g. connection between precentral and postcentral gyrus, superior temporal gyrus and middle temporal gyrus
 - 3. Long association fibers E.g. Uncinate Fasciculus, cingulum
 - iii. Projection Fibers – connects brain and spinal cord
 - 1. Corticofugal (Ascending)
 - 2. Corticopetal (Descending)
 - a. Corticospinal tract
 - b. Corticobulbar tract

- ix. Brodmann's area (47)
 - a. Function of the gray cerebral cortex has been mapped out into areas by Brodmann.
 - b. Two major types:
 - i. Primary cortical area – regions directly related to a specific function
 - ii. Secondary cortical/ Association areas – Lie adjacent to the primary area and are concerned with a higher level of organization and integration

x. Frontal lobe

Frontal Lobe	Location	Area	Function	Condition Asc.
Area 4	Precentral gyrus/ Paracentral lobule	Primary Motor	Contralateral Motor component	Early - Flaccid Late - Spasticity
Area 6	Superior frontal Gyrus	Pre Motor	Motor association Initiation	Spasticity
Area 8	Frontal lobe	Frontal Eye Field	Scanning mov't of Eye (saccade)	Eye deviation (Ipsilateral)
Area 9, 10, 11, 12		Pre-frontal Cortex	Insight & Judgement Emotions Control Personality & Behaviour	Behavioral changes & change in Cognitive function
Area 44 & 45	Frontal gyrus	Broca's Area	Motor speech area	Broca's Aphasia (incomprehensible)

- xi. Conditions
 - a. Conduction Aphasia –
 - i. Mnemonics:
 1. A – Arcuate Fasciculus
 2. R – Repetition difficulty
 3. C – conduction aphasia
 - b. Broca's/ Expressive/ Motor/ Non-Fluent/ Anterior Aphasia
 - c. Wernicke's / Repetitive/ Sensory/ Fluent/ Posterior Aphasia
 - d. Global aphasia – Wernicke's and Broca's
 - e. Anomia – inability to name
 - f. Agraphia – inability to write
 - g. Alexia – inability to read
 - h. Echolalia – Parrot like
 - i. Palilalia – repetitive use of words (without intent)
 - i. Perseveration (with intent)
 - j. Syntactic aphasia – “telegraphic” speech (conjunctions)
 - k. Aprosody – (-) melody, “monotonous speech”, without emotions
 - l. Aphasia – inability to construct phrases
 - m. Neologism – coining new words
 - n. Jargon – Incomprehensible words
 - o. Logorrhea – inability to stop talking
 - p. Paraphasia – word substitution
 - i. Semantic – E.g. Spoon/ Fork; Table/ Chair
 - ii. Phonemic – Sound E.g. Pork/ Cork

xii. Parietal lobe

Parietal Lobe	Area	Function	Condition Asc.
Area 3, 1, 2	Primary Somesthetic area	Primary Sensory	Decreased touch, pressure & proprioception Preservation of Pain & temp
Area 5, 7	2° Somesthetic area	Sensory association	Decrease tactile and proprioception Causing astereognosis and Graphesthesias
Area 39	Angular	Complex language functions	Agnosia
Area 40	Supramarginal	Language perception & processing	Apraxia
Area 43		Primary gustatory area	

Areas 5, 7, 39, 40 is a common integrative area also called the *Gnostic Area*

xiii. Conditions:

a. Gertsman Syndrome

- i. (L) and ® disorientation
- ii. Finger agnosia
- iii. Agraphia
- iv. Acalculia

Left CVA	Right CVA
Ideomotor (Area 40)	Dressing
Ideational Parietal)	Constructional
Gertsman	Hemineglect

b. Agnosia – inability to perceive

- i. Tactile, Parietal
- ii. Visual, Occipital
- iii. Auditory, Temporal

c. Prosopagnosia – inability to recognize familiar faces

d. Apraxia (inability to do work, without weakness)

- i. Ideomotor – inability to perform task upon command, but able to perform if automatic
- ii. Ideational – (-) command, inability to perform task at all
- iii. Dressing apraxia – inability to dress
- iv. Constructional Apraxia – Inability to draw geometric shapes

e. Notes

- i. Corticospinal tract is made up of BA 4 & 6 + 3, 1, 2
- ii. Corticobulbar BA 4 & 8

xiv. Occipital lobe

Occipital Lobe	Area	Function	Condition Asc.
Area 17	Primary visual area	Vision	(1) Homonymous hemianopsia (2) cortical blindness
Area 18 & 19	2° visual area	Visual association	18: Dyschromatopsia Color blindness

a. Conditions

- i. 18 & 19 visual agnosia
- ii.

xv. Temporal lobe

Temporal Lobe	Area	Function	Condition Asc.
Area 41 & 42	Heschl's (1° hearing)	Auditory	41 (1): partial deafness 41 (2): cortical deafness
Area 22	Wernickes area (L) only	Auditory association	Wernicke's Aphasia

xvi. Homunculus Vs Somatotropic organization

- a. Somatotropic organization: refers to the *arrangement of the cell bodies in the cerebral cortex by body part*
 - i. Arrangement: lateral from sylvian fissure
 - ii. Oropharyngeal > face > upper extremity (hand just above the face) > trunk > lower extremity (lower leg and foot in the medial aspect)
- b. Homunculus: proportion of the cell bodies assigned to each body part is not equal
 - i. Hand and face have the largest proportion of cell bodies/ neurons assigned to them

xvii. Basal Nuclei

- a. Function: smoothen's motion and initiating and directing voluntary motor activity
- b. Parts

- i. Claustrum
 1. Unknown function
- ii. Amygdala
- iii. Caudate
- iv. Putamen
- v. Globus pallidus

Caudate	Neostriatum		Corpus Striatum
Putamen			
Globus Pallidus		Lentiform	

E.g. Caudate + Putamen = Neostriatum



- c. Function
 - i. Caudate & Putamen: large subconscious movement of skeletal muscles (arm swing)
 - ii. Globus pallidus: Regulates muscle tone
- d. Conditions
 - i. Affection of lentiform = Dystonia (sustained posturing of a body part)
 - ii. Parkinsons disease
 1. Degeneration of the substantia nigra resulting to decrease in dopamine
 - a. Inhibitory in corpus
 - b. Excitatory in cortex
 2. Quad:
 - a. Rigidity
 - b. Bradykinesia
 - i. Management: Rhythmic Initiation towards D2 flexion
 - c. Resting tremor(most disabling)
 - i. Serotonin induced(Sullivan)
 - d. Postural Instability
 3. Pharmacologic Mx:
 - a. Sinemet + carbidopa

- iii. Chorea
 - 1. Rapid, jerky, irregular movement
 - 2. Affectation of Putamen
- iv. Huntingtons
 - 1. Choreaform movement + dementia
 - 2. Affectation of chromosome 4 (produces a protein called huntingtins)
- v. Sydenhams (saint Vitus dance)
 - 1. Choreaform movement
 - 2. Streptococcal infection, (associated with rheumatic fever)
- vi. Hemiballismus
 - 1. Affectation of subthalamic nucleus
 - 2. Arm flailing
- vii. Athetosis
 - 1. Slow, writhing, worm-like movement
 - 2. Affectation of globus pallidus
- xviii. Diencephalon
 - a. Thalamus
 - i. *Sensory relay station* of the brain – (All sensory modalities except olfaction/smell)
 - ii. Lateral Geniculate bodies – Vision
 - iii. Medial Geniculate bodies – Auditory
 - iv. Condition
 - 1. Thalamic pain syndrome: immediate hemianesthesia with increase threshold of sensitivity to pin prick, heat and cold> thalamic hyperpathia (sensation felt is unpleasant/ disagreeable)
 - a. Aka dejerine Roussy
 - b. Affectation of PCA
 - b. Hypothalamus
 - i. Antero-Inferior to the thalamus
 - ii. Thermoregulatory center of the brain
 - 1. (N) body temperature 36.5 – 37.5° C

Anterior: Heat loss	Posterior: Heat production
Sweating	Shivering
Vasodilation	Vasoconstriction
Panting	Piloerection
Increase HR	Decrease HR
 - iii. Feeding Center
 - iv. Thirst Center
 - 1. Regulation of water balance
 - v. Hunger
 - 1. Conditions:
 - a. Anorexia nervosa
 - b. Bulimia> Anorexia
 - vi. Satiety center

- xix. Mesencephalon
 - a. Tectum (roof) (MB)
 - b. Cerebral peduncles: main connection for tracts between upper parts of the brain and BS/ SC
 - i. Corpora Quadrigemina
 - 1. Superior colliculus – Reflex center for eyeball & head movement in response to visual stimulation
 - 2. Inferior Colliculus – Reflex center for head & trunk movements in response to auditory stimulation
 - 3. These both contribute formation to the Tectospinal tract (involved in blinking and head turning reflexes in response to sound and visual images).
 - c. Anatomical parts
 - i. Basis
 - 1. Base of midbrain
 - 2. Anterior to the tegmentum
 - 3. Contains the
 - a. CST, CBT, CPT
 - b. Crus cerebri & substantia nigra
 - ii. Tegmentum
 - 1. Anterior to the colliculi
 - 2. Contains
 - a. all ascending tracts, most motor
 - b. Red nucleus: motor coordination
 - c. Trochlear & oculomotor nucleus
 - iii. Tectum
 - 1. Roof of midbrain/ dorsal
 - 2. Formed by the corpora quadrigemina
 - 3. Periaqueductal gray matter
 - 4. Contains:
 - a. Descending autonomic tracts
 - b. Superior cerebellar peduncles
- xx. Rhombencephalon
 - a. Metencephalon
 - i. Pons
 - 1. Reticular formation: contains the apneustic and pneumotaxic centers (respiration)
 - b. Myelencephalon - MO
 - i. BP regulation/ Force of contraction of the heart
 - ii. *Major Respiratory control*
 - iii. Vasomotor center(vasoconstrictor)
 - iv. Location of the decussation of pyramids (cervicomedullary junction)
 - v. Vomiting Center
 - 1. Motion, Food/ GIT, Kicked in the balls
 - vi. Cough

- c. Conditions
 - i. Pin point pupils – Pons affectation
- xxi. Cerebellum
 - a. Parts
 - i. Two hemispheres
 - ii. Vermis (connects the two hemisphere)
 - b. Layers of the cerebral cortex
 - i. Molecular layer
 - ii. Layer of purkinje cells
 - iii. Granular layer
 - c. Deep cerebellar nuclei
 - i. Fastigial
 - ii. Globose
 - iii. Emboliform
 - iv. Dentate nucleus
 - d. 3 lobes
 - i. Anterior / Paleocerebellum/ spinocerebellum
 - 1. Function: stereotypic movements (arm swinging ang walking), muscle tone
 - 2. Conditions
 - a. Truncal ataxia
 - b. Gait ataxia
 - ii. Posterior / middle / neocerebellum
 - 1. Largest
 - 2. Function: Coordination of fine movement
 - 3. Conditions
 - a. Intention tremor
 - b. Dysdiadochokinesia – inability to perform rapid alternating movement
 - c. Disturbance of reflexes (Affectation of coordination of reflex arc)
 - i. Pendular knee jerk
 - d. Dysarthria
 - e. Dysmetria
 - iii. Flocculonodular/ archicerebellum/ Vestibulocerebellum
 - 1. Function: Equilibrium
 - 2. Conditions
 - a. Disequilibrium
 - b. Nystagmus
 - iv. mnemonics
 - APNP*
 - Anterior: paleo*
 - Posterior: neocerebellum*

- e. 3 Cerebellar Peduncles
 - i. Superior cerebellar peduncle (brachium conjunctivum)
 - 1. Connection with cerebellum with brain
 - ii. Middle cerebellar peduncle (brachium pontis)
 - 1. Connection with pons
 - iii. Inferior cerebellar peduncle (brachium restiform)
 - 1. Connects with medulla
 - xxii. CSF/ Cerebral circulation:
 - a. From the choroid plexus
 - b. 500ml/ day
 - c. Energy of the brain: Sugar
 - d. Largest storage of CSF: Lateral Ventricles
 - e. Function
 - i. Shock absorption, Nutrient (contains proteins), Homeostasis
 - ii. Drains unwanted substance
 - f. R/L Condition
 - i. Hydrocephalus
 - 1. Obstructive
 - a. Communication – subarachnoid space block
 - b. Non-communication – Ventricle block
 - 2. Non-obstructive
 - a. Adults, Brain atrophy (Hydrocephalus Ex Vacuo)
 - 3. S/Sx
 - a. Cracked pot sign (fontanelles are underdeveloped)
 - b. Sunset sign (Parinaud's)
 - i. Impaired upward gaze
 - c. Cushing Sign(increased ICP)
 - i. Bradypnea
 - ii. Bradycardia
 - iii. Systolic HPN/ Increased BP
 - ii. Hydromyelia -spine
 - iii. Syringomyelia – manifestation: cavitations of the spine
 - iv. Laminectomy – avoid ultrasound
 - g. Pathway of CSF
- Cum**
- | | |
|-----------------|---|
| Come | Choroid plexus |
| Lets | Lateral ventricle |
| Formally | Foramen of Monro |
| Take | Third Ventricle (<i>Choroid plexus</i>) |
| Sylvia | Sylvian Aqueduct |
| For | Fourth Ventricle (<i>Chroid plexus</i>) |
| Lushka | Lushka (2) |
| Mamaya | Magendie |
| Sa | Subarachnoid space |
| Aristocrat | Arachnoid Villi |

- xxiii. Peripheral Nervous system
- Ganglia
 - Plexuses
 - Cranial nerve (12 pairs)
 - Spinal nerves (31 pairs)
 - Autonomic Nervous system (Visceral organs)
 - Sympathetic
 - Parasympathetic
 - Somatic nervous system (Skeletal Ms, Skin, Joints)
- xxiv. Autonomic nervous system
- Sympathetic
 - T1-L3 (ThoracoLumbar)
 - "Fight or flight" response
 - Diffuse
 - Adrenergic (Epinephrine Response)
 - Parasympathetic
 - CranioSacral
 - Cranial nerves 10, 9, 7, 3
 - Sacral nerves S2 – S4
 - "Rest & Digest" Response > Conserve energy
 - Localized
 - Cholinergic (Ach)
 - Most important CN is: Vagus n.
 - 75%
 - Longest CN
 - Heart: Stimulation will cause decrease in heart rate (brady cardia)
 - Abdomen: Increase peristalsis

Sympathetic	Parasympathetic
Tachycardia (increase in HR)	Bradycardia (Decrease in HR)
Constipation	Increase peristalsis, Increase in digestion, gastric juices, saliva
Vasoconstriction Sphincter Constriction	Bronchoconstriction Pupillary constriction (CN III)
Increase in BP (increase HR and RR)	Decrease in BP (Decrease in HR and RR) * Syncope (Fainting) d/t decrease in CO
Bronchodilation	Asthma (2° to bronchoconstriction)
Mydriasis (papillary dilation)	Miosis
Decrease urine output	Increase urine output
Sweating (Hyperhydrosis)	
Decrease GFR	Increase GFR
	Defecation
Ejaculation	Erection

Erection

Psychogenic - sympathetic

Reflexogenic – parasympathetic

Eto

xxv. Spinal nerves (31 pairs)

Cervical 8 pairs
Thoracic 12 pairs
Lumbar 5 pairs
Sacral 5 pairs
Coccyx 1 pair

- a. Exit point: Cervical – above the corresponding vertebra level
- b. Thoracic to lumbar (below
- c. Conditions
 - i. HNP- (B) sensory and motor
 - ii. Rhizotomy – removal of roots (ventral or dorsal)
 1. Indicative of Pain, spasticity
- d. Dermatomes
 - i. Strip of skin supplied by one dorsal roots

C1	Vertex of the skull	C7	Middle finger	T10	Umbilicus	S1	Lateral Malleolus
C2	External Occipital Protuberance	C8	Little finger	T12/L1	Inguinal line	S2	popliteal fossa
C3	Supra Clavicular fossa	T1	Medial arm	L2	Mid anterior thigh	S3	Ischial tuberosity
C4	Acromion Process	T2	Apex of axilla	L3	Medial tibial plateau/ knee/ fat pad	S4/ S5	Peri anal area
C5	Lateral arm & forearm	T4	Nipple	L4	Medial malleolus		
C6	Thumb	T6	Xiphoid process	L5	Dorsum of the foot		

e. Myotomes

C1/ C2	Neck Flexion/ Extension	C7	Elbow Extension & Wrist Flexion	L4	Dorsiflexion
C3	Lateral Flexion	C8	Finger Flexion	L5	Big Toe Extension
C4	Shoulder Shrug	T1	Finger Abduction	S1	Plantarflexion
C5	Shoulder abduction & Elbow Flexion	L2	Hip Flexion		
C6	Wrist extension	L3	Knee Extension		

- f. DTR: Asses the integrity of the reflex arc
 - i. Biceps **C5**, C6
 - ii. Brachioradialis C5, **C6**
 1. Tap over the **Radial styloid**
 - iii. Triceps **C7**, C8
 - iv. Patellar L2, L3, **L4**
 - v. Achilles **S1**, S2

- xxvi. Cranial nerve (12 pairs)
- a. Brainstem
 - i. Midbrain
 1. CN 3-4
 - ii. Pons
 1. CN 5, 6, 7, 8
 - iii. Medulla Oblangata
 1. CN 8, 9, 10, 11, 12
 - b. Brain
 - i. CN 1, Temporal lobe
 1. Telencephalon
 2. Not a true CN
 - ii. CN 2, Occipital lobe
 1. Diencephalon CN
 2. Not a true PN
 - c. Brainstem stroke syndrome

Weber	Medial basal of MB
Benedikt	Tegmentum of MB
Locked-in	Bilateral Basal pons
Millard Gubler	Lateral pons
Wallenberg	Lateral Medulla
 - d. Note: there are only 10 pairs – true peripheral nerve
 - e. Trigeminal branches (CN V, Largest cranial nerve)
 - i. Ophthalmic
 - ii. Maxillary
 - iii. Mandibular
 - f. Exit Points

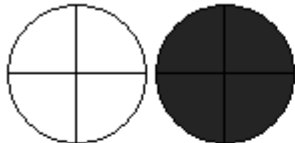
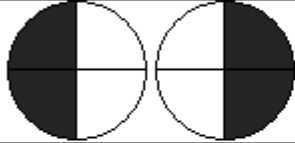
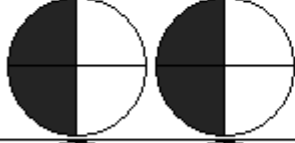
Site of Exit	Cranial Nerve
Cribriform plate	I
Optic Canal	II
Supra Orbital Fissure	III, IV, V ₁ , VI
Foramen Rotundum	V ₂
Foramne Ovale	V ₃
Internal Acoustic Meatus	VII, VIII
Jugular foramen	IX, X, XI
Hypoglossal canal	XII
* Cavernous Sinus	VI

- xxvii. Cranial Nerves
 - a. Vagal system
 - i. 10, 9, 7, 11 (Vagotomy – increase HR)
 - b. Parasympathetic
 - i. 10, 9, 7, 3
 - c. Mixed
 - i. 10, 9, 7, 5
 - d. Sensory
 - i. 1, 2, 8
 - e. Motor
 - i. 3, 4, 6, 11, 12
 - f. Longest CNX, Extracranial CN
 - g. Thinnest CN IV, intracranial CN
 - h. Largest: CN V
- xxviii. Most common
 - a. DM 3,6
 - b. MS 2, 5, 7, 8
 - c. PD 9, 7, 3
 - d. MG 3, 4, 6
 - e. TBI 7, 3, 1
 - f. GBS, BELL's, MOBIUS, RAMSAY-HUNT(bell's + H. Zoster, dermatomal distribution), - 7
 - g. *Pseudoptosis -7
 - h. Leprosy – 7, 5
 - i. Sjogrens – 7, 9
 - j. Weber/ bene/ benedikt – 3
- xxix. Specifics
 - a. CNI – Olfactory
 - i. MC Contused
 - ii. Basilar skull Fx (BAttl's Sign)> CSF Rhinorrhea > CN I affectation
 - iii. Function: smell
 - iv. Lesion:
 - 1. Anosmia
 - b. CN II – Optic
 - i. Nucleus: Edinger Westphal
 - ii. Assesment
 - 1. Pupillary light reflex
 - a. Afferent: CN II
 - b. Efferent: CN III
 - iii. Lesions
 - 1. Anopsia
 - 2. Anisocoria – unequal pupil size
 - 3. Argyl Robertson pupil (Syphilis)
 - a. No reaction to light
 - 4. Amaurosis fugax

Eto

Eto

iv. Lesion sites

<u>Visual Field</u>		<u>Lesion Site</u>	<u>Condition</u>
		Optic nerve	Monocular Blindness
		Optic Chiasma	Bitemporal Hemianopsia
		Optic tract	Homonymous Hemianopsia
		Optic Radiation (Upper)	Quadrantanopsia

c. CN III – Oculomotor

- i. Muscles of the eye
- ii. Strabismus
 1. External strabismus/ Exotropia
 - a. CNIII
 2. Internal Strabismus/ Esotropia
 - a. CN VI

iii. Condition:

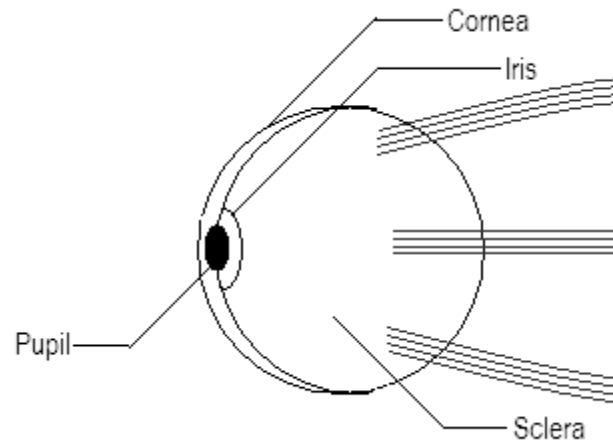
1. Ptosis (Lid Lag)

d. CN IV & VI – Trochlear, abducens

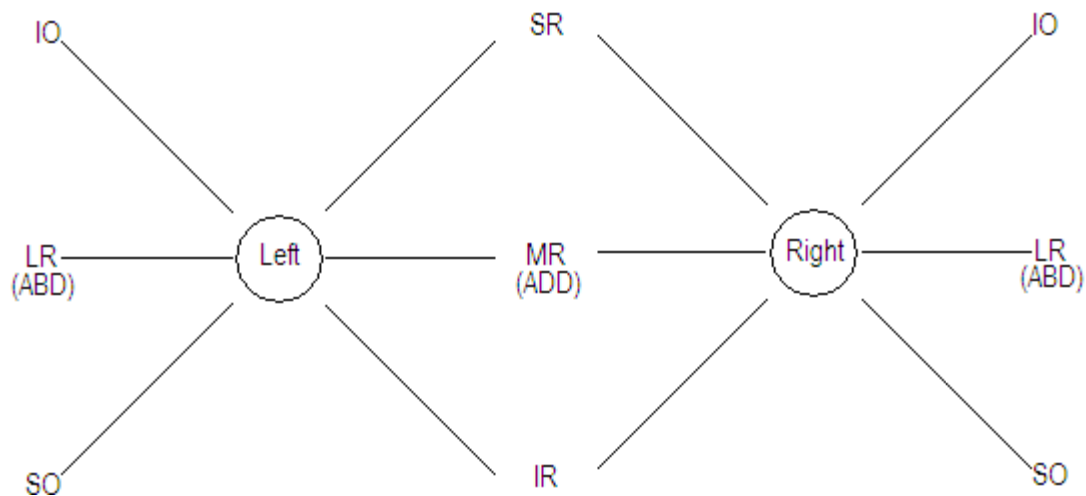
- i. Condition:
 1. Vertical Diplopia
 - a. CN IV
 - b. Up and down difficulty
 - c. Compensation: Tilt contralateral
 2. Horizontal Diplopia
 - a. CN VI
 - b. Side to side difficulty
 - c. Rotation to ipsilateral

e. Notes:

- i. EOM₃ LR₆ SO₄
- ii. SO₄ aka cheating muscle
- iii. Levator palpebrae superioris – CN III
- iv. Eye Opening:
 1. Mueller muscle 20%
 2. LPS 80%



Insertion of Extraocular muscles – back of the sclera



f. CN V – Trigeminal

- i. Muscles of mastication
- ii. Sensory
 1. V₁ - Ophthalmic
 2. V₂ - Maxillary
 3. V₃ - Mandibular
- iii. Corneal
 1. Afferent: CN V
 2. Efferent: CN VII
- iv. Tic douloureux
 1. Trigeminal neuralgia
 2. Vitamin B complex
- v. Conditions
 1. Jaw deviations
 - a. E.g. Left side jaw deviation (left lateral and right medial pterygoid)

- g. CN VII – Facial nerve
 - i. Sensory – anterior 2/3 of the tongue
 - ii. Motor
 - 1. Facial muscles
 - iii. Autonomic
 - 1. Lacrimal
 - 2. Salivary
 - a. Sublingual, Submandibular
- h. CN VIII – Vestibulo-Cochlear
 - i. Vertigo, BPPV:
 - ii. Conduction deafness –
 - 1. Reversible
 - 2. Usual cause: Impacted cerumen
 - iii. Sensorineural deafness
 - 1. Lesion to CN 8
 - iv. Cortical deafness
 - 1. Lesions to (B) BA 41
 - v. Parts
 - 1. Outer ear – conduction deafness
 - 2. Middle – conduction deafness
 - a. Infection – Otitis Media
 - 3. Inner ear – sensorineural deafness
 - a.
 - b. Meniere's Disease
 - i. Sensorineural deafness
 - ii. Vertigo
 - iii. Tinnitus
 - iv. Fullness of ear
 - vi. Organ of hearing: organ of Corti
- i. CN IX - Glossopharyngeal
 - i. Motor – stylo pharyngeus ms
 - ii. Sensory- Posterior 3rd of the tongue
 - iii. Autonomic – parotid gland
- j. CN X Vagus
 - i. Motor: pharynx and larynx
 - ii. Sensory: Pinna of the ear
 - iii. Autonomic: 1973
 - 1. 75% of parasympathetic activity
- k. CN XI – Spinal Accessory
 - i. Affectation by radical neck dissection
 - ii. SCM (C2-C3)
 - iii. Trapezius (C3- C4)

- I. CN XII -Hypoglossal
 - i. Tongue muscles
 - 1. Palatoglossus -elevation
 - 2. Genioglossus - protrusion
 - 3. Hyoglossus - depression
 - 4. Styloglossus – curls the tongue CN IX

INTEG SYSTEM

Integumentary System

- i. Skin
 - a. Largest organ in the body
 - b. 15-20% of BW
 - c. Functions
 - i. P - Protection
 - ii. I - Insulation
 - iii. R – Receptors in dermis (sensation)
 - iv. A – Aids in homeostasis
 - v. T – Temperature regulation
 - vi. E – Elimination of waste product
 - vii. S – Synthesizes vitamin D
 - viii. O – Oil production
 - ix. F – Fluid retention
 - x. C – Cosmesis
- ii. Layers of the skin (3)
 - a. Epidermis
 - b. Dermis
 - c. Subcutaneous tissue/ hypodermis
- iii. Epidermis
 - a. Outermost layer of the skin
 - b. 5 layers/ strata (Superficial to deep)
 - i. Stratum Corneum
 1. Responsible for water-proof characteristic
 2. Protection against infections
 - ii. Stratum Lucidum
 1. Located on palms and soles
 - iii. Stratum Granulosum
 1. Water retention and heat regulation
 - iv. Stratum Spinosum
 1. Protects the basale
 - v. Stratum Basale/ Germinativum
 1. Contains melanocytes (produces melanin)
 2. Epidermal regeneration
- iv. Dermis aka (corium)
 - a. True skin
 - b. 20-30x thicker than the epidermis
 - c. Contains lymphatics, blood vessel, nerve and nerve endings, sweat and sebaceous glands
 - d. Contains elastin, collagen, mucopolysaccharides

- e. Rete Pegs
 - i. Structure located between epidermis and dermis
 - ii. Function:
 - 1. Increases surface area to overcome friction
- f. 2 layers
 - i. Superficial/ papillary layer
 - ii. Deep/ reticular
- v. Subcutaneous tissue/ hypodermis
 - a. Contains fats and loose connective tissues
 - b. Function
 - i. Support, insulation, cushion, storage of energy (SICS)
- vi. Appendages of the skin/ accessory structures
 - a. Hair
 - i. Layers
 - 1. Cortex
 - 2. Medulla
 - 3. Cuticle
 - ii. Types
 - 1. Terminal hair – Coarse thick and pigmented e.g. scalp eyebrows (eyebrows: supercilli)
 - 2. Vellus – short and fine
 - a. Chest
 - b. Nail
 - i. Parts
 - 1. Nail plate
 - 2. Lateral nail fold
 - 3. Proximal nail fold (cuticle, eponychium)
 - 4. Lunula
 - ii. Function; protection
 - iii. Spoon nails – fungal infection
 - iv. Club nail – respiratory problems
 - v. White spots – trauma
 - c. Sweat glands/ sudoriferous glands
 - i. Drains at the skin pore
 - ii. 2 types
 - 1. Eccrine/ merocrine
 - a. Located on the entire surface of the skin
 - b. thermoregulation
 - 2. Apocrine
 - a. Located on Axillary and genital regions
 - b. Active during emotional stress
 - c. Secrete Vit C

- d. Sebaceous gland (oil gland)
 - i. Produces and secretes sebum (if blocked = white heads, if oxidized it becomes "black heads, (+) bacteria = acne (acne vulgaris)
- vii. Arector pili ms
 - a. Goose bumps/ goose flesh or piloerection
- viii. Common d/o skin disorders
 - a. Dermatitis/ Eczema
 - i. Inflammation of the skin with itching redness, and skin lesions
 - ii. 3 causes
 - 1. Allergic (contact dermatitis), poison ivy, harsh soaps, chemicals, adhesive tape
 - 2. Actinic – sun exposure, photosensitivity, problem with UVR
 - 3. Atopic – unknown, may be associated with allergy, hereditary, psychological stress
 - iii. Stages:
 - 1. Acute – red, crusting rash, oozing
 - 2. Subacute – skin erythema, scaling, scattered plaques
 - 3. Chronic – thickened skin, increase skin markings 2° to scratching post-inflammatory pigmentation
 - iv. Management: hydration, lubrication, systemic drugs
 - b. Bacterial infections – hot weather
 - i. Impetigo – cause by staph and strepto
 - 1. Inflammation of skin with skin lesion and itching
 - 2. Management antibiotics
 - ii. Cellulitis – suppurative inflammation fo cellular and connective tissues
 - 1. Caused by staph and strepto
 - 2. Hot, edematous, red, (+) fever
 - 3. Management: antibiotics
 - iii. Abscess – cavity – containing pus, surrounded by inflamed tissue
 - iv. Management: incising and draining
 - v. Caused by staph
 - c. Viral infections – cold weather
 - d. Herpes 1/ simplex – vesicular eruptions on face and mouth
 - i. Cold sore or fever blister
 - e. Herpes 2 – vesicular eruptions on genitalia
 - f. Herpes zoster/ Shingles – caused by varicella zoster or ganglia of posterior nerve roots
 - i. Pain and tingling sensation on CN and spinal dermatomes
 - ii. Associated with chills, fever, malaise, SI disturbances
 - iii. CN III eye pain, corneal damage CN V
 - iv. CN V loss of vision
 - v. Management: no cure, anti-viral drugs (to slow progression)
 - g. *Warts –caused by HPV
 - i. Seen on hands, fingers and pressure points of the feet
 - ii. Management: cryotherapy, acids, electrodessication, culretage, OT drugs
 - h. Fungal infections

- i. Ringworm – forms a ring-shaped patches with vesicles or scales
 1. Tinea capitis – scalp ring worms
 2. Tinea corporis – body ringworm
 3. Tinea barbae – beard ringworm
 4. Tinea unguium – nail ring worm
 5. Tinea cruris – jock itch ring worm
 6. Tinea pedis – athlete's foot
 - ix. Parasitic infections
 - a. Scabies/ mites – burrow into skin causing itching, redness and inflammation
 - i. Management: scabicide
 - b. Lice/ pediculosis – causing itching, bitemarks, redness and nits
 - i. Management: special shampoo and soap.
 - x. Common immune disorder
 - a. Psoriasis – erythematous plaque covered with silvery scale
 - i. Associated with arthritis
 1. Auspitz sign, pencil in a cup deformity
 2. Management: long wave UVER + psoralens
 - b. Lupus erthematosus – chronic progressive connective tissue disease
 - i. Discoid LE: hypopigmentation, hyperpigmentation
 - ii. SLE: collagen vascular disease
 1. Young women, involves skin, joints, multiple organs (kidneys, heart NS)
 2. Butterfly rash/ malar rash/ wolf –like rash, arthralgias, arthritis
 3. Anemia, fatigue, photosensitivity, and Raynaud's phenomenon
 4. Management: corticosteroids
 - c. Polymyositis – skin lesions associated with degeneration or atrophy of proximal muscles – shoulder and pelvis
 - d. Dermatomyositis – skin lesions with dermatitis
 - e. Scleroderma – thickening/ hardening of skin associated with involvement of joints, heart, lungs, blood vessels, kidneys and GI tract
 - i. CREST Syndrome
 1. Calcinosis – increase calcium deposits
 2. Raynaud's phenomenon
 3. Esophageal Dysfunction – GERD
 4. Sclerodactyly
 5. Telangiectasias – reds pots on skin due to abnormal dilatation of blood vessels
 - xi. Skin cancer
 - a. Benign
 - i. Seborrheic keratosis – proliferation of basal cells leading to raised lesion
 - ii. Actinic Keratosis – proliferation of squamous cells, flat, round irregular,
 - iii. Common mole/ benign nevus – proliferation of melanocytes
 1. Round or oval, <6mm

- b. Malignant
 - i. Basal cell carcinoma – slow –growing, sun exposure, ivory appearance, rolled border with indented center, rarely metastasize
 - ii. Squamous cell carcinoma – fast growing, sun exposure, red flat area, ulcer or nodule, poorly defines border, can metastasize
 - iii. Malignant melanoma – tumor arising from melanocytes > 6mm, bleeding, redness, oozing, scaling, and swelling
 - 1. A - Asymmetry
 - 2. B - Border
 - 3. C - Color
 - 4. D - Diameter
 - 5. E – Evolving
- xii. Skin trauma
 - a. Contusion – injury with intact skin; bruise
 - b. Ecchymosis – extravulsion of blood into subcutaneous tissue due to fragile blood vessels
 - c. Petechiae – tiny, red, or purple hemorrhagic spots on skin
 - d. Abrasion – scraping of skin due to mechanical injury
 - e. Laceration – irregular tear of skin producing a torn, jagged wound
- xiii. Integumentary assessment
 - a. Pruritus – itching
 - b. Urticaria/ hives – smooth, red elevated patches
 - c. Rash – local redness and skin eruptions
 - d. Edema – due to anemia, venous/ lymphatic obstructions, cardiac circulatory or renal decompensation
 - i. Pitting: persistent indentation
 - ii. Non- pitting: hard texture edema
 - e. Skin color
 - i. Cherry red – due to carbon monoxide poisoning
 - ii. Cyanosis – bluish discoloration due to CHF, CHD, advanced lung disease
 - 1. Central: lips, gums
 - 2. Peripheral: hand, fingers, foot, toes
 - iii. Pallor/ pale – due to anemia, lack of exposure to sunlight, shock, nervousness,
 - iv. Yellow/ jaundice – due to liver disease
 - v. Liver spots – brownish –yellow spots, due to aging, liver or utenne malignancis, pregnancy
 - vi. Brown – increase pigmentation due to venous insufficiency
 - f. Hydrosis
 - i. Hyperhydrosis – moist skin, pneumonic crisis, fever, hot drinks, exercise
 - ii. Hypohydrosis – dehydration, hypothyroidism, dry skin
 - iii. Cold sweats – due to fear, anxiety, depression, AIDS
 - 1. Hypothyroidism – thinning of hair
 - 2. Hyperthyroidism – silky hair
 - 3. Xeroderma – excessive dryness of skin

- g. Skin lesions
 - i. Flat spots
 - 1. Macule: <1cm
 - 2. Patch: >1cm
 - ii. Palpable elevated solid mass
 - 1. Papule <1cm
 - 2. Plaque: >1 cm
 - 3. Nodule: marble like lesion
 - 4. Wheal: irregular, localized skin edema/ hives
 - iii. Elevated lesions with fluid cavities
 - 1. Vesicle <1cm
 - 2. Bulla/blister: >1cm
 - 3. Pustule: containing pus
- h. Patterns of integument management
 - i. A – primary prevention, risk reduction of integument disorders
 - ii. B – Skin involvement, superficial
 - iii. C – skin involvement, partial thickness with scar formation
 - iv. D – Skin involvement, Full thickness with scar formation
 - v. E – Skin involvement, Full thickness extending to fascia, muscle, bone, with scar formation

PSYCHIATRY

- i. Psychiatry
 - a. Anxiety – feeling of apprehension, worry, or uneasiness
 - i. Normal reaction to conflict or stress
 - ii. Can be constructive (for the greater good)/ neurotic (pathologic, anxiety will get worse)
 - iii. S/Sx
 - 1. Sympathetic response
 - a. Increase in heart rate
 - b. Palpitations
 - c. Dyspnea
 - b. Depression – Altered mood characterized by a morbid sadness
 - i. Clinical manifestations
 - 1. Anhedonia – loss of pleasure
 - 2. Insomnia/ hypersomnia
 - 3. Excessive/ loss of appetite
 - 4. Psychomotor sign – irritable, excessive fatigue
 - 5. Impaired concentration (decreases neurotransmitters)
 - 6. Feeling of worthlessness/ hopelessness
 - 7. Recurrent act or thought of suicide
 - ii. MC drug: Tricyclic Anti-Depressant
 - iii. MC S/E Postural hypotension
 - c. Defense Mechanism/ Coping mechanism
 - i. Protection from ego(part of psychic apparatus)
 - 1. Id: pleasure principle
 - 2. Ego: Reality principle
 - 3. Super Ego: Conscience
 - 4. Types:
 - ii. Unconscious behavior
 - 1. By which an individual tends to resolve conflict
 - iii. Types
 - 1. Compensation
 - a. Covering up the weakness by stressing the strong trait
 - 2. Denial
 - a. Cannot accept reality
 - 3. Regression
 - a. Immature pattern of functioning
 - b. E.g. nail biting, thumb sucking, fetal positioning
 - 4. Repression
 - a. Memory inhibition
 - b. You cannot remember/ recall facts or events

5. Projection
 - a. Attributing of your own undesirable behavior to another
6. Rationalization
 - a. Justification of behavior using reason other than the real reason
7. Displacement
 - a. Channeling/ transferring your negative emotions to a less dangerous subject
8. Reaction Formation
 - a. Behavior is exactly opposite on what to expect
- d. Pathologies
 - i. Anxiety disorder
 1. Panic attack: loss of mental control
 - a. Phobia – Excessive or unreasonable fear
 - i. Arachnophobia – fear of spider
 - ii. Acrophobia – fear of heights
 - iii. Pyrophobia - fear of fire
 - b. Obsessive Compulsive Disorder – Repetitive activity that already interferes with social functioning
 - c. Post-Traumatic Stress Disorder (PTSD) – exposure to a traumatic event
 - i. S/Sx: fear, anxiety, impaired concentration, irritability, sleep disturbance
 - ii. Types
 1. Acute - Duration S/Sx is present less than 3 months
 2. Chronic - Duration S/Sx are present more than 3 months
 - ii. Psychosomatic disorder/ Briquette disorder
 1. Have physical signs and disease that are related to emotional stress
 - a. Conversion Disorder/ Hysterical paralysis: Neurologic signs: numbness, paralysis, blindness without neurologic condition
 - i. *S/Sx are real but s the disease*
 - ii. Etiology: Unknown
 - iii. E.g. Hysterical pregnancy (S/Sx of pregnancy s fetus)
 - b. Hypochondria: A false belief that they are suffering from one condition, overconcern.
 - c. Mood disorder:
 - i. Major depressive disorder
 1. Depression >6 months
 2. S/Sx: impaired concentration, loss of appetite, irritability
 - ii. Bipolar
 1. Manic depressive (hyper)
 2. Extreme emotions followed by depression

- d. Schizophrenia:
 - i. Disrupted thought pattern
 - 1. Hebephrenic: most severe
 - 2. Paranoia: Over suspiciousness
 - 3. Catatonia: Flexed posture, cataplexic posture
 - ii. Clinical manifestations:
 - 1. Disordered thinking, Poor judgement
 - 2. Error's of logic
 - 3. Disordered speech
 - 4. Disordered perception (Hallucination)
 - a. T – Tactile (Fornication)
 - b. A – Auditory (MC site of hallucination)
 - c. V - Verbal
 - d. O – Olfactory
 - 5. Withdrawal to the real world
 - 6. Loss of self-identity
- e. Personality disorder
 - i. Schizoid – Prefer to be alone
 - ii. Schizotypal – Magical thinking “ can see into the future”
 - iii. Histrionic – Center of attention
 - iv. Borderline – White and black thinking
 - v. Anti-Social – Does not want rules and regulation
 - vi. Narcissistic – Concentration of all effort is to self
 - 1. Power & Glory to self
 - vii. Paranoia – Over suspiciousness
- f. Eating disorder
 - i. Anorexia Nervosa – Fear of eating, becoming fat, and maintaining normal body weight.
 - ii. Bulimia – Binge eating followed by purging behavior(self – induced vomiting)
 - iii. Binge-eating disorder – binge eating with loss of control
- g. Childhood Disorder:
 - i. Autism: Marked impairment of (Triad)
 - 1. communication
 - 2. social relatedness
 - 3. Repetition of activity.
 - ii. ADHD: Persistent activity or inattention
 - 1. S – Short attention span
 - 2. H - Hyperactive
 - 3. E – Easily Distracted
 - 4. Medication: Ritalin

- h. Sleeping disorder:
 - i. Insomnia – Inability to sleep
 - ii. Hypersomnia - Excessive sleepiness
 - iii. Narcolepsy – Excessive daytime sleeping
 - iv. Sleep walking – Somnambulism
 - v. Sleep talking – Somniloquy
 - vi. Parasomnia – Nightmare
- i. Stages of recovery
 - i. Kubler Ross
 - 1. Denial – will act as a buffer to an impending death
 - a. Shock, healthy
 - 2. Anger – Blame other people
 - 3. Bargaining – Exchange for time / opportunity
 - 4. Depression – To be alone
 - 5. Acceptance
- j. PsychoPharmacology
 - i. Antipsychotics (decrease neurotransmitters)
 - 1. Chlorpromazine
 - a. MC drug for SLE
 - 2. Olanzapine
 - 3. Haloperidol
 - 4. Too low NT (Tardive dyskinesias)
 - ii. Antidepressants
 - 1. Tricyclic anti-depressant
 - 2. MAO- inhibitors
 - 3. SSRI – (Serotonin Selective Reuptake inhibitor)
E.g. Prozac
 - iii. Antimanic (For bipolar)
 - 1. Lithium carbonate
 - iv. Anxiolytics: Can sedate or calm the patient
 - 1. Buspirone
 - 2. Benzodiazepines
 - 3. Lorazepam
 - 4. Phenobarbitol
- k. Management on psychiatric conditions
 - i. Do's:
 - 1. Allow the patient to express his or her feelings
 - 2. Provide message of hope (Tempered with realism)
 - 3. Help re-establish self-dignity and self-worth
 - 4. Recognize losses
 - 5. Involve him or her in goal setting
 - ii. Don't's
 - 1. Do not sympathize but empathize

MOV'T D/O

- i. Movement Disorder
 - a. Basal nuclei – a group of nuclei that is situated at the upper brainstem and deeper part of the cerebrum
 - b. Function
 - i. R – Refining of movement
 - ii. I – Initiation of movement
 - iii. P – Planning of movement
 - c. Neurotransmitters

Structures	Excitatory NT	Inhibitory NT
Cortex	Dopamine Ach Glutamate	*GABA
Striatum	*Ach	*Dopamine
Substantia Nigra	Pars Comapacta Pars Reticulata	Dopamine GABA
Thalamus	Glutamate	GABA
Subthalamic nucleus/ Globus Pallidus	(-)	GABA

- ii. Corpus striatum
 - a. Neostriatum (Striatum)
 - i. Caudate nucleus
 - ii. Putamen
 - b. Lentiform Nucleus
 - i. Putamen
 - ii. Globus Pallidus
- iii. Classification of movement disorders
 - a. Hypokinetic
 - i. Primary Parkinsonism
 - 1. Parkinson's disease
 - ii. Secondary Parkinsonism
 - 1. Drug induced (MC)
 - 2. Toxin
 - 3. Infectious disorder
 - iii. Parkinson's Plus syndrome
 - 1. Progressive Supranuclear Palsy
 - 2. Lewy body syndrome
 - 3. Alzheimer's
 - 4. Shy Dragger Syndrome

- iv. Heredo-familial disorder:
 1. Huntington's Disease
 2. Gertsmann-Staussler Scheinker's disease
 3. Hallorverden – Spatz disease
 4. Wilsons Disease
- b. Hyperkinetic
 - i. Tics
 - ii. Chorea
 - iii. Athetosis
 - iv. Dystonia
 - v. Hemiballismus
 - vi. Myoclonus
 - vii. Asterixis
 - viii. Stereotypy
- iv. Primary Parkinsonism
 - a. Parkinson's disease/ Paralysis Agitans/ idiopathic parkinsonism)
 - i. Most common hypokinetic disorder
 - ii. Progressive degenerative disease that affects the extrapyramidal system that results to decreased in dopamine.
 - iii. Epidemiology:
 1. Peak onset: 6th decade of life
 2. Males > Females
 - iv. Etiology: Idiopathic
 - v. Manifestations:
 1. Cardinal signs and symptoms
 - a. Resting Tremor
 - i. Frequency: 4-7Hz
 - ii. MC initial symptom
 - iii. Distal > Proximal (Hands)
 - iv. Pill Rolling
 - v. Evident: Oscillatory pronation and supination
 - vi. Factors that triggers tremor
 1. T – Thyrotoxicosis: Excessive thyroid hormones
 2. H - Hypoglycemia
 3. E - Emotional
 4. F – Fever
 5. F - Fatigue
 6. A – Alcohol withdrawal
 7. D – Drug- Induced
 8. E – Exercise
 9. S – Stress

b. Bradykinesia

- i. Slowness of movement
- ii. Associated with
 - 1. Akathisia: Inner restlessness
 - a. Inability to stand/ sit still
 - 2. Akinesia: Absence of movement
 - 3. Hypokinesia: Small amplitude of movement
- iii. Evident:
 - 1. Alternating movements: alternating movement (pronation/ supination)
- iv. Kinesia Paradoxa
 - 1. Pt feels for a rapid movement when there is a surge of emotional energy

c. Rigidity

“RAGS: Rigidity: Alpha, Gamma: Spasticity”

- i. Increase firing of alpha motor fibers
- ii. Non-velocity dependent
- iii. Types
 - 1. Cogwheel: Ratchet like movement
 - 2. Leadpipe: Constant sustained resistance throughout ROM. *Proximal Ms affected, Asymmetrical*

d. Postural Instability

- i. Increased risk of fall
- ii. *Most disabling manifestation of PD*
- iii. Posture
 - 1. Forward head
 - 2. Kyphotic
 - 3. Protracted shoulder
 - 4. Hip & knee flexed
 - 5. Striatal hands
 - a. Wrist flexion + Phalangeal extension
 - 6. Striatal foot
 - a. Ankle inversion and dorsiflexion
 - b. Big toe extension

2. Other manifestations

a. Cranial nerve affection

- i. III – Levator palpebrae superioris
 - 1. Blepharospasm: Involuntary blinking of the yes
 - 2. Blepharoclonus: upon closing the eyelid flutters
 - 3. Normal blinks per minute: 20
 - 4. PD: 5-7 blinks/ min

- ii. VII –
 - 1. Masked Fascie – Expressionless face
 - 2. Hypomimia – Expressionless face + Animation of face
 - a. Seen in Chronic PD
 - iii. IX –
 - 1. Sialorrhea – increased drooling of saliva
 - 2. Seborrhea – increase oil mostly affects the face (Not CN IX)
 - 3. Dysphagia – difficulty swallowing
 - b. Micrographia: Small hand writing
 - c. Bradyphrenia – Slowness of thought
 - d. Gait abnormality
 - i. Festinating gait- Decrease step length, increased cadence
 - ii. Shuffling gait- Small steps
 - 1. Anteropulsive
 - 2. Retropulsive
 - e. Motor En Bloc – “Freezing phenomenon”
 - i. Absence of heel off to toe off and difficulty in turning
- vi. Primitive reflexes
 - 1. Glabellar Tapping
 - a. Myerson’s Sign
 - b. Persistent blinking
 - 2. Palmomental reflex
 - a. Pressure on the palm
 - b. Result: Pouting
 - 3. Hypokinetic Dysarthria
 - a. Hypophonia
 - b. Aprsody: monotonous in thought
- vii. Diagnostic tools
 - 1. Apomorphin test
 - a. Levadopa therapy
 - b. Result: worsens: Secondary Parkinson’s
 - i. Dissipates: Primary Parkinson’s
 - 2. Radiographic
 - a. MRI
 - b. PET Scan (Positron Emission Tomography)
- viii. Hoen & Yahr classification
 - 1. Minimal or absent. *Prominent tremor*
 - 2. Minimal, bilateral/ midline involvement. *Bradykinesia*
 - 3. Impaired Righting reflex/ Balance impairment.
 - 4. All S/Sx are severe
 - 5. Confined to bed/ wheelchair

ix. Medical management

1. Levodopa (L-Dopa)
 - a. **Gold standard** for PD
2. Sinemet
 - a. Carbidopa and Levodopa
3. Anti-Cholinergics
 - a. Amantadine
4. Dopamine Agonist
 - a. Bromocriptine
 - b. Pergolide
 - c. Ach- blocker

x. Prognosis

Positive	Negative
Resting Tremor	Bradykinesia
Rigidity	Postural instability
Family history	Gait Abnormalities

xi. Clinical Fluctuations

1. Natural effectiveness of the Levadopa
 - a. Honeymoon Period: Acute beneficial effect
 - b. Lasts 2-5 years
 - i. Wearing – Off
 1. End-Of-Dose Deterioration
 2. Fading off of the levadopa, there is a need of levadopa treatment
 - ii. On – Time
 1. Patient feels the effectiveness of the Levadopa
 - iii. Off-Time
 1. Patient feels unsafe and feels no effect of Levadopa
 - iv. On/ Off Time
 1. Sudden change of S/Sx on:off
 - c. Adverse effects
 - i. Nausea & Vomiting
 - d. Paradise- Lost Period: Chronic
 - i. Long term use of Levadopa
 - ii. Result: Tardive Dyskinesia:
 1. Manifestation:
 - a. Smacking lips
 - b. Protruded tongue
 - c. One shoulder shrug

- xii. Clinical management
 - 1. Strengthening of the extensors
 - 2. Stretch the flexors
 - a. (GPPS) Gentle Progressive Passive stretching
 - 3. PNF
 - a. UE: Bilateral Symmetrical D2 Flexion
 - b. LE: Bilateral D1 Extension
 - c. Rhythmic initiation: initiates movement
 - d. Rhythmic Stabilization: Promote stability
 - e. Rhythmic Rotation: Trunk movement (Rotation)
 - 4. Shoe modification
 - a. Anteropulsive: Toe wedge
 - b. Retropulsive: Heel Wedge
- v. Secondary Parkinsonism
 - a. D – Drug induced
 - i. Anti-hypertensive drugs(long term use)
 - 1. Reserpine & Methyldopa
 - ii. Anti- Depressant drugs
 - 1. Tricyclic Anti-depressant
 - iii. Neurotic Drugs
 - 1. Haloperidol
 - b. Toxin
 - i. Mg (Manganese) in Miners (MC)
 - ii. Carbon monoxide & disulfide
 - iii. Cyanide
 - iv. Methanol
 - c. Post- Infectious disorders
 - i. AIDS
 - ii. Post-Encephalitic Lethargica (Von Economo Disease)
 - 1. Statue like appearance
 - 2. Masked fascie
 - iii. CJD (Creutzfeldt Jakob's disease)
 - 1. Human Mad Cow disease
 - a. "prion" - Protein abnormality
 - 2. Image: Spongy like appearance
- vi. Parkinsonism/ Dementia Pugilistica
 - a. PD among boxers
 - b. Dementia

- vii. Parkinsonism Plus Syndrome (variants)
 - a. Progressive Supranuclear Palsy (PSP)
 - i. MC type of Parkinson's plus
 - ii. Most distinguishing feature
 - 1. Problem with vertical gaze
 - 2. CN III & IV affection
 - 3. Pseudobulbar effect:
 - a. Emotional/ labile effect
 - b. Lewy Bodies: protein aggravation
 - c. Alzheimer's Disease
 - i. Dementia
 - d. Shy-Drager syndrome
 - i. Age of Onset: 3rd and 4th decade of life
 - ii. Autonomic Dysfunction
 - iii. Impotence
 - iv. Sphincter problem
 - v. Orthostatic Hypotension
 - 1. Best medication: Water
- viii. Heredo-familial disorders
 - a. Wilson's Disease
 - i. Autosomal Recessive, Chromosome 13
 - ii. Copper overloading in
 - 1. Liver, Cornea, Brain
 - iii. (+) Kayser- Fleisher Ring: Copper in Cornea
 - b. Hallervorden – spatz disease
 - i. 7-15 y/o
 - ii. Survival Rate: 3rd decade of life
 - iii. Mortality: Pulmonary Compromise (MC)
 - iv. Reveals: Iron deposition in the basal nuclei
 - v. Globus pallidus and substantia nigra (Pars reticulate)
 - vi. MRI: Rust discoloration
 - c. Gertsmann – Stradslers Scheinker's Disease
 - i. Autosomal Dominant (chromosome 20)
 - ii. Age Onset: 20-60y/o
 - iii. Manifestations
 - 1. Dysarthria
 - 2. Ataxia
 - 3. Dementia

- ix. Hyperkinetic
 - a. Chorea
 - i. Rapid, jerky, irregular movements
 - ii. Purposeless movements
 - iii. Putamen & caudate nucleus
 - iv. Types
 - 1. Sydenhams chorea
 - a. Saint Vitus Dance
 - b. In childhood 2° acute rheumatic fever
 - c. Streptococcal bacterial infection
 - 2. Huntington's chorea
 - a. Chromosome 4 Autosomal Dominant
 - b. Presence of Huntingtin
 - c. Triad:
 - i. Choreaform
 - ii. Cognitive deficits
 - iii. Familial History
 - d. Manifestations
 - i. Ataxia
 - ii. Myoclonus
 - iii. Bulbar effect
 - 1. Dysarthria
 - 2. Dysphagia
- b. Athetosis
 - i. Affectation: Globus Pallidus
 - ii. Slow writhing, worm-like movement
 - iii. Increased GABA
- c. Hemiballism
 - i. Violent flailing of the body parts
 - ii. Affectation: Subthalamic nucleus
 - iii. Increased GABA
- d. Dystonia
 - i. Slow sustained contractions of muscle that cause twisting and abnormal turning
 - ii. Types
 - 1. Segmental: Entire limb
 - a. Cranial dystonia – head segment
 - b. Brachial dystonia – UE segment
 - c. Crural dystonia – LE segment
 - 2. Focal: localized
 - a. Torticollis
 - b. Blepharospasms
 - c. Writer's cramp

3. Other
 - a. Multifocal dystonia
 - b. Hemidystonia
 - c. Generalized type of dystonia
- e. Myoclonus
 - i. Contraction of a single muscle
 - ii. Sudden jerky irregular contraction
 1. Hiccups, Hypnic jerk
- f. Tics
 - i. Brief, sudden, and intermittent movement(motor tic) or sounds (vocal tics)
 - ii. Only one group of muscles
 - iii. May be repetitive & stereotypic uniform movement of the whole body part that is purposeless.
 1. E.g. head banging
 - iv. Types
 1. Gile De La Tourette syndrome
 - a. Combination of one/ two motor tics with one or two vocal tics
 - b. One criteria: Tics should be for more than 1 year
 - c. Most common cause of tic
 - d. Age of onset: 21 years old

2. Vocal Tics

Simple Vocal Tic	Complex Vocal Tic
Coughing	Coprolalia: Obscene language
Blowing	Echolalia: Repetition of other's utterance
Screaming	Palilalia: Repetitions of someone's phrase
Sneezing	
Sucking	
Throat clearing	

3. Motor Tics

Simple Motor Tic	Dystonic Motor Tic	Complex Motor Tic
Blinking	Blepharospasm	Coprophagia: obscene gesture
Head jerking	Bruxism	Ecopraxia: Imitation of gesture
Nose twitching	Torticollis	Jumping & Kicking
	Shoulder shrug	Touching

TBI

- i. Traumatic Brain Injury
 - a. Injury to the brain that is caused by external force
 - b. External force
 - i. MVA, MC direct cause of TBI
 - ii. Alcohol, MC indirect cause of TBI
 - iii. Falls, Elderly and infants E.g. Shaken baby syndrome
 - iv. Sports injuries: Athletes
 - v. Violence: Antisocial
 - vi. Gunshot
 - c. Epidemiology
 - i. Male > Female
 - ii. 18-25 y/o
- ii. Classification
 - a. Closed TBI, skull intact
 - b. Open TBI, skull is open with penetrating object
- iii. Pathophysiology
 - a. Primary Injury, Direct and immediate cause of injury
 - i. Diffuse axonal injury:
 - 1. Wide spread stretching of axons
 - 2. Most distinguishable factor for TBI
 - 3. MOI:
 - a. Acceleration – Deceleration: The head is hit by an object that is considered to be stationary.
 - 4. Types
 - a. Linear force (Translational), displacement of the skull and the brain
 - b. Angular Force (Rotational), Rotation/ displacement of the skull but the brain is in position
 - 5. Affects
 - a. Midbrain - Pons
 - b. Corpus Callosum (MC)
 - c. Subcortical White matter
 - 6. Primary cause of LOC
 - 7. Risk factor:
 - a. Alzheimer's dementia

- ii. Cerebral Contusion/ Focal brain injury
 - “FITA”
 - 1. Common area affected
 - a. Frontal: Inferior
 - b. Temporal: Anterior
 - 2. Coup: Site of impact & site of injury
 - 3. Contre-coup: Site of impact & damage to the opposite side
 - b. Secondary injury: any damage to the brain tissue that takes place after the primary injury
 - i. Increased ICP (normal: 5-15° mmHg)
 - 1. Herniation & hydrocephalus
 - a. Uncal: Uncus & hippocampus
 - b. Central: Midbrain & Pons
 - c. Tonsilar: MO & cerebellum
 - 2. Hydrocephalus:
 - a. Acquired
 - b. Congenital
 - i. Communicating: After 4th ventricle
 - ii. Non-communicating: Before 4th ventricle
 - c. Normal pressure
 - i. Caused by infections, intrinsic factor
 - d. Hydrocephalus Ex Vacuo
 - i. Extrinsic factor
 - 1. Stroke, TBI
 - 2. Associated with elderly
 - ii. Toxicity
 - 1. Excitatory Amino Acid
 - a. Brain function
 - b. Glutamate & Aspartate
 - c. Increase EAA:
 - i. Intracellular – Increase CA (Most amount of fluid)
 - ii. Extracellular – Increase NaCl
- iii. Hematoma
 - 1. Epidural: Skull and Duramater
 - 2. Subdural: Duramater & Arachnoid
 - 3. Subarachnoid: Arachnoid & Piamater
- c. Manifestations
 - i. Cranial nerve affectation
 - 1. CN1: anosmia/ CSF Rhinorrhea
 - a. MC contused
 - 2. CN 2: Scotoma (Blind spot at the center of the cornea)
 - a. Amaurosis Fugax/ monocular blindness:
 - 3. CN3, 4, 6:
 - a. III: Anisocoria: Unequal size of pupils
 - b. IV (Horizontal diplopia) & VI (Veritcal diplopia)

4. CN7: MC affected
 - a. Panda eye sign
 - i. Mastoid Echymosis
 - ii. Temporal fraction
 - b. Bell's palsy
 - i. Central: Can raise eyebrow
 - ii. Peripheral: Cannot raise
5. CN 8:
 - a. Balance & Incoordination
 - i. Ataxia
- ii. HO: bone growth over the soft tissue
 1. Common Sites:
 - a. TBI: Shoulder
 - b. SCI: Hip> Knee> Shoulder
 - c. Burn: Elbow
- iii. DVT
 1. (+) Homan's sign
 2. MC cause of morbidity
 3. Increased risk
 - a. Prolonged immobilization
 - b. Older age
 - c. Fracture
 - d. Clotting disorders
- iv. Spasticity
 1. Velocity dependent
 2. Increased firing of gamma motor neurons
 3. Modified Ashworth Scale (MC used in the clinics)
 - a. 0 – No increase in muscle tone
 - b. 1 - Slight increase in muscle tone with catch and release throughout the beginning of the ROM
 - c. 1+– Slight increase in muscle tone with catch followed by resistance throughout the ROM
 - d. 2 – Marked increase in muscle tone
 - e. 3 – Considerable increase in muscle tone
 - f. 4 – Rigid flexion and extension
 4. Tardieu scale
 - a. A true measure of spasticity among patients with TBI
 - b. A scale that notes the difference of the velocities that is muscle stretched from the angles where the catch is noted
 - c. V1: Slow speed
 - d. V2: Slow and noting where the catch is felt
 - e. V3: Fast velocity

- f. Grading:
 - i. 0 – No resistance throughout the ROM
 - ii. 1 – Slight resistance throughout the ROM
 - iii. 2 – Clear catch at a precise angle, resisted PROM followed by release
 - iv. 3 – Fatigable clonus , < 10 seconds
 - v. 4 – Non- Fatigable clonus >10 seconds
5. Memory impairment
 - a. Anterograde amnesia – cannot recall *new* memory
 - b. Retrograde amnesia – Cannot recall before the incident
 - c. Post traumatic amnesia – between time of injury and when Pt is able to remember ongoing events.
 - i. Declarative memory – ability to recall facts and previous events.
 - ii. Procedural memory – Knowledge how to do motor tasks.
6. Pulmonary affectation
 - a. Pneumonia, MC
7. Bowel and bladder incontinence
 - a. MC social disabling
- v. Diagnostic tools
 1. CT (MC)
 2. MRI
 3. PET (positron Emission tomography)
 4. SPECT (Single-photon emission computerized tomography)
 5. fMRI
- d. FIM (126)
 - i. LO CO MO S3
 - ii. Locomotion
 - iii. Communication
 - iv. Mobility
 - v. Sphincter control
 - vi. Self care
 - vii. Social –cognitive functions
 - viii. Rating
 - 7- independence
 - 6- moderate independence
 - 5 – supervision
 - 4 – Minimal assist
 - 3 – moderate assist
 - 2 – maximal assist
 - 1 – total assist

- e. GOAT (Galvestone Orientation Amnesia Test)
 - i. For post-traumatic amnesia (adults)
 - ii. Normal: 76 – 100 points
 - iii. <75: Amnesic
 - iv. Normal : >75 for 2 consecutive days
 - v. COAT (children) 4-15 years old
- f. GOS – extended
 - i. Death
 - ii. Vegetative state, presence of sleep wake cycle (+) eye movement (-)cortical function maximal assistance
 - iii. Severe disability lower, moderate assistance in ADL
 - iv. Severe disability upper, minimal assistance in ADL
 - v. Moderate disability lower, (-) vocational, can go shopping
 - vi. Moderate disability upper, (+) vocational, with modification
 - vii. Good Recover lower, (+) vocational, without modification with reports of S/Sx
 - viii. Good Recovery upper, (+) vocational, without modification, without reports
- g. GCS
 - i. Teasdale and Jenett
 - ii. MC used to measure level of consciousness, define and classify the level of injury
 - iii. Mild: 13-15. Moderate 9-12. Severe <8

Eye opening		Best motor response		Verbal response	
Adult & Pedia		Adult	Pedia	Adult	Pedia
4	Spontaneous	6	Follows	5	Obeys (n) Movement
3	To speech	5	Localizes	4	Disoriented
2	To pain	4	Withdraws	3	Inappropriate
1	NR	3	Decorticate	2	Incomprehensible
		2	Decerebrate	1	NR
		1	NR		

iv. Scoring

	Severe	Moderate	Mild
GCS	3-8	9-12	13-15
LOC	>6 hours	<6 hours	<20 minutes
PTA	>24 hours	<24 hours	1 – 2 hours

iv. RLA-LOCF

- a. Examines cognitive and behavioral recovery in individuals with TBI
- b. This + GCS is widely used

I	No Response	Completely unresponsive
II	Generalized Response	Gross motor movement Nonspecific movement Nonpurposeful movement
III	Localized Response	Specific response to pain
IV	Confused – Agitated	<i>Heightened state of activity</i> Bizarre / nonpurposeful movements Inappropriate behavior
V	Confused – Inappropriate	Follows Simple commands consistently <i>Impaired selective attention</i>
VI	Confused – Appropriate	<i>Goal directed responses</i> Follows simple commands Consistently with carryover
VII	Automatic – Appropriate	Automatic daily routine, <i>Robot like</i> Social interaction <i>Judgement impaired</i>
VIII	Purposeful - Appropriate	Oriented, Responds to environment Impaired abstract reasoning,

i. Predictors of outcome (6 months)

Good	Poor
GCS: >8	GCS: <8
Young adults	Late adults
(+) OVR (dolls eye sign)	(-) OVR
(+) Caloric	(-) caloric
PTA <4 weeks	PTA >12 weeks
Coma < 1 week	Coma > 2 weeks

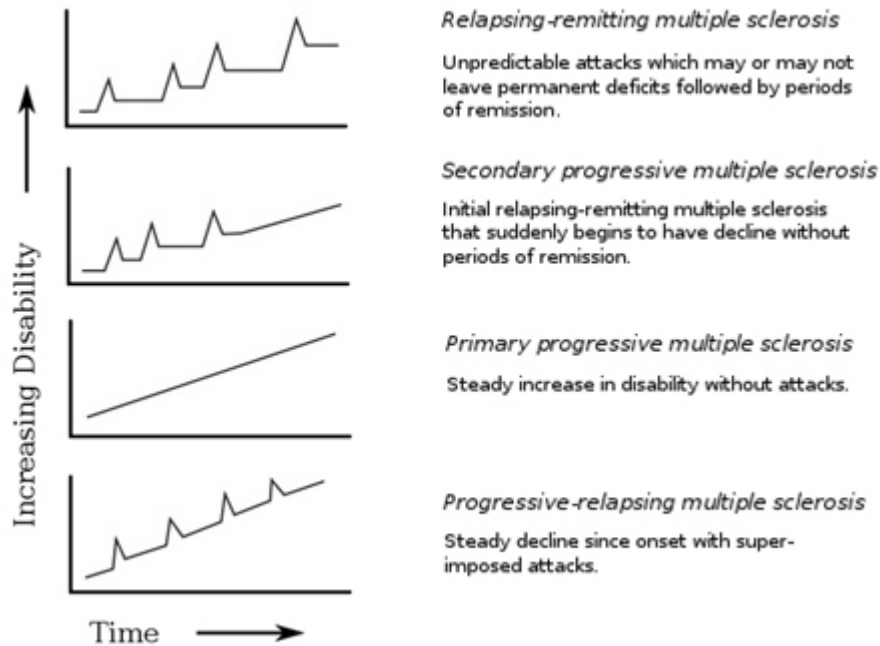
MS

- i. Multiple Sclerosis
 - a. *Great crippler of the young adults*
 - b. Autoimmune disorder, inflammatory that affects the myelin sheath to be demyelinated
 - c. Epidemiology:
 - i. Age: 20-40 y/o
 - ii. Race: white > blacks > Asian
 - iii. Gender: Female (Estrogen promotes production of T & B cells)
 - iv. Money: Rich Countries
 - v. Location: Temperate climates (4 Seasons); Pole Areas (North & South)
- ii. Pathophysiology
 - a. Causes
 - i. Autoimmune
 - ii. Viral: Epstein- Barre Virus
 - iii. Idiopathic
 - b. Pathology:

1 →	2 →	3 →	4 →	5 →	6 →	7
Inflammation	Activation of T-Cells	Demyelination	Glios (Proliferation of Glial cells)	Glial Scarring	Hardens/ Thickens Scars	Plaque formation

- c. Areas of predilection:
 - "PBO Cinema Central"*
 - d. Periventricular white matter/ Subcortical Area
 - e. Brainstem (Cranial Nerves)
 - f. Optic nerve
 - g. Cerebellum (Inner, Arbor Vitae)
 - h. Cervical spinal column
- iii. Variants
 - a. Optic neuritis: Sudden transient loss of vision
 - b. Transverse myelitis
 - c. Devic syndrome (Neuromyelitis Optica) – severe form of the 2 combination above

iv. Patterns of MS



Relapsing Remitting is MC

v. Manifestations

a. Periventricular/ Subcortical Areas

i. CSF findings

1. Increase myelin fragments
2. Increase Oligoclonal bands
3. Increase IgG

b. Brainstem/ Cranial Nerve affection

i. CN II - Optic

1. MC affected
2. Scotoma (Central Blindness)
 - a. Glaucoma (Peripheral blindness)
3. Optic neuritis/ Retrobulbar Neuritis (Sudden visual loss)
4. Note: Axons are intact without blindness

ii. CN V - Trigeminal

1. Trigeminal Neuralgia/ Tic Doreux
 - a. Hypersensitivity to pain on face
 - b. Affected sensory only

iii. CN VII – Facial

1. Facial Myokymia: Vernicular contraction of facial muscles
 - a. worm-like movement of the face
 - b. Motor affected only

- iv. CN VIII – Vestibulocochlear
 - 1. Vomiting
 - 2. Vertigo
 - 3. Nystagmus
 - 4. Incoordination
 - c. Cerebellar
 - i. Charcot's Triad
 - 1. S – Scanning speech
 - 2. I – Intention tremor
 - 3. N – Nystagmus
 - d. Gait disturbances
 - i. Staggering gait
 - e. Motor affectation
 - i. Spasticity
 - ii. *Fatigue* MC symptom
 - iii. Hypotonia
 - f. Sensory affectation
 - i. Numbness/ tingling sensation (Paresthesias)
 - ii. Position/ vibratory sense disturbances
 - iii. Pallesthesia
 - g. Other manifestations
 - i. Bowel & Bladder Function
 - 1. Conflicting/ *Dyssynergic bladder*
 - a. Incoordination between sphincter and bladder (Detrussor Ms)
 - b. Sphincters and bladder contracts (increase in pressure)
 - 2. Cognitive Sx
 - a. Euphoria: Excessive happiness
 - b. Emotional lability/ dysregulation
 - i. Extreme emotions
- vi. Special test
 - a. Lhermitte's sign
 - b. Brudzinski – Kernig
- vii. Exacerbating factors
 - a. Heat
 - i. *Uhthoff's Phenomenon* – Increase in heat causes, Nerve conduction block
 - 1. Causing fatigue and increase in Sx (E.g. during exercises, exposure to heat)
 - b. Trauma
 - c. Fatigue
 - d. Stress
 - e. Pregnancy
 - f. Infection
 - g. Heavy Metal (Zinc)
 - h. Diet
 - i. Note: First sign of exacerbation in MS: *Hyperventilation*

viii. Prognosticating factor

Poor	Good
Male	Female
Late onset	Early Onset
Polysymptomatic	Monosynaptic
Rapid progression	Full remission
	*Optic Neuritis

ix. Diagnostic Criteria (Duration of Sx: >24hours)

- a. Possible MS
 - i. One lesion undocumented
- b. Probable MS
 - i. One documented attack with 2 lesions
 - ii. Two documented attack with 2 lesions
- c. Definite MS
 - i. 2 attacks; 2 lesions 1 month
- d. Ancillary procedure:
 - i. MRI (plaques)
 1. Size of lesions (plaques): > 3mm
 - ii. NCV Test

x. Kurtzke Expanded Disability Status Scale (KEDSS)

- a. Focuses on mobility more than sensory and other functional impairments
- b. Note: *Motor function is not included*
- c. (7) system
 - i. Pyramidal
 - ii. Cerebellar
 - iii. Brainstem
 - iv. Sensory function
 - v. Mental function
 - vi. Bowel/ bladder function
 - vii. Visual
- d. Grading
 - i. 0 – normal
 - ii. 1-1.5 no disability
 1. (1) Functional system involved
 2. (2) Functional system
 - iii. 3.0 to 2.5 minimal disability
 1. (1) Functional system
 2. (2) Functional system
 - iv. 3.0 moderate disability
 1. Functional scale
 - v. 3.5 mild disability
 1. 3 functional scale

- vi. 4.0 to 4.5 fully ambulatory
 - 1. 500m
 - 2. 300m
 - vii. 5.0 to 5.5 ambulatory
 - 1. 200m
 - 2. 100m
 - viii. 6.0 to 6.5 Assistance
 - 1. Unilateral, 100m
 - 2. Bilateral, 20m
 - ix. 7.0 to 7.5
 - 1. Wheelchair bound: manual
 - 2. Wheelchair bound: electrical
 - x. 8.0 to 8.5
 - 1. Bed bound (can transfer)
 - 2. Bed ridden
 - xi. 9.0 to 9.5
 - 1. (+) Eat and swallow, communication
 - 2. (-) Eat and swallow, communication
 - xii. 10.0
 - 1. Death
- xi. Treatment
 - a. Do NOT fatigue the patient
 - i. Medications
 - 1. AM: Amantadine (anticholinergics)
 - 2. PM: Pemoline (CNS stimulant, *Withdrawn d/t hepatotoxicity*)
 - ii. Time of exercise
 - 1. Early Morning
 - 2. Late Afternoon
 - iii. Diabetic Pt's
 - 1. Mid-day
 - 2. IDDM: Insulin (-) exercises 3-4 hours
 - iv. Do not five resistance exercise
 - 1. Priority: Endurance Exercise
 - v. Frenkel's Exercise: For coordination
 - vi. Energy conservation techniques

MOTOR NEURON DSE

- i. Motor neuron disease (traditional)
 - a. Amyotrophic Lateral Sclerosis
 - b. Spinal Muscular Atrophy
 - c. Poliomyelitis
- ii. Current definition
 - a. Diseases or conditions that produce dysfunction of the motor neurons resulting in weakness and muscle wasting
- iii. MND: Definition
 - a. Pathology confined to the motor pathway
 - b. Classification by site of lesion
 - i. Upper motor neuron disease
 - ii. Combined UMN – LM disorders
 - iii. Lower motor neuron disease
- iv. Upper motor neuron disorders
 - a. *Primary lateral sclerosis*
 - b. Tropical spastic paraparesis
 - c. Lathyrism
 - d. Epidemic spastic paraparesis
 - e. Familial (Hereditary) spastic paraplegia
- v. Combined UMN – LMN disorders
 - a. ALS
 - b. Western pacific ALS – parkinsonism
 - c. Demetia Complex (Guam, Rota & Tinian of the Mariana Islands)
 - d. Groote Eylandt MND (affect Australian aborigines)
 - e. *Post encephalitic (Encephalitis Lethargica)* ALS (occur 10 years after encephalitis)
 - f. Juvenile Inclusion body ALS (onset at 12-16 y/o with 1- 1 ½ years survival)
- vi. Lower motor neuron disorders
 - a. SMA
 - b. Bulbar and Bulbospinal muscular atrophies
 - c. Poliomyelitis
- vii. ALS: Epidemiology
 - a. Onset
 - i. Late middle life (about 58 y/o)
 - ii. Average age at time of diagnosis (63 y/o)
 - iii. Rare before 40 and after 70
 - b. Slight male predominance (1.2-1.6:1)
 - c. (+) Family history in 5-10% of the cases
 - d. Prevalent in Guam and Kii island
 - e. Incidence: 1.6 to 2.4 cases/ 100,000 population

f. Pathology

- i. *Anterior horn of spinal cord (LMN, MC)*
- ii. Brainstem (LMN & UMN)
- iii. Spinal cord corticospinal tract (UMN)
- iv. Internal capsule posterior limb (UMN)
- v. Motor cortex (UMN)
- vi. Etiology: Unknown
 - 1. Traced to the degeneration of the motor neuron
 - 2. Increased amount of glutamate
 - 3. Pharmacological Mx: Riluzole (glutamate antagonist, commonly given to patient with ALS, to slow down the process of the disease)

g. Pathophysiology

- i. *LMN signs: Amyotrophy* (muscle weakness and atrophy) & fasciculation
 - 1. Due to loss of anterior horn cells in SC and brainstem motor nuclei
 - 2. Motor nerve cells replaced by astrocytes
- ii. *Loss of LMN cell bodies is the principal lesion in ALS thus LMN signs dominate* (Late phase)
- iii. *UMN signs: generalized hyperreflexia, mild spasticity & babinski*
- iv. Due to degeneration of the CST especially at the lower part of cord
- v. May also have degeneration of UMN in posterior limb of the internal capsule, corona radiata, & brainstem
- vi. Also have degeneration of the large pyramidal neurons in the primary motor cortex and of the pyramidal tracts
- vii. Onuf's Nucleus (controlling striated muscles of pelvic floor and bowel/ bladder sphincters) is preserved
- viii. Sensory neurons unaffected
- ix. *No Bowel, bladder, and sensory deficits*
- x. *Summary*
 - 1. Combined UMN & LMN lesions
 - 2. Combined brainstem & spinal involvement
 - 3. * Classic ALS is the benchmark of the motor neuron diseases

h. Findings

- i. Progressive motor weakness/ paralysis
- ii. Fasciculations and atrophy
- iii. Hyperreflexia, extensor plantar reflex, spasticity
- iv. Dysarthria & dysphagia
- v. Most striking feature: focal asymmetrical onset of weakness that eventually progresses

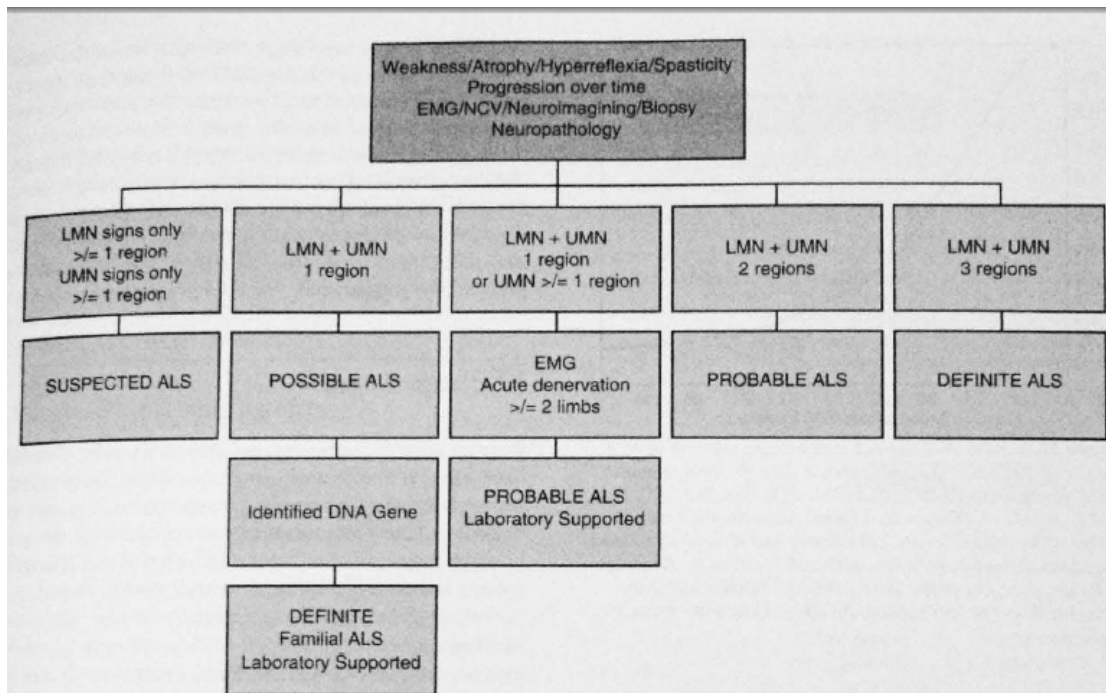
i. Course:

- i. Earliest sign: difficulty in doing fine motor tasks due to hand weakness
- ii. Atrophy, frequent cramping and UE fasciculations
- iii. Weakness unilateral at start
- iv. Same findings seen in opposite after several weeks and months
- v. Abductors, adductors, and extensors of hand affected more than long flexors

- vi. Subsequently amyotrophied accompanied by slight spasticity of arms and legs generalized hyperreflexia without any sensory changes
- vii. Babinski present in early course of the disease
- viii. Eventually weakness progresses proximally to involve upper arm and shoulder girdle
- ix. After UE become extremely weak atrophy and weakness spread to neck, tongue pharyngeal and laryngeal muscles
- x. At end stage trunk and LE muscles also weak and atrophic > LMN signs dominate & umn signs disappear
- xi. However, abdominal, deep tendon and plantar reflexes continue to be present
- xii. Summary
 - 1. Unilateral, distal UE weakness with LMN signs > opposite to distal UE > UMN signs > proximal UE weakness > brainstem and bulbar involvement > trunk and LE weakness > UMN signs disappear > LMN signs dominate but reflex present
- xiii. Other findings
 - 1. Brisk muscle stretch reflex in areas of muscle atrophy
 - 2. Muscle cramping frequent complaint
 - 3. Constipation
 - 4. May have paresthesia and decrease vibratory sense in 25% of cases
 - 5. 3.5% with signs of dementia
 - 6. 1.5% with parkinsonism
- j. Prognosis
 - i. Poor (2-5 years survival rate)
 - 1. Average survival from time of diagnosis 2.5 years
 - 2. Predominantly bulbar > 2.2 year
 - 3. Predominantly spinal > 3.3 years
 - ii. Cause of death: respiratory failure
- k. Poor prognostic indicators
 - i. Early onset
 - ii. Dominantly bulbar involvement
- l. Variations
 - i. Familial ALS (Note: Classic ALS is NOT hereditary)
 - 1. Same clinical features
 - 2. Younger age of onset (mean age of diagnosis 46 y/o)
 - 3. Survival 2-12 years
 - 4. Autosomal dominant inheritance pattern
 - ii. Primary lateral sclerosis (UMN only)
 - 1. *LE first*
 - 2. Rare, non familial disorder
 - 3. M=F involvement
 - 4. Onset: 20-60 years old
 - 5. Affect corticobulbar & CSR
 - 6. Start with LE spasticity > UE & bulbar spasticity > urinary incontinence at late stage of disease
 - 7. No atrophy or fasciculation

8. No EMG findings of denervation
9. Survival: 2-3 decades or longer
10. Pathogenesis is unknown
11. Find reduce number or absence of Betz cells in Area 4 with degeneration of cortico spinal pathways
- iii. Classical ALS (LMN+ UMN, B & S)
- iv. Brainstem (UMN + LMN + predominantly bulbar involvement)
 1. Primary bulbar palsy
 2. Rapidly progressive
 3. 2-3 years onset
- v. Spinal (UMN+ LMN + predominantly spinal involvement)
 1. Primary muscular atrophy
 2. Slower progression
 3. 15 years from onset
- vi. Other UMN disorders
 1. Tropical spastic paraparesis: in countries in the tropics, affecting mainly black adult males or female
 2. Lathyrism
 3. Epidemic Spastic paraparesis
 4. Familial or Hereditary Spastic Paraplegia
- viii. Lathyrism
 - a. UMN disorder produced by excessive consumption of chickening pea (Lathyrus Sativus)
 - b. Toxic agent: Beta-N- oxalyamino-L-alanine (BOAA)
 - c. Endemic to Indian subcontinent
 - d. Onset from 2-70 years old
 - e. Affect both sexes but more common in males (onset 5-40 years old)
 - f. Females onset 10 years old
- ix. Epidemic Spastic paraparesis
 - a. Like tropic spastic paraparesis but different in etiology
 - i. Affects children and adults that are seronegative for human Tcell lymphotropic virus type1
- x. Familial/ hereditary spastic paraplegia (Strumpell –Lorraine syndrome)
 - a. Autosomal dominant mainly
 - b. Any age but often onset in childhood or early adult life
 - c. Stiffness and unsteadiness of legs progressing to spastic paraplegia
 - d. Primary pathology
 - i. Degeneration of descending and ascending spinal tracts (especially CST to legs, fasciculus gracilis and spinocerebellar tracts)
 - e. Manifestation
 - i. Leg stiffness, weakness of hip flexors and ankle dorsiflexors
 - ii. May begin as early as during infancy, or later in childhood
 - iii. Have progressive difficulty in walking
 - iv. Alter> paresthesia> decrease vibratory sense and urinary dysfunction
 - v. (+) UMN signs * hyperreflexia, babinski, clonus
 - vi. Some with (+) pes cavus deformity

- vii. Complicated cases present with spasticity, optic neuropathy, retinopathy, extrapyramidal disturbance, dementia & ataxia
- xi. ALS treatment (braddom)
 - a. Phase 1: patient independent
 - i. Stage 1: (Ambulatory & ADL independent)
 - 1. ROM & strengthening exercises or unaffected muscle
 - 2. No strenuous exercise
 - 3. Psychological support
 - ii. Stage 2: (ambulatory but decreased ADL independence)
 - 1. Difficulty stair climbing, raising arms over head, buttoning
 - 2. Modification of commonly used devices, clothes etc
 - 3. AFO, hand splints
 - 4. Selective strengthening and stretching exercises
 - iii. Stage 3: (ambulatory but easily fatigues)
 - 1. Have severe selective weakness of ankles, wrist, hands with moderate decrease in ADL independences
 - 2. Add deep breathing exercises
 - 3. Wheelchair/ scooter for long distance
 - b. Phase 2: partially independent
 - i. Stage 4 (wheel chair bound)
 - 1. Severe LE weakness but still with some UE ADL function
 - 2. Thermal agents for shoulder pain
 - 3. Anti-edema measure for hand
 - 4. PROM and stretching exercises
 - 5. Isometrics for uninvolved muscles (compensation)
 - ii. Stage 5
 - 1. Moderate to severe UE weakness
 - 2. Family trained on transfer techniques, pressure sore prevention, proper positioning
 - 3. Home environment modification
 - c. Phase 3: completely Dependent
 - i. Totally dependent
 - ii. ROM-stretching exercises
 - iii. Soft diet, tube feeding
 - iv. Respiratory assistance



- xii. Spinal Muscular Atrophy
 - a. Synonym: hereditary form of progressive muscular atrophy
 - b. Subtypes differentiated based on
 - i. Inheritance pattern (although the most common is autosomal recessive)
 - ii. Age of onset
 - iii. Rate of progression
 - iv. Prognosis
- xiii. Infantile SMA
 - a. Pathophysiology: have severe loss of motor neurons throughout brainstem and spinal cord
 - b. Primary clinical features
 - i. Progressive muscle weakness
 - ii. Atrophy of trunk & limbs
 - iii. Hypotonia (floppy infant)
 - iv. Feeding difficulties
 - c. Floppy at birth with generalized weakness and areflexia
 - d. Frog position in supine
 - e. Feeding difficulty and poor breathing soon apparent
 - f. Weakness initially proximal > affects pelvic girdle, trunk and shoulder girdle
 - g. Later affect all skeletal muscles except EOM
 - h. Intercostal fasciculation & paralysis > paradoxical respiratory movements & weak cry
 - i. Oropharyngeal paralysis > poor sucking and swallowing
 - j. Facial weakness > expressionless with open mouth
 - k. No head control & never sits
 - l. Fasciculations of tongue pathognomonic
 - m. Death before 1 year of age

- n. The earlier the manifestations the shorter the course
 - o. Cause of death *typically respiratory failure*
- xiv. Intermediate & late forms
 - a. Child born normal
 - b. Later have symmetrical, proximal weakness
 - c. LE before UE, proximal before distal
 - d. Respiratory weakness very late
 - e. Bulbar muscles often spared
 - f. Good prognosis: late onset proximal weakness
- xv. Type II SMA
 - a. More slowly progressive than Type I
 - b. Clinical signs usually present by 3 y.o occasionally by 3 months old
 - c. Same pattern: proximal weakness: LE before UE
 - d. Also with atrophy & areflexia/ hyporeflexia
 - e. Scoliosis, thoracic deformities and equines deformities of feet due to slow progression
- xvi. Juvenile & Adult SMA
 - a. Type III – autosomal recessive
 - b. Type IV juvenile – AD
 - c. Type V adult – AD
 - d. Onset anytime between childhood and 7th decade of life but usually between 2-17 years old
 - e. Duration from 2-40 years
 - f. Predominantly males
 - g. Starts as symmetrical atrophy & weakness of pelvic girdle & proximal LE > then proximal UE > leg and forearm later
 - h. Fasciculations in 50% of the cases
 - i. Late onset mild dysphagia and dysarthria
 - j. Adult onset > AD pattern poorer outcome than AR type
- xvii. Variants of SMA
 - a. Distal SMA
 - b. Two forms of progressive bulbar paralysis of childhood
 - i. Fazio-Londe Disease
 - ii. Brown - Vialleto-van Laere Syndrome
 - c. Scapuloperoneal SMA (an adult bulbar muscular atrophy)
- xviii. Distal SMA
 - a. Also called
 - i. Distal hereditary motor neuropathy
 - ii. Spinal form of Charcot-Marie Tooth Disease
 - b. Various inheritance patterns
 - i. AR – Juvenile mild DSMA (onset (2-10y/o)
 - ii. AR – Juvenile Severe DSMA (onset 4mos to 20 y/o)
 - iii. AD – Juvenile DSMA (onset 2-20 y/o)
 - iv. AD – Adult DSMA (onset 20-40 y/o)

- c. Clinical features
 - i. Weakness and atrophy initially in distal lower extremity especially anterior tibial and peroneal muscles
 - ii. Upper limbs spares and rarely predominantly affected
 - iii. Sensation is intact
 - iv. Motor and sensory NCV is normal
 - v. Life expectancy: generally normal
- xix. Bulbar disease of childhood
 - a. Slowly progressive weakness of facial muscles, tongue, & pharyngeal muscles
 - b. Two types (both autosomal recessive)
 - i. Fazio-londe (more cranial nerves involved)
 - 1. Onset at 2-12 year old
 - 2. Duration of 9 months to 8 years
 - 3. All bulbar motor neurons affected
 - ii. Brown-Vialetto (longer survival)
 - 1. First symptom is bilateral deafness between 18 mo's to 31 years old
 - 2. Average onset 12 years old
 - 3. Cranial nerve palsies occur 405 years later
 - 4. Survival rate may exceed 2 decades from onset
 - 5. CN VII to XII affected in most cases
 - 6. CN II, V, and VI may be affected but not in all cases
- xx. Adult SMA
 - a. Scapuloperoneal (fascioscapuloperoneal) Muscular atrophy
 - i. Autosomal dominant; purely LMN
 - ii. Atrophy begins between 30-50 years of age
 - iii. Slow progression with normal life span
 - iv. Disability only after 10-20 years from onset
 - v. Weakness and atrophy begin in the legs sparing intrinsic of feet
 - vi. Several years later affect shoulder girdle muscles and then thigh, pelvic girdle, upper arm, neck, and then face
 - b. Chronic bulbospinal muscular atrophy of late onset
 - i. Also called Kennedy's disease
 - ii. Onset at 3rd to 6th decade of life
 - iii. Weakness and atrophy in shoulder and pelvic girdle > less severe distal muscle weakness > dysarthria, dysphagia and tongue atrophy
 - iv. No UMN signs
 - v. Bowel-bladder & sensation preserved
 - c. Monomelic (Segmental SMA)
 - i. Not inherited
 - ii. Mainly in far east: India, Japan, and Malaysia
 - iii. Male juveniles or young adults
 - iv. Segmental in distribution
 - v. Affect only a portion of one limb like forearm and hand, shoulder and upper arm or thigh
 - vi. Contralateral limb may also be affected

- vii. Progressive over 1-3 years but remains focal
 - viii. Cranial nerves, UMN, bowel, bladder, sensory functions spared
 - ix. Pathogenesis unknown
- xxi. Poliomyelitis
 - a. Previously caused by 3 types of poliovirus (RNA virus, enterovirus group, and picornavirus family)
 - i. Have oral vaccine using live attenuated virus
 - b. Presently due to other enteroviruses (Coxsackie A & B echovirus)
 - i. Has benign presentation
 - ii. Paralysis is rarely significant
 - c. Characteristics
 - i. Highly communicable but mild effect
 - ii. Main reservoir of infection > human intestinal tract
 - iii. Route of infection > orofecal route
 - iv. Multiply in pharynx & intestines
 - d. Effect of infection
 - i. Incubation period > 1 to 2 weeks
 - ii. 90-95% of patients asymptomatic or in apparent infection
 - iii. 1-2% will have CNS involvement
 - e. Clinical features
 - f. Onset: malaise, muscle aches, low grade fever of 1-3 days then asymptomatic
 - g. In apparent, minor or abortive polio
 - h. Pharyngitis or gastroenteritis
 - i. In paralytic type
 - i. At onset have fever, generalized headaches and neck and back stiffness
 - ii. Paralysis on 2nd to 5th day
 - j. Paralytic type
 - i. Have muscle soreness and sensation of tightness
 - ii. Shooting pains and hyperesthesia
 - iii. Sensory loss rare: may have paresthesia
 - iv. Weakness appears and evolves over hours to a few days
 - v. Progression stops after temperature normalize within 48 hours
 - vi. Lumbar area more frequently involved than cervical area or the cranial nerves
 - vii. Severe bulbar involvement in 10-15 cases (more common in young adults)
 - viii. Muscle atrophy within first week (peak by 12th-15th week)
 - ix. Coarse fasciculation's frequent
 - x. Decreased MSR
 - xi. Autonomic dysfunction with cardiac arrhythmia, HPN, hyperhidrosis, urinary retention and constipation can occur
 - xii. May also have change in mental status
 - xiii. Death usually due to bulbar or respiratory involvement

- k. Bulbar paralysis
 - i. Most frequently affected CN 9, 10, 11
 - ii. Nucleus ambiguus affecting deglutition
 - iii. Other possible effects: hiccups, shallow & progressive slowing of respiration, cyanosis, restlessness, anxiety (air hunger), HPN, later hypotension & circulatory collapse)
- l. Polio prevention
 - i. Vaccination
 - 1. Initially Salk Vaccine (formalin inactivated virulent strains)
 - 2. Presently Sabin vaccine (attenuated live virus) oral, 2 doses 8 weeks apart with booster at 1 year and before schooling
 - ii. Paralytic polio with 5-10% mortality
- m. Prognosis
 - i. Neurologic recovery depends on # of surviving motor neurons, # of recovered neurons & # of neurons that undergo axonal sprouting
 - ii. Recovery fastest in children
 - iii. Period of recovery usually 3-6 months but may continue for several years
- n. Sharrard's Index
 - i. LE
 - 1. 1 month = +2.0
 - 2. 2 months = +1.5
 - 3. 4 months = +1.0
 - 4. 6 months = +0.75
 - ii. UE
 - 1. 1 month = +2.5
 - 2. 2 months = +2.0
 - 3. 4 months = +1.5
 - 4. 6 months = +1.0
- xxii. Post-polio syndrome: typical patient
 - a. Had acute polio with onset at 5-10 years of age
 - b. Gradually improving function over 5-8 years after acute illness
 - c. Clinically stable for 25-30 years
 - d. Note onset of new health or ADL problems 5-8 years before consulting a post-polio clinic
 - e. Most common new symptoms (at least 2)
 - i. *Fatigue, MC*
 - ii. Muscle or joint pain
 - iii. Weakness
 - iv. Difficulty walking and stair climbing
 - v. Others
 - 1. Cold intolerance
 - 2. Atrophy
 - 3. Dressing difficulties

- f. Halstead and Rossi's Criteria
 - i. (+) history of paralytic polio
 - ii. (+) partial to complete neurologic & functional recovery
 - iii. Stable for at least 15 years neurologically & functionally
 - iv. No other medical explanation
 - v. At least 2 new health problems during period of stability
 - 1. Unaccustomed fatigue
 - 2. Muscle and/or joint pain
 - 3. New weakness in muscles previously affected and/or unaffected
 - 4. Functional loss
 - 5. Cold intolerance
 - 6. New atrophy
- g. Management
 - i. Primarily related to overuse & over activity
 - ii. New or modified aids
 - iii. Energy conservation techniques
 - iv. Change in exercise program – should avoid excessive fatigue, muscle and joint pain
 - 1. Aquatic exercise best type for polio
 - v. Change in orthosis
 - vi. Weight loss
 - vii. New or modified wheel chair
 - viii. Gentle exercises program
 - 1. Aerobic exercises
 - 2. Stretching
 - 3. Strengthening
- h. MND: common contracture
 - i. Neck flexion
 - ii. Shoulder adduction
 - iii. Elbow flexion
 - iv. Forearm pronation
 - v. Finger adduction
 - vi. Finger extension
 - vii. Hip flexion
 - viii. Hip internal rotation
 - ix. Knee flexion
 - x. Ankle plantarflexion
 - xi. Foot inversion

- i. MND: Primary goals
 - i. Assist patient in the maintenance of
 - 1. Function
 - 2. Independence
 - 3. And quality of life
 - ii. For as long as possible
 - iii. Preventative and therapeutic intervention for pain, soft tissue tightness, deformity, scoliosis management, motor weakness and respiratory dysfunction to minimize complications and maximize function
 - iv. Functional training for locomotion, dressing, eating, other ADL
 - v. On exercise
 - 1. Avoid vigorous, fatiguing, progressive resistive exercises, contraindicated in most motor neuron disease because of risk of overuse weakness
 - 2. Low intensity, non-fatiguing exercise beneficial

CVA

- i. CVA VS Stroke VS CVD
 - a. CVD
 - i. Brain impairment due to a non-traumatic pathology involving vertebral blood vessels
 - b. Stroke or apoplexy
 - i. Non-convulsive, focal neurologic deficits of sudden onset
 - c. CVA
 - i. Sudden onset neurologic deficits due to brain infarction (ischemia or hemorrhage)
- ii. Temporal classification
 - a. Transient Ischemic Attack (TIA) – deficits disappear within *24 hours*
 - b. Reversible Ischemic Neurologic Disease (RIND) – Deficits resolve within a week
 - c. Stroke in evolution – stepwise increase in deficits; *progressing stroke*
 - d. Crescendo TIA – Series of transient, sudden onset, deficits over several hours or days (on/off)
 - e. Completed stroke – neurologic deficits *stabilize*
- iii. Etiologic Classification
 - a. Ischemic stroke
 - i. Thrombosis
 - ii. Embolism
 - iii. Hypoperfusion
 - b. Hemorrhagic stroke
 - i. Intracerebral hemorrhage
 - ii. Subarachnoid hemorrhage (aneurysms)
- iv. Causes of CVD
 - a. Atherosclerosis (MC)
 - b. Narrowing of cerebral blood vessels with dysplasia or hyperplasia of its walls
 - c. Congenital aneurysms
 - d. Vasculitis (E.g. Panarteritis Nodosa)
 - e. Coagulation problems
 - f. Immunological abnormalities
 - g. Systemic hypoperfusion (due to heart failure, severe hypovolemia, severe respiratory disease)
- v. Risk factors for stroke
 - a. Non modifiable
 - i. Age (65 y/o)
 - ii. Sex (Male> female, 19% higher incidence in males)
 - iii. Race (blacks)
 - iv. Previous stroke

- b. Modifiable
 - i. By medical means
 - 1. HPN
 - a. Major risk factor in stroke
 - b. Systolics > 165 mmHg & Diastolics > 95 mmHg increase stroke risk by a factor of six
 - c. Decrease systolic BP by 10 mmHg reduces stroke risk by 35-40%
 - d. Lowering diastolic BP by 7.5mmHg reduces risk by 46%
 - e. Controlled though anti-hypertensive drugs, diet, aerobic exercise
 - 2. TIA
 - a. Anti-coagulants like aspirin
 - 3. DM
 - a. *Double stroke risk*
 - b. Prevent with hypoglycemic drugs, diet, aerobic exercise
 - 4. Carotid artery stenosis
 - a. Asymptomatic carotid bruit due to carotid stenosis – surgical removal of the atheromatous plaques
 - ii. Through lifestyle changes
 - 1. Direct risk factors
 - a. Smoking,
 - i. Heavy smokers (>40 cigarettes/ day) 2x more likely to have stroke
 - ii. Reversed after 5 years of quitting
 - iii. Increased risk for hemorrhagic stroke
 - b. heart disease
 - i. increased risk by 2-6x normal
 - ii. causes:atrial fibrillation, LVH, CHF
 - 2. Indirect risk factors
 - a. Hypercholesterolemia (Low HDL, High LDL) > CAD, atherosclerosis
 - b. Obesity > HPN
 - c. Potential risk factors
 - i. Alcohol
 - ii. Inactivity
 - iii. Oral contraceptive use
 - iv. Migraine
 - v. Drug abuse
 - vi. Conditions cause blood hyperviscosity
- vi. Laterality: Right or Non dominant hemisphere
 - a. Parosody, amelodia, amusia
 - b. Sparse use of gestures
 - c. Auditory agnosia
 - d. Impaired music and art appreciation
 - e. Dressing & constructional apraxia
 - f. Neglect syndromes

- g. Prosopagnosia
 - h. Astereognosis
 - i. Visiospatial or perceptual deficits
 - j. Falls more frequent
 - k. Safety problem
 - l. Poor judgement
 - m. Poor retention for new tasks
- vii. Laterality: Left/ dominant hemisphere stroke
 - a. Aphasia
 - b. Can understand gestural commands
 - c. Alexia without agraphia (word blindness)
 - d. Visual agnosia
- viii. Etiology of stroke
 - a. Thrombotic
 - i. Most common
 - ii. Slow progression
 - b. Embolic associated with cardiac disease especially arrhythmia
- ix. Hemorrhagic
 - a. Usually with headache, vomiting and altered sensorium
 - b. With risk of rebleeding
 - c. High mortality at onset but has potential for good neurologic recovery if patient survives
- x. Basis of clinical presentation
 - a. Laterality – right or left hemisphere
 - b. Etiology
 - i. Thrombotic
 - ii. Embolic
 - iii. Hemorrhagic
 - c. Blood vessel
 - i. Anterior vs posterior circulation stroke
 - ii. ICA, ACA, MCA, PCA, vertebrabasilar stroke
 - d. Associated medical problems
- xi. Sites of intracranial aneurysm
 - a. Posterior inferior cerebellar artery
 - b. Basilar artery
 - c. Posterior communicating artery
 - d. Internal carotid artery
 - e. Anterior communicating artery
 - f. Bifurcation of MCA

- xii. Clinical findings
 - a. ICA, PCoA
 - i. 3rd nerve palsy
 - b. MCA
 - i. Contralateral hemiparesis (mainly face and hands, aphasia; contralateral visual neglect)
 - c. ACoA
 - i. Bilateral leg paresis; bilateral babinski
 - d. Apex of BA
 - i. Vertical gaze paresis; coma
 - e. Intracranial vertebral artery, PICA
 - i. Vertigo, elements of lateral medullary syndrome
- xiii. Intracerebral hemorrhage
 - a. MCA > Stem > lenticulostriate arteries > supplies basal nuclei
 - i. Lacunar type of stroke
- xiv. Clinical presentation by vessel involved
 - a. Anterior vs posterior

Basis	Anterior circulation strokes	Posterior circulation strokes
Involved arteries	Intrenal carotid artery, middle cerebral, anterior cerebral arteries anterior communication arteries	Vertebrobasilar artery branches
Incidence	More frequent	Less frequent
Site of ischemia	Cerebral hemispheres	Brainstem and cerebellum
Laterality	Usually unilateral, contralateral to the side of hemispheric lesion	Bilateral signs frequently present
Suggestive clinical feature	Cortical dysfunction like aphasia, apraxia, agnosia Monocular blindness (amaurosis fugax) Facial weakness sparing the frontalis and corrugators muscle contralateral to the side of the lesion Eye deviation directed away from the side of hemiparesis (or toward the side of hemispheric involvement)	Bulbar signs like dysarthria, dysphagia, dysphonia Dizziness, diplopia Facial paralysis involving all muscles of facial expression ipsilateral to the side of lesion Eye deviation directed towards the side of the hemiparesis ipsilateral to the side of the lesion
Prognosis	Neurologic recovery often less complete because the areas of ischemia is often extensive	More complete neurologic recovery for small infarct Large infarct are often fatal because of the vital centers

- xv. Review of cerebral circulation
 - a. ACA: Superior frontal, and superior parietal
 - b. MCA: Most of lateral part
 - c. PCA
 - d. Coronal section: Reference would always be the Sylvian fissure
- xvi. ACA vs MCA
 - a. ACA
 - i. LE > UE
 - ii. No facial palsy
 - iii. No aphasia

- b. MCA
 - i. UE weaker than LE
 - ii. May have facial palsy
 - iii. May have language impairment
- xvii. Summary presentation by artery
 - a. ICA – transient monocular blindness
 - b. ACA LE > UE weakness
 - c. MCA – aphasia
 - d. PCA peripheral > visual deficits
 - e. PCA central > thalamic syndrome
 - f. Vertebrobasilar artery stroke > cerebellar > cranial nerve, brainstem manifestations
- xviii. Brainstem syndromes
 - a. Sympathetic tract
 - i. T1 > preganglionic neuron > postganglionic neuron > carotid > n to sweat gland > nerve to dilator muscle
 - ii. Damage to descending sympathetic pathways to upper thoracic cord:
 - 1. Horner's syndrome
 - a. Ptosis (damage to involuntary eye opening: mueller's muscle)
 - b. Miosis: n to dilator muscle
 - c. Anhydrosis: n to sweat gland
 - iii. Medial alternative inferior hypoglossal hemiplegia
- xix. Cerebellar syndromes
 - a. PICA syndrome
 - i. Ipsilateral weak palate, horner's deacreaqs pain & temperature ipsilateral face, ipsilateral limb ataxia, decreased pain and temperature contralateral body
 - b. AICA syndrome
 - i. Ipsilaterally > deafness, facial weakness, decreased sensation face, limb ataxia
 - ii. Contralaterally > decrease pain and temperature in half of body
 - c. SCA syndrome
 - i. Nystagmus, ipsilateral deafness and limb ataxia, diplopia, contralateral decrease in all sensation to one half of the body
- xx. PCA syndromes
 - a. Dejerine-roussy sundrome
 - i. Thalamus/ (left pulvinar)
 - 1. Contralateral sensory loss to all modalities, severe dysesthesias on involved side (thalamic pain), vasomotor disturbance, transient collateral hemiparesis, and choeroathetoid or ballistic movements
 - b. Balint's syndrome
 - i. Bilateral occipital or parieto occipital infarctions
 - ii. Optic ataxia, psychic paralysis of fixation with inability to look to the peripheral field and disturbance of visual attention

- c. Thalamic syndrome
 - i. Postero ventral thalamic nucleus
 - ii. Sensory loss (all modalities), spontaneous pain and dysesthesias, choreaathetosis, intention tremor, spasm of hand, mild hemiparesis
 - d. Thalamo perforate syndrome
 - i. Crossed cerebellar ataxia with ipsilateral 3rd nerve palsy
 - e. Anton's syndrome
 - i. Bilateral occipital or occipitoparietal infarctions
 - ii. Cortical blindness often with denial or unawareness of the blindness
- xxi. Other syndromes
 - a. Gertsman syndrome
 - i. Dominant hemisphere parietal lesions
 - ii. Right-left confusion, finger agnosia, alexia/agraphia, acalculia
 - b. Frontal opercular syndrome (associated with Broca's)
 - i. Facial weakness with motor aphasia with or without arm weakness
 - c. Brachial syndrome
 - i. Partial syndrome due to embolic occlusion of a single branch
 - ii. Include hand or arm and hand weakness alone
 - d. Medial lenticulostriate artery infarction (artery of Heubner)
 - i. Positions of anterior limb of internal capsule
 - ii. More pronounced weakness of face and arm without sensory loss
- xxii. Basis of clinical presentation
 - a. Laterality
 - i. Right or left hemisphere
 - b. Etiology
 - i. Thrombotic
 - ii. Embolic
 - iii. Hemorrhagic
 - c. Blood vessel
 - i. Anterior vs posterior circulation stroke
 - ii. ICA, ACA, MCA, PCA cerebrobasilar stroke
 - d. Associated medical problems
- xxiii. Basic deficits in stroke
 - a. Cortical deficits
 - i. Speech vs language disorders
 - ii. Aphasia
 - iii. Parietal
 - iv. Agnosia
 - v. visuospatial

- xxiv. Clinical course (MCA ischemic stroke)
- 24Hours – complete flaccid hemiplegia areflexia
 - 2nd day – MSR – DTR return
 - 2nd week – spasticity
 - Limb synergy
 - Motor recovery

xxv. Brunnstroms

Stage	Stage	Tone	Manifestation
1	Flaccid	None	
2	Beginning spasticity	Minimal voluntary movement	Basic limb synergies or some of their components appear as associated reactions
3	Increases spasticity may be severe	Voluntary control of movement synergies	Full range of all synergy components does not necessarily develop
4	Spasticity begins to decline	Can master some movement combinations deviate from synergies	
5		Most difficult movement patterns learned	Basic limb synergies lose their dominance over motor acts
6	Spasticity disappears	Individual joint movement possible	

- xxvi. Poor prognosticating indicators
- Perceptual deficits
 - Prolonged flaccidity
 - Persistent areflexia
 - No motor recovery after 6 weeks
 - Severe spasticity
 - Bicortical stroke (bowel and bladder incontinence)
 - Multiple stroke
 - Older age
 - Severe memory problems
 - Inability to follow verbal and gestural commands

HGD

- i. Cognitive/ Perceptual/ speech language and communication
 - a. Piaget (4 stages, Piaget)
 - SPCF
 - G02711
 - "Go Two 7-11"
 - i. Sensorimotor Stage (0 – 2 years)
 - ii. Preoperational Stage (2 – 7 years)
 - iii. Concrete Operational Stage (7 – 11 years)
 - iv. Formal Operational Stage (11+ years)
- ii. Psychosocial
 - a. Erikson (Erikson)

					(4-6)	Preschool	"Ini"	Initiative vs. Guilt
					(7-12)	Elementary school	"Indu"	Industry vs. Inferiority
"Power of 3"	1	3	6	65				
	(0-1)	(2-3)	(4-6)	(35-65)	(13-19)	Serious college	"Iden"	Identity vs. Role Confusion
					(20-34)	Settling down	"Inti"	Intimacy vs. Isolation

HOPE WILL PURPOSE COMPETENCE (HWPC)

FIDELITY LOVE CARE WISDOM (FLCW)

- i. Trust vs Mistrust (0 -1 years)
- ii. Autonomy vs Shame and Doubt (2-3 years)
- iii. Initiative vs Guilt (4 – 6 years)
- iv. Industry vs Inferiority (7-12 years)
- v. Identity vs Role Confusion (13 – 19 years)
- vi. Intimacy vs Isolation (20-34 years)
- vii. Generativity vs Stagnation (35-65 years)
- viii. Integrity vs Despair (65+ years)

iii. Physical/ Motor Gross/ Fine Developmental milestones

Developmental Milestone Highlights

Month/s	Cognitive	Speech	Social	Gross Motor					
				Head	Supine	Prone	Sit/ Kneel	Stand	Fine Motor
1	Eye Tracking	"Coo"	Parent Identification	(+) Head lag	(+) Head lag Partial Segmental rolling	Lift head and look side to side		Automatic Stepping	(+) Palmar grasp Hand Fist
3	Visually detected reacting								Raking Full hand
5					Head lifts Full segmental rolling	Prone on Hands			
6		"Babble"	Stranger Anxiety	(-) Head lag	Sidelying to Prone	Wt. Shift posteriorly & Reach with hand	Sitting Supported		Holds Bottle, Raking with few fingers, Hand to hand transfer objects
9		"mama & papa" w/o sense	Playing					Cruising Crawling	Hand Gestures "High five"
10								Creeping	
11									Inferior pincer grasp Holds Crayon
12		"mama & papa" w/ sense	Parent Imitation				Sitting & throwing ball Half kneel to stand	Walk fast, side to side	Superior pincer grasp w/ Graded pressure
15								Creeping on stairs	Controlled release
18									Scribbles
24 (2 years)	2 step command	2 word phrase						2 step climb	Stack 6 Blocks Tricycle Single hop
36 (3 years)									Draw circle
48 (4 years)								Catch & throw standing	Draw square, cross, stick figure Use scissors
5 years									Draw Triangle Dress
6 years									Draw diamond Alphabets Tie shoe
7-8 years									Bicycle Use fork & knife

iv. Self-help/ Adaptive

a. Refers to how independently a person function in three main skill areas:

- i. Practical skills, such as personal hygiene and keeping a living area clean
- ii. Social skills, such as interpersonal communication and the ability to establish and maintain friendships
- iii. Conceptual skills, such as language and literacy, the ability to plan and organize and the ability to understand abstract concepts like time and money

b. Age in months, Self-Help skills

(Adapted from Cook, Klein, & Tessier (2008). *Adapting early childhood curricula for children with special needs and other sources*)

	24-36 months	36-48 months	48-60months
Feeding	Feeds self with little spilling Holds spoon independently	Can pour from a small pitcher Can spread soft substances with a knife	Can cut with a knife independently
Dressing	Undresses with no help Can take of coat and put on with assistance	Unties bows, unbuckles belt, unzips sippers Buttons and unbuttons large buttons	Can put shoes on right feet; tie shoes Knows clothing from front from back
Hygeine	Incomplete toothbrushing Can wash and dry hands with adult assistance	Can wash hands and body independently Wipes nose without being told	Brushes teeth independently
Toileting	Expresses verbal anticipation;control during the day; needs reminders	Control at night, may need help with wiping; goes to bathroom independently	Fully independent
Other	Can turn a door handle		

PNI

- i. Traumatic/ non-traumatic injury involving motor, sensory, or autonomic nerves.
 - a. Sensory
 - i. (-) phenomenon, numbness
 - b. (+) phenomenon, paresthesias (pins and needles), dysesthesias (continuous burning pain), hyperalgesia (decreased pain threshold)
 - c. Autonomic changes
 - i. Causalgia
 - 1. H – Hyperpathia
 - 2. A – Autonomic Phenomenon (E.g. Hyperhidrosis)
 - 3. T – Trophic skin changes (E.g. Hyperhidrosis, Hypertrichosis, Hirsutism (Increased hair growth))

ii. Types

- a. Traumatic (70%)
- b. NonTraumatic (30%)
 - i. Metabolic, DM (MC caused on NT)
 - 1. MC CN affected 3 & 6
 - 2. MC PN affected: Median nerve

Type I	Type II
IDDM Juvenile Onset MC hypoglycemia Diabetic Ketoacidosis (kussmaul's) Somogyi (Rebound hyperglycemia)	NIDDM Adult Onset MC

- ii. Infection: Leprosy (oldest infection known to man)/ Hansen's Disease
 - 1. Transmission: Droplet
 - 2. Bacteria: M. Leprae
 - 3. MC CN affected: 5, 7
 - 4. MC PN: Ulnar nerve
 - 5. Leonine Fascie
- iii. Toxic neuropathy
 - 1. MC cause: Lead
 - 2. MC PN affected: Radial nerve
 - 3. Develop Bilateral wrist drop
 - 4. Other substances
 - a. Mercury
 - b. Gold
 - c. Magnesium
 - d. Arsenic
 - e. Copper

- iv. HMSN (Hereditary motor & sensory neuropathy)
 1. Hallmark: Peroneal/ distal leg atrophy (Inverted Champaign bottle sign/ Stork leg)
 2. MC foot deformity: Pes Cavus
 3. CMTD other name PMA (Peroneal muscular atrophy)
 4. Types
 - a. CMTD I– (MC)
 - i. Juvenile onset
 - ii. Hypertrophic type
 - iii. (+) onion bulbs (hypertrophy of nerves)
 - iv. Decreased NCV
 - b. CMTD II
 - i. Adult onset
 - ii. Neuronal/ axonal type
 - iii. Normal NCV
 - c. Landouzy – Dejerine/ Dejerine Sottas
 - d. Refsums
 - e. Spinocerebellar Deg.
 - f. Optic Atrophy
 - g. Retinitis Pigmentosa
- v. Vitamin Deficiency
 1. Vitamin B- complex (MC)
 - a. Remyelination and regeneration of the nerves

Vitamins

H₂O – B & C

Fat – A, D, E, K

A – Retinol (Retin –A)

Nyclopia (Night blindedness)

D – Calciferol

Osteomalacia (rickets)

E – Tocopherol

K – Menadione (hemorrhagic manifestations, for clotting factors)

C – Ascorbic Acid

Scurvy

Most important for wound healing

B – Complex

B₁ – Thiamine

Beri-Beri

B₂ - Riboflavin

B₆ – Pyridoxine

B₁₂ – Cyanocobalamin

Pernicious anemia

- vi. Idiopathic
 - 1. Self limiting
 - 2. Brachial plexitis/ Parsonage turner syndrome/ Brachial plexus neuritis/ Neuropathic amyotrophy
 - 3. Tic Doloieux/ Trigeminal Neuralgia
 - 4. Bell's Palsy
 - a. Crocodile tears:
 - i. Patient chews and tears at the same time
 - ii. CN7 neuronotmesis, aberrant regeneration goes to the lacrimal gland
 - 5. GBS
- iii. Seddon's Classification

Neurolemma – found in the PNS only, allows recovery of nerve (Stronger than the nerve)

 - a. Neuropraxia – MC (+) conduction block, impingement
 - b. Axonotmesis – Wallerian degeneration (5-7 days), complete reinnervation, (axon is cut)
 - i. Axon regeneration: 1-2mm/ day
 - ii. Tinel's test: Determines length of regeneration
 - 1. Last innervated by:
 - a. Radial: Extensor Pollicis
 - b. Ulnar: Adductor Pollicis
 - i. Determine severity: Froment's Sign
 - c. Neuronotmesis – Least common, Most severe, Aberrant regeneration (Naligaw)
- iv. Sunderland Classification

Endoneurium: covering of nerves

Perineurium: covering of bundles of nerves

Epineurium: covering of nerve bundles

Neuropraxia

 - a. 1st degree - (-) Axon is cut

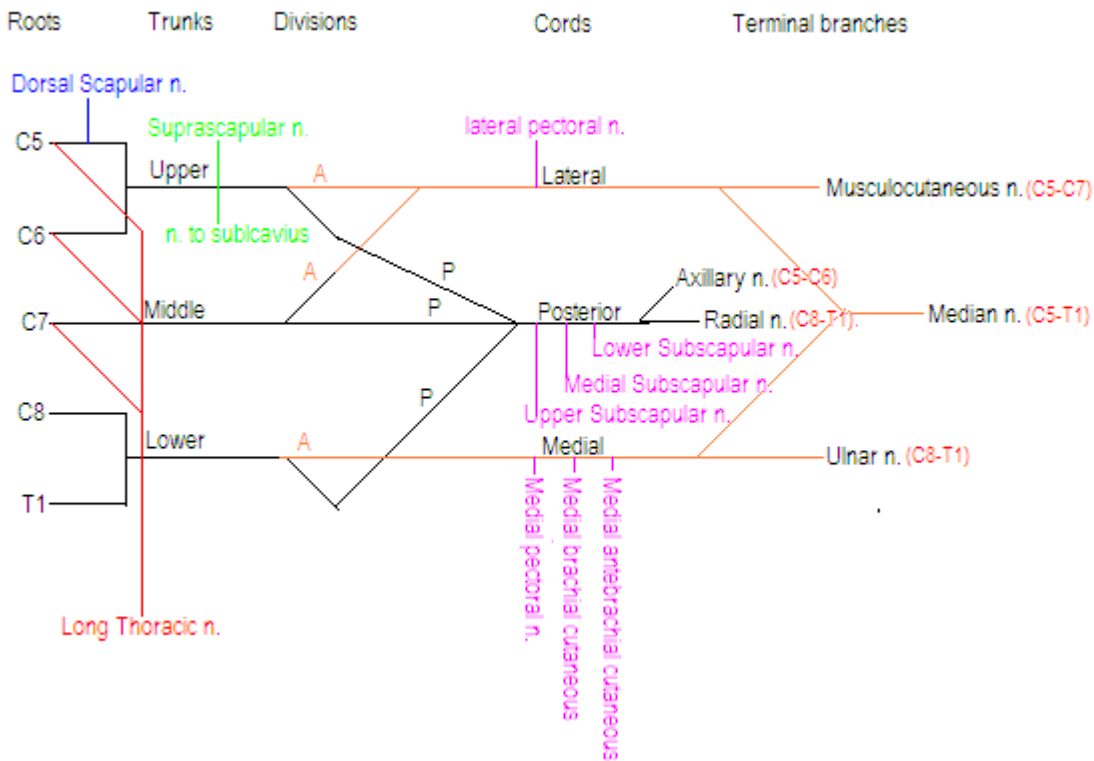
Axonotmesis

 - b. 2nd degree – Axon is cut

Neuronotmesis

 - c. 3rd degree – Axon and Endo
 - d. 4th degree - Axon + Endo + Peri
 - e. 5th degree Axon + Endo + Peri + Epi

v. Brachial plexus



- C5-T1 Ventral rami
- Total number of branches: 16
- Terminal branches: 5
- Roots:
 - Lateral cord – C5, C6, C7
 - Medial Cord – C8, T1
 - Posterior Cord
- Branches
 - Median nerve (C5-T1)
 - Volar: Rule of 3 ½
 - And tips of volar & dorsal of fingers
 - Ulnar nerve (C8-T1)
 - Volar & Dorsal Rule of 1 ½
 - Radial (C8-T1)
 - “LAHAT NG EXTENSORS , RADIAL”
 - Axillary nerve (C5-C6) (just a branch form radial nerve that's why its C5-C6 only)

- f. Intrinsic
- i. Median
 1. Abductor pollicis brevis
 2. Flexor pollicis brevis
 3. Opponens pollicis
 4. ½ of lumbricals
- g. Other names
- i. Long thoracic Nerve: Nerve of Bell
 - ii. Axillary nerve: Circumflex nerve
 - iii. Radial nerve: Musculospiral nerve
 - iv. Thoracodorsal: Medial subscapular nerve
 - v. Musculocutaneous nerve: Lateral Cutaneous nerve
- h. Dual innervations
- i. UE: (6)
 1. Pectoralis Major: Lateral and medial pectoral nerve
 2. Subscapularis: Upper and lower subscapular nerve
 3. Brachialis: Musculocutaneous & Radial nerve
 4. FDP, FPB & Lumbrical : Median and ulnar nerve
 - ii. LE: (2)
 1. Adductor magnus: Obturator & Sciatic nerve
 2. Pectineus: Obturator and Femoral Nerve
- i. Deformities
- i. Classic claw hand
 1. Ulnar and median nerve
 - ii. Partial claw hand
 1. Ulnar nerve
 2. Active papal: Median (moving)
 3. Passive papal: Ulnar (not moving)
 - iii. Precision: Median, Power: Ulnar
- vi. Root level entrapment
- a. Plexus : Traction

Upper Plexy Palsy		Lower Plexy Palsy
Erb-Duchenne Palsy	AKA	Klumpke Palsy
C5-C6	Affected Roots	C8-T1
Downward traction	Injury	Upward traction
Head first	Delivery	Breach
Shoulder abductors & External rotators Elbow flexors Supinators Wrist extensors	Weak	Shoulder Adductors & Internal rotators Elbow extensors Pronators Wrist flexors
Policeman's tip/ waiters tip/ porter's tip	Presentation	Classic claw hand Horner's Syndrome (T1)

vii. Differential Diagnosis

a. TOS (lower trunk, MC)

- i. Goes through the thoracic outlet at the apex of axilla
- ii. Pain in the fifth digit
- iii. (-) radial pulse
- iv. Subclavian artery > outer border of the first rib > axillary > brachial a. > ulnar & radial artery
 1. R – Roo's
 2. A – Allen's
 3. H - Halstead
 4. A – Adson's (most reliable)

b. Cervical Rib Syndrome

- i. Mx: Stretching
- ii. Special test: Costoclavicular test/ Military brace test

c. Scalenus Anticus Syndrome

- i. Mx: Stretching
- ii. E.g. COPD > difficulty breathing with use of accessory muscle of respiration > hypertrophy of scalene muscles entraps BP
- iii. Action: Accessory muscle for breathing
 1. Anterior & Medius: Insert first rib
 2. Posterior: Insert 2nd rib

d. Pancoast Syndrome

- i. Tumor at the apical aspect of the lung
- ii. Root's affected C8-T1
- iii. Can compress the brachial plexus

e. Winging/ Tipping

	Medial Winging	Lateral Winging	Posterior winging
Aka	Open book paralysis/ Active/dynamic winging of the scapula	Sliding door paralysis	Posterior Tipping
Affected	Serratus Anterior (Long Thoracic Nerve/ Radical mastectomy)	Trapezius CN 11/ Radical Neck Dissection	Levator Scap Rhomboids (Dorsal scapular nerve)
Actions	Upward rotation Protraction	Upward rotation retraction	Scapular downward Rotation and retraction

Common site of tumor Upper outer quadrant for Pancoast

viii. Radial nerve entrapment

Level	Axillary	Spiral Groove (MC)	Elbow	Wrist
Nerve	Radial	Radial	PIN	Superficial branch
Cause	Crutch palsy Anterior Shoulder D/L Humeral neck fracture	Honeymoon palsy Saturday Night palsy Bridegroom's palsy Sleep palsy	Radial tunnel Syndrome Arcade of Frohse PIN syndrome Resistant Tennis elbow Supinator Syndrome	Tight wrist band/ watch Cherlagia parasthetica Wartenberg disease Handcuff neuropathy DDx: DQT
Motor loss	Deltoids Teres Minor	Below Spiral groove	Brachioradialis ECRL	Sensory only
Spared	None	Triceps	EDC, EIP, EPC, EPL, EDM	All
Wrist drop	(+)	(+)/ weakest	(-)	(-)

Cubital fossa: Median nerve, brachial artery, biceps tendon, Radial nerve and Deep branch (medial to lateral)

Radial affection: Lead Toxic neuropathy

ix. Ulnar nerve entrapment

Level	Elbow (MC)	Wrist
Nerve	Ulnar	Ulnar
Cause	Cubital tunnel syndrome (acute onset) Tardy Ulnar palsy (chronic onset) 2° to C. Valgus (10-15 years of rep. stress) <i>Ligament: Osborne fascia</i>	Guyon's Canal/ PISO Hamate canal (Ulnar nerve/ artery) Stick palsy Biker's palsy/ handle bar Bowler's thumb Guttering
Spared	None	FCU FDP (Ulnar 1 ½)
Motor loss	All below lesion	Hypothenar Interossei
Special test	(+) Froment's test (+) Elbow flexion test	(+) Froment's test

Ulnar affection: Hansen's Disease/ Leprosy

x. Median nerve

Level	Supracondylar	Pronator Teres	Forearm	Wrist (MC)
Nerve	Median	Median	AIN	Median
Cause	Supracondylar syndrome Ligament: Struther's (-) Radial pulse	Pronator Teres syndrome	AIN syndrome Kiloh Nevin Syndrome Impinging Ms: Gantzer Ms	Ape/ Simian hand
Motor loss	All below lesion	All below lesion	PQ, FPL, FDP	AbPB FPB OP
Spared	None	Pronator teres	All above lesion	All above lesion
Special Test	All	Resis pronation	(-) OK sign/ Kiloh Nevin sign (+) Pinch grip test	(+) Flick sign Phalens/ Reverse Most reliable: NCV Carpal compression

Median affection: CTS (MC focal neuropathy), DM, typist, RA, Gout, pianist, guitarist

LE

- xi. Nerve roots
 - a. LFCN (L2, L3) – Meralgia paresthetica
 - i. Causes:
 1. P – Prolonged Sitting
 2. E – Extremely tight garments
 3. T – Tight Corset
 4. S – Sudden Hyperextension
 - ii. Management
 1. Lose weight
 - iii. Avoidance of causes
 - b. SGN (L4, L5, S1) –G. Med - Trendelenburg
 - c. IGN (L5, S1, S2) – G max – backward lurch
 - d. FN (L2, L3, L4) – Quads – forward lurch
 - e. ON (L2, L3, L4) – Adductors
 - f. SN (L4, L5, S1, S2, S3) – Hams

Nerve	Sciatic (MC)	Common Peroneal(90%)	Tibial (10%)	Deep Peroneal
Causes	Piriformis syndrome Fat wallet syndrome Injection palsy	Crossed leg palsy Fx head of fibula PTB Cast Bed-ridden patient c legs ER	Baker's cyst Popliteal bursitis Wheelchair too long seat depth Tarsal tunnel syndrome (<i>Lancinate ligament</i> stretch by pronation)	Posterior compartment (MC)
Presentation	Spindle leg atrophy		(-) Push off Decreased step length	Pulselessness, Pallor, Pain, Paresthesia, Paralysis

Nerve	Lateral plantar	Medial plantar
Cause	Interdigital: Morton's Neuroma (Bet. 3 rd & 4 th)	Jogger's Foot

- xii. Fracture related:
 - a. Suprascapular notch fracture: Suprascapular nerve
 - b. Humeral Neck fracture/ Surgical neck fracture: Axillary nerve
 - c. Fracture on spiral groove/ mid shaft: Radial nerve
 - d. Supracondylar fracture: Cubitus varus: Median nerve, Cubitus valgus (traction): Ulnar nerve
 - e. Scaphoid Fracture: Radial nerve
 - f. Lunate fracture/ dislocation: Median nerve
 - g. Hook of Hamate: Ulnar nerve

GMSC

- i. Immune system
 - a. 2 structures
 - i. Central immune Structure: Immune cell production from the bone marrow and thymus
 - ii. Peripheral immune structure: immune cell production from the spleen and lymphatic system
 - b. WBC's
 - i. Neutrophils – First line of defense
 - ii. Monocytes – Slow, but arrive in larger numbers
 - iii. Basophils – allergic reaction, releases histamine (vasodilator)
 - iv. Eosinophils – allergic reaction, releases, histaminases (break down of histamine)
 - v. Lymphocytes
 - 1. B – cells, kills bacteria
 - 2. T- cells, Virus and fungi
 - a. Cytotoxic
 - b. Helper
 - c. Memory
- ii. AIDS
 - a. Virus: HIV
 - b. Retrovirus: RNA code (DNA code), releases reverse transcriptase (Viral DNA code)
 - c. Pathophysiology: reduction of CD4 helper T-cells
 - i. T cells: Helper T cells are targeted first
 - d. Opportunistic infection:
 - i. Pneumocystis Carini Pneumonia (MC infection)
 - ii. Herpes – Simplex 1 & 2
 - iii. Cytomegalovirus
 - iv. Toxoplasmosis
 - e. Mode of transmission:
 - i. Body and blood fluids
 - ii. Maternal- Fetal
 - iii. High risk of transmission:
 - 1. Contaminated needles
 - 2. Unprotected sexual intercourse
 - iv. Low risk of transmission
 - 1. Contact: Kissing
 - v. No risk
 - 1. Hugging

- f. Laboratory evidence
 - i. ELISA (Enzyme Linked ImmunoSorbent Assay)
 - 1. (+) 30-50K HIV copies
 - ii. Absolute CD4 count
 - 1. (n) – 800-1200/mL
 - 2. (+) HIV (+) 200-500/mL
- g. Medical Intervention
 - i. NRTI (Nucleoside Reverse Transcriptase Inhibitor)
 - ii. Azidothymidine
- iii. Chronic fatigue syndrome
 - a. Disabling fatigue with other various complain
 - b. Etiology: Unknown
 - c. Female > male 20-30 years old
 - d. Diagnostic criteria (2 major)
 - i. Relapsing fatigue > 6 months, Exclusion of a chronic disease
 - ii. 6-8 symptoms criteria
 - 1. Fatigue
 - 2. Low-grade fever
 - 3. Sore throat
 - 4. Lymph node pain/ tender
 - 5. Muscle weakness
 - 6. Muscle discomfort
 - 7. Sleep disturbances
 - 8. Head ache
 - 9. Cognitive impairment
 - 10. Arthralgias
 - e. C/i: Bedrest
 - f. Management: Light to moderate aerobic exercises
- iv. Hepatitis
 - a. Inflammation of liver
 - b. Types
 - i. A – Hepa A Virus (HAV), contaminated food and water (Fecal –Oral route)
 - ii. B – HBV (Serum hepa) Body & blood fluids
 - iii. C – HCV – body and & blood fluids (liver cirrhosis, liver degeneration)
 - c. Phase:
 - i. Pre Icteric Phase
 - 1. Low grade fever
 - 2. Nausea/ vomiting
 - ii. Icteric Phase
 - 1. High grade fever
 - 2. (+) Jaundice

- v. Cancer
 - a. Definition: Rapid proliferation of anaplastic cells
 - b. Multiple risk factor:
 - i. Carcinogen
 - ii. Hereditary
 - iii. Diet – high in fat/ cholesterol, insufficient vitamin A, C, E
 - iv. Psychological factors: Chronic Stress
 - c. Early warning signs:
 - i. Bleeding: cancer is well established in the body 2° to ulceration on the central part of the tumor
 - ii. Tumor: Abnormal growth of tissue that already competes with a vital blood supply
 - iii. Lump: (+) 1 months in more than 1 location (Red flag sign)
 - iv. Pain: Sign that cancer is growing
 - v. Proximal muscle weakness
 - vi. Absent/ Diminished DTR's (can inhibit reflex arc)
 - vii. Integumentary lesions (E.g. Moles > 6mm)
 1. Alopecia Neoplastica: Hair loss 2° to leukemia
- vi. Types of cancers
 - a. Carcinoma – tumor is originating from epithelial tissues (skin, breast, lungs, colon, stomach)
 - i. Skin cancer (MC)
 1. Basal cell carcinoma: involves the bottom layer of the epidermis (hair bearing areas that are exposed to the sun, E.g. face, arms)
 2. Squamous cell carcinoma: involves the top layer of the epidermis
 3. Melanoma:
 - a. Most serious skin cancer
 - b. Arises from the pigmented cells in the skin (Melanocytes)
 - c. Primary sign: Changing mole
 - b. Sarcoma
 - i. Arising from mesodermal – connective tissue
 - ii. E.g. bone, muscle, fasts cartilage
 - iii. Types
 1. Soft-tissue sarcoma:
 - a. Caused by clorophenol (wood preservative)
 - b. Do not cause signs and symptoms (soft tissue elastic, allows the tumor to grow)
 - c. MC site: lower extremity >trunk > retroperitoneal
 2. Bone tumor
 - a. Osteosarcoma (Tumor from osteoblast cells)
 - i. Males > Female (between 10 & 25 years)
 - ii. MC site: (knee joint)
 1. Lower end femur
 2. Upper end of tibia and fibula
 3. Upper end of humerus
 - iii. Manifestation (knee pain)

iv. Types

1. Ewing's sarcoma

- a. Defect gene chromosome #22)
- b. Male>female 15 & 16 y/o
- c. X-ray: (+) Onion skin (layers of periosteum)
- d. (+) intramedullary tumor > erodes to bone cortex

b. Chondrosarcoma

- i. Rises from cartilage cells
- ii. MC malignant tumor in the sternum and scapula
- iii. MC in adults
- iv. 40 y/o

c. Cancer of blood and lymphatic system

i. Tumor originating from blood and blood forming organs

ii. Types

1. Leukemia: originating from the bone marrow

- a. Release immature WBC (decrease mature WBC, prone to infections)
- b. Decrease blood cell production
- c. Decrease platelets (prone to hemorrhages)
- d. Manifestation:
 - i. Anemia

2. Hodgkin's Disease

- a. Painless enlargement of lymph nodes
- b. Caused by: Epstein barr virus

3. Non-Hodgkins lymphoma

- a. Painless enlargement of lymph nodes
- b. Acquired or congenital (High levels of radiation)

d. Endocrine pathophysiology

i. Pituitary glands (lies in the sella turcica)

1. Diabetes Insipidus

- a. Lack of secretion of ADH
- b. Reabsorption of H₂O
- c. Manifestations: Polyuria, Polydipsia, Dehydrated

2. SIADH, Syndrome of Inappropriate ADH

- a. Excessive secretion of ADH
- b. (+) edema in the body

3. Acromegaly

- a. Increase human growth hormone at adult life

4. Gigantism: increased human growth hormone at childhood

5. Dwarfism: decreased human growth hormone at childhood

- ii. Thyroid glands (anterior to larynx)
 - 1. Goiter: Enlargement of thyroid gland, deficiency in iodine (Salt) in the diet
 - 2. Thyroiditis: Inflammation of the thyroid gland
 - a. MC form: Hashimoto's thyroiditis
- iii. Hyper function of thyroid, Graves Disease
 - 1. Hyperthyroidism
 - 2. Exophthalmos (+) edema behind the eyes
- iv. Hypofunction of thyroid gland
 - 1. Cretinism, decrease thyroid gland hormone at birth (+) Mental Retardation
 - 2. Myxedema
 - a. Hallmark: Edema
 - b. Decrease thyroid hormone at adult life
 - c. Manifestation
 - i. (+) lethargic
 - ii. Weight gain
- v. Adrenal gland

Addison's Disease	Cushing's
Decrease Aldosterone	Hyperfunction of Adrenal gland/ Cortex
(+) Bronze (Stimulate, MSH)	Increase Cortisol
Sun Tan (JFK)\	(+) Buffalo's Hump
S/Sx	Pendulum Abdomen
Palpitation	(+) Striae
Dehydration	Moon Facie
Decreased Cardiac output	Moody

- vi. Parathyroid gland

Hypoparathyroidism	Hyperparathyroidism
Decrease blood calcium levels (Hypocalcemia)	Excessive osteoclastic activity
(+) Neuromuscular irritability	Increase Blood Calcium levels (hypercalcium)
Tetany	Prone to:
Twitches	Fracture
	Osteoporosis (brittle bones)

- vii. Hemophilia
 - 1. Bleeding disorder
 - 2. X - linked disorder
 - 3. Male = manifest XY (Y does not clotting factor)
 - 4. Female = carrier XX (2nd X produces clotting factor)

PHARMACOLOGY

- i. Pharmacology
 - a. Study of substances that interact with the living system through chemical processes
 - b. Pharmacokinetics: action of the body on the drugs
 - i. Govern: absorptions, distribution, metabolism, and elimination
 - c. Pharmacodynamics: action of the drugs in the body
 - d. Absorption
 - i. Oral – Highly/ Fairly lipid soluble
 - ii. Sublingual – Under the tongue E.g. Oral Nitrates,
 - iii. Buccal – between gums and cheeks
 - iv. Inhalation
 1. Bronchodilator (Side-effects are minimized)
 - v. Intramuscular
 1. Deltoids and V. Lateralis, Gluteus Maximus (L5-S2) (slower absorptions)
 - vi. Intravenous
 1. Fastest in absorption (into systemic circulation)
 2. Controlled by drip rate
 - vii. Topical/ Transdermal
 1. Patch/ ointment
 2. Absorption depends: Skin hydration
- ii. Distribution
 - a. Drug molecule bind to protein (Albumin, synthesized in the liver)
 - i. E.g. Warfarin (Coumadin) 99% protein bound
- iii. Metabolism
 - a. First Pass effect
 - i. Liver
 1. Enzymes:
 - a. decrease effect of the drug (Inderal, Beta blockers > 70%)
 - b. Inactive > active form (prodrug E.g. Inactive cortisol)
- iv. Elimination
 - a. Main site: Kidney
- v. Pharmacodynamics
 - a. Actions of drugs in the body
 - b. 2 types drug receptors
 - i. Agonist drug: bind to specific receptors + stimulate/ activate
 - ii. Antagonist drug: deactivates/ blocks
- vi. Anti inflammatory
 - a. Inhibit cyclooxygenase >prostaglandins
 - i. COX1: protective lining
 - ii. COX 2: pain and inflammation

- b. NSAIDS
 - i. Non-selective NSAIDS, inhibits both COX1 and COX2
 - 1. E.g. Aspirin, Ibuprofen, Ketoprofen, Naproxen(flanax), Mefenamic Acid
 - 2. Side-effect: Ulceration, GI bleeding
 - ii. Selective NSAIDS
 - 1. Inhibit COX2 (COX2 NSAID)
 - a. E.g. Celecoxibs (Celebrex, Arcoxia, Dextra, Etodolac, Rofecoxib, Viox)
 - b. Side- Effect: Cardiac Arrest
- c. Glucocorticoid/Glucocorticoid steroids
 - i. Stimulates leukocytes and macrophages
 - ii. E.g. Prednisone, Prednisolone, Cortisone
 - iii. Endogenous steroid: aldosterone
 - iv. Inhibit's breaking down of Vitamin -D
 - v. Side-effect: Fracture, infection, cataracts
- d. DMARDS
 - i. Disease modifying rheumatic drugs
 - ii. Prevent chronic inflammation and prevent joint destruction
 - iii. 6 weeks to 6 months
 - iv. E.g. **Methotrexate**, Leflunomide (Arava), Anakinra (Kineret)
- e. Anti-Osteoporosis
 - i. Non pharmacological agents
 - 1. Calcium and vitamin D
 - 2. Weight bearing E.g. Walking
 - 3. Decrease intake of alcohol and caffeine
 - ii. Bisphosphonates – decrease osteoclastic activities
 - 1. Alendronate (Fosamax) (becareful with stretching, prone to fracture)
 - 2. Residronate (Actonel)
 - 3. Etidronate (didronel) (HO, Paget's)
 - 4. Teriparatide (Forteo)
 - 5. Calcitonin (Hormone) (nasal spray)
- f. Opioid-Narcotics (powerful analgesia in the body)
 - i. Chronic Pain
 - ii. Endogenous opioids: Endorphins (Suppress all pain) (Agonist)
 - iii. E.g. Vicodin, Codeine, Demerol, Morphine
 - iv. Side effects: euphoria, dysphoria, constipation(**Decrease in GI tract, MC side effect**)
- g. Non-opioid-narcotics
 - i. Mild to moderate pain
 - ii. Prodrug: Phenacetin> Acetaminophen (Tylenol)
 - iii. Side effect: liver damage

- h. Muscle relaxants
 - i. Centrally acting muscle relaxant
 - 1. Diazepam (Valium)
 - 2. Baclofen, inhibit neuronal activity on spinal cord (Lioresal)
 - ii. Peripheral acting muscle
 - 1. Inhibit Ca influx
 - 2. E.g. Dantrolene (Dantrium)
 - iii. MC side effect: Sedation
- i. Antihypertensive
 - i. Diuretics “water pills”
 - 1. Inhibit water reabsorption
 - 2. Ultimate result: Decrease in peripheral resistance
 - 3. E.g. Thiazide, furosamide (Lasix), bumex,
 - ii. Beta-blockers
 - 1. Beta receptors
 - a. 1 – heart: increase in contraction
 - b. 2 – lungs: Relax bronchioles
 - 2. Non selective Beta Blocker
 - a. Propranolol (Inderal)
 - b. Nadulol (Corgard)
 - c. Penbutolol (Levitol)
 - d. Cannot be given to all kinds of COPD
 - 3. Selective Beta Blocker
 - a. Inhibit Beta 1 receptor
 - b. Atenolol
 - c. Lopressor
 - iii. Calcium channel blocker
 - 1. Inhibits Ca^{2+} influx directly on the myocardium
 - 2. “dipine”
 - 3. E.g. Amlodipine, Felodipine, Nisodipine, Nifedipine (adalat) Verapamil
 - 4. Side effect: Dry cough
 - iv. Alpha blockers
 - 1. Alpha receptors:
 - a. Alpha 1: constrictor
 - b. Alpha 2: vasodilator
 - 2. E.g. zocin
 - 3. Terazosin, Dexazosin, Prazosin
 - v. Side effect: dizziness and hypotension
- j. Heart Failure
 - i. ACE inhibitor
 - 1. Convert Angiotensin I to Angiotensin II
 - 2. Degrade Bradykinin (vasodilator)
 - 3. E.g. Lisinopril (Accupril), Captopril (Capoten), Fosinopril (Monopril)

- ii. Digoxin (Lanoxin)
 - 1. Mainstay drug for CHF
 - 2. Increase contraction of the heart
 - 3. E.g. Atropine, Digitalis
 - 4. Side effect: Confusion (MC), hallucination, cardiac arrhythmia
- iii. Quinidine
 - 1. Slows down hearts contraction
- k. Antiangina
 - i. Vasodilator
 - ii. Nitrates: nitroglycerin, Nitrostat, Nitrobid
 - iii. Isorsorbide Dinitrate: Isordil, Isorbid
- l. Anticoagulant:
 - i. Warfarin (Coumadin)
 - 1. Antagonist to vitamin K
 - 2. 3-6 days
 - 3. Oral
 - ii. Heparin (Fragmin)
 - 1. Clotting factors
- m. Anti-platelet
 - i. Inhibit platelet aggregation
 - ii. E.g. Aspirin, clopidogrel (Plavix), Ticlopidine (Ticlid)
 - iii. MC: Hematuria, echymosis, bleeding

PHYSICAL AGENTS

- i. Physical Agents
 - a. Intro and terms
 - b. Heat modalities
 - i. HMP
 - ii. PWB
 - iii. Hydro
 - c. Cryomodalities
 - i. Cold pack
 - ii. Ice massage
 - iii. Vaso sprays
 - iv. Contrast bath
 - d. Mech agents
 - i. ILT & ICT
 - ii. ICU
 - iii. CPM
 - iv. TT
 - e. Massage
- ii. Intro & terms
 - a. Heat = increase in temperature
 - i. Physiologic effects of heat (Therapeutic range: 40-44° C/ 104-111°F)
 - 1. M – Muscle relaxation:
 - a. Inhibition of glycolytic process (Type II)
E.g. condition Multiple sclerosis, exacerbation of S/Sx (Uthoff's Phenomenon)
 - b. Inhibition of muscle spindles
 - i. Alpha – Rigidity
 - ii. Gamma – Spasticity (Affected by heat B > C > A)
 - 2. I – Increase vasodilation
 - a. Hypothalamus
 - i. Heat production: Posterior
 - ii. Heat removal: Anterior (Vasodilation)
 - b. Hormones
 - i. Vasodilation
 - 1. B – Bradykinin (most potent)
 - 2. H – Histamine
 - 3. L – Lactic Acid
 - 4. K – Kinin like proteins
 - 5. A – Acetylcholine

- ii. Vasoconstrictors
 1. C - Catecholamine
 2. E - Epinephrine
 3. S - Serotonin
 4. N - Norepinephrine
 5. A – Angiotension II (most potent)
 3. A – Acceleration of metabolism
 - a. Vant Hoff's Law, for every 10° C increase in temperature there is 2-3x in metabolism
 - b. For every 10° F increase = 10bpm
 4. H – Hyperemia
 - a. Rubor: Redness in the presence of inflammation
 - b. Hyperemia: Redness when heat is applied
 - c. Burn (non blanchable erythema, (n) = < 3 secs)
 - i. Fair skin: becomes pink, becomes red
 - ii. Dark pigmentation: becomes white to whiter
 5. I – Increase cell membrane permeability
 - a. ↑ interstitial fluid → Edema
 6. A – Analgesia:
 - a. ↑ circulation = wash away pain metabolites
 - b. Counter – irritation = “masking effect”
 7. I – Increase collagen extensibility: Stretching (Can relieve wrinkles)
- b. Cold = absence of heat (Therapeutic range: 0-10°)
- i. Intense cold → burning → aching → numbness
 - ii. Lewis – Hunting Response: Initial constriction followed by vasodilation
 - iii. Decrease pain: by numbness
 - iv. Increase pain threshold
 - v. Vasoconstriction: Best modality for swelling, reduce lactic acid formation, cramps (Icing + stretching), DOMS
 - vi. *Increases spasticity(to induce fatigue)*
 - vii. Techniques
 1. Slow icing (Best)
 2. Fast icing
- c. Transfer hot and cold
- i. C – Conduction: Transfer heat via direct contact
 - ii. C – Convection: Moving fluids
 1. Air: Fluido therapy
 2. H₂O: Hydrotherapy
 - iii. C – Conversion: Transformation of energy into heat E.g. Diathermies, US
 - iv. R – Radiation, Electromagnetic travel in space E.g. IRR, UVR, LASER
 - v. E – Evaporation: Drying up of liquids E.g. Vapocoolant sprays

iii. Heating Modalities

a. HMP (Hot moist pack, HydroCollator Pack)

- i. Silica Gel = 2-3x (Can contain: Hot and cold)
- ii. 6-8 layers of towels
- iii. Sizes:
 1. Standard: 10 x 12 " (MC)
 2. Half: 5x12"
 3. Cervical: 24x 6.5"
 4. Lumbar: 24 x 15"
 5. Jumbo: 24 x 30"
- iv. Treatment temperature: 40-44°C
- v. Water: Wadsworth = 65-90°C (IER, PEAT, GILES = 70-80° C), (Braddom 76-80°)
- vi. Towel: (Turkish/ Woolen)
 1. Bony areas: ↑ layers
 2. ↓ Temperature: ↓ layers

Advantages	Disadvantages
Cheap Available Comfortable	Heavy ↑ risk burn (high thermal conductivity) Contaminate Seepage

vii. Contraindication:

1. Acute inflammation
2. Osteoporosis
3. Open wounds
4. DVT; Cancer
5. Heat hypersensitivity

b. PWB

- i. 1:6/7 oil: wax
- ii. Treatment temperature: 40-44°C
- iii. Melting point: 50-54° C/ 118-130°F
- iv. Self sterilizing temperature: 60-65°C/ 165-180°F
- v. Clean/ Change: Presence of sediments
- vi. Advantage:

Advantage	Disadvantage
Distal Extremities Recyclable Putty Ex. (E.g. CTS & DQT) Low risk burn	Messy Easily Contaminated Prepare

vii. Technique

1. Brushing: Least effective; used in pedia
2. Pouring
3. Dipping MC used (6-10)
4. Immersion: Most effective
5. Combination
6. Drafting/ Pancake

viii. C/i

1. Open and freshly healed wounds
2. Thin skin
3. Tuberculosis
4. Acute Dermatological conditions E.g. Eczema

c. Hydro:

i. Hydrology: Study of water in its different states (E.g. Solid, Liquid, Gas)

ii. Medical hydrology: in medical field

iii. Water: pH7.0

1. Solid → ice → cryotherapy
2. Liquid → water → hydrotherapy
3. Gas → Vapor → Vapocoolant

iv. Principles

1. Archimedes Principle

a. The amount of water displaced is the amount of weight reduced from a submerged body.

b. Proponents:

i. DeWeight (DeLisa), amount of weight that is lost level:

1. **Symphysis pubis** 40%
2. **Umbilicus** 50%
3. **Xiphoid process** 60%

ii. Kisner/ weight bearing level:

1. **ASIS/ Hip** 50%
2. **Xiphoid process** 33%
3. **C7/ Neck** 10%

2. Bouyancy Principle

a. Upward force/ Float

b. Increase surface area = increase bouyancy

c. Exercise

i. B. Assisted: Movement up (Distraction) = inhibition

ii. B. Resisted: Movement Down (Compression) = Facilitation

1. Wt. Bearing E.G S/P knee arthroplasty

iii. B. Supported: Movement Horizontal

1. Indication: Post Ligamentous Reconstruction, Arthritic conditions

3. Hydrostatic pressure
 - a. The deeper: ↑ pressure
 - b. Pressure is equal on sides
 - c. Most affected: Feet
 - d. Best for: Edema by increasing venous return, increasing circulation, increase lymph flow
4. Hydrodynamic pressure
 - a. Increase pressure/ resistance = ↑ surface area, ↑ speed, ↑ viscosity, ↓ Temperature
5. Cohesion: water will attract other water molecules
6. Adhesion: water will stick to others
7. Surface tension: water will resist anything on surface (wt. will break surface tension).
8. Turbulence: Disturbance in the water
- v. Scenarios
 1. Seizure/ Epileptic attack: Bring to the center and support the head & maintain the airway clear.
 2. MI: Bring to the side, and active EMS
 - a. Check VS
 3. (+) HIV, Join in water without open wounds
 4. Multiple sclerosis: 0-4° C
 5. Bowel and bladder incontinence: C/i
 6. Chronic conditions: less than core body temperature
 7. Painful conditions: Hot and very hot
 8. General exercise: Tepid (22-33.5°C)
 9. Full body immersion: Less than 38°C or <100°F
 10. PVD, cardiac conditions, open wounds <37°C
 11. Debridement: Povidone Iodine, NSS, Bleack , Hydrogen peroxide
- vi. Mnemonics temperature

	Celcius	Farenheit
Very Cold	1-13	35-55
Cold	13-18	55-65
Cool	18-27	65-80
Tepid	27-33.5	80-92
Neutral	33.5 – 35.5	92-96
Warm	35.5 – 36.5	96-98
Hot	36.5 – 40	98-104
Very Hot	40-60	104-115

Point of thermal indeference: Neutral 92°F

vii. Tanks

1. Parts

- a. B - Bottom
- b. O- overflow
- c. A - Agitator
- d. T - Thermostat
- e. E – Edges
- f. D- Drainage (dirtiest)

2. Types

- a. Extremity Tank:
 - i. (2) UE, (1) LE, Edematous Volumetric Measurement
- b. Highbuoy tank UE, LE
 - i. Deepest Tank
 - ii. Does not allow extension
 - iii. Indication: Debridement
- c. Low buoy LE, and lower trunk
 - i. Long sitting by hydraulic lift or manual
 - ii. 56-66"
- d. Walk tank: For ambulation
- e. Hubbard tank: Butterfly tank, Figure of 8
 - i. Indication: OH, Bedridden, Burns, Pressure ulcers on the back

viii. Standard Pool size: 15 x 30feet

- 1. $F = 9/5 \times C + 32$
- 2. $C = 5/9 \times F - 32$
- 3. $K = C + 273$

iv. Cryomodalities

a. C/i

- i. Raynaud's Disease/ phenomenon
- ii. Sick cell anemia
- iii. Cryoglobulinemia = ischemia, becomes like gel
- iv. Cancer
- v. Cold Hypersensitivity (E.g. Polio)

b. Cold packs

- i. conduction
- ii. Indicated: Acute inflammation, swelling, cramps, spasticity

c. Ice massage

- i. Ice towels
- ii. Types
 - 1. Fast: 3-4 inches per seconds, Indication: Facilitation
 - 2. Slow : 2 inches per second, Indication: Inhibition

- d. Vapocoolant sprays
 - i. H₂O Vapor + Ethyl Chloride, + *Flourimethane*
 - ii. Evaporation
 - iii. Not effective: Swelling (lasts only <5 mins)
 - iv. Indication: Immediate relief of pain, cramps, Acute conditions,
 - v. Protocols
 - 1. Distance: 18-24 inches
 - 2. Angle: 30- 45°
 - 3. Rate: Intermittent 2-4inches/ sec (until numbness)
 - vi. Precaution: increase VCS = burns
- e. Contrast bath
 - i. Alternating hot and cold
 - ii. Goal: Vascular exercise

Reference	Temp	Time	Parameter	Indication
Braddom	Hot	40-45°C	4mins	Swelling
	Cold	15-20°C	1min	
Wadsworth	Hot	100-110°F	3mins	Promote circulation
	Cold	50-55°F	1min	

- iii. C/i: DVT, cancer, Unstable BP, Atherosclerosis/ Arteriosclerosis, pregnant patients, renal disorder
 - iv. Disadvantage: Cannot elevate limb (JICU + cold better for swelling)
- v. Mechanical agents
 - a. ILT
 - i. Goal: promote extension
 - ii. Parts:
 - 1. Traction machine
 - 2. Table: Split or Non Split
 - 3. Harness: Pelvic or Thoracic
 - 4. Foot stool
 - iii. Position
 - 1. Supine: hips and knee to 90°: Williams position
 - 2. Prone: Post HNP
 - a. Tolerate: No pillow
 - b. Cannot tolerate: with pillow
 - 3. Sidelying: Unilateral impingement (predisposes to dislocation)
 - iv. Targets
 - 1. L3-L4: 75-90° flexion
 - 2. L5-S1: 45-60° flexion

v. Traction force

1. Intro: 25-50lbs
2. Initial:
 - a. Split: 25% BW
 - b. Non-split: 50%BW
3. Maximum:
 - a. 50% or as tolerated
 - b. Conditions
 - i. DJD = 50%
 - ii. Mech LBP, Muscle spasm, stretching= 25%
4. Male $\frac{1}{2}$ of body weight
5. Female: $\frac{1}{3}$ of body weight
6. Hold relax duty cycle
 - a. (n) = 1:1 (1 on:1 off)
 - b. DJD = 3:1
 - c.
7. *Always give patient emergency button
8. C/i:
 - a. Spinal stenosis, Spondylolisthesis, RA, Pregnant, Spinal tumor

b. ICT

i. Parts

1. Traction unit
2. Harness
 - a. Occipital (Primary pull)
 - b. Mandibular
3. Table
4. *Square knot*

ii. Positions

1. Supine>Sitting (relaxed muscle without actions of gravity)
2. Sunder's CTD: TMJ dysfunction
3. Dentures
 - a. Kisner: place cotton in the mouth
 - b. IER: Remove dentures, use mouth piece

iii. Targets

1. C1-C5: 0-5° flexion
2. C5-C7: 25-30° flexion
3. Facet impingement: 25° flexion
4. Disc Dysfunction: 0° or neutral
5. Cervical conditions: 25°

iv. Traction force

1. Initial: 10lbs or 7% body weight
2. Max: 35lbs or 10-15%
3. Conditions
 - a. OA/ DJD: 35lbs

- v. C/i:
 1. RA (C1-C2) & JRA (C2, C3)
 2. Down syndrome
 3. Skin tumors
 4. VA syndrome
- vi. Scenarios
 1. Dizziness; nausea (stop test and refer, if test for VA pt will exhibit LOC)
 2. Claustrophobia, Continue treatment as (N)
 3. Sleeping: rouse the patient
- c. Intermittent compression unit
 - i. Indication: Edema
 - ii. Best combination for edema/ swelling ICU + ice + swelling
 - iii. VS: Diastolic BP (if greater will cause implosion of blood vessels)
 - iv. Parts
 1. Compression unit
 2. Sleeve
 - v. Dosage

Dosage	Pressure
UE	50-60mmHg
LE	60-70mmHg
Pressure increase	10-20mmHg per session
Pressure difference per segment	20mmHg
 - vi. Position of limb
 1. UE: Abduction 20-70°
 2. LE: Abduction 45°
 3. (B) elevated above the heart level or >45°
 - vii. Declog: Proximal first
 - viii. Inflation: distal to proximal
 1. Ratio: 60mmHg
 - a. Distal: 60
 - b. Middle: 40
 - c. Proximal: 20
 - ix. Indication:
 1. Lymphedema :2-3 hours, 2-3 treatment sessions per day
 2. Traumatic edema: 2-3 hours per day
 3. Residual limb reduction: 2.5 hours/ day 2-3x per week
 - x. Inflation –deflation ration
 - a. (n) 1:1
 - b. Lymphedema 3:1
 - c. Stump shaping 4:1
 - xi. Contraindications:
 1. Unstable blood pressure
 2. DVT, Cancer
 3. Fracture and dislocation

d. CPM

- i. Goal: increase ROM other goal: relieve edema (primary goal after swelling), swelling, prevent contracture/ adhesions, increase healing rate, lubricate the joint

- ii. Dosage

Doctor prescribed	20-40°
Pain free range	Pt's tolerance

- iii. Aggressive knee

1. Day 1: 20-40
2. Day 2: 15-45
3. Day 3: 10 – 50
4. Day 4: 5- 55
5. Day 5: 0 – 60 (rare, use PJM to reach, last 10° use patella instead of tibiofemoral)

- iv. Contraindications

1. Acute or freshly healed scars and sutures
2. Life threatening scenarios
3. Fractures/ dislocations

- v. *Knee mimic: active knee flexion*

e. Tilt table

- i. Indication: Orthostatic hypotension, Bed ridden, SCI, Postural reflexes, stretching, increasing bone density,

- ii. VS:

1. BP:
 - a. systole <20mmHg
 - b. Diastole: slight same
 - c. HR: <20bpm

- iii. Parts

1. Table
2. Goniometer
3. Straps chest, pelvic, knee, forehead strap (head lag)

- iv. Scenarios

1. Autonomic dysreflexia:
 - a. Sit patient upright
 - b. Check for noxious stimulus
 - c. Monitor Vs
2. Orthostatic hypotension

Gradual OH	Sudden OH
Slow ↓ BP	↓BP extreme levels
Lower tilt table 15-20°	Completely lower
Monitor VS	Tilt table
Elevate tilt table by 5-10°	Elevate the legs
	Perform Ankle pumps
	Next treatment:
	Add compression stockings
	Abdominal binder
	(↑ intrathecal P ₀)

vi. Massage

- a. Manual manipulation of the human body using the different parts of the human body
- b. Medium
 - i. Oil
 - ii. Lotion
 - iii. Liniment
 - iv. Powder
 - v. Honey, milk chocolate
- c. Goals
 - i. Stimulate muscle
 - ii. Relieve tension
 - iii. Prevent/ destroy adhesion
 - iv. Relieve edema
 - v. Relaxation
- d. Massage techniques
 - i. Effleurage/ stroking/ caress massage:
 - 1. Superficial/ light: build rapports, spread medium, start; end massage
 - 2. Deep/ heavy: Hematoma, abscesses,
 - ii. Petrissage/ compression:
 - 1. Kneading: small to medium sized muscles
 - 2. Squeezing: medium to large size muscles
 - a. Milking effect: Lactic acid
 - b. Before and after exercise
 - 3. Picking-up
 - 4. Rolling
 - 5. shaking
 - iii. Friction
 - 1. Circular: centripetal, centrifugal
 - 2. Cross fiber massage: perpendicular
 - a. Trigger points, taut bands, MPS, adhesions
 - iv. Tapotement/ percussion – stimulate muscles
 - 1. Tapping
 - 2. Clapping/ cupping/ slapping – postural drainage
 - 3. Pounding/ beating
 - 4. Hacking/ chopping
 - 5. Vibration – postural drainage
 - 6. Raindrops

Random fact: Witmack Eckborn reflex–Sign of sexual Repression

THERA EX 1

- i. Range of motion –
 - a. Functional excursion – distance travelled by the contacting muscle.
 - b. Insufficiency: 2 jointed muscles
 - i. Active: Shortens
 - ii. Passive: Lengthens
 - c. Active: joint is moving in his pain free range by means of muscle contraction (3 and above)
 - d. Passive: External force E.g. gravity, patient (2- and below)
 - e. Active Assisted: Weak
 - f. Rationale: Counter the effects of immobilization. (E.g. Contractures, Hemostasis, Edema)
 - g. LOM
 - i. Cannot prevent atrophy/ can delay
 - ii. Cannot increase muscle strength
 - iii. Cannot improve skill
 - h. C/i
 - i. Disruptive to healing process (E.g. Inflammatory phase)
 - ii. Patients response or condition is life threatening
 - i. Continuous Passive Motion (CPM)
 - i. Immediately after the surgery (While anesthesia is still in effect)
 - ii. 20-30 arch
 - iii. Duration: 1 hour – 24 hours
 - j. Equipment
 - i. Shoulder wheel
 - ii. T-bar/ wand
 - iii. Finger ladder
 - iv. Overhead pulley
 - v. Powder board/ skate board
 - k. Exercises
 - i. Pendulum/ Codman's/ Sperry's
- ii. Stretching
 - a. Flexibility: Ability of a joint/ series of joints to be moved
 - b. Extensibility: Ability of the muscle to be elongated
 - c. Contractures:
 - i. Myostatic/ myogenic: Adaptive Shortening of the muscles, (-) Ms Pathology
 - 1. # of sarcomeres remain the same but adhere together
 - ii. Pseudomyostatic/ Pseudomyogenic: (+) Ms pathology d/t CNS problem: spasticity, Ms spasm, Ms guarding
 - iii. Arthrogenic contracture: (+) adhesions of joint capsule E.g. Adhesive Capsulitis
 - iv. Irreversible/ Fibrotic Contracture: Permanent contracture Tx: Surgery
 - 1. E.g. Volkman's Ischemic Contracture

- d. Types
 - i. Manual Stretching: Minimum is 30 seconds to 160 seconds
 - ii. Mechanical stretching: Machine, Cuff weights, adjustable orthosis
 - iii. Selective stretching: E.g. SCI patients, Tenodesis effect
- e. Neuromuscular Facilitation techniques: Inhibition
 - i. Autogenic: Fatigue the tight muscle, range limiting muscle
 - ii. Reciprocal
- f. Techniques
 - i. Hold relax → autogenic inhibition
 - 1. Quick stretch
 - 2. Isometric contractions 6-10 seconds
 - 3. Relax
 - 4. Passive elongation
 - ii. Hold relax with agonist contraction → autogenic/ reciprocal inhibition
 - 1. Quick stretch
 - 2. Isometric contractions 6-10 seconds
 - 3. Relax
 - 4. Active MS contraction of agonist
 - iii. Agonistic contraction → Reciprocal inhibition
 - 1. Isotonic
- iii. Resistance Exercises
 - a. To increase muscle strength → increase intensity and decrease number of repetitions
 - b. Strength: Ability of the muscle to produce tension
 - c. Power: Tension over time → speed or rate of work
 - d. Endurance: Sustained movement for a prolonged period of time
 - i. Decrease intensity, increase reps for prolonged period
 - e. Muscle performance: Strength + Power + Endurance
- iv. Principles
 - a. Overload principle: Imposing challenge to the muscle, gradual and progressive
 - b. Specific Adaptation to Imposed Demands (SAIDS)
 - c. Specificity of exercise: increase muscle strength = resistance, Increase endurance = aerobic exercises
 - d. Reversibility principle
 - i. Detraining principle: 1-2 weeks of no exercise
 - 1. Decrease strength / endurance
 - ii. Hypertrophy: 6-8 weeks of regular exercise, 2 wks → max resistance
 - 1. Increase muscle size
 - iii. Hyperplasia: increase in number of muscle cells

- v. Precautions
- Valsalva – increase intra abdominal, ICP, arterial P^o → increase BP
 - Expiratory effort against a closed glottis
 - Substitute motions – Proper stabilization, or decrease the intensity
 - Soreness/ Exercise induced muscle soreness d/t accumulate muscle acid/ muscle tears
 - Acute → After exercise
 - DOMS → 24 hours after exercise
 - Overwork vs Overtraining
 - Overwork: Permanent deterioration of muscle strength
 - Overtraining: Fatigue “burnout”, normal decrease energy of the body.

- vi. Types or resistance exercise
- Isometric exercise/ Muscle setting
 - (+) muscle contraction
 - (+) resistance
 - (-) change in joint position
 - Isotonic
 - Eccentric – lengthening
 - Concentric – shortening
 - Isokinetic
 - Rate of muscle contraction depends on rate limiting machine
 - E.g. Cybex

- vii. Regimen
- Repetition maximum (RM): Maximum amount of repetition/ activity.

Delorme↑	Oxford ↓	DAPRE
10 reps 50% of ROM	10 reps 100% of ROM	6 reps of 50% 6RM
10 reps 75% of ROM	10 reps 75% of ROM	10 reps 75% 6RM
10 reps 100% of ROM	10 reps 50% of ROM	Max reps 100% of 6RM
		Max reps

Oxford: Indicated for easily fatiguing muscles

- DAPRE

Reps	Day 1	Day 2
0-2	↓ 5-10lbs	↓ 0-5lbs
3-4	↓ 0-5lbs	Keep same
5-6	Keep the same weight	↑ 5-10lbs
7-10	↑ 5-10lbs	↑ 10-15 lbs
11 or more	↑ 10-15lbs	↑ 15-20lbs

- viii. Aerobic Exercise
- Increase muscle endurance
 - Cardiopulmonary patient
 - 20-30mins
 - Large muscle group

- ix. PJM
 - a. Increase joint play (Combination of motions)
 - i. Sliding, Rolling, Spinning
 - b. Indications:
 - i. Pain
 - 1. Mechanical: ↑ synovial fluid
 - 2. Neurologic: Inhibitory signal → SC → Nociceptors
 - ii. Hypomobility
 - iii. LOM
 - c. Contraindications
 - i. Hypermobility e.g. Lax lig; (+) pathology
 - ii. Joint effusion (inflammation)
 - d. Joint Shapes
 - i. Ovoid (one convex; concave)
 - ii. Saddle/ Sellar
 - 1. 1st CMC
 - a. ABD/ ADD (opposite)
 - b. Flex/ Ext (same)
 - 2. SC joint
 - a. Elev/ Dep (Opposite)
 - b. Pro/ Ret (Same)
 - iii. PRU → Pronation – posterior; supination – anterior
 - iv. DRU → Pronation – anterior; supination – posterior
 - e. PJM grading
 - i. Rhythmic Oscillation: 2-3 cycles per seconds

Grade	
I SARO	At the beginning of the range
II LARO	Within the range
III LARO	Up to the limit of the range
IV SARO	At the end of the range
V LARO	Snap adhesions

Gr 1 & 2 Acute pain

Gr 3 & 4 Chronic pain, stretch

- ii. Sustained

Grade	
I	Loosen/ Distraction
II	Tighten/ Taking up the slack Initial grade
III	Stretch 7-10SH with 3-5 second rest

- f. Oscillation & Sustained
 - i. Mechanical
 - ii. Physiologic
- g. Order of manipulation
 - i. Start at grade II

x. Aerobic Exercise

- a. Low intensity, ↑ number of reps and duration
- b. Indication: cardiopulmonary patients
- c. Duration: 20-30 minutes

d. Intensity

i. THR, using Karvonens Formula

$$THR = MHR - RHR (40-60\%) + RHR$$

$$MHR = 220 - age$$

$$HR = (n) 60-100bpm$$

- 1. 40-60%
- 2. 60-80%
- 3. 80-90% *Athletes*

$$Max HR 60-90\% = 40-59\% \text{ Heart rate reserve}$$

ACSM guideline

ii. Borg's Scale

- 1. Rate of perceived exertion
- 2. *15 = Hard

7	Very very light	0	Nothing at all
9	Very light	.5	Very very weak
11	Fairly light	1	Very weak
12-13	Somewhat hard 60%	2	Weak
15	Hard 85%	3	Moderate
17	Very hard	4	Somewhat strong
19	Very very hard	5	Strong
		7	Very strong
		10	Very very strong

iii. Metabolic Equivalent

- 1. 1MET = 3.5 ml of O₂ / kg of BW/ min
- 2. VO₂ Max: Maximum oxygen consumption

iv. Talk Test

- 1. (n) talk without SOB
- 2. (+) talk

e. Stage of exercise

i. Warm-up

- 1. 5-10 minutes
- 2. Increase temperature, build circulation
- 3. Calisthenics, stretching

ii. Exercise Proper

iii. Cool-down

- 1. 5-10 mins
- 2. Prevent venous pooling
- 3. Prevent lactic acid accumulation
- 4. Stretching

THERA EX 2

Obstetric PT

i. POSTURE

- a. Increased upper cervical lordosis, lumbar lordosis
- b. Rounded upper back & shoulders
- c. Knee hyperextension (d/t ligamentous laxity)

ii. PROBLEMS & MX

Problem	Management
1. Pelvic floor mm & laxity a. Pubococcygeus b. Iliococcygeus c. Coccygeus	✓ Kegel's ex – hold/contract pelvic mm for 5 SEC ✓ Graded isometrics ✓ Pelvic clock & pelvic rotation ex – Pt in bridging, pelvis move side to side & rotation
2. Diastasis recti – separation or split of abdominals at the midline (linea alba) - Measure: use 2 fingers above umbilicus - Sig: > 2 cm	Light ex: Head lifts or Partial sit-ups • up to level of scapula only • 5-10 sec hold (initial: 5 sec) • c approximation of arms
3. Ligamentous Laxity – d/t inc RELAXIN	Mm strengthening ex (light weights)

iii. SEQUENCE OF OBSTETRIC EX

- a. Warm-up Activities – to inc mm temp
- b. Gentle selective stretching – small mm only; if larger mm: one at a time lang
- c. Cardiovascular training or Aerobic ex – 15 mins or less; e.g swimming, walking, biking
- d. UE & LE strengthening
- e. Cool-down activities – prevent venous pooling
- f. Abdominal ex (partial sit-ups) – prevent diastasis recti
- g. Pelvic floor ex (Kegel's, isoms, pelvic clock)
- h. Relaxation technique
- i. Pt educ. (pre- & post-partum)
- j. Home ex program

iv. ABSOLUTE CONTRAINDICATIONS TO OBSTETRIC EX – “Executive Vice Pres. ApPle RIM”

- a. Early cervical dilatation – incompetent cervix
 - b. Vaginal bleeding – may be (+) rupture of BV
 - c. Placenta previa – nauna yung placenta (dapat sabay)
 - d. Abruptio Placenta – naiwan yung placenta
 - e. Premature labor
 - f. Rupture of a membrane – rupture of amniotic sac
 - g. Intrauterine growth retardation
 - h. Maternal hypertension = ECLAMPSIA
 - i. Maternal DM – Pregnancy-induced DM (maa-acquire ng baby yung DM)
- *egg cell – largest cell that can be seen in female human body

- v. **CRITERIA TO DISCONTINUE OBSTETRIC EX** (“PaBaSa ID FaT BoDy”)
 - a. Pain (any pain!)
 - b. Bleeding
 - c. SOB
 - d. Irreg heart beat
 - e. Dizziness
 - f. Faintness
 - g. Tachycardia (**max** pulse rate must be: **140 bpm**)
 - h. Back & pubic pain
 - i. Difficulty in walking
- vi. **UNSAFE EXERCISE FOR PREGNANT WOMEN:** (“FUBA”)
 - a. Firehydrant ex → compress SI jt → LBP
 - b. Unilat. LE wt bearing → compress pubic symphysis → pubic pain
 - c. Bilat SLR → compress baby & inferior vena cava
 - d. All 4s hip extension → compress back area & SI jt → LBP
- vii. **PRECAUTIONS**
 - a. No valsalva maneuver! Always count.
 - b. Supine position
 - c. Max supine positioning: 5 mins
- viii. **RELAXATION** – conscious effort to relieve tension in mm; c 4 ELEMENTS (CIQS)
 - a. **C**onscious relaxation
 - b. Incorporate deep breathing
 - c. **Q**uiet environment
 - d. **S**oft tone of voice – also applicable for CP/ADHD Pt
- ix. **COORDINATION**
 - a. Def: correct timing & sequencing
 - b. ex: Frenkel’s, finger-to-nose
- x. **BALANCE**
 - a. Def: ability to align body segments against gravity to maintain or move c in the available BOS s falling
 - b. ex: wobble board, kitchen sink ex., theraball/vestiball/physioball/gym ball
 - c. measurement of theraball:
 - i. sitting: feet flat, hip & knee 90deg
 - ii. supine: distance bet hip & knees
 - iii. quadruped: distance bet shoulders
- xi. **STABILITY** = Rhythmic Stabilization
- xii. **ATAXIC Pt** = Slow Reversal Hold
- xiii. **AQUATIC EX**
 - a. **Precautions** (“IFA Paki Check Cung Secure”)
 - b. Inability to swim
 - c. Fear of water (hydrophobia)
 - d. Ataxic Pt c postural instab.
 - e. Pt c heat intolerance
 - f. Controlled seizures

- g. Controlled cardiovascular dysfunction
- h. Small open wounds and lines
- xiv. **Absoulte C/I** ("BoSSS BaWaL")
 - a. Bowel & bladder incontinence
 - b. Severe kidney dse (e.g UTI)
 - c. Severe epilepsy
 - d. Severe cardiovascular & respiratory dysfxn
 - e. Severe PVD
 - f. Bleeding
 - g. Water/air-borne infections (better swim in OPEN AREAS; sunlight = sterilization)
 - h. Large open wounds
 - i. Colostomy
- xv. **Physics**
 - a. Buoyancy – upward thrust og water on immersed body or body parts
 - b. Hydrostatic Pressure – circumferential pressure on immersed body/body parts
 - c. Viscosity – friction of olecules of fluid resulting to resistance to flow
 - d. Surface tension – water molecule on surface has greater tension or viscosity
- xvi. **Summary to INCREASE RESITANCE in Aquatic Ex.:** ("SISFloat")
 - a. Short lever arm
 - b. Increased surface area (taasan ang water)
 - c. Speed of movement
 - d. Floatation device (e.g. collars, rings, vest, pelvic)
- xvii. **Temperature**
 - a. Cardiovascular/aerobic ex: between 26 & 28 deg C
 - b. Flexibility, relaxation, gait training, strengthening: bet 26 & 33 deg C

SOFT TISSUE INJURIES

- xviii. **TERMS**
 - a. Strain – mm
 - b. Sprain – ligament
 - c. Subluxation – mm
 - d. Dislocation – ligament
 - e. Synovitis – inflamm of synovium
 - f. Bursitis
 - g. Hemarthroses – bleeding in a jt d/t direct trauma
 - h. Ganglion – ballooning of a wall of jt capsule (mc: bilat on wrist, usually sa ulna)
 - i. Overuse syndrome
- xix. **Tendinous lesions**
 - a. Tenosynovitis – inflamm of tendon & synovial sheath
 - b. Tendinitis – inflamm of tendon, c scarring & calcium deposits
 - c. Tenovaginitis – inflamm c thickening of tendon
 - d. Tendinosis – degeneration

xx. **SEVERITY OF SOFT TISSUE INJURIES**

- a. **Grade 1** (first degree)
 - i. Mild pain at the time of injury usually within 24hrs
 - ii. Mild swelling
 - iii. Pain when stressed
 - iv. Local tenderness
- b. **Grade 2** (second degree)
 - i. Moderate swelling that requires stopping of activity
 - ii. Stress & palpation increases pain
 - iii. Results to jt hypermobility
- c. **Grade 3** (third degree)
 - i. Near complete or complete avulsion/tear of ST (tendon/ligament)
 - ii. Severe pain (ceiling pain)
 - iii. Palpation reveals the defect
 - iv. Stress does not increase pain (kasi severe na nga)
 - v. Results to jt instability

xxi. **STAGES OF INFLAMMATION, REPAIR & MATURATION**

- a. **Inflammatory stage**
 - i. Acute stage or Inflammatory reaction
 - ii. 4-6 days
 - iii. Characs: "VEPEQ"
 - 1. Vascular changes
 - 2. Exudation of cells
 - 3. Phagocytosis
 - 4. Early fibroblastic activity
 - 5. Plaque formation
 - iv. 5 cardinal signs
 - 1. Rubor –redness
 - 2. Callor – heat
 - 3. Dolor – pain
 - 4. Tumor – swelling
 - 5. LOF
 - v. Clinical pres.:
 - 1. (+) inflammation
 - 2. Pain before tissue resistance
 - vi. Rehab phase: protection phase
 - 1. PRICEMEMS
 - 2. PROME

- b. **Sub-Acute**
 - i. Repair & healing
 - ii. Duration: 10-17 days, 14-21 days from onset, up to 6 mos
 - iii. Characs: “ReGrow Collagen GraVe”
 - 1. Removal of noxious stimuli
 - 2. Growth of capillary beds
 - 3. Collagen form’n (for elasticity)
 - 4. Granulation form’n
 - 5. Very fragile or easily bruised
 - iv. Clinical presentation
 - 1. Decreasing inflammation
 - 2. Pain synchronous c tissue resistance
 - v. Rehab phase: Controlled Motion phase
 - 1. Multiple Angle Isoms
 - 2. Done every 10 deg
 - 3. Submaximal Isoms
 - 4. AROME
 - 5. CKC & proprioceptive ex
 - c. **Chronic**
 - i. Maturation & healing
 - ii. Duration: >6mos
 - iii. Characs: “MaCo ReCo”
 - 1. Maturation of scar tissue
 - 2. Contracture of scar
 - 3. Remodelling of scar
 - 4. Collagen aligns to stress
 - d. Rehab phase: Return-to-function
 - i. Multiple Angle Isoms
 - ii. Plyometrics
 - iii. Strengthening
 - iv. Stretching
 - v. Skill & agility training
- xxii. **4TH STAGE: CHRONIC INFLAMM.**
- a. **E.g.**
 - i. overuse syndrome
 - ii. cumulative traumatic d/o
 - iii. repetitive strain injury
 - iv. mm weakness
 - v. poor mobility, cxr
 - b. **Acute**
 - i. Non-stressful activities
 - ii. Non-stressful intensity

- c. **Sub-Acute**
 - i. Strengthening
 - ii. Stretching
 - iii. Skill & agility training
- xxiii. **Soft tissue friction massage**
- xxiv. **Work conditioning**
- xxv. **Work hardening**

	Work-hardening	Work conditioning
Discipline model	Single (Pt just kicking a ball towards the goal)	Multi (playing c teammates but not in actual game)
Hrs/day	4	8
Common Denominator	5 days/wk	5 days/wk

- xxvi. **LOW BACK PAIN EX**
 - a. Williams flexion ex
 - b. McKenzie
 - c. Kendall's
- xxvii. **SCOLIOSIS EX: CLAPP'S EX**
 - a. move towards CONVEX side
 - b. T3 – lowered (heel-sitting on elbows)
 - c. T6 – semi-lowered (heel-sitting on hands)
 - d. T8 – quadruped
 - e. T11 – kneeling, leaning forward (trunk flexion)
 - f. L2 – tall kneeling
 - g. L4 – kneeling, leaning backward

ELECTRO 1

i. Terms

- a. Voltage – Electro Motive Force (EMF): Force that drives electrons-
 - i. Symbol: (V)
 - ii. Unit: Volts
- b. Current : Net flow of electricity
 - i. Symbol (I)
 - ii. Can be: (+), (-), or (+/-)
 - iii. Unit: Ampere
- c. Resistance: oppositions to electron flow
 - i. Symbol (R)
 - ii. Unit: Ohm (omega symbol)
- d. Ohm's law
 - i. $V=IR$
 - ii. $I=V/R$
- e. Conductor/ Resistor
 - i. Menomics
 1. B – Bone (Resistor)
 2. C - Cartilage
 3. T - Tendon
 4. S – Skin (Dirty: Resistor, Moist: Conductor)
 5. M - Muscle
 6. B - Blood
 7. N – Nerve (Conductor)

f. Ions vs Electrodes

Ions	Electrode
(+) Cation	(+) anode
(-) Anion	(-) cathode

(+) (-) = *opposites attract*

(+) (+) = *like repel (lontophoresis)*

	Anode	Cathode
pH	↓ acidic	↑ alkaline
Color	Pinkish hyperemia	Marked red discoloration
Burn	↓ risk for burn	↑ risk for burn
Polarity	Hyperpolarization	Depolarization
NCV	↓	↑
Effect	Sclerotic (hardening)	Sclerolytic (softening)
Function	Healing & closure	Disinfection

Electrotonus	
Increase activity	Catelectrotonus
Decrease activity	Anelectrotonus

- g. Polar Formula, Erb's formula, Pleuier's law
 - $C = \text{Cathode/ Closing}$ $O = \text{Opening}$ $A = \text{Anode}$
 - i. Innervated = **CCC** – ACC – AOC – **COC** (UMNL)
 - ii. Denervated = ACC – **CCC** – **COC** – AOC (LMNL)
- h. 3 main electrical properties of nerve
 - i. E – Electrical Excitability (RMP, resting membrane)
 - Nerve -70mV
 - Muscle -90mV
 - Heart -88mV
 - GIT -56mV
 - Electrical excitability proportional speed, intensity, duration*
 - ii. R – Refractory Period: Period when membrane excitability is depressed
 - 1. Absolute: 2nd potential cannot stimulate
 - 2. Relative: Supramaximal stimulus to stimulate (Staircase phenomenon/ Treppe)
 - Continuous treppe = tetany*
 - iii. A – Accomodation: adapts itself slowly increase intensity of stimulus
 - 1. *Dubois Raymonde Law*: To stimulate a nerve there must be a sudden change/ variation in the *current intensity* is needed (REMBE)
- ii. DC/AC Currents
 - a. DC: monophasic current (+) or (-); not (+/-)
 - i. Rheobasic current for dennervated muscle, Galvanic current
 - ii. Best for wound healing
 - iii. Types
 - 1. Uninterrupted (DUC): without off time (off time = pulse interval, on time = pulse duration) E.g. Iontophoresis = medical galvanism\
 - a. Increased risk for burns
 - 2. Interrupted (DIC): with off time
 - a. B₁ – Short DIC – faradic type
 - i. Pulse duration: 0.02- 1ms
 - ii. Frequency: 50-70hz
 - iii. Sensation: Marked stabbing (Causing titanic contraction)
 - iv. Indication: innervated muscles
 - Mnemonic:*
 - Short DIC Stabs Far*
 - b. B₂ – Long DIC
 - i. B2.1: Rectangular long DIC (quick rise)
 - 1. Pulse duration: 1-600ms
 - 2. Indication: innervated & dennervated muscles
 - ii. B2.2: Sharp/ exponential progressive long DIC (slow rise)
 - 1. Pulse duration: 300-1,000ms
 - 2. Indication: dennervated muscle

b. AC: alternating current

i. Faradic current

1. Indication: innervated muscles

Faradic Current	Sinusoidal current
Uneven AC	Even AC
Asymmetrical	Symmetrical
PD: 1ms	PD: 10ms
F: 50hz	F: 50Hz
Mild tingling	Mild tingling

 $1FUSE_{10}$

50

iii. Mono & bipolar electrode application

	Monopolar	Bipolar
pairs	1	1
Pole	1 active: nerve trunk, motor points, muscle belly 1 dispersive: Anywhere	2 active: muscle belly . Muscle bulk
Indication	Innervated muscle	Innervated & dennervated Muscle

Bells palsy: 7 – 14 muscle, pen-type, 30 contraction x 3 sets

iv. LFC vs MFC vs HFC (Hz = pps = decibels (dB))

	LFC	MFC	HFC
Frequency(Wadsworth)	1-2000hz	3000-6000Hz	>500,000Hz
Frequency (Sullivan)	1-1000hz	1000-10000Hz	>10000Hz
Nerves	√	√	x
Innervated muscle	√	√	x
Dennervated muscle	√	x	x
Indication	Motor purposes	Sensory	Thermal
Example	ES, FES, TENS	TENS, IFC, Russian	Diathermy, US, UVR, IRR, LASER

v. Iontophoresis (Transdermal)

a. DUC Medical Galvanism

b. No first pass effect

c. Principles

i. Electrolysis: separation of (+) and (-) ions using electricity

ii. Hydrolysis: separation using water

iii. Ionization: Charge

iv. Like: repel

d. Dosage: intensity x time= mA/min

e. Medications

(-) anion (-) cathode	Indication	(+) cation (+) anode	Indication
NaCl	Adhesion	-caine	Pain
Salicylate	Anti-inflammatory; pain	Cortisone	Anti-inflammatory
Acetic Acid	Calcium deposit	-mecholy	Chronic arthritis Varicose ulcers, Scleroderma
Iodine	Wounds, keloids	Hyaluronidase	Edema
Dexamethasone	Anti-inflammatory	Renotin	Adhesion
H ₂ O	Hyperhydrosis	MgSO ₄ CaCl	Muscle spasm
		ZnSO ₄	Hay Fever, allergic rhinitis Indolent ulcer
		CuSO ₄	Fungal infection
		Glycopyronium bromide/ Glycopyrolate	Hyper hydrosis
		Vinca Alkaloid	Neuralgia

Wyadase: (-) dexamtheasone, Decadron (+) dexamethasone

- vi. ES, FES, NMES
 - a. FES: functional activity
 - b. NMES: used for impairments and deformities
 - i. Stimulation: nerves
 - c. Condition
 - i. Shoulder subluxation
 - 1. Anterior: Subscapularis
 - 2. Inferior: Supraspinatus + Posterior Deltoids
 - ii. Edema & Swelling
 - 1. FUP (Faradism under pressure)
 - 2. On:Off ratio: 1:1
 - 3. Target: Muscle bulk
 - a. Arm: Biceps
 - b. Forearm: Anterior
 - c. Hand: Palmar
 - d. Thigh: Quads> Hams
 - e. Leg: Posterior
 - f. Foot: plantar
 - iii. Scoliosis
 - 1. Place electrodes: Above and below apex
 - 2. Sidelying on convex, gravity
 - 3. S curve: along with klapps or ambling walk
 - iv. Relaxation/ Spasticity
 - 1. Autogenic inhibition
 - a. Goal: Fatigue the spastic muscle
 - 2. Reciprocal inhibition: Reticence
 - a. Goal: Stimulate antagonist
 - v. Changing to type II to type I
 - 1. Type II: Lower intensity, higher duration

- vi. Strengthening
 - 1. Atrophy:
 - a. Mild: 1: 5
 - b. Moderate: 1:10
 - c. Severe: 1:15
 (longer off time, prevent fatigue)
 - 2. (n) muscles:
 - a. Initial: 1:5
 - b. Progress: 1:3
 - vii. SCI: to facilitate Wallerian regeneration
 - 1. Placement: paraspinal ms
 - viii. Sexual rehabilitation:
 - 1. Male: E.D. (surgical implant a pen type of electrode)
 - 2. Placed: corpora cavernosa
 - 3. Stimulate: Vasodilation
 - ix. Respiratory assistance
 - 1. Target: Accesory muscles
 - x. Bowel & bladder dysfunction:
 - 1. Bowel: induces valsava
 - a. Placement: rectus/ transverses abdominis & internal obliques
 - 2. Bladder: detrussor muscles
 - xi. Adjunct to exercise
- vii. TENS
- a. Indication: relief of pain
 - b. Rationale: Gate control/ Spinal gating theory (Melzack & wall)
 - i. Pain
 - 1. A delta: acute/ fast; E.g. itch
 - 2. C: chronic/ slow; somatic
 - 3. TENS uses A beta = vibration, light touch, LT, myelinated, large
 - c. Gate control:
 - i. Conventional & accommodation TENS only
 - ii. Open Gate theory
 - 1. A delta → SC → SGII (T-Cell)
 - 2. C fibers → SC → SG II (T-Cell)
 - 3. T-Cell (opens or closes) open gate to STT → thalamus → Area 3,1,2 → pain
 - iii. Closed gate theory
 - 1. TENS → A Beta → SC → SG (RLII) → T-cell → close STT
 - iv. T-cell (First come, first serve)
 - v. How to
 - 1. Open gate: *Intense stimulation of small diameter fibers* or Presynaptic facilitation of T-Cells
 - 2. Close Gate: Mild stimulation of large diameter fibers or *Presynaptic inhibition of T-cells*

- d. Descending pathway inhibition technique:
- i. Area 3,1,2 (detects pain) → send signal to periaqueductal gray matter (Raphe magnus nucleus) → Release endorphins & enkephalines (endogenous opiates) → Spinal cord (Substantia Gelatinosa) → produces serotonin & norepinephrine → blocks pain (by calming, sedatives)
- e. Counter-irritation
- i. Stronger stimulus will override weaker stimulus (masking effect)

TENS	Other name	Pulse duration (ms)	Frequency(pps)	Effect	Function
Conventional	High rate tens	50-100	50-80	Temporary	Acute & chronic
Acupuncture	Strong low rate	150-300	1-5	Long lasting	Chronic; MPS
Burst mode	Pulse train	50-200	50-100 (1-4 per burst)	Long lasting	Chronic; Diffuse
Brief intense	-	50-250	80-150	Temporary	Painful procedure, During labor & Delivery; stretching PJM, debridement
Hyperstimulation	Point stim	150-300	1-5	Long lasting	Noxious stimuli, Trigger point (Best for MPS)
Accommodation	Modulation	Varies	Varies	Long lasting	Pt's that accommodate

- f. Contraindications:
- i. Pacemaker
 1. Demand (absolute): (-) battery
 2. Fixed: (+) battery
 - ii. Pregnant Uterus
 - iii. Decreased sensation
 - iv. Very young, Very old
 - v. DVT & Cancer
 - vi. Unstable blood pressure
 - vii. Abdominal organs
 - viii. Mucus membrane

Mnemonics:

TENS Virgin

I. Conventional ay High Rate

50 to 100 ang pulse duration

Ang pulse rate per second ay 50 to 80

II. Acupuncture... Strong Low Rate

Hyperstim... aka Point stim

150 to 300... 1 to 5 pps

Pareho sila (Hoo!)

III. Train burst mode

50 to 200 (1 to 4 per burst

Oooh Pulse Train Burst Mode

50 to 100 ang pps

IV. Brief intense wala syang AKA

50 to 250 ang pulse duration

Ang pulse rate per second ay 80-150

[Accomodation& modulation Varies]

Chant:

For innervated muscles	For dennervated muscles
Fa radic type	G alvanic
Fa radic current	E xponential progressive
R ₂ Rectangular/ Russian	R ectangular
A lternating current	D irect current
S inusoidal Current	B ipolar
I nterferential C	
M onopolar	
B ipolar	

viii. Russian/ IFC

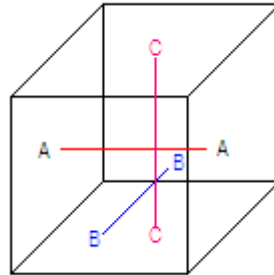
a. Russian

- i. Most painful
- ii. Frequency: 2500 Hz
- iii. Function: Stregthening, relaxation, Fatiguing muscles (5:1),
- iv. Duty cycle: On/ On+off x 100%

b. Interferential current

- i. Superficial: 1 -2cm
- ii. Combine 2 medium frequency to produce 1 LFC
- iii. E.g. 4,000Hz & 4100Hz = 100hz (Beat frequency/ dosage)
- iv. Beat summation/ Intensity/ total:
 1. Add: 4,000Hz + 4100Hz =8100Hz
- v. Beat average:
 1. 4,000Hz + 4100Hz / 2 = 4050Hz
- vi. Indication
 1. 100Hz: constant IFC: Inhibit sympathetic system (E.g. RSD)
 2. 100Hz rhythmic IFC: Swelling & Edema
- vii. Techniques:
 1. Bipolar/ Diagonal Technique
 - a. 1 pair: 2 active
 2. Quadripolar/ Cross-Diagonal
 - a. 2 pairs: 4 active
 - b. Indication:
 - i. nondiscrete pain (unable to say PS)
 - ii. non localized pain (unable to pinpoint)

3. Stereodynamic effect/ star electrodes
 - a. 3 pairs:6 active
 - b. Y-shaped on both sides
 - c. 3D effect



- d. Indication: visceral pain

4. Electrokinesy
 - a. Glove type electrode
 - b. Circular approach
 - c. Indication:

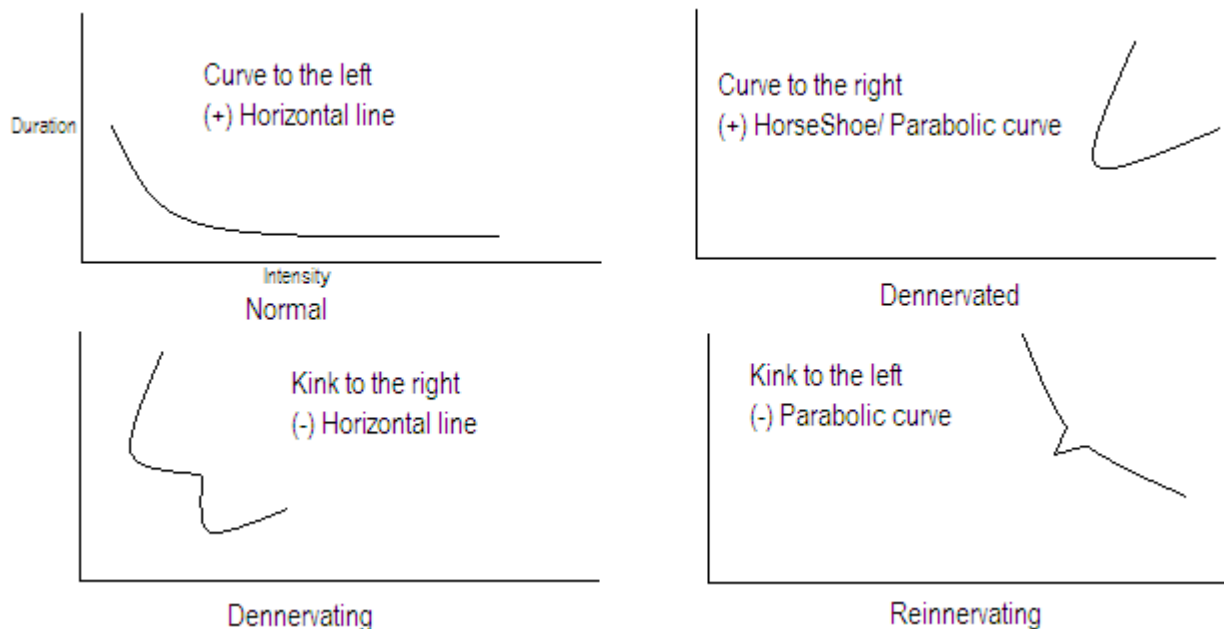
- ix. Electro Diagnostic techniques
 - a. Rationale: Confirm diagnosis, differentiate different levels of denervation
 - b. NCV
 - i. Indication: Check latency
 1. Increased: Signals are slow, PNI (Denervation)
 2. Decrease: Signals are fast, (n)
 3. Signals are 30-40% lower in infants vs adults
 - ii. Check
 1. Latency
 2. Location
 - c. ROD (reaction of Degeneration)
 - i. Differentiate partial, full, absolute

	Partial	Full	Absolute
FaT	↑	↓	(-)
FaC	↓	(-)	(-)

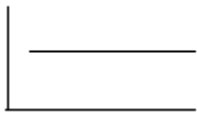
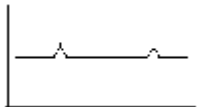
Inversion of polar formula: seen in full, ACC – CCC – COC – AOC (Denervated indication)/ Fibrosis: Absolute

- x. Strength-Duration curve
 - a. Indication: Denervate or (N), Denervating, Reinnervating
 - b. Rheobase: minimum intensity required for a peak muscle contraction
 - i. (n) 2-18mA, 5-35 volts
 - ii. Factors that increase rheobase
 1. C - Cold
 2. R - Reinnervation
 3. E - Edema
 4. D – Degeneration
 - iii. Factors that decrease rheobase
 1. Heat

- c. Chronaxie: minimum time for a peak muscle contraction
- i. (n) 0.05-0.5ms <1ms
 1. Increase chronaxie
 - a. Difficult to stimulate (denervated)
 2. Decrease chronaxie
 - a. Normal
 - ii. Factors that increase chronaxie
 1. D - Denervation
 2. I - ischemia
 3. E - Edema
 4. P - Partial reinnervation
 5. N - nerve root lesion
 6. F - fatigue (can be seen in (n) person)
 - a. Signs:
 - i. Fibrillation: single
 - ii. Fasciculation: groups, visible to eye
 - iii. Factors that decrease chronaxie: reinnervation



- xi. EMG
- a. 10-14 days (wait to finish Wallerian degeneration)
 - b. Electrodes:
 - i. needle (MC used), accurate (physician)
 1. Dadv: painful, requires skill
 - ii. surface/ flat electrodes, easy to apply (physical therapist)
 1. Dadv: cross-talk

	Normal	Dennervation
Rest	Electrical silence 	Spontaneous fibrillation/ fasciculation  Positive Sharp waves  Myokimic Discharges
Contraction	Interferential signals: Diphasic Triphasic	Polyphasic Signal: Neurogenic VVWVWVW Myogenic vwww

c. Steps

- i. Acquire a Raw EMG channel
- ii. Retification: Fix Quality of Signal
 1. Half wave: all waves on the bottom will be placed on the top
 2. Full wave: Fix signals to make it more even
- iii. Pass filtering
 1. Low: to lower
 2. High: to higher
- iv. Integration:
 1. Quantification of waves
 - a. Increase waves: Spastic or contract
 - b. Decrease wave: Relaxing or very weak

d. Protocols

	Relaxation	Strengthening
Initial	Close spaced Low sensitivity detection	Wide spaced High sensitivity
Progressive	Wide spaced High sensitivity	Close spaced Low sensitivity detection

Mnemonic CHILLAX (initial: Closed space, Low sensitivity detection, Relaxation :progressive)

Random note: Fracture: low intensity tens, low dose US

SPINAL CORD INJURY

- i. Spinal Cord Injury
 - a. Epidemiology
 - i. 80% are young males, 18-25 years old
 - ii. Causes:
 - 1. Vehicular accidents 44%
 - 2. Falls 18%
 - 3. Acts of violence 17%
 - 4. Sports injuries 13%
 - iii. Most common sport causing SCI – diving
 - iv. Falls most common cause in individuals >45 y/o
 - v. 10-15% have associated head injury
 - vi. 10-15% have multilevel injury
 - vii. Most common in July
 - viii. Least common in February
 - ix. Most commonly occurs on Saturday
 - x. *Cervical involvement (C5→C4→C6) more common than thoracolumbar involvement*
- ii. Adult vs pediatric SCI
 - a. Pediatric SCI
 - i. Secondary to pedestrian accidents & falls
 - ii. High cervical injury
 - iii. Injury due to subluxation high
 - iv. SCI with normal X-ray
 - v. Delayed onset injury common
 - vi. Complete injury more common

Characteristics	Adult	Pediatric
Mechanism of injury	Motor vehicular accidents	Pedestrian accidents and falls
Level of injury		
C1-C3:	1-2%	60%
C3-C7:	85%	30-40%
Thoracolumbar:	10-15%	5%
Type of injury		
Fracture –dislocation:	>70%	25%
Subluxation alone:	<20%	>50%
SCI without radiological abnormalities	Rare	Up to 50%
Delayed neurological deficits	Rare	Up to 50%
Proportion of complete injuries	Less	More

- iii. Basic guidelines in SCI evaluation
 - a. Neurological level (Physiatrist)
 - i. NORMAL segment just above the most rostral abnormal segment
 - ii. Key muscle with Grade 3 or 4 may be NORMAL provided that the next most rostral muscle is 5
 - b. Functional level (Physical Therapist)
 - i. Most distal segment with *at least a grade of 3*
 - ii. Provided that no segment rostral to it has a grade less than 3
 - iii. Basis for predicting the functional activities an SCI patient can still do
 - 1. L4-L5 - community ambulatory
 - 2. T12 – home ambulation with KAFO
 - 3. T1 – independent wheelchair ambulation C5 & higher – complete ADL dependence

Example:

Case 1	Case 2
C5 -5	C5 - 5
C6 -5	C6 - 4
C7 - 4	C7 - 2
C8 - 3	C8 -1
T1 - 3	T1 & below - 0
L2 & Below -0	
N. Level: C7	N. Level: C6
F. Level: T1	F. Level: C6

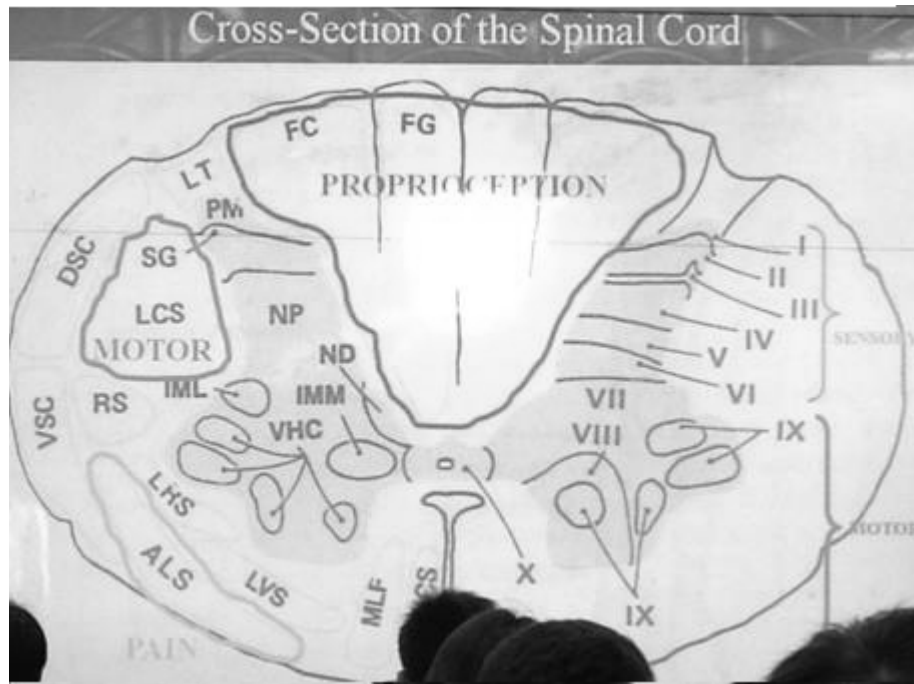
- iv. Basic guidelines in SCI evaluation
 - a. Complete injury
 - i. All motor and sensory function below level of injury absent
 - b. Incomplete injury
 - i. Motor or sensory function preserved below the zone of injury
 - ii. Have sacral sparing
 - c. Both motor and sensory levels should be determines
 - d. Motor level is determines using the myotome
 - i. Grading system > 0-5
 - ii. Not all spinal cord segments have a key muscle
 - iii. Muscles are always tested in supine
 - e. Sensory level is determined using the dermatome
 - i. Sensory test uses a specific sensory grading scale of 0 to 2
 - ii. Test sites use specific key sensory points
 - iii. Test for both sharp pain (using a safety pin) and light touch (using cotton)
 - iv. Hyperesthesia = 1
 - v. Dull rather than sharp pin = 0
 - f. Use of – paresis discourages

g. Degrees of completeness or incompleteness should be described using the ASIA impairment scale

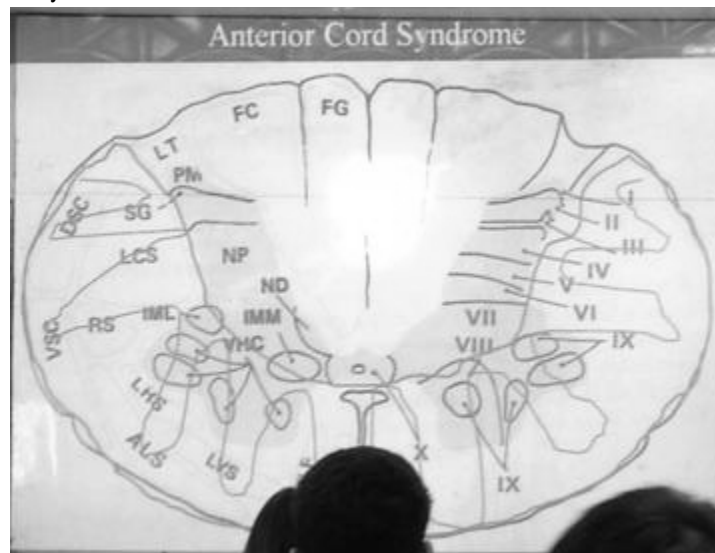
Scale		Sensory	Motor
A	Complete	No sensory function in the sacral segments s4-s5	Complete motor paralysis
B	Incomplete	Sensory preserved below the neurologic level including sacral segments s4-s5	Complete motor paralysis
C	Incomplete	Same as above	Motor function preserved below the neurological level and more than half of the key muscles below the neurologic level have a muscle grade of <3
D	Incomplete	Same	As above but with at least half of the key muscles below the neurologic level have a muscle grade of >3
E	Normal	Normal	Normal

- h. Complete diagnosis should include
 - i. Cause of injury
 - ii. Complete or incomplete & ASIA classification
 - iii. Neurologic level (motor & sensory; R and L sides)
 - iv. Functional level
 - v. Motor and sensory index scores
- i. E.g., Spinal cord injury, complete, ASIA class B, C8 motor bilaterally, C8 sensory right, T1 sensory left, secondary to vehicular accident with fracture-dislocation C6-C7 vertebrae, motor index score 40, sensory index score 67, functional level of T1
- v. Sacral sparing
 - a. Most peripherally located sacral tracts are spared
 - b. Clinical signs
 - i. Intact perianal sensation
 - ii. (+) rectal sphincter contraction
 - iii. (+) cutaneous sensation in saddle area
 - iv. (+) active contraction of sacral innervated toe flexors
 - c. Often the first sign that a cervical lesion is incomplete
- vi. Causes Spinal injury
 - a. Trauma (compression fracture, fracture-dislocation)
 - b. Transsection from stab wound & gunshot wounds
 - c. Compression tumors
 - d. Infiltration from bacteria (Pott's disease) or malignancy (metastasis)
 - e. Infection (ex. Transverse myelitis)
 - f. Congenital (myelomeningocele)
- vii. Incomplete syndromes
 - a. Anterior cord syndrome
 - b. Posterior cord syndrome
 - c. Central cord syndrome
 - d. Brown-Sequard syndrome (Hemisection of the spinal cord)
 - e. Conus medullaris syndrome
 - f. Cauda equina syndrome

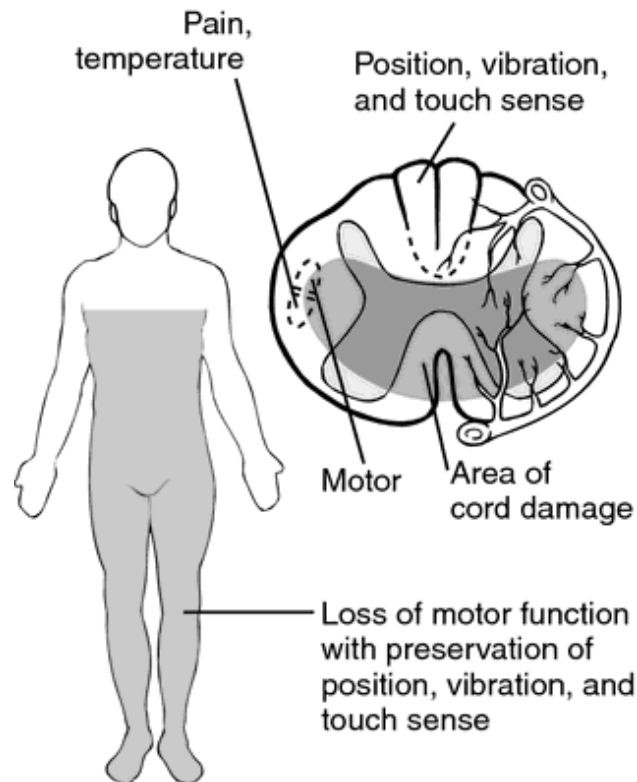
- viii. Cross section of the spinal cord



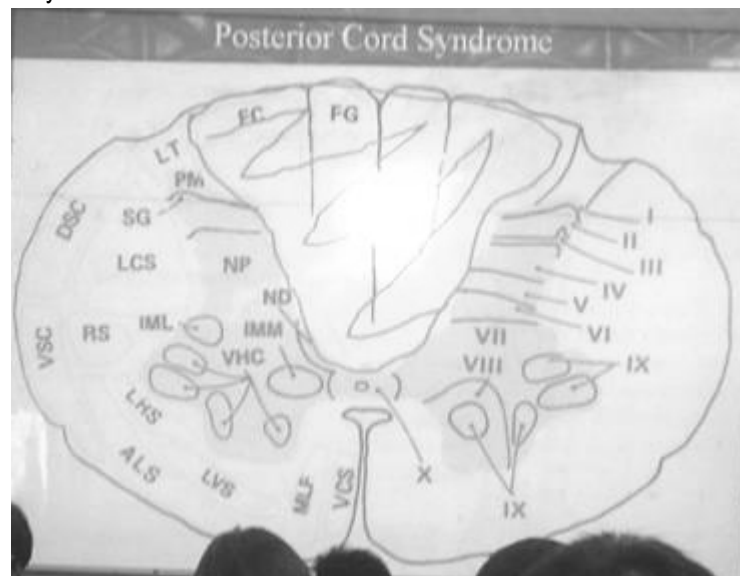
- ix. Anterior Cord syndrome



Can cause occlusion of anterior spinal artery anterior 1/3 supply



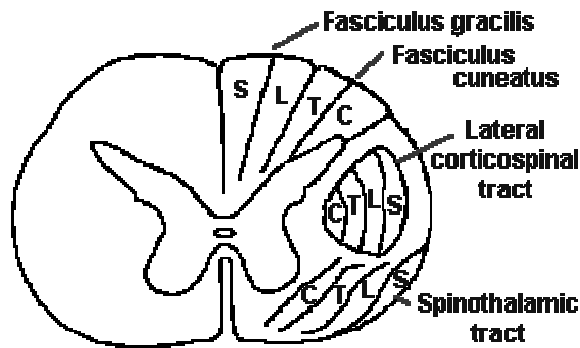
- x. Posterior cord syndrome



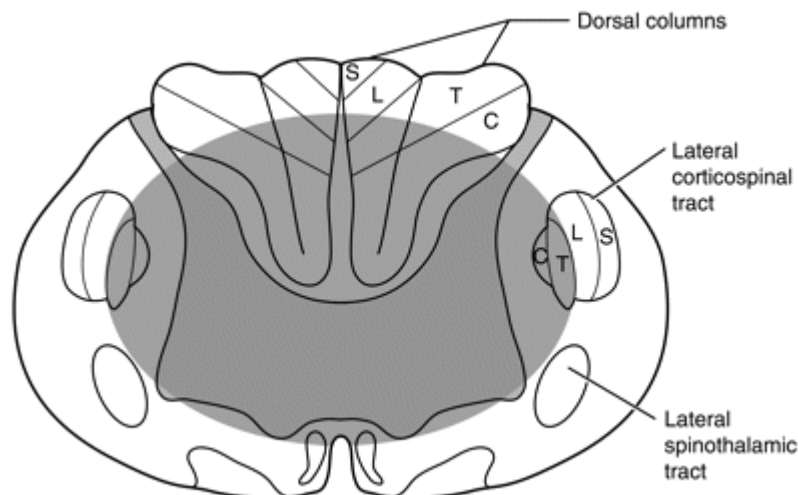
Can cause occlusion to posterior spinal artery 2/3 of blood supply to spinal cord

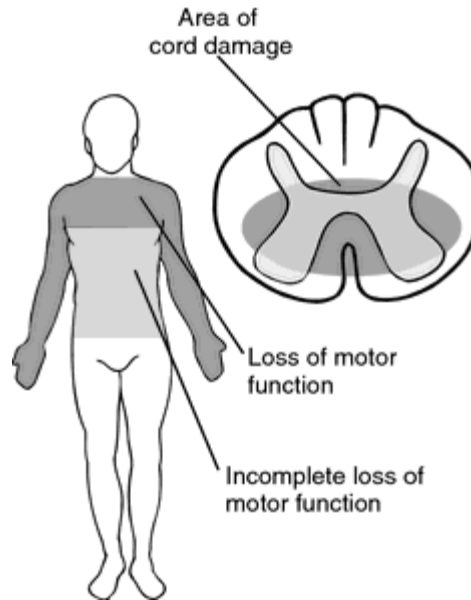
- xi. Anterior vs posterior cord syndrome
- a. Anterior cord syndrome
 - i. (+) motor paralysis
 - ii. Loss of pain and temperature sensation
 - iii. Intact proprioception

- b. Posterior cord syndrome
 - i. Absent proprioception
 - ii. Intact motor function
 - iii. Intact pain and temperature
- xii. Somatotopic organization of SC white matter
 - a. Cervical tracts are primarily central in location
 - b. Lumbosacral tracts are primary peripheral in location



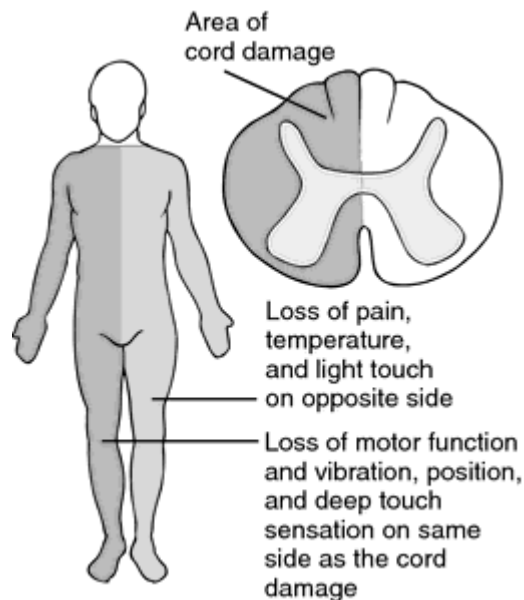
- xiii. Central cord syndrome
 - a. Have central area of necrosis where damage most severe
 - b. Surrounding edema least in periphery so subsides in periphery first
 - c. May have sparing of sacral tract
 - d. Motor recovery ascending UE severely affected than LE
 - e. Due to hyperextension injury in an osteoarthritic spine > result in acute cord compression between bony spurs Anteriorly & thickened ligamentum flavum posteriorly > so more injury to medial corticospinal tracts
 - f. More common in elderly following fall
 - g. Good prognosis for recovery
 - h. Quadripareisis with UE > LE distal muscle more affected than proximal
 - i. Motor recovery follows ascending pattern
 - j. May have sacral sparing at onset





xiv. Brown Sequard syndrome

- a. Ipsilateral motor paralysis
- b. Ipsilateral loss of proprioception
- c. Contralateral loss of pain & temperature below level of lesion; ipsilateral at level of lesion
- d. Due to hemisection of cord usually from stab wounds or gunshot wounds
- e. Since motor & proprioception tracts decussate at the cervicomedullary junction > lesions below this (in the SC) would be ipsilateral
- f. Since decussation of pain and temperature pathways is segmental > pain & temperature loss is generally contralateral



- xv. Syringomyelia
- Formation of cyst in the area of the central part of the cord
 - Diameter of central canal is normal
 - UE>LE
 - Affects spinothalamic tract, decussates runs through middle of spinal cord
 - Sensory dissociation
 - Significant loss of pain & temperature
 - No similar deficit in proprioception
 - Ascending level of injury

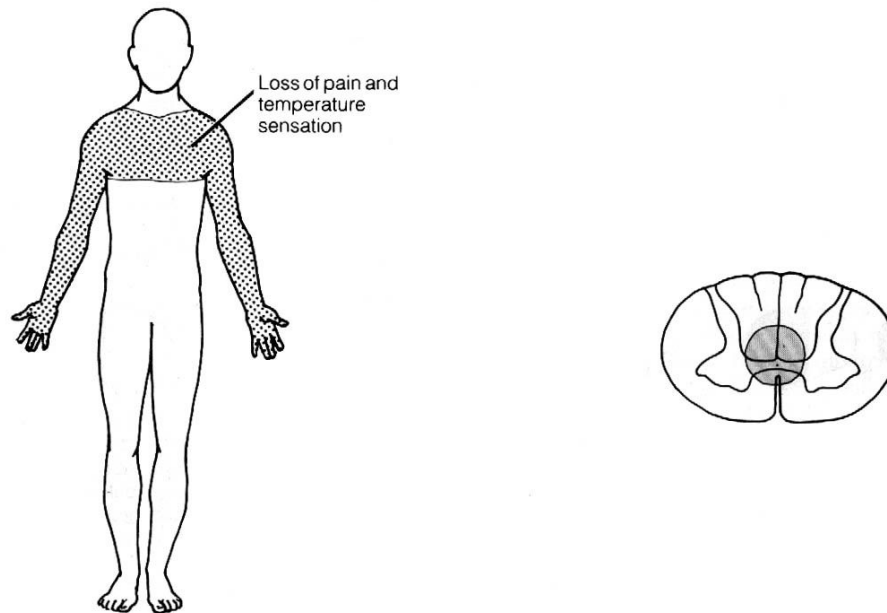


Figure 5-17. Syringomyelia involving the cervicothoracic portion of the spinal cord.

Key muscles

C1-4	use sensory level and assess respiration for involvement of diaphragm
C5	elbow flexors (biceps, brachioradialis)
C6	wrist extensors (ECRL, ECRB)
C7	elbow extensors (triceps)
C8	FDP to middle finger
T1	small finger abduction (ADM)
T2-T5	use sensory level, intercostals
T6-t10	upper abdominal
T11-t12	lower abdominals
L2	hip flexors (iliopsoas)
L3	knee extensors (quadriceps)
L4	ankle dorsiflexors (tibialis anterior)
L5	long toe extensors (EHL)
S1	Ankle plantar flexors (gastroc soleus)
S2-S5	Use sensory level and sphincter ani

	Wheelchair transfer	Wheelchair propulsion	Ambulation	Orthosis
C3-4		Chin controlled		UE externally powered, dorsal cock-up splint BFOs
C5		Manual WC short distances, indoors		
C6	With transfer board	Moderate distance indoors		Wrist driven, universal cuff, built-up handles, writing device
C7	Assisted to/ from floor	Outdoors (Stairs dependent)		
C8-T1	<i>Independent</i>	Curbs, elevators (stairs assisted)	Exercises assisted	
T2-T10		<i>Independent</i>	Exercise only	KAFO FA crutch, walker
T11-T12			<i>Functional indoors</i>	<i>KAFO or AFO with crutch</i>
L3-S3			<i>Community</i>	<i>AFO with crutch or cane</i>

	Feeding	Grooming	Dressing	Bathing	Bowel and bladder	Bed mobility
C3-4						
C5	Independent with equipment	Independent with equipment				
C6	Drinks from glass		UE	With equipment	Independent for bowel only	With equipment
C7	Independent	Independent	With equipment		Independent for both bowel and bladder	independent
C8-T1			Independent	independent		
T2-T10						
T11-T12						
L3-S3						

	Pressure relief	Communication	Pulmonary hygiene	Transportation
C3-4		Adapted phone, typing		
C5				Adapted van
C6	Independent	Adapted hand writing, turns pages	Independent with seated	
C7				Hand controlled car/ special vans, can put WC into car
C8-T1		independent		
T2-T10			T6-T10 independent	
T11-T12				
L3-S3				

Patient Name _____

Examiner Name _____ Date/Time of Exam _____



STANDARD NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY



MOTOR

KEY MUSCLES
(scoring on reverse side)

	R	L	
C5	☐	☐	Elbow flexors
C6	☐	☐	Wrist extensors
C7	☐	☐	Elbow extensors
C8	☐	☐	Finger flexors (distal phalanx of middle finger)
T1	☐	☐	Finger abductors (pinky finger)
UPPER LIMB TOTAL (MAXIMUM)	☐ + ☐ = ☐	(25) (25) (50)	

Comments:

L2	☐	☐	Hip flexors
L3	☐	☐	Knee extensors
L4	☐	☐	Ankle dorsiflexors
L5	☐	☐	Long toe extensors
S1	☐	☐	Ankle plantar flexors

Voluntary anal contraction
(Yes/No) ☐

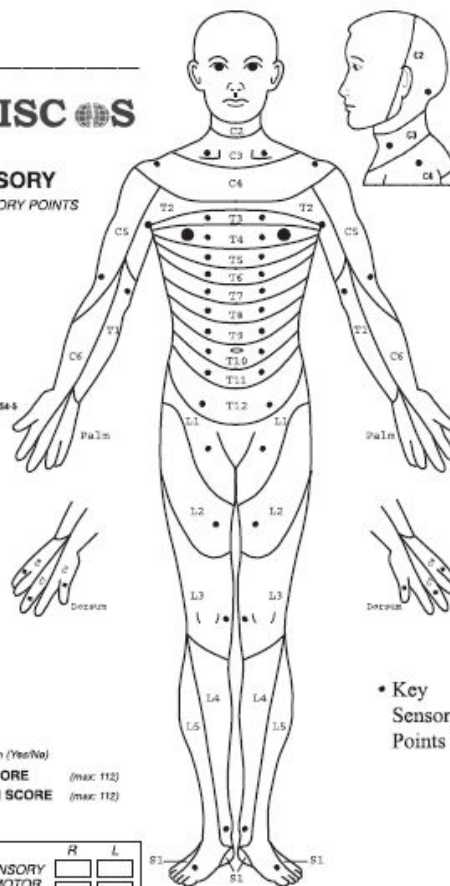
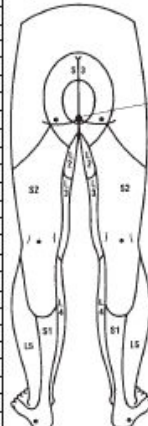
LOWER LIMB TOTAL (MAXIMUM)	☐ + ☐ = ☐	(25) (25) (50)
----------------------------	--	----------------

	LIGHT TOUCH		PIN PRICK	
	R	L	R	L
C2				
C3				
C4				
C5				
C6				
C7				
C8				
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
T10				
T11				
T12				
L1				
L2				
L3				
L4				
L5				
S1				
S2				
S3				
S4-S5				
TOTALS	☐ + ☐ = ☐	☐ + ☐ = ☐	☐ + ☐ = ☐	☐ + ☐ = ☐
	(MAXIMUM) (58) (58)	(58) (58)	(58) (58)	(58) (58)

SENSORY

KEY SENSORY POINTS

0 = absent
1 = impaired
2 = normal
NT = not testable



• Key
Sensory
Points

Any anal sensation (Yes/No) ☐
PIN PRICK SCORE (max: 112)
LIGHT TOUCH SCORE (max: 112)

NEUROLOGICAL
LEVEL
The most caudal segment
with normal function

	R	L
SENSORY	☐	☐
MOTOR	☐	☐

COMPLETE OR INCOMPLETE?
Incomplete - Any sensory or motor function in S4-S5

ASIA IMPAIRMENT SCALE

ZONE OF PARTIAL
PRESERVATION
Gross extent of partially
derogated segments

	R	L
SENSORY	☐	☐
MOTOR	☐	☐

RESEARCH & EBP

i. Research

- a. Evidence based problem:
 - a. Identify the problem
 - b. Review of related literature
 - c. Summarization
 - d. *Application into clinical practice*
- b. Research
 - a. Statement of the problem
 - b. Collect Data
 - c. Methodology
 - d. Conclusion
- c. Panel; Institutional review board

a. G -eneral Knowledge	P- Protection from harm
b. R - isk's Precaution	I – Informed consent
c. <i>E - thical Disclosure</i>	C - Confidentiality
d. B - enefits	K – knowledge of outcome

i. Designs

- a. Historical Design
 - i. Past events
 1. Primary Resource: Directly involve; eyewitness (E.g. Direct Recording)
 2. Secondary Resource: TV Radio, Internet, Social Media (Summary of the event)
- b. Descriptive
 - i. Describing traits/ Characteristics (E.g. Survey Questionnaires)

1. Case study vs Case report

Case Study	Case Report
<i>Objective</i>	<i>Subjective</i>
Reliable	Opinion
	Prognosis
	PT Impression
	Treatment

2. Developmental study: Studying development/ progression of a disease/ growth
 - a. Limited Time frame (E.g. A study of a male patient spina bifida at age of 2, 4 ,5)
3. Longitudinal: Also study of development
 - a. Extended Time frame (E.g. Milestones, *Developmental sequence*)
4. Normative studies
 - a. Normals, Standards, Trends (E.g. Blood pressure, (n) Values)
5. Qualitative studies: Traits
 - a. Human *experience*, Inductive reasoning (E.g. Free taste, Describe pain)

c. Correlative studies: Correlation

Near +1.00	Positively correlated	Frequency directly correlated to Absorption
Near 0.00	No relationship	≠
Near -1.00	Negative correlated	Frequency inversely proportional to wavelength

i. Correlation Coefficient

No	No	0.00
Loving	Low	0.01-0.25
For	Fair	0.25-0.50
Me	Moderate	0.50-0.75
Honey	High	0.75-0.99

Close numbers: Rule of Homogeneity

d. Experimental studies: Cause and effect

i. Variables

1. Independent: Cause, Treatment/ Intervention/ Disease
 - a. True Independent variable: The *adding* independent variable, usually the unique variable
2. Dependent: Effect; Target of treatment
3. Intervening/ Extraneous/ Covariate: unaccounted variables
4. Discrete Variable: Countable/ Finite (E.g. number of finger)
5. Continuous Variable: Uncountable: Infinite (E.g. number of grains on a beach)
 - a. Unrealistic

ii. True Experimental study/ Randomized Control Trial (MC used)

1. Random Sampling: Control and treatment group
2. Blinding
3. ↓ Risk for bias

iii. Cohort/ Quasi-Experimental(Slight experimental)/ Prospective study: Future

1. Opposite: Historical & Case control
2. Only have Treatment group
3. ↑ Risk for bias
4. E.g. Long term assessment of MI patients that smoke *after 10-20 years*

iv. Within a subject/ Repeated Measure

1. Pre & Post test (Checks for improvement)

v. Between Subject

1. Checks for efficiency

vi. Single-Subject Experimental

1. A = Baseline, B= Treatment
2. Cases:
 - a. A- B: Single treatment; Single baseline
 - b. A – B – A: *MultiBaseline (For improvement)*
 - i. E.g. What happen's when treatment is withdrawn?
 - c. A – B – A – B: MultiBaseline treatment approach

3. Scenario:
 - a. $A > B$: Treatment is effective
 - b. $A < B$: Treatment is lacking E.g. Increase intensity of treatment
 - c. $A = B$: Treatment is ineffective E.g. Change treatment if ineffective
- vii. Epidemiology
 1. Study the prevalence of a disease
- ii. Hypothesis
 - a. Testable tentative assumption/ educated guess
 - i. A: Directional/ Straight/ Researchers hypothesis (Expects, assuming)
 1. E.g. There is
 - ii. B: Null hypothesis (Does not expect, doesn't assume; used by researchers)
 1. E.g. There is no

	Conclusion	Error
<i>Alpha/ Type I</i>	Reject the null hypothesis (-) + (-) = (+)	Reject Null hypothesis; Null hypothesis is TRUE
<i>Beta/ Type II</i>	Accept the null hypothesis (+) - (-) = (-)	Beta/ Type II error (Failure) Accept null Hypothesis; Null hypothesis is FALSE

E.g. There is no relationship between studying & passing the board exam: Reject Null hypothesis

- iii. Data types
 - a. N – Nominal E.g. Name, gender, places (Weakest) uses Mode
 - b. O – Ordinal E.g. Order, grading/ scale, level, rank uses Median
 - c. R – Ratio uses True/ Absolute Zero value 0= Nothing E.g. Age, height, length, width (Dimensions)
 - d. I – Interval uses Not true/ Absolute value 0=something E.g. Time, Temperature, IQ, profound mental retardation
- iv. Sampling
 - a. Representative from the whole population
 - b. Types
 - i. Random: All have equal chances of chose MC used(E.g. Lotto)
 - ii. Stratified: According to criteria
 - iii. Systematic: Uses numerical or intervals
 - iv. Generalizability: applying the results of the sample to the general population E.g. Drug Trial studies
 - v. Blinds:

Blinds	Indication
No	Terminal Ill patients
Single	Subject or researchers
Double	Both Subject & Researcher (MC)
Triple	Subject , Researcher, Rate (↓ Bias)

- vi. Cluster: Divide into clusters then random sampling, More effective (E.g. large populations)
- vii. Snowball: Chain referral, word of mouth "sample of one", rare conditions
- viii. Convenience: Locals, volunteers, available subjects
- ix. Purposive: Strictly following according to criteria

v. Validity: Appropriateness

Type of validity	Definition	Example
Internal	Independent is the direct cause of change in the dependent, no intervening variables	DNA test, Gold standard
External	Apply the results of the sample to the general population	
Face	Judging according to appearance, subjective, not reliable	
Content	Checking if right Measurement tool is used	Wrong tool, wrong measurement
Concurrent	Comparing a gold standard vs another standard	Ligament: MRI vs CT scan Lachman vs Ant. Drawer
Construct/ Overall	Abstract concept (not tangible, not countable) Construct measure	Love, Hate Measuring something That is not usually measured, Pain
Convergent	2 constructs are directly related Near +1.00 Coefficient: Mod + high	Increase love, increase desire
Divergent	2 constructs are not related	Increase love, decrease hate
Predictive	Instruments ability to establish future outcome	Mockboards

- a. Threats to validity:
 - i. Sampling bias E.g. Convenience sampling
 - ii. Experimental Bias E.g. Doctoring
 - iii. Hawthorne effect: Awareness, causes a change in behavior
 1. E.g. Knowing someone that notices you, and changing behavior
 - iv. Placebo effect: Positive response to a sham treatment (purely psychological)
 - v. John-Henry effect: Competition causes increase
 - vi. Pygmalion/ Rosenthal effect: Positive Reinforcement/ Rewarding
 1. Opposite: Devil's effect:: underestimation
 - vii. Carryover effect: "Learning effect"
 - viii. Multiple treatment uncontrolled variables
 1. E.g. HMP – TENS – US to treat pain
 - ix. Instrument inaccuracy: Unreliable assessment
 - x. Experimenter Mortality: Researcher gets lazy
 - xi. Subject-Experimenter Fondness/ Relationship: ↑Bias
- vi. Reliability
 - a. Consistency, Variability, Reproducibility
 - b. How Useful (Consistency: +/- 5 units)

Types	Definition	Example
Intrarater	One person	one person measuring 40, 44, 45 (+/- 5) Consistent
Interrater	Two or more	Three people performing Same assessment
Test-Retest	Measures the subject	Aki hip abduction Sober: 0-40° Drunk: 0-180°
Split Half reliability	Divide the group to check for Internal consistency	Two groups measuring

- c. Threats to reliability
- i. Errors in measurement
 - ii. Rater bias: Memory of first test
 - iii. Carryover effect/ learning effect
- vii. Sensitivity vs Specificity

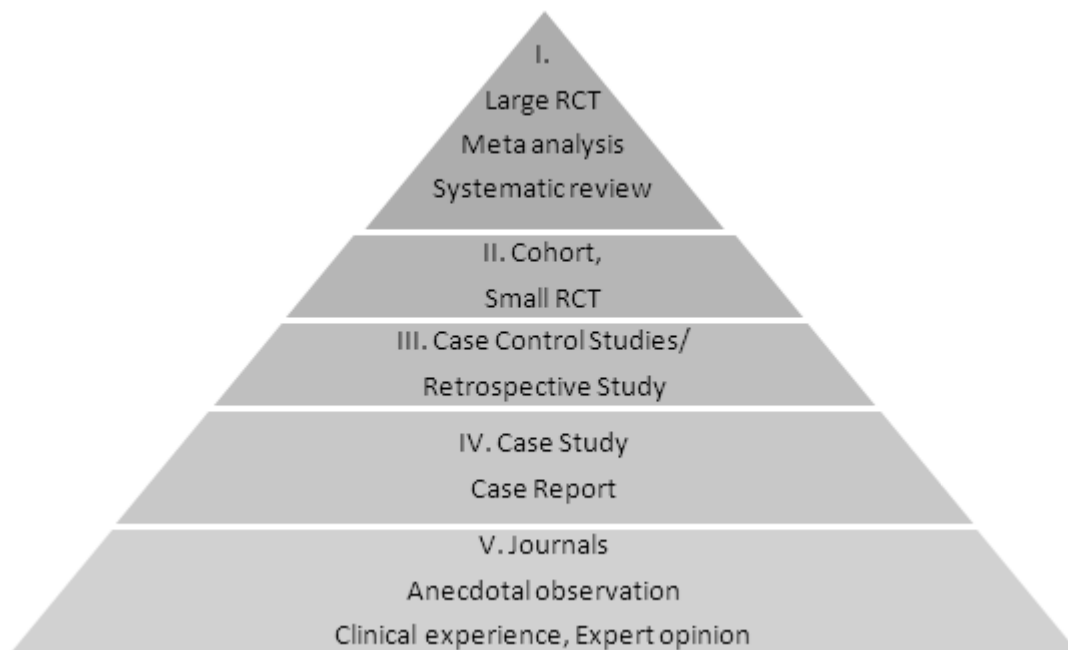
Sensitivity	Specificity
Detect True positive Increase sensitivity: rule in the patient as having the condition "have the disease" Decrease sensitivity: Rule out the patient as having the condition "does not have the disease" E.g. 0.77 sensitivity = True positive 0.13 sensitivity = False Negative	Detect True negative Increase specificity: True Negative Rule patient as not having the condition "Does not have the disease" Decrease specificity: Rule out the patient as not having the condition "Has the disease" E.g. 0.84 specificity = true negative 0.16 specificity = false positive

False Negative: Decrease in sensitivity

	True positive	True negative	False positive	False Negative
Diagnosis	(+)	(-)	(+)	(-)
Follow up statement	(+)	(-)	(-)	(+)

E.g. alzheimers in the gene, but (-) presentation , false positive

- viii. Sackett's



viii. Data Interpretation

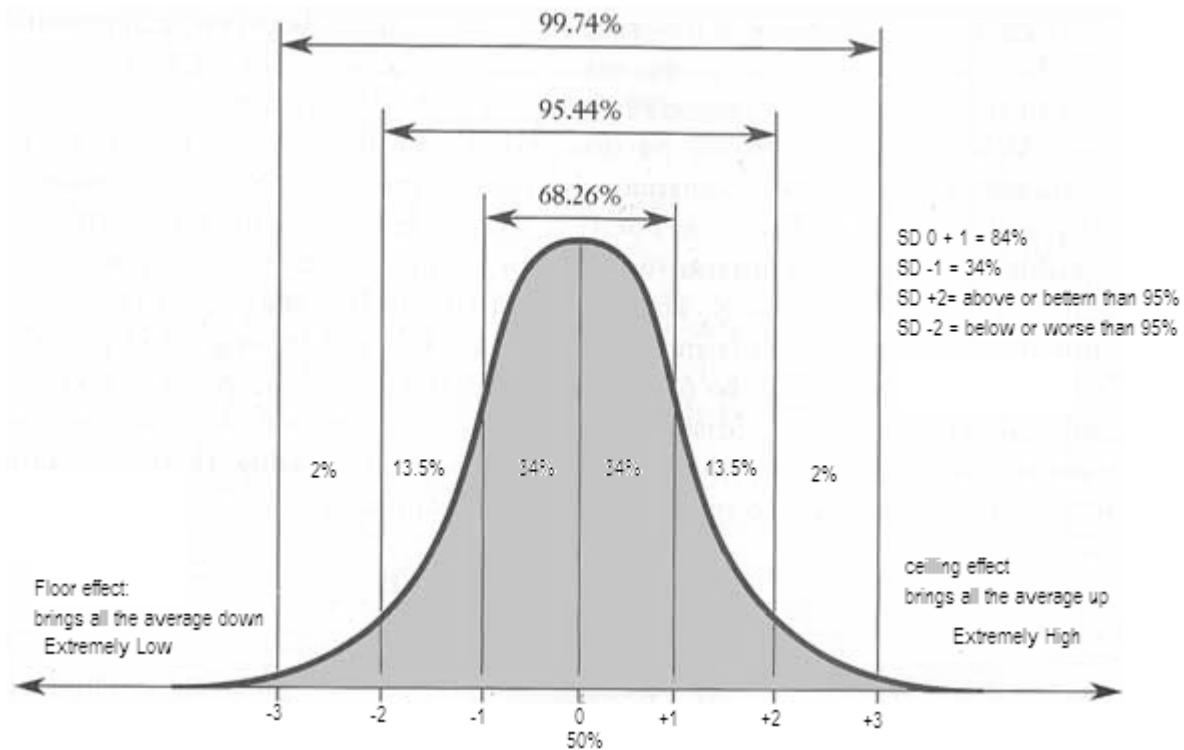
a. Measures of central tendency

- i. Mean: average
- ii. Median: middle
- iii. Mode: MC used value

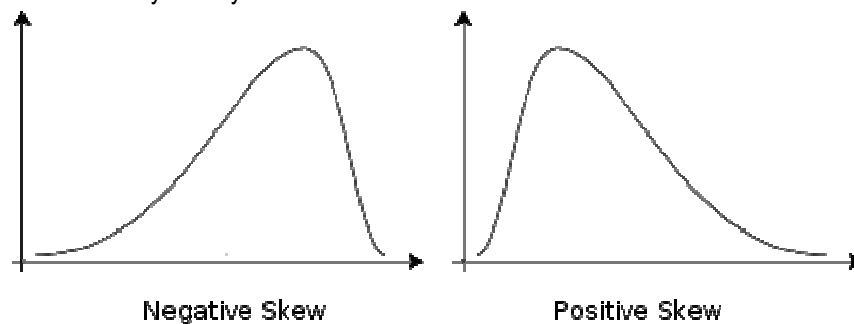
b. Measures of Variation

- i. Range: Difference of highest to lowest E.g. 1 to 5, difference = 4 (range)
- ii. Variance: Square of the average deviations around the means
- iii. Standard Deviation: Average deviation of values around the mean

c. Gaussian, Mesokurtic (values are evenly distributed)



d. Skewness: Asymmetry



i. Positively skewed, below average

- 1. Tail to the right
- 2. Skew to the right
- 3. Majority of the values are below the average

- ii. Negativity Skewed, above average
 - 1. Tail to the left
 - 2. Skew to the left
 - e. Kurtosis-Sharpness
 - i. Mesokurtic: values are evenly distributed
 - ii. Platykurtic: values are spread far apart (flat)
 - iii. Leptokurtic: values are concentrated at the center (sharp)
- ix. Confidence & Error

Confidence	Error
95%	5% : probability (p)=0.05
99%	1% p=0.01
- x. Parametric Test
 - a. Data types
 - i. Interval & Ratio
 - b. Requires normal distribution, random sampling, ↓bias
 - i. T-Test: For 2 means, pre versus post, one group is tested twice
 - 1. Used for samples less than or equal to 30
 - ii. Z- Test: same but > than 30 samples
 - iii. ANOVA: Analysis of variance
 - 1. Simple/ one way:
 - a. one independent vs multiple dependent
 - i. E.g. 77 (Panacea = cures all) used for DM, Tumors, Skin diseases
 - b. Multiple independent vs 1 dependnet
 - i. E.g. HMP, TENS, US > for pain
 - 2. Multiple ANOVA/ Factorial
 - a. Multiple independent vs multiple dependent
 - iv. ANCOVA: analysis of covariance
 - 1. Independent vs Dependnet vs intervening variable
- xi. Non Parametric test
 - a. Data types: ordinal & nominal
 - b. "Unknown distribution"
 - i. Chi-Square (or chis-square Goodness of fit test)
 - 1. Comparison of 2 or more independent nominal data types
 - a. E.g. used in epidemiology
 - ii. Mann-Whitney's U's Test (man Whitney Houston)
 - 1. Comparison of 2 ordinal independent data types
 - a. E.g. Check for treatment & result or treatment & ranking
 - i. Ranking modalities according to subjective effectiveness
 - iii. Wilcoxon Sign Rank test: Comparison of 2 ordinal dependent data types
 - a. E.g. Top notcher scores ranking & score
 - iv. Kruskal Wallis: Comparison of 3 or more ordinal independent data types
 - 1. E.g. check the origin of scores
 - 2. "Where did the come from?"

- xii. Co relational test
- Pearson Product Moment Coefficient (r): For ratio and interval data types (pLRson)
 - Spearman Rho: Ordinal (Spearman RhOrdinal)
 - Transformation test
 - Brush: Transforming interval to Ordinal data type
 - Rasch: Transformation of Ordinal to interval data typ
- "BIOROI"

xiii. Mnemonics

2	DR	N	McNomer	3	DR	N	Cochrane Q test
		O	Wilcoxon Sign rank			O	2-way anova
		I/R	Correlated T-test			I/R	Repease T-Test
	IU	N	Chi-square		IU	N	Chi-square
		O	Man-Whitney U's			O	Kruskal Wallis
		I/R	T-Test(Z-Test)			I/R	A-anova M- multi anova A-ancova

DR= dependent related (2 groups from same group) IU= independent unrelated (2 groups from not same group)

Marc 2
 Wilson's Recieve
 Corny Cheesy
 Cheesy Kisses
 Mother AMA
 Traded
 coke

THERA EX 3

- i. Facilitation & remediation
 - a. Facilitation: Making a task easier
 - b. Remediation: to correct
 - c. Old Techniques
 - i. Doman-Delecatto (Delikado)/ patterning
 - 1. Remediation
 - 2. Cholic (Parasympathetic) Tx: Sympathetic hang upside down + whirl
 - 3. Respiratory acidosis: CO₂ inhalation, paper bag
 - ii. Temple-Fay
 - 1. Ontogeny recapitulates phylogeny
 - 2. Human development mimics animal evolution
 - 3. Fish > reptiles> amphibians> mammals > man
 - 4. Prone> homolateral stage> Contralateral stage > all fours> standing
 - 5. Ontogenetic motor patterns
 - a. Mobility > stability > controlled mobility (CKC + weightbearing)> skill
(combined opened and closed)
 - 6. Eight ontogenetic motor stages
- | | Stages | Motor patterns |
|---|--------------------|---------------------------------|
| 1 | Flexor withdrawal | Mobility |
| 2 | Roll over | Mobility |
| 3 | Pivot prone | Mobility + stability |
| 4 | Neck cocontraction | Stability |
| 5 | prone on elbow | Stability |
| 6 | quadruped | Stability + controlled mobility |
| 7 | standing | Skill |
| 8 | walking | Skill + mobility + stability |
- iii. Ayres
 - 1. *Sensory integration*
 - 2. Associate > action
 - a. Different from Roods
 - i. Sensory input
 - ii. Motor input
 - iii. Gamma efferent activity
 - iv. Phelps - CP
 - 1. 15 modalities (E.g. Massage, ROM, PJM etc.)
 - v. Deaver
 - 1. Extensive use of bracing to promote ADL
 - 2. Bed & W/c mobility & activities

- vi. Vojta & Peto: *Reflex Locomotive Patterns & positions*
 - 1. Reflex creeping: prone
 - 2. Reflex crawling: prone
 - 3. Reflex rolling: sidelying
 - 4. Seen in Eclectic technique, for pediatric patients
- vii. Movement therapy, Brunnstrom
- viii. Bobath, Berta
 - 1. Active & dynamic
 - 2. Inhibition of abnormal tone and posture
 - 3. Key points of control
 - a. Prox: Shoulder, pelvic girdle, spine
 - b. Distal:
 - i. UE: Wrist, hand and thumb
 - ii. LE: Ankle, knees, foot
 - c. E.g. proximal moved, ease of movement of distal, Distal stabilization
ease of movement of proximal
 - d. Marie foix/ Bechterew: DF + knee flexion + hip flexion
 - e. RIP
 - 4. Synergy:

--	--

Strong**Weak**

UE	Flexion	Extension	Typical	RIP
Scapula	Retracted & elevated	Protracted		
Shoulder	FABER	EXADIR	ADIR	ABER
Elbow	Flexed	Extended	Flexion	Extension
Forearm	Supinated	Pronated	Pronation	Supination
Wrist & finger	Flexed	Flexed	flexion	Extension

RIP: Weak thumb over good thumb

LE	Flexion	Extension/ Typical	RIP
Hip	FABER	EXADIR	
Knee	Flexion	Extension	Flexion
Ankle	Dorsiflexion + Inversion	Plantarflexion + inversion	Dorsiflexion
Toes	Big toe extension Toes flexion	Big toe flexion Toe flexion	

RIP: Best way to break synergy is bridging, tall kneeling, quadruped, hooklying

- 5. Stages: (3)
 - a. Flaccidity:
 - i. Goal: From bed ridden to bed bound, bed positioning
 - ii. Initial: Sidelying on good (To push self up)
 - iii. Progression: Sidelying on bad
 - iv. Orthotic: Bobath sling
 - v. Roll
 - 1. Initial: Roll to good
 - 2. Progression: Roll to bad side (guard)

- b. Spastic
 - i. Goal: Bedfast
 - ii. Sit to stand (in preparation for ambulation)
 - iii. Scoot (Head and hip rule, movement of head opposite to hip movement)
 - iv. Standing
 - 1. Initial: bad front
 - 2. Progression: bad back
 - v. Wt. Shifting: CKC exercises (squats & lunges)
- c. Relative recovery
 - i. Only 70-90% d/t residual deformities E.g. Hand, Foot
 - ii. Orthotic devices: E.g. PLS, ADA

ix. Brunnstrom, Signe

- 1. Movement therapy, Advocates postural reflexes and associated reactions
- 2. Associated reactions

Associated reaction	
Raimiste's Phenomenon	Adduction > abduction (Extensor synergy) Resist one side adduction, opposite will adduct UE: sterling's phenomenon LE: Raimiste's phenomenon
Souque's Phenomenon	Reflex finger extension after elevation >90°
Homolateral Synkinesis/sia	E.g. right upper ex moves, right lower ex will follow Can be performed with other movements
Global spasticity/ La Syncinesie Global Ou Spasmodic	E.g. right elbow flexion all other limbs flex
Imitation synkenesia/ Imitatores syncinesio	Normal copies the abnormal cannot do
Coordination synkinesis	Flexion/ shortening synkenesia/ La syncinesie de raccourcissement: (process of shortening): Extensio/ lengthening synkenesia/ La Syncinesie d' allongement (process of lengthening)
Marie Foix/ Bechterev	Passive toe flexion with induced ankle dorsiflexion, Hip & knee flexion
Huntington's Phenomena	Yawn, sneeze, cough will increase spasticity of upper extremity
ATNR	Rotation, same side will extend, opposite, flex. (archer/ fencer's position) D/L: at skull site
STNR	Neck flexion: UE: flexion LE: extension Neck extension: UE: Extension LE flexion
STLR	Supine: Extension Prone: Flexion
Listing/ Pusher syndrome	Leaning towards affected side

3. Brunnstrom's Stages (7)

	Stage	Manifestation	Hand
I	Flaccid	No voluntary movement	No hand function
II	Beginning spasticity	Increase spasticity	Mass grasp
III	Peak spasticity	No movement outside of synergy	Mass grasp carryover(MGCO) Hook prehension
IV	Waning spasticity	Movements available into synergistic pattern	Lateral prehension Minimal thumb movement MGCO
V	Further decline in spasticity	Outo synergy	Palmar prehension, Spherical & cylindrical grasp Hand release
VI	Isolated joint movement	Balance/ equilibrium reaction	All Hand functions
VII	Normal motor function	Normal	Normal

Stage IV vs V:

Stage IV	Stage V
Handslide to the back Supination & pronation Only if elbow is flexed Reach forward	Abduction >90° (forming a "V") Supination & pronation in the Extended position Reach overhead

d. Roods

- Sensory input
- Motor input
- Gamma efferent activity

Inhibitory	Facilitator
Light joint compression (Distraction) Light strokes/ slow brushing Slow/ prolonged stretch Slow icing Pressure on tendon/ insertion Slow rocking/ neutral warmth	Heavy Joint compression Quick stroking Quick stretch Quick icing Pressure on muscle belly Osteopressure (joint pounding) Vibration Gravity intention

e. PNF

- Diagonals/ spirals
- Maximum resistance will produce maximum contraction
- Inconjunction with roods
- Promotion of ADL's
- Principles
 - Progressive milestones
 - Proximal to distal
 - Cephalo-caudal
 - Flexion > extension
 - Adduction > abduction *FeAdab*

2. Overflow/ Irradiation

- a. Strength of a strong action moves to a weaker action
- b. PNF pattern: Timing for emphasis,

vi. Patterns

Strengthening patterns		Promotes
Rhythmic stabilization	Alternating contractions of opposite muscles on opposite sides	Trunk control
Alternating isometrics	Alternating isometric contractions of opposite muscle on the same side	Trunk stability
Repeated contractions	Starts with Quick stretch → contraction of muscle up to its limit → Passively lengthen muscles → continue to contract muscle	Strengthening
Timing for emphasis	Deliberately slowing down of movements to make the action, stronger to increase resistance/ force on the muscle	Strengthening

Alternating Isometrics > Rhythmic stabilizations AIRS

	Chop	Lift
	Trunk flexion + Rotation + Expiration Lead : Weak arm D1 flexion > D1 extension Assist: Good arm Chop towards the weak side	Trunk extension + rotation + inspiration Lead: weak arm D2 extension > D2 flexion Assist: Good arm Lift towards the weak side
Indication	turn supine to prone More functional Break synergy	Turn prone to supine Kyphotic

vii. Diagonals

UE	D1 Flexion	D1 Extension	D2 Flexion	D2 Extension
Shoulder	FADER	EXABIR	FABER	EXADIR
Elbow	Flexion	Extension	Flexion	Extension
Forearm	Supination	Pronation	Supination	Pronation
Wrist and finger	Flexion	Extension	Extension	Flexion
Deviation	Radial	Ulnar	Radial	Ulnar
Functional	Eating, grooming <i>Most functional</i>	Seatbelt Wheelchair	Brushing, wearing Cap	Zipper Tie shoe lace Wear pants Washing perineum
Muscle	Anterior deltoids	Posterior deltoids	Supraspinatus Serratus anterior	Lats Dorsi Teres Major Pects major
LE	D1 flexion	D1 Extension	D2 Flexion	D2 Extension
Hip	FADER	EXABIR	FABIR	EXADER
Knee	Flexion	Extension	Flexion	Extension
Ankle	DF + INV	PF + EVE	DF + EVE	PF + INV
Functional	Crossing leg Heel strike	Skateboarding Bike Pedal Jumping	Getting on bike Squats	Stretch ITB & TFL Push off
Muscle	Tibialis Anterior	Peroneus L. & B. GMED & MIN	Peroneus T.	Tibialis Posterior GMAX

viii. Other PNF patterns

1. Rhythmic Initiation
2. Indication: PD, Apraxia
3. Passive → Active Assisted → Resisted → Active
4. E.g. PD: Bilateral symmetrical D2 Flexion with rhythmic initiation + inspiration

Directed to Agonist	Reversals of Antagonist	Relaxation
Repeated contractions	Slow reversal	Hold Relax:
Rhythmic initiation	Quick Reversal	(Tightness with pain, E.g. RA)
Hold Relax Active Motion	Slow Reversal Hold	Contract Relax
Hold Relax Active Contraction	Rhythmic Stabilization	(Tightness s pain, E.g. Adhesions)
	Alternating isometrics	Slow Reversal Hold Relax

ii. Motor control & learning

a. Goal: repetition

b. Practice

- i. Serial: arranged in proper order A > B > C > D (rest intervals 3-5 mins)

1. Goal: increase endurance

- ii. Serial Distributed AB – CD , decrease rest interval

- iii. Serial Block

- iv. *Random practice* C – D – B – A

1. No particular order

2. Best for memory retention

- v. Random Distributed/ Block C – DB – A

c. Feedback (during and after the exercises)

i. Types

1. Intrinsic Feedback: Patient

- a. E.g. Pain, proprioception, Kinesthesia, Memory, Experience

2. Extrinsic Feedback: from therapist

- a. E.g. Tactile, *Verbal/ Auditory*, Visual feedback

iii. Task Specific/Related Approach

a. Activity

- i. E.g.: Stairs: lead up activity, marching, step ups/ downs

b. Position

- i. E.g. walking: standing, rolling: supine, sidelying, prone

iv. Compensatory training approach

- a. BWSTT (balance weight supported treadmill training approach)

- b. Forcing the patient to perform activities unable to perform

- c. CVA> mock stand, using bike with bandage (constraint induced movement therapy)

TESTS & MEASURES

i. Subjective

a. Pain (location, type, severity, mechanism of onset, aggravating & relieving)

i. Severity

1. Verbal Pain Scale(0-10)
2. Visual Analog Scale (Pedia)
3. McGill Pain questionnaire (Emotional or Affective component)

ii. Mechanism of Onset

1. Provocative factors/ Aggravating (position/ direction)
2. Palliative factors/ Relieving/ Alleviating

iii. Centralization: Localized, (-) radiating pain, good prognosis

iv. Peripheralization: (+) radiating pain, poor prognosis

v. Referred: Internal organ pain

Type of pain	Quality	Origin
Vascular	Throbbing, pounding, diffuse	Blood Vessels
Cramping	Dull, aching, cramping	Muscle
Bone	Dull, deep, nagging	Bone
Sympathetic nerve	Burning	ThoracoLumbar nerves
Nerve Root	Sharp shooting,	Nerve Root
Nerve	Sharp, lightning like	Nerve
Fracture/ Dislocation	Sharp, intolerable pain	Bone/ Joint

b. Red flag signs

	Manifestation
Cancer	Unexplained weight loss Consistent at night Loss of appetite Unusual lumps or growths
Cardiovascular	SOB Dizziness Chest Pain Fatigue Shoulder & Arm pain
Neurological	Hearing/ Visual changes Frequent headache w/o history of injury Faint spells Sudden weakness
Genitourinary/ Gastrointestinal	Frequent abdominal pain Frequent nausea & Vomiting GERD/ Heartburn Menstrual Irregularities

ii. Objective

a. Vital signs

i. Blood pressure

1. <120/80
2. Stages

	Systolic	Diastolic
Prehypertensive	120-139	80-89
Stage 1	140-159	90-99
Stage 2	>160	>100

ii. Respiratory Rate (1 cycle = 1 inhalation/ 1 expiration)

1. (n) = 12-20cpm, Eupnea
2. >20cpm Tachypnea
3. <12cpm Bradypnea

iii. Pulse rate

1. (n)= 60-100bpm
2. >100bpm tachycardia
3. <60bpm bradycardia
4. Grading of pulse:

	Grade	Characteristic
0	Absent	No pulsed
1	Thready	light pressure
2	Weak	Slight pressure, easily obliterated
3	Normal	Normal
4	Bounding	Strong pulse

iv. Irregular pulses

1. Pulsus Tardus: weak amplitude, delayed/ decreased SV/ CO
2. Pulsus Bisfriens: 2 abnormal systolic beats , Aortic Stenosis, Regurgitation
3. Pulsus Alterans: Alternating weak and strong beats, (L) sided heart failure
4. Pulsus Paradoxus: Abnormal respiration during respiration (inspiration), pericarditis, cardiac tamponade (excessive pericardial fluid)

v. Temperature

1. (n) = 37° Celcius +/- one, normothermia
2. >37° Celcius, hyperthermia
3. <37° Celcius hypothermia
4. Conditions

a. Hyperpyrexia (fever)

b. ROM

i. Tools: Goniometer, tape measure

ii. Schober's Test

1. Landmarks: C7 spine > S2
2. (n) within 10 cm)

Joint	OPP	CPP	Capsular patterns
TMJ	Slightly Ajar	Clenched teeth	Limitation of mouth opening
Acromioclavicular	Arms resting by side	90° abduction	Pain at extremes end range movement
Sternoclavicular	Arms resting by side	Full elevation	Pain at extremes end range movement
AO	-	-	Extension and side flexion equally limited
Spine	Midway between flexion and extension	Full extension	Cervical spine: Side flexion and rotation equally limited, extension Thoracic & Lumbar: Side flexion and rotation equally limited, extension
Glenohumeral	55° Abduction 30° horizontal adduction	ABER	ER > AB > IR
Ulnohumeral	70° flexion 10° supination	Full Extension	Flexion > Extension
Radiohumeral	Full extension, full supination	90° flexion 5° supination	Flexion, extension, supination, pronation
Proximal radioulnar	70° flexion, 35° supination	5° Supination	Supination, pronation
Distal radioulnar	10° supination	5° Supination	Full range of movement, pain at extremes of motion
Radiocarpal	Neutral with slight ulnar deviation	Extension Radial deviation	Flexion and extension equally limited
Acetabulofemoral	30° flexion 30° abduction slight ER	Full Extension & IR Or EXABIR	IR > F > AB
Tibiofemoral	25° knee flexion	Full knee extension and ER of tibia	Flexion > Extension
Talocrural	10° plantarflexion	Full dorsiflexion	Plantarflexion > Dorsiflexion
Subtalar	Midway between extremes ROM	Supination	Limitations of varus range of movement
Midtarsal	Midway between extremes ROM	Supination	DF, PF, Adduction, IR
MTP	Neutral	Full extension	1 st MTP: extension, flexion 2-5 th MTP: Variable
IP	Slight flexion	Full Extension	Flexion, Extension
MCP	Slight flexion	Finger: Full flexion Thumb: Full opposition	Flexion and Extension

c. Superficial reflexes

- i. Upper abdominal Reflex (T7-T9)
- ii. Lower abdominal Reflex(T11-T12)
 1. Stimulus: brush abdomen
 2. Response: movement of umbilicus towards same direction
- iii. Cremasteric Reflex (T12-L1)
 1. Stimulus: Brush/ blunt object of the medial thigh
 2. Response: Elevation of the scrotum to the ipsilateral side
- iv. Plantar Reflex: (S1-S2) Response: Flexion of toes
- v. Gluteal (L4-L5, S1-S3) Response: Skin tenses in gluteal areas
- vi. Bulbocavernosus Reflex (S2-S4)
 1. Stimulus: Tugging the catheter
 2. Response: finger in the anus, Anal contraction

vii. Pathologic Reflexes

Reflex	Elicitation	Positive Response	Pathology
Babinski	Stroking of lateral aspect of sole of foot	Extension of big toe and fanning of four small toes	Pyramidal tract lesion Organic Hemiplegia
Chaddock's	Stroking of the lateral side of foot beneath lateral malleolus	Same response as above	Pyramidal tract lesion
Oppenheim's	Stroking of anteromedial tibial surface	Same response as above	Pyramidal tract lesion
Gordon's	Squeezing of calf muscles Firmly	Same response as above	Pyramidal tract lesion
Bing's	Pinprick on the plantar surface of foot	Same response as above	Pyramidal tract lesion
Stransky's	Flicking		

1. Pathologic reflexes:

- a. Schaeffer's: pinching the middle third of Achilles tendon (not a Babinski-like reflex)
 - i. Response: flexion of toes
- b. Piotrowski
 - i. Stimulus: Tapping the Tibialis anterior
 - ii. Response: dorsiflexion of the foot
- c. Rossolimo's
 - i. Stimulus: Tapping the plantar surface of toes
 - ii. Response: plantarflexion of toes
- d. Brudzinski:
 - i. Stimulus: passive flexion of one lower limb
 - ii. Response: Similar movement of the contralateral limb
- e. Hoffman
 - i. Stimulus: Flicking of terminal phalanx
 - ii. Response: flexion of DIP'

d. DTR

- i. Biceps C5 C6
- ii. Brachiorad C5 C6
- iii. Triceps C7 C8
- iv. Hamstrings lateral: S1 – S2
- v. Hamstring medial L5- S1
- vi. Patella: L2 – L4
- vii. Achilles: S1-S2
- viii. Tibialis posterior L4-L5
- ix. Maneuver to elicit reflex: Jendrassik Maneuver
 1. Clench teeth, clasp hands together, look away

e. MMT

i. Grading

Grading		Description
5	Normal	Full ROM, against gravity, Maximal Resistance
4	Good	Full ROM, against gravity, Moderate Resistance
3+	Fair Plus	Full ROM, against gravity, Minimal Resistance
3	Fair	Full ROM, against gravity, No Resistance
3-	Fair Minus	Less than half ROM, against gravity, without resistance
2+	Poor Plus	Initiates movement against gravity
2	Poor	Full ROM, Gravity eliminated, No resistance
2-	Poor Minus	Initiates movement in a gravity eliminated position
1	Trace	Minimal palpable/ visible contraction
0	Absent	No contraction

ii. Testing

1. One jointed muscle: Apply resistance at the end of the range
2. Two jointed muscle: Apply resistance at the mid range
3. Break testing: apply resistance at end- available range

f. Non pathologic endfeels

g. Pathologic endfeels

- i. Early muscle spasm: Starting range, Protective muscle spasm after injury
- ii. Late muscle spasm: End range, instability
- iii. Mushy: Muscle tightness
- iv. Hard: Heterotopic Ossifications
- v. Hard capsular: Frozen shoulder
- vi. Soft capsular: Synovitis, soft tissue edema
- vii. Springy block- Meniscal Tear
- viii. Empty: Pain
- ix. Boggy: Effusion / Edema

h. Senses

i. Superficial

1. Pain: Pin, paperclip
2. Light touch: brush or cotton
3. Temperature: test tube
 - a. Temp: Hot: 40-44°C Cold: 5°C

ii. Deep

1. Proprioception: Position sense
2. Kinesthesia: Movement Sense
3. Vibration: Tuning fork on the joints

iii. Cortical

1. 2 point discrimination: discrimination of 2 points Veniere Caliper,
2. Bargonosis: Discrimination of different weights
3. Graphesthesia: Interpretation of traced letters on skin
4. Stereognosis: Interpretation of objects with palpation

- i. Anthropometric measurements
 - i. Leg length Discrepancy
 - 1. Significance: >1.5cm difference
 - 2. True leg Length: ASIS to medial malleolus
 - 3. Apparent Leg length: Umbilicus to medial malleolus
 - ii. Stump measurement refer to page 223
- iii. Special test:
 - a. Geriatric patients
 - i. Performance Oriented Mobility Assessment (POMA) & Tinetti Test
 - 1. Maximum score: 28
 - 2. High risk for fall: <19
 - ii. Berg balance scale
 - 1. Items:14
 - 2. Maximum score: 56
 - 3. 0-20: wheelchair dependent
 - 4. 21-40: ambulation with assistive device
 - 5. 41-56: Patient has good prognosis in walking independently
 - iii. Get up and go test
 - 1. Maximum score: 5
 - 2. 3 or more: high risk for fall
 - iv. Timed up and go test
 - 1. From sitting then stand up and walk 3 meters
 - 2. (n) time: <10 seconds
 - 3. 11-20 seconds: (n) for elderly
 - 4. 21-30 seconds: moderate risk for falls
 - 5. >30 seconds: high risk for fall
 - v. Functional reach:
 - 1. Reach:
 - 2. 10 inches: normal
 - 3. 6-10 inches: moderate risk for fall
 - 4. <6 inches: high risk
 - vi. Dynamic gait index:
 - 1. Maximum score: 24
 - 2. <21 high risk for fall

PEDIA CONDITIONS

- i. Cerebral Palsy (MC type)
 - a. Any lesion to an immature brain→Neurological deficits
 - b. 5-7 years old (immature brain)
 - c. ↑ Risk: First trimester of pregnancy: Formation of CNS
 - d. Causes
 - i. Prenatal
 1. Prematurity (Most important factor) < 37, < 32
 - a. (n) age of gestation 36-40; 38-42
 - b. (n) birth weight: 3.4kg
 - i. LBW= 3.5kg
 - ii. VLBW= 1.5kg
 - c. (n) height: 50cm
 - d. (n) head circumference: 35cm
 - i. < microcephaly
 - ii. > macrocephaly
 2. Rh Compatability – Rhesus Factor
 3. Kernicterus: increase bile in the basal nuclei & spinal cord
 - a. (n) bile: 16-18mg/dl
 - b. Triad:
 - i. Hearingloss
 - ii. Athetosis
 - iii. Parinaud's (Paralysis of upward gaze)
 4. Infections
 - a. Syphilis
 - b. Toxoplasmosis
 - c. Rubella
 - d. Cytomegalovirus (Bursting of RBC's)
 - e. Herpes
 - ii. Perinatal
 1. Cord coil
 2. Breech delivery
 3. Placenta previa
 4. Abruptio placenta
 - iii. Postnatal
 1. Late developmental defects
 2. Trauma
 3. Infection

e. Pathophysiology

- i. Parasagittal injury: Spastic Diplegia → Quadriplegia
- ii. *Periventricular Leukomalacia*: Highest & greatest prediction of CP; Spastic Quadriplegia
- iii. Focal/ Multifocal: Spastic hemiplegia; MC blood vessel affected: MCA, (+) Seizure
- iv. *Status Marmoratus*: Rare, basal ganglia → *Choreoathetosis*
- v. Selective Neuronal Necrosis

f. Grades of Intraventricular Hemorrhage

Grading	
I. Isolated Germinal Matrix	(-) Deficits
II. Ventricular hemorrhage	Without dilatation of ventricles; (n) ventricle size
III. Ventricular hemorrhage	With dilatation of ventricles
IV. <i>Parenchymal hemorrhage</i>	<i>Greatest prediction of having CP</i>

g. Classification:

- i. Hereditary
- ii. Topography
 1. Monoplegia: 1
 2. Diplegia: LE > UE; 1st MC
 3. Triplegia: 3
 4. Quadriplegia: UE > LE; 2nd MC
 5. Hemiplegia ½ of the body
- iii. Tonal:
 1. Spastic: Pyramidal tract lesion BA 4 & 6 (75%)
 - a. Diplegia:
 - i. (n) Intelligence
 - ii. (+) Strabismus
 - iii. (+) Synergies
 1. Flexion: Frog like positioning > Bunny hopping movement
 2. Extension: Scissoring gait > Combat crawl
 - MC type of hip deformity: Coxa Vara
 - MC type of spinal deformity in pedia: kyphoscoliosis
 - b. Quadriplegia
 - i. (+) mental retardation
 - ii. (+) Straphanger sign
 - iii. (+) Opisthotonus posturing
 - iv. (+) Petit mal type seizure
 - v. Poor prognosis
 - c. Hemiplegia
 - i. Good Prognosis
 - ii. (+) Scoliosis 2° to muscle imbalance
 - iii. Limb asymmetru
 - iv. (+) Ambulation

2. Dyskinetic: Extrapyrarnidal lesions (25%)
 - a. Chorea: Jerky movements
 - b. Athetosis: Slow, writhing movement of distal extremities (fingers)
 - c. Ballismus: Flinging of UE
 - d. Dystonia: Sustained posture of trunk & neck muscles
 3. Ataxic: (5%)
 - a. Cerebellar manifestations
 - b. Hypotonia
 - c. Dysmetria
 4. Mixed: Rare
 - a. *Spastic Athetosis: MC Mixed type*
 - b. *ChoreoAthetosis: MC Dyskinetic type*
- ii. Spinal dysraphism: Any malformation in neural tube
- i. Spinal cord
 - ii. Meninges
 - iii. Vertebra
- b. Spina Bifida: aka divided spine/ incomplete fusion of spinous process
- i. Occulta: Divided spine seen in CT scan & MRI
 - ii. Cystica/ Aperta
 1. Skin no longer intact; exposure of neural tube
 2. Faun's beard: "Tuft of air"
 3. Café – Au – Lait spots: Brownish pigmentation
 - iii. Types:
 1. Meningocele: herniation of meninges; w/o herniation of spinal cord & meninges
 2. Myelomeningocele: Herniation of spinal cord and meninges
 3. Myeloschisis: "Phaque" formation on the SC
 4. Myelocystocoele: Cystic formation, communication w.in the central canal
 5. Lipomeningocele: Cystic formation, Lipomas at the periphery of SC
 6. Hydromyelia: Dilatation of the central canal. E.g. Transverse Myelitis
- c. Diastatomyelia/ Split Cord Malformation: Splitting of the SC L1 – L2 level (Termination of SC)
- d. Tethered Cord syndrome: Traction of the conus medullaris: tip/ end of spinal cord
- e. Arnold chiari malformation: (Type II) posterior cranial fossa > Depression of occipital bone
- i. Kinking of 4th ventricle
 - ii. Herniation of MB & pons
 - iii. Occipitocervical myelomeningocele
 - iv. Atrophy/ hypoplasia of cerebellum
- f. Levels & Manifestation

Level	Manifestation
T6-T12	Kyphoscoliosis
L1, L2, L3	Severe hip dislocation
L4, L5	Mild hip dislocation with calcaneovalgus
S1	Calcaneovarus
S2	Claw of toes
S3-S5	Sphincter control

Best therapeutic position for hip dislocation: Abduction & Extension

iii. Congenital Anomalies

a. Trisomy: Type of polysomy, three instances of a particular chromosome instead of the normal two)

Trisomy	Syndrome	Hallmark	Manifestations
21	Down syndrome	Simian Crease	Brachycephalic skull Palpebronasal Lax Odontoid ligament ($\neq$ neck flexion) Hypotonia Protruding tongue Pes planus Flabby fingers
13	Patau's Syndrome	Death (6 months)	Visual deficits Cleft palate Hearing deficits Mobile joints (thumb rotation 360°)
18	Edward's syndrome	Cone-shaped head (Microcephaly)	Dies immediately
5	Cri - du - chat "Cry of the cat"	High pitched monochromatic cry	Poor suck and swallow
15	Prader - Willi Syndrome	Hyperphagia	(-) Satiety center $\rightarrow$ Obese Dysmorphic face Death 2° to obesity complications
Sex: XXY47	Klinefelter's syndrome	Azoospermia (no sperm in semen)	Supermale Problems with erection
Sex: X45	Turner's Syndrome/ Gonadal Dysgenesis	Short stature	Superfemale Infertility (Primary amenorrhea)

Warkany syndrome Chromosome 8, very rare
Cat-Eye syndrome Chromosome 22

iv. Muscular Dystrophies

a. DMD

- i. MC type of MD
- ii. Death: <20 y/o
- iii. Absence of dystrophin
- iv. (+) Pseudohypertrophy of calf muscle
- v. (+) Gower's sign $\rightarrow$ Weakness of Quads and G Max
- vi. 15 y/o = WC borne; encourage walking as long as patient can still stand / walk

b. BMD

- i. Milder form of MD
- ii. Lacks dystrophin
- iii. Lifespan: 30 y/o

c. Emery Dreifuss

- i. Emerin
- ii. Triad
 1. Cardiac conduction deficits
 2. Progressive muscle weakness
 3. Early contractures

- d. Fascioscapulohumeral
 - i. Facial muscles & shoulder girdle weakness
 - ii. Difference in reaching/ overhead activities
 - iii. Primary disability: Scapular winging
- e. Limb girdle dystrophy
 - i. Shoulder & pelvic girdle weakness
 - ii. Major feature: scapular winging
 - iii. Disability: difficulty rising from chair
- f. Myotonic Dystrophy/ "Steinert's"
 - i. "Myotonic Phenomenon":
 - 1. Delayed muscle reaction
 - 2. APB → Flexion of the thumb
- v. Congenital Myopathies
 - a. Aka Elilenberg's disease → Paramyotonia congenital; milder form of Myotonia Congenita
 - b. Myotonia Congenita: Thomsen's disease, "Infantile herculism"
 - c. Nemaline Myopathy: Chromosome #1, poor suck and swallow
 - d. Central core Myopathy
 - i. Decrease ATPase → congenital dip dislocation
- vi. Metabolic
 - a. Glycogen storage
 - i. Type II: Acid Maltase deficiency → Pompe's disease
 - ii. Type III: Debranching Enzymes
 - iii. Type V: MC Ardlle Disease
 - iv. Type VII: Phosphofructokinase
 - b. Menke's disease
 - i. Mental retardation
 - ii. X- linked
 - iii. Copper
 - c. Kwashiorkor: Caloric
 - d. MAramus: Caloric + Protein
- vii. TBI in children

3 - Crying	4 – Flexion Extension	4 – Pursuit
2 – Hinga (2x)	3 – Withdraws	3 – Intact reactive
1 – Apneic	2 – Hypertonic	2 – Pursuit impaired
	1 - Floppy	1 – Fixed paralyzed

Maximum Score: 10

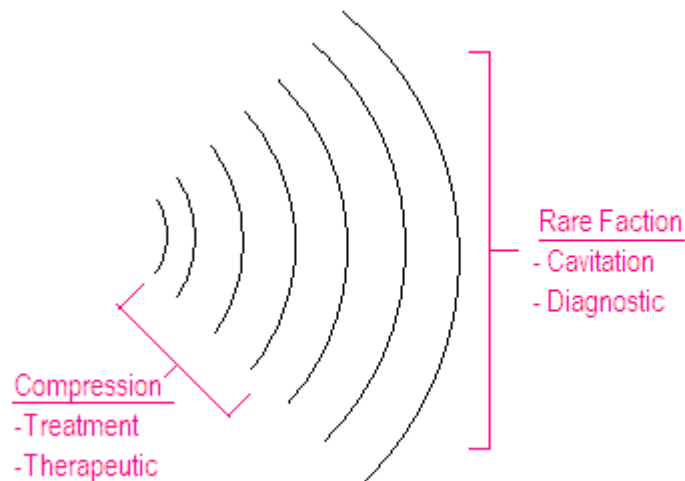
ELECTRO 2

i. High frequency currents

- a. Frequency >500,000 Hz
- b. For thermal effects
- c. Indication
 - i. Muscle relaxation
 - ii. Increase vasodilation
 - iii. Acceleration of Metabolism
 - iv. Hyperemia
 - v. Increase cell membrane permeability
 - vi. Analgesic effect
 - vii. Increase collagen extensibility

ii. Ultrasound

- a. Sound – audible mechanical property seen in most vibrating objects as a result of vibration between the particles of matter between the object
 - i. Infrasonic: <20mHz (Cats, dogs, whales)
 - ii. Audible sound: 20-20000 Hz
 - iii. Ultrasonic= >20,000 Hz



Frequency of noise: >100Hz (125Hz)

Frequency that will injure the eardrum: >1000Hz

b. US frequency

- i. Diagnostic: 10-15mHz
- ii. Surgical: 5-8mHz
 - 1. E.g. Vibrating Scalpel
- iii. Therapeutic: 0.5-3mHz
 - 1. 1mHz: 4-5cm, deep
 - 2. 3mhz 1-2cm, superficial

- c. Principles of production
 - i. Piezoelectric: mechanical forces applied on a crystal will produce electricity
 - ii. Reverse piezoelectric effect: high frequency current applied on a crystal will produce vibration (sound)
 - iii. Crystals in the transducer head:
 - 1. Quarts
 - 2. Molybenium
 - 3. Ceramic
 - iv. Factors affecting depth of penetration
 - 1. Frequency
 - 2. Wavelength
 - 3. Intensity
 - 4. Mode
- d. Principles and properties
 - i. **Absorption Vs. Penetration**
 - 1. Absorption: energy goes into the cell (superficial)
 - 2. Penetration: Energy goes through the cell (deep)
 - a. Absorption is inversely proportional to penetration
 - 3. Factors affecting Absorption and penetration
 - a. Frequency and wavelength
 - i. Relationships: Absorption is directly proportional to frequency, while penetration is inversely proportional to Wavelength
 - ii. Penetration is directly proportional to wavelength, while absorption is inversely proportional to wavelength
 - iii. Therefore, the relationship between frequency and wavelength is inversely proportional
 - iv. Thus, the higher the frequency, the more energy is absorbed, less penetration (3mHz)
 - v. Thus, the lower the frequency, the less energy absorbed, more energy penetrates (1mHz)
 - b. Protein content
 - i. More protein content, more energy is absorbed (Absorption)
 - ii. Less protein content, less energy is absorbed (Penetration)

“Be Careful To See My Newly Furnished Bed”

(absorption) Bone>Cartilage>Tendon>Skin>Muscle> Nerve>Fat/Fascia>Blood (penetration)

Bone receives the most amount of ultrasound energy therefore, it is prone to cavitations

 - c. Fluid content
 - i. More fluid content, more energy penetrates
 - ii. Less fluid content, less energy penetrates

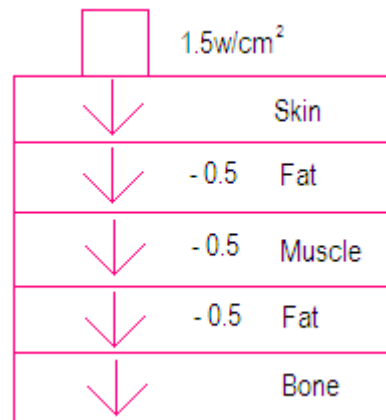
“Be Careful To See My Newly Furnished Bed”

(absorption) Bone<Cartilage<Tendon<Skin<Muscle< Nerve<Fat/Fascia<Blood (penetration)

Blood receives the least amount of ultrasound energy therefore, it is the least prone to cavitations

ii. Attenuation

1. Progressive loss of US energy



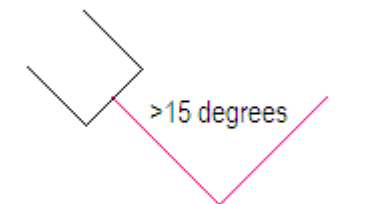
Deeper structures require more intensity

2. Factors affecting attenuation

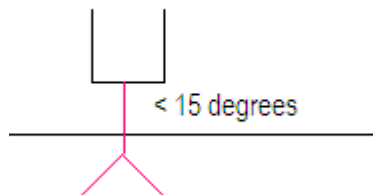
- Absorption: the major cause of attenuation is absorption; more energy is absorbed, more attenuation occurs
- Reflection vs Refraction
- Scattering: the amount of dispersion from a fixed energy source; the further the energy source from its target, the greater the dispersion.
"Rayleigh's Law"

iii. Reflection vs. Refraction

- Reflection: bouncing of energy, occurs when an energy source is tilted by more than 15 degrees from its target area; no energy is absorbed by the target area



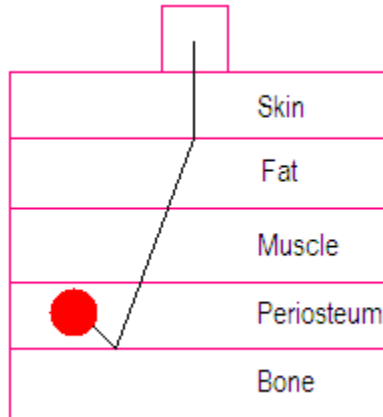
- Refraction: bending of energy occurs when an energy source is tilted by less than 15 degrees from its target area; minimal energy is absorbed by the target areas



3. Factors affecting Reflection and refraction

- Lambert – Cosine Law: Maximum energy absorption occurs only if the energy source is perpendicular to the target area; critical angle is 15 degrees

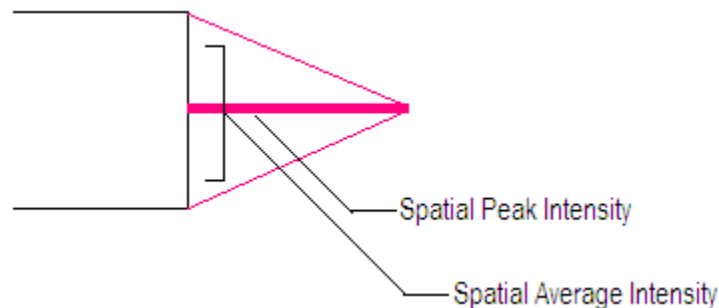
- b. Acoustic Impedance: the property of matter to resist acoustic energy, varies with each substance
 - i. Acoustic mismatch: occurs when two substance of different acoustic impedance coefficients are layered on top of each other thereby resulting into reflection or refraction
 1. Most commonly occurs as periosteal heating resulting in cortical thinning and increased predisposition to fractures



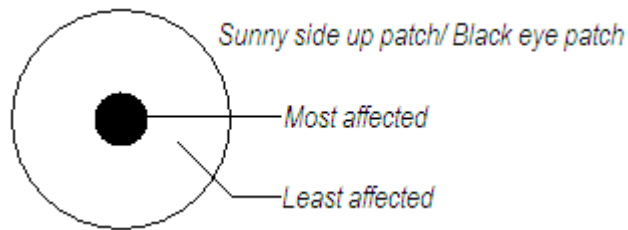
2. Solved by:
 - a. Moving the US head
 - b. US holiday/ Recess (after 12-15 sessions)
 - <60 y/o: 1-2 weeks of rest
 - >60 y/o: 2-4 weeks of rest
- ii. Transducer- Air Interface: occurs when the US head is lifted from its target area, air passes in between and reflects the US energy back onto the head, resulting in heating and damage to transducer head.\

Air is a major reflector of US energy

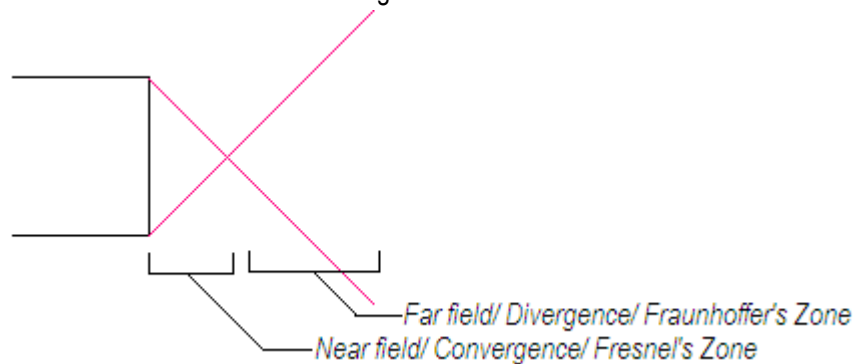
 1. Solved by
 - a. Increasing the coupling medium
 - b. Maintaining perpendicular contact
- iv. Beam Non-Uniformity Ratio
 1. The ratio between the Spatial Peak Intensity (6 or 7) and the Spatial Average Intensity (1 or 2)
 2. Most commonly used rate is 6:2



3. E.g. $6 \times 1.5\text{w/cm}^2 = 9\text{w/cm}^2$ (Burns)
4. $2 \times 1.5\text{w/cm}^2 = 3\text{w/cm}^2$ (Does not burn)
5. The higher the BNUR, the higher the risk for BURNS



6. Solve by:
 - a. Moving the US head
 - b. Increase the treatment area
- v. Beam Spread
 1. The amount of scattering and the direction of the energy source
 2. There are two types of fields
 - a. Near field/ Convergence/ Fresnel's Zone: Immediately next to the transducer head
 - b. Far field/ Divergence/ Fraunhofer's zone: farther from the transducer head



- e. Effects of US
 - i. Thermal effects (MAHAI mnemonic)
 1. Only seen in 100% duty cycle or Continuous Mode US
 2. Muscle = temp increase by $2-3^{\circ}\text{C}$
 3. Tissue around the joint = temp increase $3-5^{\circ}\text{C}$
 4. Joint itself = temp increase $7-8^{\circ}\text{C}$
 5. Indication
 - a. Prior to stretching and exercise
 - b. Prevent formation of contractures and adhesions
 - c. Treat edema and pain (Edema pulsed)
 - d. Used to treat muscle spasms and spasticity

ii. Non – Thermal effects

1. Only seen in 20-50% duty cycle or Pulse Mode US
2. Cavitation: spontaneous bubble formation by minute gas or air pockets inside the body (Dependent on intensity & speed of application)
 - a. Has good and bad effects
 - b. Good (Stable): when present in adhesion and contracture (destroys contractures); when in the presence on interstitial fluid and edema (returns fluid into blood vessels)
 - c. Bad (Unstable: when it occurs in bones and destroys the surrounding tissue when it explodes; when it occurs near nerves or the spinal cord itself, causing severe neural lesions; when it occurs near internal organs)
 - d. E.g. Liver> Gall bladder>Bile (toxin) > Destruction of protein
3. Acoustic streaming: the property of sound to push objects, especially fluids, along its path
 - a. Useful for returning edema (Pulsed mode is best for edema)
 - b. Contraindicated when there is acute and active infection in the area
4. Micromassage/ Krasny-Ergen effect: sound vibrates so structure at the molecular level also vibrate
5. Effects on Nerves
 - a. Can be used to increase NCV (1MHz & 3-5w/cm² setting)
 - b. Can be used to decrease NCV (3MHz & 1-2w/cm² setting)
 - c. Can be used to treat spasticity by acting on the gamma fibers (B gamma> C gamma> A gamma)

f. Parameters for using US:

i. Duty cycle

$$(\text{On} / \text{On} + \text{Off}) \times 100\%$$

100% Duty Cycle (continuous) = no off time (Thermal effect)

Indication: Chronic pain, after stretching, chronic infection

20-50% Duty Cycle (pulsed) = with off time

<50% non – thermal effect

>50% thermal effect

Indication: Acute pain, Edema

ii. Duration: 2-3x of Effective Radiation Area (diameter of transducer head 2-3")for 5-10mins

1. Small area: 2-3mins E.g. Carpal Tunnel
2. Medium area: 3-5mins E.g. Achilles Tendon
3. Large area: 5-10 mins E.g. Quads, lower back (5mins each side)

iii. Intensity (w/cm²)

Stage	Intensity	Time frame
Acute	0.1-0.5w/cm ²	3 days
Subacute	0.5-1.0 w/cm ²	3days to 1week
Chronic	Chronic: 1-2 w/cm ²	1 month

For Pedia/ Geria patients reduce the intensity!

- iv. Speed of application
 - 1. Circular: 2-3 seconds/ cycle and overlap 50%
 - 2. Parallel: 4cm/sec then overlap 50%
 - a. Small head: 2cm/ second
 - b. Big head: 4cm/ second
- v. Size of transducer head
 - 1. Big head: deep penetration E.g. Supraspinatus
 - 2. Small head: superficial penetration E.g. Bicipital groove, CTS
- vi. Techniques
 - 1. Direct contact: most convenient for regular area; if applied on a irregular surface, just add more coupling medium
 - 2. Underwater/ immersion technique: used for distal and irregular surfaces
 - a. Uses degassed water that has lesser bubbles and less chances for reflection of US energy
 - b. Transducer head should be 1 inch or 2-3 *fingerbreadths* from the target area
 - 3. Water bag technique: for irregular proximal areas and open wounds
 - a. E.g. plastic bag, glove, condoms, hydrogel
- vii. Sensation: no sensation should be felt

Sensation	Action
Electric shock-like	Unplug the machine and label it defective
Slight burning	Add coupling medium to air interface
Sharp stabbing/ aching pain	Decrease the intensity

- viii. Joint and muscle positioning
 - 1. Joints should be positioned in the open packed position
 - 2. Muscles should be position in the fully lengthened position
 - E.g. Biceps in bicipital groove: extended
 - Supraspinatus: EXABIR
- ix. Transfer of drugs (Phonophoresis)
 - 1. Transdermal route (>5cm) using pulsed mode
 - 2. Advantage: No first pass effect

g. Advantages/ disadvantages/ Contraindications

Advantage	Disadvantage	Contraindication
Can be used in metallic implants Fast & effective Less time consuming	Cavitation Effect Expensive	Growing endplates Fractures (Low dose US) Cemented prosthesis Pregnant uterus Demand type pacemakers Abdominal organs SCI and post laminectomy

iii. Laws of Radiation

Law	Definition	Example
Grothus drapier	Max absorption is proportional to max effect	↑ Absorption = ↑ Effect
Cosine	Max absorption only if $\perp 90^\circ$	Critical angle 15°
Joules	$Q \text{ (Heat)} = \text{Intensity}^2 \times \text{Resistance} \times \text{time}$	$Q = I^2 \times R \times T$
Kirchoff's	Q (current density)	↑ Density = ↑ Heat
Vanthoff's	For every $\uparrow 10^\circ\text{C}$ in $T^\circ = 2\text{-}3\times \uparrow$ cell metabolism/ oxidation	
Wien's	$\lambda \propto 1/T^\circ$	$\text{Freq} \propto T^\circ$
Bunsen – Roscoe Law of Reciprocity	Time $1/\propto$ Intensity	$\uparrow \text{Intensity} / T^\circ = \downarrow \text{Time}$
Inverse Square	Intensity $1/\propto \text{Distance}^2$	
Rayleigh's	Amount of scattering is $1/\propto \text{Distance}^4$	Flashlight
Stefan- Boltzmann	Radiation $\propto T^{\circ 4}$	
Arndt-Shultz Rule	Excessive radiation is harmful, Lacking radiation is not effective	

iv. Short-wave Diathermy

Frequency (MHz)	Wavelength (Meters)
13.56MHz	22m
27.12MHz	11m
40.68MHz	7m

a. 2 types of fields

i. Electrostatic

1. Vibratory friction: in free ions moving back and forth E.g. Na, K, Cl, Mg, Ca
2. Rotator friction: in dipolar molecules (H_2O) as polarities change making them rotate
3. Disruptions: in non polar molecules (fat)

ii. Electrodynamic

1. Produces Eddy currents and magnetic friction

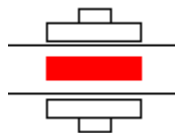
b. Methods of application

i. Condenser field/ Capacitor field (Electrostatic)

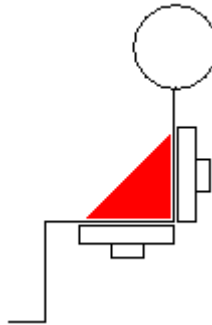
1. Uses flexible plates and pads
 - a. Coplanar: arranged in parallel (Superficial)



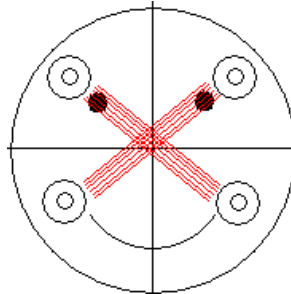
- b. Contraplanar: Arranged in series (Deep)



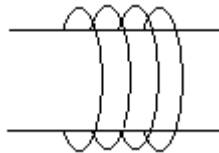
- c. Cross-fire: for the pelvis and sinuses
 - i. Pelvis: use contraplanar and perpendicular
 - 1. Indication: dysmenorrheal, Orchitis



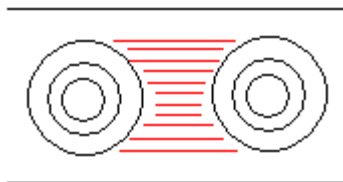
- ii. Sinuses of the face: combinations
 - 1. Indication: sinusitis, Allergic Rhinitis



- ii. Inductotherapy (Electrodynamic > electrostatic)
 - 1. Uses coil, cables, and drums
 - 2. Coil Technique: wrapping extremities with 1 inch space coils
 - a. Indication: swelling/ inflammation



- 3. Helix/ Pancake: coils on flat surfaces
 - a. Indication: referred pain



- 4. Spacing:
 - a. Increased spacing: deep heating
 - b. Decreased spacing: superficial heating
 - c. Sensation: mild comfortable warmth
 - d. Mode: Pulsed SWD better than continuous SWD

c. Indications & Contraindications

Indications	Contraindications
Pain relief	Malignancy & Thrombophlebitis
Decrease muscle spasm	Ischemic Tissue
Inflammation	Edema
Delayed healing	Wet Dressings
Infection	Metallic implants
Fibrosis	Pacemakers (Demand & Fixed)
Sprain, Strain	Hemorrhagic areas
DJD (RA & OA)	Tuberculous joints
Chronic Inflammatory	Hypertensive/ Unstable BP
Non-infective condition	Pregnant patients

v. Microwave Diathermy

a. Uses a magnetron that oscillates for millions of cycles per seconds to generate microwave radiation

- Can penetrate up to 30mm/ 3cm (2450MHz frequency setting)
- More effective on fluid tissues since water quickly absorbs microwave
- Can be used for pain, muscle spasm, inflammation, delayed healing, infections
- Rarely used nowadays because of EMP

b. Techniques

- Large circular field: Outer heating > Inner heating



- Small circular field: Outer heating > Inner heating



- Longitudinal Director: Heating concentrated at the center



c. Frequency, Wavelength

- 2450MHz, 122.5nm

vi. Infrared radiation (IRR)

a. Any object with a core temperature of greater than 0°C will emit radiation

- Humans emit non-luminous IRR

IRR type	Name	Wave length	Penetration	Target
IRR A	Short/ Near	70-4000nm	1-3mm	Subcutaneous tissue
IRR B	Long/ Far	4000-15000nm	0.5-mm	S. Corneum
IRRC	Very long/ Very far	15000-40000nm	Several μ m	Not used

b. Principles

- Production: Wien's Law
- Propagation: Inverse Square law
- Absorption: Cosine Law

- c. Pigmentation
- IRR: Motled
 - UVER: Homogenous
"HUMI"
- d. Types of generators

Luminous	Non-Luminous
<i>Tungsten/ carbon filament</i>	<i>Resistance wire</i>
Very hot/ glowing bodies	Resistant metals produce heat
Short IRR/ IRR A	Long IRR/ IRR B
Has visible light 4.8%	Far IRR (90%)
Has UVR (1%)	Near IRR (10%)
Far IRR (24%)	<i>Requires pre-heating 5-10'</i>
Short IRR (70%)	More hotter
<i>Distance: 18-24"</i>	<i>Distance: 29-36"</i>
Uses localizer	

- e. Effects/ Indications/ Contraindications

Effects	Indications	Contraindications
Nerve stimulation	Decrease pain &	Impaired sensation
Vasodilation	Muscle spasm	<i>Fungal infection,</i>
Phagocytosis	<i>Edema</i>	<i>Dermatitis, Eczema</i>
(↑ immune response)	Healing of wounds	Eyes (Bell's Palsy use
Reflex heating		Cotton/ eye patch)
Pigmentation &		<i>After deep X-ray</i>
Erythema		Topic creams & analgesics
Seating		Unstable BP
Decrease BP		Skin tumor
		Acute skin infection

After Deep X-ray, there is decrease in immune response & increase in dead cells. Effect of IRR is increase immune response that result to phagocytosis of dead cells. Once recovered, continued phagocytosis of the same cell type of the dead cells (Autoimmune)

- vii. Ultraviolet Radiation

UVR type	Name	Wavelength
UVR A	Long/ Far	290-390nm
UVR B	Short/ Near	180-290nm
UVR C	Very short/ Very near	<100nm

UVR C can cause Cancer

UVR A, Apple (Girl) likes long objects older men 29-30y/o

UVR B, Boy likes short skirts young women 18-29y/o

UVR C, children are very short. C for Cancer

- a. UVR Generators use *mercury vapour lamps*
- Air cooled lamps: UVB
 - Hanovia: Acne and Psoriasis
 - Hanau-hohensonne: Acne and psoriasis
 - Birtcher: Pressure ulcers
 - Water cooled: UVA & UVB (Kromayer lamp: Pressure ulcers and phototherapy)
 - Flourescent tubes: UVA (Therarkin Lamp: psoriasis of large body parts)

b. Effects of UVR

Local	General
Pigmentation (homogenous) NAntibiotic (UVB, Bactericidal) Growth of epithelial cells Erythema (Primary visible effect, Erythema Ab Igne -) Desquamation	Vit. D synthesis(UVB) Esophylactic effect (UVA Bacterial resistance) Natural protection to harmful UVR

Reticulo-endothelial effect: 2900 – 4000 A

Erythema is first seen at this wavelength: 240-300nm

c. Indications/ Contraindications

Indications	Contraindications
Acne: Face and neck E2, Back and shoulder E3 Insipient Pressure Areas E1/E2 Pressure Ulcers - Non – infected, E2 - Infected c yellow pus, E3 - Infected c green pus, E4 Counter-Irritation, E4 Psoriasis - Leed's Regime: Coal tar bath + UVR + Dithranol cream then bandage (Most Effective) - Photochemotheraopy/ PUVA UVA + Gamma methoxypsoralens	Acute dermatological disorders Photosensitivity (Lupus, albinism) Febrile After Deep X ray Tx After IRR Tumors and TB

	Dosimetry	Progression	Appears	Disappears	Pigmentation	Desquamation	Description	TBS	Frequency
MED			6-8 hrs	After 24hrs	(-)	(-)	Mild Erythema	100%	Daily
E1	1 x MED	25% of Previous Tx	6-8 hrs	After 24hrs	After Repeated Exposure	After repeated Exposure	Mild Erythema	100%	Faily
E2	2.5 x E1	50% of Previous Tx	4-6hrs	After 48 hours	Definite	Powdery	Mild sunburn	20%	5x/ wk
E3	5 x E1	75% of Previous Tx	2-4hrs	After 72-96hrs	Marked	Marked	Severe Sunburn/ Marked Erythema	250cm ²	3x/ wk
E4	10 x E1	Not Progressed	Within 2hrs	After 2 wks	Marked	Marked	Blisters/ Intense Erythema	25cm ²	1x/ wk

TBS= Total Body Surface

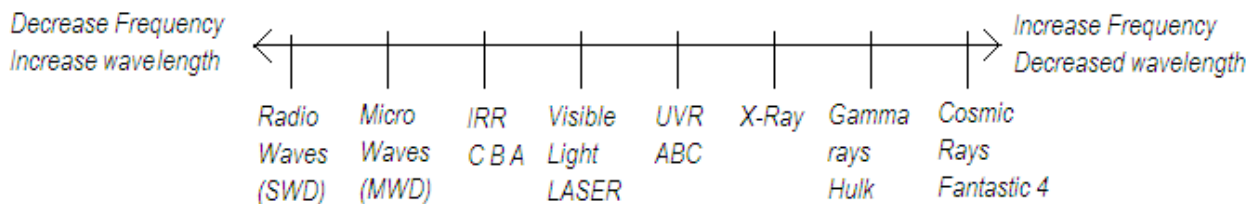
- viii. LASER (Light Amplification by Stimulated Emission of Radiation) / Cold LASER
 - a. Characteristics
 - i. Monochromaticity: 1 color
 - ii. Collimation: minimal divergence
 - iii. Coherence
 - 1. Temporal: same phase or wavelength
 - 2. Spatial: Same direction
 - b. Types
 - i. Helium-Neon (HE-NE) – red: Noble Gas Laser; MC used
 - ii. Gallium – Arsenide (GA-AS) – green: Used in semi-conductors; more powerful
 - c. Dosage: Joules/cm²
 - d. Indications
 - i. Escharotomy
 - ii. Pin-point muscle heating
 - iii. Softening of adhesions
 - iv. Vasodilation
 - v. Collapsing of obstructed blood vessels/ varicose veins
 - e. Contra indications: similar to UVR

ix. Mnemonics/ others
Radiation bells
I. **Grothus drapier** law
Maximal absorption daw
Ay proportional doon
Sa maximal effect

II. Cosine law naman
Maximal absorption if
Perpendicular at 90°
Critical ay $15'$ (Hey!)

III. **Joules law** daw
Q equals I squared R times T
Kirchoff's law Q equals
Current Density! (Hey!)
Vant Hoff's Law
Every 10, 2-3 increase
Wien's Law wavelength daw
Kabiliktaran ng T°

IV. Bunsen Roscoe law.... of reciprocity
Time is the reverse.... of your intensity
Inverse square law daw...
Madali lang po iyan....
Ilayo ninyo at hihina na ang intensity! (Hey!)
(Repeat verse III)



VESTIB PHYSIO & REHAB

i. Anatomy of Ear

a. Outer:

- i. Auricle/ Pinna, collects sound waves from external environment
- ii. External Auditory Meatus: Transmits sound waves from external environment
- iii. Ear drum/ Tympanic Membrane: Thin sheath; absorbs sound waves
 1. Demarcation between outer and middle ear

b. Middle:

- i. Auditory Ossicles
 1. 3 small bones (Transmits electrical impulse)
 - a. Incus
 - b. Malleus
 - c. Stapes
- ii. Organ in the ear that transduces sound waves to electrical impulses? Organ of Corti
- iii. Eustachian tube
 1. Maintains pressure inside the ear
 2. Connection with throat
 3. Swallow: Eustachian tube opens, External and internal environment pressure equal

c. Inner

- i. Oval & round windows: demarcation between middle & inner ear
- ii. Cochlea: main sensory organ *for hearing*
- iii. Semicircular canals(SCC): Dynamic equilibrium (balance)
- iv. Otolith – Vestibule: Static equilibrium

ii. Vestibular system

a. Vestibular apparatus

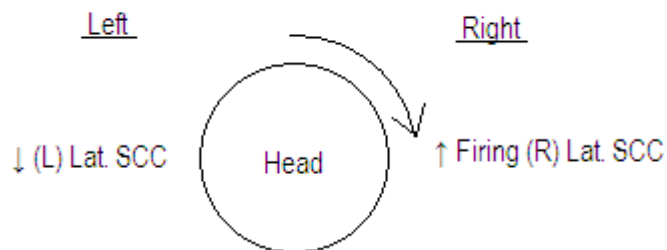
- i. Central (balance inside the brain)
 1. Cortex, cognition
 2. Cerebellum

Names & Lobes	Paleocerebellum	Archicerebellum	Neocerebellum
	Spinocerebellum	Vestibulo	
	Anterior	Flocculonodular	Posterior/ Middle
Function	Posture	Balance	Coordination

1. Thalamus (Sensory function)
 - a. All sensations pass through the thalamus except smell (uncus)
2. Reticular formation (Reticular Activating System)
 - a. RAS-Sleep wake cycle
 - b. Wakefulness
 - c. Arousal

- ii. Peripheral (balance outside the brain)
 - 1. SCC: Gaze stability with head movement
 - a. Sensory receptor: Cupula
 - b. Roll eyes:
 - i. Anterior SCC/ Superior: Head moves down
 - ii. Posterior SCC/ Inferior: Head moves up
 - iii. Horizontal SCC/ Lateral: Head side to side
 - 2. Otolith: postural stability
 - a. Sensory receptor: Macula
 - b. Utricle: Horizontal
 - c. Saccule: Vertical
 - d. Otoconia: Crystals, dictates posture of patient
- b. Vision
 - i. Vestibulo Ocular Reflex (VOR): Eye stability with head movement
 - ii. Vestibulo Spinal Reflex (VOS): Body stability with dynamic movement
 - iii. Smooth pursuit, Eye velocity: Head velocity with slow movement
 - iv. Vestibular Gain, Mechanism:
 - 1. $<60^\circ/\text{sec}$
 - 2. $60\text{--}20,000^\circ/\text{sec}$ – VOR
 - 3. $>20,000^\circ/\text{sec}$ – Nystagmus: Eye manifestation with rapid head movement
- c. Somatosensory
 - i. Proprioceptive joint receptors
 - 1. Ankle joint
 - 2. AO joint (Type 1 lever) upper cervical facets
 - 3. Muscle spindles: posture, sensitive to stretch
- iii. Vestibular physiology
 - a. Tonic firing rate: 80pps – continuously fire
 - b. Push pull system – Firing of SCC (opposite)

Example:



- c. Inhibitory cut-off: Lowest possible tonic firing is 0
- d. Velocity Storage system: >10 Sec after stimulus

iv. Assessment

a. Symptoms:

- i. Dizziness: whirling sensation
- ii. Vertigo: Spining sensation
- iii. Lightheadedness:
 1. Vague sensation, non-localized, systemic E.g. Hypoglycemia, drunk
- iv. Oscillopsia: Blurred vision with bouncing images
- v. Nystagmus: any direction oscillation of eyes (up, down, left, or right)
 1. 4 types
 - a. Gaze evoked: Eye position changes
 - b. Positional nystagmus: Changes in body position E.g. BPPV
 - c. Pendular: Equal oscillation of eyes
 - i. Congenital
 - ii. CNS lesion (inside brain)
 - d. Spontaneous: Vestibular pathology E.g. Tumors

b. Tempo: Frequency of attack

c. Circumstances: Aggravating & relieving factors

d. Physical exam: Special tests

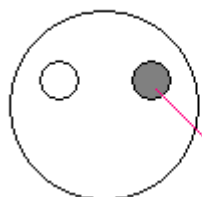
- i. Head thrust test: VOR
- ii. Head shaking induced test:
 1. Horizontal nystagmus: Left to right; unilateral lesion
 2. Vertical nystagmus: up, down; CNS lesion
- iii. Positional testing: Dix Hallpike maneuver (Gold standard, BPPV)
 1. Benign paroxysmal positional vertigo
 2. Sit to supine; (+) nystagmus & vertigo
- iv. Gait & balance testing
 1. Ambulation (Form of balance training)
 - a. 4 levels:
 - i. Static balance
 - ii. Wt. shifting
 - iii. Postural reaction (with challenge)
 - iv. Ambulation

v. Caloric testing

1. Indication: unilateral lesion
2. C - Cold
3. O - Opposite
4. W - Warm
5. S – Same

Right

Left



*Cold - Lesion on right
Nystagmus on Left

- vi. Rotary test
 - 1. Piano stool test
 - 2. Sudden stop after spinning the stool
 - 3. Positive test: nystagmus
- v. Conditions
 - a. BPPV (Benign Paroxysmal Positional Vertigo)
 - i. MC cause of vertigo
 - ii. Otoconia dislodge → SCC (MC SCC affected: Post SCC)
 - iii. Duration < 60 seconds 1- 2mins
 - iv. Special test: Dix hallpike maneuver
 - b. Perilymphatic Fistula (PLF)
 - i. Abnormal communication between middle & inner ear (Delisa)
 - ii. Damage Oval & round window
 - iii. Abnormal pressure inside ear (Sullivan)
 - iv. Damage to Eustachian tube
 - c. Acoustic Neuroma
 - i. Benign tumot inside ear
 - ii. Affect CNVIII (Vestibulocochlear)
 - d. Endolymphatic Hydrops (Menierre's Disease)
 - i. Oversecretion of endolymph
 - ii. Sensation of fullness
 - iii. E.g. Tinnitus
 - e. Herpes Zoster Oticus (Ramsay hunt syndrome)
 - i. Facial weakness
 - ii. Burning sensation inside the ear
 - iii. Caused by Zoster virus
 - f. Ototoxicity
 - i. Poisoning of the ear (systemic)
 - ii. MC: Bilateral Vestibular lesion
 - iii. Cause: Overmedication, kidney disorder
 - g. CNS lesion
 - i. Cause: TBI, MS, VertebroBasilar Insufficiency (VBI),CerebroVascular Insult (CVI)
 - ii. Type of nystagmus: Pendular & vertical

iii. Ocular tilt reaction: (3 stages)

1. Head tilting
2. Ocular torsion



3. *Skew Deviation*: upper eye rotates superior; lower eye rotates inferior



vi. Treatment

a. BPPV

- i. Canalith repositioning treatment/ Epley's reposition treatment
 1. Most extensive
 2. Canalithiasis → Crystals goes to canal
- ii. Liberatory Semont Maneuver
 1. Cupulolithiasis → Crystals goes to cupula
- iii. Brandt Daroff Ex
 1. Midlest form of exercise
 2. Recommended for HEP

b. Unilateral Vestibular lesion (one sided), Sullivan

- i. Gaze stability:
 1. Goal: Promote VOR
 - a. X1: Focus on target, only head moves
 - b. X2: Both head and target moves
 - c. Progression: ↑Distance, background color
 - d. Duration: 8weeks/ 2 months
- ii. Postural stability
 1. Goal: Promote balance
 2. Vestibular/ Swiss ball
 3. Ambulation: Catch & throw
- iii. Motion stability
 1. Goal: habituation training
 2. Repetitive movement

- c. Bilateral vestibular lesion (two sided)
 - i. Same treatment
 - ii. Duration: 2 years
 - iii. Priority/ Goals: ADL's
 - iv. Alternative treatment: Swimming (Buoyancy)
- d. CNS Lesion
 - i. Same treatment
 - ii. Duration: > 6 months
 - iii. Contraindication: TBI RLA IV, Confused Agitated
- vii. Contraindications for vestibular rehabilitation
 - a. Unstable conditions (PLF, Menierre's disease)
 - b. Sudden loss of hearing → stop exercise
 - c. Sudden increase pressure (ICP) → stop exercises
 - d. Fluid discharge in the ears & nose (CSF) Glucose/ sugar
 - i. Otorrhea: CSF Ears
 - ii. Rhinorrhea: CSF Nose

PRESSURE SORES

- i. Pressure Sores
 - a. aka Pressure ulcer, bed sore, bed ulcer
 - b. Decubitus Ulcer
 - i. Decumbre: to lie down
 - ii. Ulcer: wound
 - c. Definition: Area of unrelieved pressure
Bony prominence → Ischemia → Cell death → Necrosis
- ii. Predominant Sites
 - a. Ischium:
 - i. MC site, 28%
 - ii. Position: Sitting
 - iii. Patients: Wheelchair borne
Wheelchair borne patients, common site of pressure sore
 - a. Occiput
 - b. Trochanters
 - c. **Elbow** (Due to arm rest)
 - d. Heel Area
 - b. Sacrum
 - i. 2nd MC site, 27%
 - ii. Position: Long sitting
 - c. Trochanter
 - i. 3rd MC site, 19%
 - ii. Position: Sidelying
 - d. Heel area
 - i. Common in Diabetic (Glove & stocking: sensation on distal parts)
 - e. Occiput
 - i. Position: Supine
 - ii. Type of patient: Pedia & Geria Pt's (Bed Bound)
Pressure ulcer on spinous process?
→ Common in Osteoporotic patients (Kyphotic)
→ Common in Thoracic area
- iii. Etiology
 - a. Pathomechanical factors (Extrinsic)
 - i. Prolonged pressure:
 1. Mean capillary pressure → Opens blood vessels
 2. ↓<44mmHg → Vasoconstriction
 - ii. Tissue Tolerance: Most susceptible to pressure ulcer: Muscle
 1. Attached to bone
 2. Increase H₂O Content

**Shape of damage of pressure ulcers: Conical shape*

- iii. Interface pressure
 - 1. Pressure that closes blood vessels
 - 2. $\uparrow >32\text{mmHg} \rightarrow \text{Vasoconstriction}$
 - * $\uparrow \text{force} = \text{decrease surface area}$*
- iv. Shear: without moisture, 1 plane moving E.g. Sacral ulcer
- v. Friction: with moisture, 2 planes moving
 - 1. Moisture, maceration (softening of the skin), tissue breakdown
- vi. mobility:
 - 1. Common in SCI, no motor & sensory function
 - 2. Turning q 2 hours
 - 3. Move q 15mins during sleep (Physiological)
- b. Physiological (Intrinsic)
 - i. Nutrition: anemia, $\downarrow$ blood cells = $\downarrow$ hemoglobin = $\downarrow \text{O}_2$
 - ii. Age: $\downarrow$ Tissue strength
 - iii. Moisture: MC cause/ intrinsic factor (Maceration)
 - iv. $\uparrow$ Temperature: Sweating (Moisture)
 - v. Smoking: $\downarrow$ Immune system
 - vi. Cognition, Education & Psychological: Preventative factors

iv. Grading of pressure ulcers

Grade	Manifestations
I	Non blanchable erythema, (-) epidermis, reversible
II	Partial Tissue destruction, (+) epidermis, (+/-) Dermis, Blisters
III	Full tissue destruction, subcutaneous tissue
IV	Deep tissue distraction, muscles, bones, joints, fascia

v. Scales

- a. Norton: Measure incontinence
 - b. Gosnell: Most extensive scale for pressure ulcer
 - c. Braden: Only scale that uses/ measure Shear & friction
- *MC used: Photography & Diagram*

vi. Complications

- a. Infection – PVS
- *Complication of Pressure ulcer?*
- a. RA
 - b. OA
 - c. Septic Arthritis**
 - d. NOTA

vii. Treatment

a. Basic Principle

- i. Wound healing & avoid infection

b. Solutions

- i. Normal saline solution: Salt & H₂O, cleansing purposes, No germicidal effect
- ii. Povidone Iodine: Bactericidal effect
- iii. Acetic Acid: Pseudomona (Aeruginosa)
- iv. Sodium Hypochlorite: (+) necrosis

c. Dressings:

- i. Basis: Amount of exudates (liquid content)
- ii. Hydrocolloid wafer dressing: Minimal exudates
- iii. Gel dressing: Minimal to moderate exudates
- iv. Foam Polynem: Severe exudates
- v. Calcium Alginate Dressing: Severe exudates & bacteria
- vi. Transparent adhesive dressing: Gas Exchange

d. Debridement

- i. Enzymatic: Proteolytic enzymes
- ii. Mechanical
 - 1. Wet to dry dressing
 - 2. Whirlpool: important property for debridement; agitation
 - 3. Sharp debridement: Most effective (knife, scalpel, forceps)

e. Surgical considerations:

- i. Indication: pressure ulcers Gr. III & IV

f. Modalities

- i. Whirlpool (Agitation)
- ii. Ultrasound → periphery of wound
 - 1. Continuous: stretching purposes
 - 2. Pulsed: acute wound, promoted blood flow
- iii. UVR:
 - 1. (+) Heat (increase blood flow) with bactericidal effect
 - 2. Desquamation
 - 3. (+) Vitamin D
- iv. ES: Promote blood flow (non continuous)

HGD 2

I. History & Examination

a. **Prenatal History** (Before birth)

- i. Hereditary: Family History (Parents)
- ii. Radiatio: X – Ray, 1st trimester
- iii. Maternal Infection: STORCH (MC Infection during prenatal)
 1. S – Syphilis
 2. T O – TOxoplasmosis
 3. R – Rubella
 4. C – Cytomegalo Virus
 5. H – Herpes Simplex
- iv. Prematurity: (n) Gestational age: 38 – 42 weeks
 1. Premature: 37 weeks
 2. CP: Spastic Diplegia
- v. Hemorrhage: Bleeding
 1. MC type: Periventricular/ intraventricular hemorrhage
 2. Least Common type: Subdural hemorrhage
 3. CP: Spastic Diplegia
- vi. Anoxia: No O₂ inside the womb
 1. Resulting to ischemic brain injury
- vii. Kernicterus: Hyperbilirubinemia on the *basal ganglia*
 1. Triad of Kernicterus
 - a. H – Hearing Loss
 - b. A – Athetosis (slow, writhing, and worm-like movements)
 - c. P – Parinaud's syndrome (Loss of upward gaze)
 - d. choice

b. **Perinatal History** (During birth) – prolonged labor → complication

- i. Primigravida: <18hours Gravida: # of pregnancies
- ii. Multigravida <12 hours Parity: # of fetus
- iii. APGAR scoring: Taken after 1 minute & after 5 minutes

	0	1	2
Appearance	Blue	Body: pink Limbs: Blue	Pink
Pulse	Absent	<100bpm	>100bpm
Grimace(reflex)	No reaction	Grimace (Slight twitching of face)	Cries
Activity	No movement	Slight flexion	Spontaneous
Respiration	No chest mobility	Slight chest movement	Cries (best indication of Child respiration)

Max score: 10

(n) heart beat of baby: 120-140bpm

iv. Description of score 1 pulse category APGAR scoring?

1. Slow Irregular beats (1)

2. Fast regular beats (2)

v. APGAR grading

Score	Indication
7 – 10	Good prognosis
5- 6	Impending respiratory arrest
0 – 4	Needing ventilator support

c. Postnatal History (After birth)

i. Trauma (2- 6y/o) → Sensitive brain development

ii. Poison/ Toxicity

iii. Infection (MC post natal infection: Meningitis, encephalitis)

II. Growth & Development

a. Head circumference:

i. Birth: 35cm

1. After 4 months $35\text{cm} + 6 = 41\text{cm}$

2. After 12 months $+6 = 47\text{cm}$

3. Adult: $+ 10 = 57\text{cm}$

ii. Measured until 3 y/o

iii. Increased head circumference: Macrocephaly

1. Hydrocephalus: Increased H_2O / CSF inside brain

2. Hydromyelia: Increase CSF inside Spinal Cord

iv. Decreased head circumference: Microcephaly

1. Brain growth retardation

2. CP (Spastic Hemiplegia)

b. Weight

i. Birth: 3,400g/ 3.4kg

1. After 5 months: $\times 2 = 6,800\text{g}/ 6.8\text{kg}$

2. After 12 months: $\times 3 = 10,200\text{g}/ 10.2\text{kg}$

ii. Low birth weight: 1,500-2,500g → CP

1. Spastic Diplegia (MC type of CP)

a. Prematurity

b. Hemorrhage

c. Low birth weight

d.

iii. Very low birth weight: $<1,500\text{g}$

c. Height

i. Birth: 50cm

ii. Adult height: $2\times$ (2y/o)

iii. Peak Growth: Menarche (girls); After puberty (boys)

- d. Other musculoskeletal findings
- i. Pes Planus (Flat foot) (n): 0 – 18 months
 - ii. Genu Varum (Bowleg): (n): 0 - 2y/o
 1. Physiological Bowleggedness: Birth – 2 1/2 y/o
 - iii. Genu Valgum (Knock Knees) (n): 2 - 8y/o
Most inappropriate treatment for 1y/o child
 - a. Pull to stand
 - b. Balance ex.
 - c. US on plantar
 - d. NOTA
- III. Development
- a. Holistic; sequence of events through which individuals, grows, changes, evolves and matures.
- i. Developmental scales (Braddom)

Scale	Age	Purpose
Denver	0 – 6y/o	Deviation from normal development
Bayley	0 – 2 1/2y/o (30 months)	Mental & motor scale
Gessel	1month – 6y/o	Current developmental level (Milestones)

ii. Reflexes

1. Infantile/ Spinal
2. Tonic/ Brainstem
3. Cortical/ Midbrain (Highest form; persists)

Infantile/ Primitive/ Spinal: 1st set of reflexes to integrate (18months)

Reflex	Time frame	Stimulus	Reaction
Flexor withdrawal	0 – 2 months	Noxious(pinprick) on sole of foot	Mass flexion of ipsi LE
Crossed extension	0 – 2 months	Noxious(pinprick) on ball of foot (metatarsal heads)	Mass flexion of contra LE, Extension of LE
Traction/ Pull to sit	4 – 6 months	Pull on (B) UE	Mass flexion of (B) UE
Moro	4- 6 months	Sudden neck extension	Abduction & adduction of UE, then Cries
Startle	4 – 6 months	Loud sound	Cries
Palmar grasp	0 – 4months	Maintain pressure on palm	Flexion of fingers
Plantar grasp	28weeks – 9 months	Maintain pressure on ball	Flexion of toes
Galant	0 – 2 months	Stroking of paravertebral area	Lateral flexion towards the stimulus
Rooting	0 – 3 months	Stroking of lateral mouth/ Perioral area	Mouth deviates towards the stimulus

Rooting: nipple searching, important in feeding

Only primitive reflexes that persist: Startle

May affect patients health: Rooting

Last reflex that should be assessed: Moro

Tonic/ Brainstem (PASTA)

Reflex	Time frame	Stimulus	Reaction
Positive supporting 1° standing	0 – 3 months	Weight bearing on ball of foot	Mass extension of limbs (LE, cocontraction)
ATNR	6 – 8 months	Head rotation	Jaw side: extension of UE & LE Skull side: flexion of UE & LE
STNR	6 – 8 months	Head flexion & extension	Head flexed: UE flexes & LE extends Head extended: UE extends & LE flexes
STLR	0 – 6 months	Supine or prone position	Supine: ↑ extension tone Prone: ↑ flexion tone
Associated reaction		Voluntary resistance on limb	Involuntary movement on opposite limb

Reflex that should be integrated to promote:

Walking & stairclimbing: (+) Supporting reaction

Quadruped & creeping: STNR

Hand to mouth act & rolling: ATNR

Flexion of head in supine: TLR

Midbrain/ Cortical

Reflex	Time frame	Stimulus	Reaction
Body on head	4 – 6 months → persists	Supine/ prone	Supine: head flexes Prone: head extends
Body on body	6 – 8 months → persists	1 body segment rotates	Others will follow
Neck on body	0 – 6 months	Head rotation	Body will follow
Optic righting	Persists	With vision, body tilting	Head orients to vertical
Labyrinthine head	Persists	Without vision, body tilting	Head orients to vertical
Protective extension	Persists	Push the Pt	Abduct & extends (opposite)
Equilibrium	Persists	Push the Pt, narrow BOS	Abducts & extends (same)

Protective reaction/ extension/ parachute: Downward

Forward: 5 – 6 months – persists (first, forward/ anterior)

Lateral: 6 – 7 months - persists

Posterior: 9 – 10 months – persists

IV. Cognitive Development

a. Piaget stages

Stages	Age (years)	Description
Sensorimotor	0 – 2	Master reflexes
Preoperational	2 – 7	Language formation; Symbols (letters)
Concrete Operational	7 – 11	Similarities & differences; Basic logical thinking
Formal Operational	>11	Abstract thinking

Stage, capable of insight: 18 months (Sensorimotor stage)

b. IQ scoring

Score	Indication
>130	Very superior/ genius
120 – 129	Superior
110 – 119	Above average
90 – 109	Average
80 – 89	Below Average
70 – 79	Borderline/ Dull
50 – 69	Mild MR/ Moron
35 – 49	Moderate MR/ Imbecile
20 – 34	Severe MR/ Idiot
<20	Profound MR/ Idiot

* Borderline: 75, Average: 100

c. Intellect scales

- i. Stanford Binet: make use of mental age & IQ
- ii. Weschler: Verbal performance
- iii. Peabody Picture Vocabulary test (PPVT): Speech & motor impairment (Test of language)
- iv. Wide range achievement test (WRAT): Mathematics

d. Emotional: Erikson's Stages

V. Motor Development

Prognosis for ambulation (Bleck's) (MASSPEN)

Moro, ATNR, STNR, Stepping, Parachute, Extensor Thrust (Positive Supporting), Neck wrighting

a. Bleck's reflexes

- i. Good: (+) parachute (-) Primitive 1y/o
- ii. Fair: (+) Parachute 1 primitive 1 y/o
- iii. Poor: (+) Parachute 2 Primitive 1y/o

b. Onset of sitting

- i. Good: Can sit by 2y/o
- ii. Fair: Can sit by 2 – 3 y/o
- iii. Poor: Can't sit by 4y/o

c. Grading for static balance

- i. 0: Absent
- ii. 1: Can assume position
- iii. 2: Can assume and maintain
- iv. 3: Can assume, maintain, wt. shifting
- v. 4: Can assume, maintain, wt. shift can be challenged

d. Grading for tolerance

- i. Poor: <15 minutes
- ii. Poor +: 15 – 30 minutes
- iii. Fair: 30 – 45 minutes
- iv. Fair +: 45 – 60 minutes
- v. Good: >60 minutes

e. Gait pattern

- i. Mature (Orthosecrets): 3y/o, Normal (Braddom): 7/yo

Eye pursuit @ midline (90°): 1 month *child can see at extreme (R) & extreme (L) [180°]: 3 months

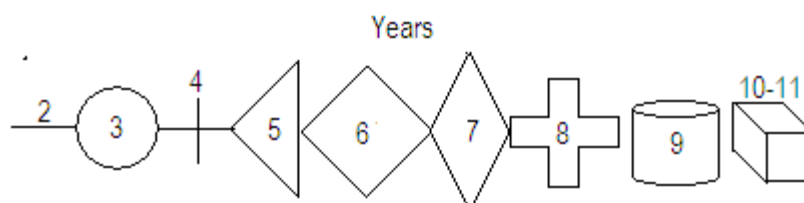
VI. Milestones

Months/ Years	Gross Motor Development	Fine Motor Development	Personal/ Social development
Newborn	Flexor tone Automatic reflex walking	Palmar grasp	Habituation/ state dependent
4 months	Rolling prone to supine	Crude palmar grasp	Recognizes a bottle
6 months	Sitting with support (Pull to sit: 5 months)		
7 months	Rolling supine to prone	Intermediate palmar grasp Transfers cubes from hand to other hand	Holds a bottle
8 months	Sitting without support (Pull to stand: 10months)		
10 months	Stand, 3 C's (Creeps, Crawls, Cruises), Pivot in sitting Lumbar lordosis Creeps: without chest contact on floor Crawl: without chest contact on floor Cruises: Walking with support	Pincer grasp (thumb to index finger grasp)	Chews with rotary motion, Plays peek-a-boo
14 months	Walking with high guard, increase hip/knee flexion Wide BOS, Genu Varus, full contact of sole on floor	Crayon holding at full length, scribbles	Removes garment (M), uses spoon with overpronation
18 months	Walking with low guard, heel strike, can walk backwards	Hold at butt end, dumps raisin in a bottle	Pulls on clothes (B), sit properly, Drink neatly from a cup Plays with doll
2 years	Starts to run, walk up downstairs with step to pattern	Draws vertical line, pencil holding with 2 fingers	Pulls on garment (M), uses spoon well, toilet training start
3 years	Runs well, walk upstairs with alternating feet, unstable tandem walking	<i>Draws circle</i>	Toilet trained
4 years	Downstairs with alternating feet, Hops	<i>Draws cross, uses scissors</i>	Dress/ undress with supervision
5 years	Skips, Stable tandem walking, tiptoe	<i>Draws triangle</i>	Dress/ undress without supervision except shoelaces
6 years	Ride bicycle & rollerskates	<i>Draws</i>	Shoelaces, uses fork & knife Teach, play with table games
7 years	Refinement of skills	<i>Fat diamond, print alphabet</i>	Combing and grooming
8 years		<i>Draws Crucifix</i>	
9 years		<i>Draws Cylinder</i>	
10-11yrs		<i>Draws Cube</i>	

Emerging Hand dominance: 18 months

Usual hand dominance: 2 y/o

Established/ Expected hand dominance: 5y/o



- a. Play:
- i. Midline hand play: 4 months
 - ii. Parallel play: 3y/o, 1 playmate
 - iii. Cooperative play: 4 y/o, 2 or more playmates
 - iv. Creative play: 5y/o, imagination

Years	Catching	Throwing
3 y/o	Catch with arms	Overhead throwing
4 y/o		Underhand throwing
5 y/o		Diagonal throwing
6 y/o	Mature catch	Mature throw

Months/ Years	Speech Development
Newborn	Crying
4 months	Squeals, laughs(3 months), coos & chuckles(2-3 months)
7 months	Utter 1 syllables, "Ma", "Da", "Pa"
10 months	Utter 2 syllables, "Mama", "Papa", "Dada"
14 months	One word
18 months	Points to named body parts, identify picture
2 years	Two-word phrase
3 years	Three-word sentences, *Stutters
4 years	Repeat 4 digit numbers, *past tense
5 years	Number concept until 10, *fluent speech
6 years	Mastery of grammar

- b. 7 months
- i. Looks for dropped objects
 - ii. Talks to his mirror image
 - iii. Cervical lordosis
- c. Speech development
- i. Good: 1 word by 2y/o
 - ii. Fair: Sound by 2y/o
 - iii. Poor: No sound by 3-4y/o